

Computational models of episodic-like memory in food-caching birds

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Birds of the crow family adapt food-caching strategies to anticipated needs at the time of cache recovery and rely on memory of the what, where and when of previous caching events to recover their hidden food. It is unclear if this behavior can be explained by simple associative learning or if it relies on higher cognitive processes like mental time-travel. We present a computational model and propose a neural implementation of food-caching behavior. The model has hunger variables for motivational control, reward-modulated update of retrieval and caching policies and an associative neural network for remembering caching events with a memory consolidation mechanism for flexible decoding of the age of a memory. Our methodology of formalizing experimental protocols is transferable to other domains and facilitates model evaluation and experiment design. Here, we show that memory-augmented, associative reinforcement learning without mental time-travel is sufficient to explain the results of 28 behavioral experiments with food-caching birds.

Food-caching birds of the crow family (*Corvidae*) have been proposed as animal models for cognitive neuroscience, because of their remarkably complex cognition¹. In the wild, nutcrackers and jays cache food items like nuts or insects in thousands of small cracks or in loose soil, for hours, days or months. Recovery of their own caches is highly probable (50–99%), dependent on visual cues, independent of olfactory cues and inconsistent with random search at preferred locations². In laboratory experiments (Fig. 1A), jays rely on episodic-like memories of what they cached where and when (henceforth called ‘memory experiments’^{3–7}) and adapt their caching strategy to anticipated future needs (‘planning experiments’^{8–12}), among other behavior¹.

The interpretation of the planning experiments is controversial. Are they evidence for ‘mental time-travel’^{10–12}, a characteristic feature of human behavior¹³? Or are they explained by a mnemonic-associative recall of previous actions at the time of cache recovery^{8,9,14}? A resolution of this controversy would determine those aspects of cognition for which corvids are representative animal models.

To formalize different interpretations, we turn to reinforcement learning^{15,16}, a computational framework to describe behavioral experiments. Reinforcement learning models are distinguished by how action selection depends on past experiences. First, in a model-free

approach, action selection depends on a policy, whose parameters (e.g. synaptic weights) are directly changed by experience. Memory about past events is stored implicitly in these policy parameters. Second, in a model-based approach, past experiences are used to build a model of the world in the form of causal information like “doing A in state B has consequences C with probability P”. This model can be used for ‘planning at decision time’ or to update a model-free policy¹⁷. Memory about past events is stored in aggregated form in the model of the world. Third, in a buffer-based approach, the sequence of past experiences is explicitly stored in a memory buffer. Using ‘experience-replay’, this buffer can be used to repeat actions with successful outcome or to update a world-model or a model-free policy.

Model-free reinforcement learning is usually associated with learning of habits. Often this simple form of learning is sufficient to explain behavior: tool-use experiments with corvids and apes can be explained by model-free reinforcement learning¹⁸ and the mnemonic-associative account¹⁴ of the planning experiments resembles model-free reinforcement learning¹⁹. On the other hand, there is evidence for simple versions of model-based reinforcement learning in different species^{20,21} and processes like planning or mental time-travel require a model- or buffer-based approach^{16,22}.

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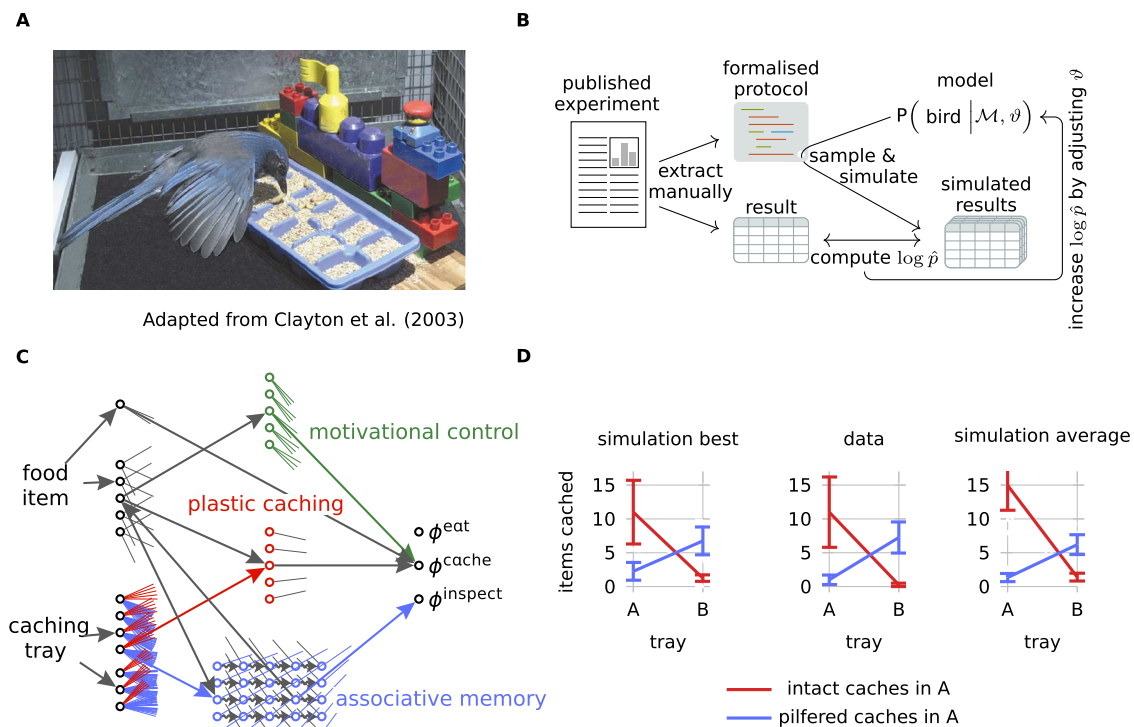


Fig. 1 | Computational modeling approach. **A** In the experiments, jays can eat different items of food, cache them in visuospatially distinct sites (ice cube trays with different arrangements of LEGO Duplo blocks) or inspect their own caches for available food. With different feeding schedules and manipulations of the cached food items the experimenter can change the birds' motivational states and caching experiences. Adapted from Clayton, N., Bussey, T. & Dickinson, A. Can animals recall the past and plan for the future? *Nat Rev Neurosci* 4, 685–691 (2003). <https://doi.org/10.1038/nrn1180>, Springer Nature Limited. **B** For 28 published experiments we formalized the experimental protocol in a domain-specific language and extracted all measured quantities from text and figures. Our models (top right) describe populations of simulated birds as distributions over dynamical systems. Simulated birds are sampled from these populations to participate in simulated runs of the experimental protocols and produce simulated results. We compute the approximate log-likelihood $\log \hat{p}$ by comparing simulated with actual results (Methods). The hyperparameters ϑ that determine the mean behavior and the inter-

individual differences are adjusted to maximize $\log \hat{p}$. **C** In the Plastic Caching Model the preferences of eating ϕ^{eat} , caching ϕ^{cache} , and inspecting ϕ^{inspect} depend on the neural activities and synaptic strengths in a structured neural network featuring motivational control (green), plastic caching weights (red) and an associative memory with systems consolidation (blue). Only some connections are fully drawn; short outgoing lines indicate connections that are suppressed in this figure. The hyperparameters ϑ set initial values of the weights and the speed of learning. **D** Example experiment 'deKort07 exp4a' (see Figs. S9–S168 for other experiments and all 78 comparison figures per model). Birds that experienced pilfered caches in tray A (blue) stopped caching food in their previously preferred tray A, whereas birds that successfully retrieved their caches in A continued to cache preferentially in A (center, error bars = SEM, $n = 4$ birds per group). These experimental results are reproduced in simulations with the Plastic Caching Model (left: best out of 10^5 simulations, error bars = SEM). Also on average (right), the simulated results match qualitatively the experimental data (error bars = average SEM).

Model-free, model-based or buffer-based approaches allow reinforcement learning agents to adapt their policies to the specificities of the environment. In addition to this kind of memory, agents need dedicated memory if they act in a partially observable domain, where perceptions inform only partially about the full state of the environment²³. Because cached food is hidden, food-caching animals need this additional kind of memory to remember their food caches. Finally, the birds' behavior is affected by internal motivational states, like the level of hunger for each type of food^{11,12,24}. Hence, in a reinforcement learning model of food-caching animals, policy and reward are functions of external stimuli (e.g. visual cues), internal motivational states (e.g. hunger) and memory of food caches.

Here, we ask whether the birds' food-caching behavior can or cannot be explained by established concepts like reward-modulated synaptic plasticity and associative memory networks²⁵. To answer this question, we translated different explanations of food-caching behavior into computational models of neural circuits of memory and decision making, formalized experimental protocols in a domain-specific language (Figs. 1B and Fig. S5) and compared simulated to measured results extracted from 28 published experiments (1568 data points).

Results

Each computational model specifies a population of simulated birds, with mean behavior and inter-individual differences characterized by some hyperparameters ϑ (Fig. 1B). To obtain simulated results we sample groups of simulated birds from these populations and let them participate in simulated experiments. For example, in one experiment⁹ the caching behavior of $N_{\text{experiment}} = 8$ birds was studied for two experimental conditions. We created 10^5 groups, each containing $N = N_{\text{experiment}} = 8$ simulated birds and analyzed their behavior with the same criteria as in the experiment (Figs. 1D and 2A and "Methods"). Like real birds, simulated birds perceive items of different kinds of food and potential cache sites and they perform different actions, like eating food items, caching food items or inspecting a cache site for available food. For each model and experiment, the hyperparameters ϑ are adjusted by likelihood-free inference (Fig. 1B, "Methods"). We also fitted each model to all experiments jointly, such that the model-specific hyperparameters ϑ are the same for all experiments, but the simple models we considered fail to capture the strong inter-experimental variability, which is potentially due to seasonal or other unobserved effects (Appendix). To investigate reproducibility, we also ran simulations with 10 times more subjects than in the real experiments (Fig. 2B). We do not discretize time but respect in all

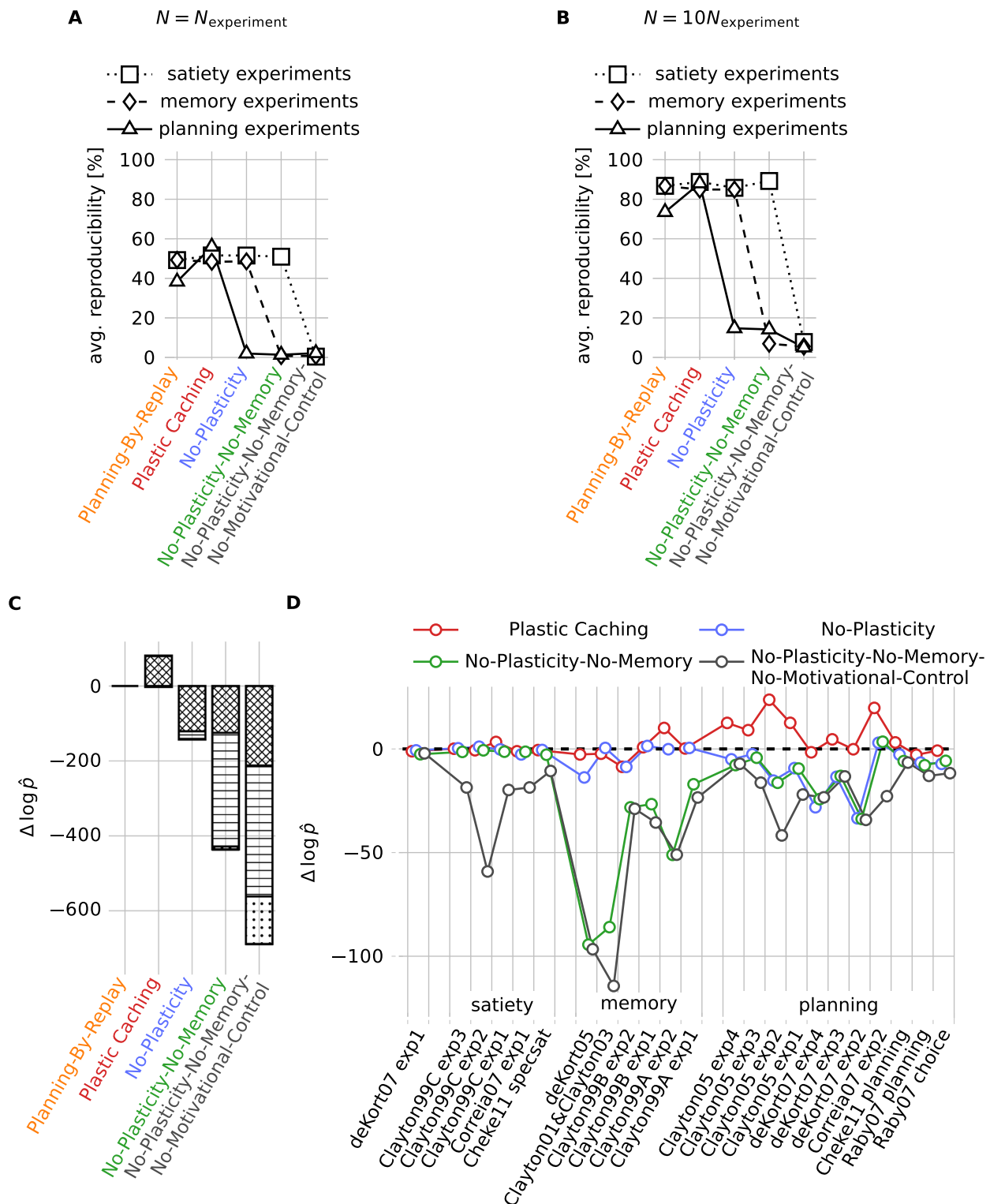


Fig. 2 | Fitting results and model comparison. **A** On average, 48–55% of experiments simulated with the Plastic Caching Model reproduce the significance level of the most relevant statistical tests, if the same number of subjects is used in the simulations as in the experiments (Methods). Without plasticity (No-Plasticity Model) performance decreases only for the planning experiments, lesioning additionally the associative memory (No-Plasticity-No-Memory Model) and motivational control (No-Plasticity-No-Memory-No-Motivational-Control Model) additionally decreases performance in memory and satiety experiments. The control model Planning-By-Replay is not better than the Plastic Caching Model. **B** If 10 times more subjects are used in the simulations than in the experiments, 85–90% of

most significant statistical tests can be reproduced. Assuming a 10% false discovery rate in the experiments, this means that the Plastic Caching Model can reproduce almost all experimentally observed effects. **C** Models with lesions perform worse than the Plastic Caching Model and the Planning-By-Replay Model as shown by the approximate log-likelihood $\Delta \log \hat{p}$ summed across all experiments and measured relative to the Planning-By-Replay Model (hatched: planning experiments, lines: memory experiments, dotted: satiety experiments). **D** Performance $\Delta \log \hat{p}$ relative to Planning-By-Replay Model for all 28 individual experiments. The 6 experiments of Clayton01& Clayton03 build upon each other and are therefore shown as a single point. DeKort07 exp1 is a control experiment.

experimental protocols the actual durations. We do not explicitly simulate movements of birds within their cage or the actual visual input; instead we assume that sensory processing in a bird's brain enables its memory and decision system to access a neuronal representation of all relevant objects.

Our main model, the Plastic Caching Model, contains modules of motivational control (green in Fig. 1C), associative memory (blue in Fig. 1C) and a caching module that relies on known synaptic plasticity rules (red in Fig. 1C, “Methods”). To study the relative importance of these modules, one or several modules can be lesioned. We also compare the Plastic Caching Model to a Planning-By-Replay Model (“Methods”), where the caching module is replaced by an explicit planning module.

Actions depend on the available objects and the birds' motivational state

We model action selection as a random process with probabilities that depend on experiences of the simulated birds. We assume an attention mechanism that allows a bird to attend to a randomly selected object in the cage. The bird performs an action on that object if the preference is higher than a random threshold; otherwise it attends to another object until some action is selected (“Methods”).

The motivational control module makes the eating and caching probabilities depend on hunger variables that are influenced by the recent eating history. We model hunger with several variables h_f to capture specific satiety. Eating one type of food with a high fat concentration may decrease the first hunger variable more than the second one, whereas another type of food with a high protein concentration decreases the second variable more than the first one. If we focus only on the five experiments that measure effects of specific satiety, the motivational control module is necessary and sufficient to assure good performance; lesioning the plastic caching module (No-Plasticity Model, i.e. fixed caching probabilities) or the associative memory module (No-Plasticity-No-Memory Model, i.e. fixed caching and inspection probabilities) does not impair performance, whereas lesioning also the motivational control leads to a strong drop in performance (No-Plasticity-No-Memory-No-Motivational-Control Model, i.e. fixed caching, inspection and eating probabilities. See Fig. 2).

Whereas the motivation for eating depends undeniably on satiety, the caching behavior may be independent of hunger, as jays are known to cache predominantly in seasons when food is abundant². Therefore, we compared the motivational control module to alternative motivational control models where caching is independent of the recent caching and eating history (unmodulated caching) or where caching depends only on the recent caching history (caching-modulated caching; “Methods”). We found that models without any motivational control reproduce the five specific satiety experiments clearly worse than models with hunger-modulated caching, whereas models with caching-modulated caching perform better than hunger-modulated caching models on one experiment (Clayton99C exp1) and worse on two experiments (Clayton99C exp3 and Cheke11 specsat; see Fig. S8).

Remembering the what-where-when of caching events with an associative memory

If we extend the tests to also include the 11 memory experiments, we cannot lesion the associative memory module without a large loss in performance (Fig. 2). The module of associative memory stores caching events in a hetero-associative neural network²⁶ with memory consolidation at the systems level²⁷. In the Plastic Caching Model (Methods), the features of the cache site (‘where’) get associated with the cached food (‘what’) by growing synaptic weights $w_{fx}^{(1)}$ (blue in Fig. 1C) between cache-site-feature neurons ϕ_x and first-layer food-type memory neurons $\phi_f^{(1)}$. Synaptic growth is governed by a neo-Hebbian three-factor learning rule²⁸. In these rules the co-activation of

a pre- and postsynaptic neuron tags a synapse as eligible for change, but strength and sign of the actual synaptic change depends on a modulatory signal which arrives at the synapse up to a few seconds later. The simultaneous focus of a simulated bird on a caching tray A and a food item f activates pre- and postsynaptic neurons of memory synapses, leading to Hebbian eligibility traces. These traces are turned into actual synaptic growth, if a subsequent modulatory signal communicates that the simulated bird has indeed cached food item f in caching tray A . Some consolidation mechanism (e.g. ref. 27) moves memories over time from synapses targeting one layer of memory neurons $\phi_f^{(1)}$ to synapses targeting the next layer of memory neurons $\phi_f^{(1+1)}$, which allows a flexible readout of how long ago a caching event happened (‘when’). At retrieval time, the perception of a cache site triggers recall of the associated food type by activating memory neurons in the layer that corresponds to the time passed since caching (Fig. 1C). Synaptic readout weights are adaptive and allow the simulated birds to adjust the preferences for inspecting different cache sites. When the simulated birds experience, for example, degraded worms that were cached for a certain duration T , these readout weights decrease via a neoHebbian reinforcement learning rule²⁸, where the modulatory third factor signals the unpleasant encounter with the degraded worm. This lowers the future preference for inspecting cache sites containing worms of age T (“Methods”).

The second experiment reported in ref. 6 (included in Clayton01&Clayton03 in Fig. 2D) is one of the most demanding memory retrieval experiment with birds. In this experiment the birds cached on days 1, 2 and 3 in caching trays 1, 2 and 3, respectively. They were allowed to cache peanuts in one half of each caching tray and crickets in the other half of each caching tray. On day four, three days after caching in tray 1, the birds recovered their caches from tray 1. On day five they recovered their caches from tray 2. Birds in the group with manipulated trays found fresh peanuts and decayed crickets. These two recovery trials informed them that crickets decayed after 3 days, which was contrary to earlier experiences where they found fresh crickets after 3 days. On day six, they were allowed to inspect tray 3. As one would expect from birds that can generalize their experience from days four and five, they searched less for crickets than for peanuts.

In all models with associative memory (Plastic Caching Model, No-Plasticity Model, Planning-By-Replay Model), the caching events lead to the formation of synaptic contacts between caching-tray neurons ϕ_x and memory neurons $\phi_f^{(1)}$ (blue in Fig. 3A). Because different trays are used, different caching tray neurons ϕ_x are active on each of the first three days (c.f. Fig. 3A, B). However, the same memory neurons $\phi_f^{(1)}$ for $f = \text{Peanut}$ and $f = \text{Cricket}$ get activated on all three days. Over several nights, memory consolidation moves the newly formed synaptic contacts to post-synaptic neurons $\phi_f^{(2)}$, $\phi_f^{(3)}$ and $\phi_f^{(4)}$. On day four, the perception of caching tray 1 activates the corresponding caching-tray neurons, which activates $\phi_f^{(4)}$ through the synaptic contacts formed by memory consolidation. The food type f of the activated memory neuron depends on which half of the caching tray the bird is looking at. The feedback after observing a decayed cricket $f = \text{Cricket}$ lowers the synaptic retrieval weight $v_{\text{Cricket}}^{(4)}$ (Fig. 3C; Eq. (5), “Methods”). This synaptic weight is further lowered on day five, where again decayed crickets are experienced. Consequently, on day six, when $\phi_{\text{Cricket}}^{(4)}$ is again activated while focusing on one half of tray 3, the retrieval weight is so low that there is only a small probability of inspecting the part of the caching tray containing crickets.

Because the weight update involves only the inputs and outputs of associative memory (blue in Fig. 3C), but not the plasticity of the caching weights (red in Fig. 3C), no differences are expected between the Plastic Caching Model, Planning-By-Replay Model and No-Plasticity Model whereas a lower performance is expected and observed for No-Plasticity-No-Memory Model and No-Plasticity-No-Memory-No-Motivational-Control Model (c.f. Fig. 2D, entry Clayton01&Clayton03).

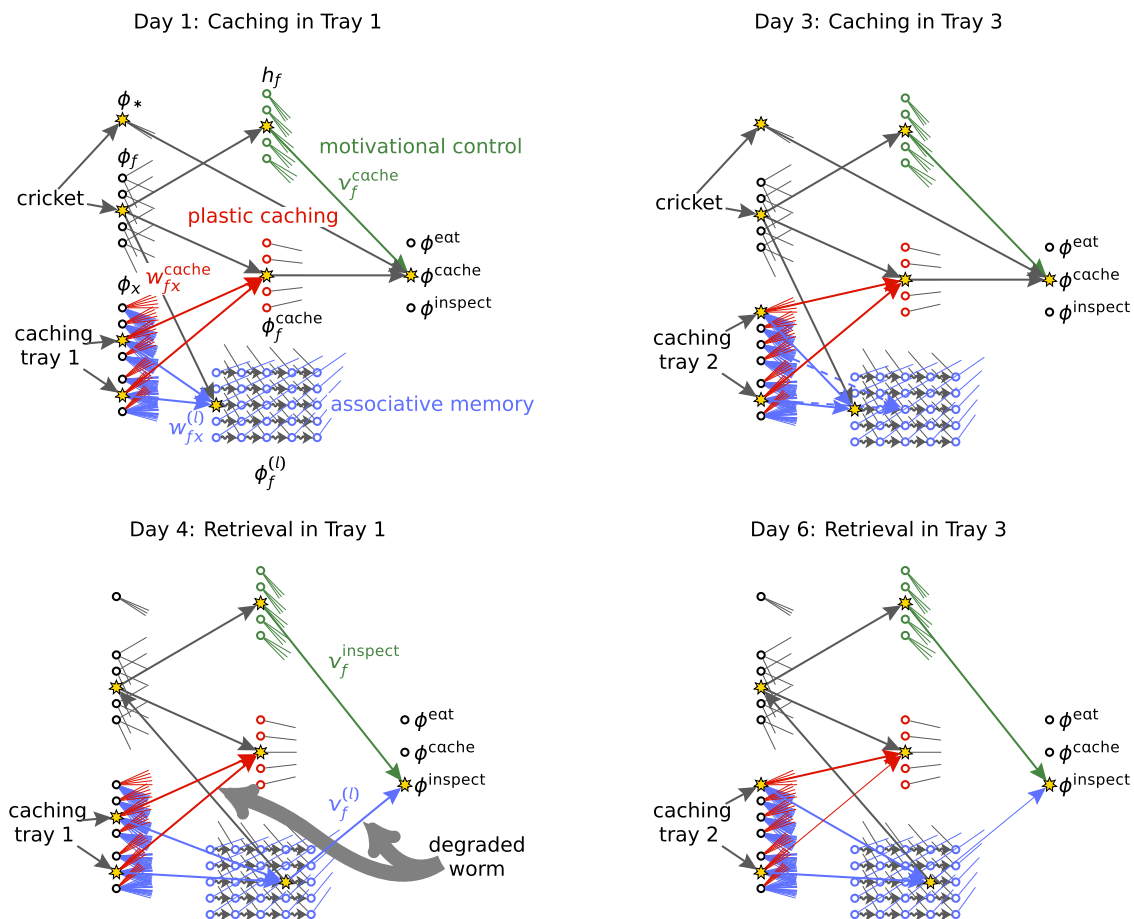


Fig. 3 | Flexible cache memories. In the second experiment of Clayton03, birds cached crickets on **day 1** in caching tray 1. In the Plastic Caching Model, an available food item activates a food-type neuron ϕ_f and the food indicator neuron ϕ_* and a caching tray activates cache-site neurons ϕ_x that code for the tray's appearance and position in the cage. The caching preference ϕ_{cache} depends on the current motivational state h_f weighted by v_f^{cache} and the caching weights w_{fx}^{cache} . Memory weights $w_{fx}^{(1)}$ (blue arrows) are tagged when the bird evaluates the caching preference and grown after successful caching. Caching on day 2 in tray 2 (not shown) and on **day 3** in caching tray 3 (same position, different appearance than tray 1) leads to the formation of new memory weights (blue arrows). Memory consolidation over night (wiggly gray arrows) transferred the memory of the caching event on day 1 to weights that target the third layer of the associative memory (dashed blue arrows).

The caching event on day 2 is stored in memory weights that target the second layer (not shown). Perceiving caching tray 1 on **day 4** activates the memory of the food type cached in this tray. This activates the foodtype neuron ϕ_f , the foodtype-specific caching neuron ϕ_{cache} and the readout of the motivational state. The food indicator neuron ϕ_* is inactive, because the food item is not actually perceived but only remembered. The inspection preference $\phi_{inspect}$ depends on the current motivational state h_f weighted by $v_f^{inspect}$ and the memory readout weights $v_f^{(4)}$. The unpleasant feedback of the degraded recovered cricket (thick gray arrows) decreases the caching weights w_{fx}^{cache} and the memory readout weights $v_f^{(4)}$. These weights are further lowered on day 5 (not shown). Consequently, on **day 6** the bird has a lower preference of inspecting the tray where it cached a cricket three days ago.

Model-free reinforcement learning is sufficient to explain the adaptation to anticipated future needs

The 11 planning experiments require additionally to motivational control and associative memory a mechanism to adapt the caching strategy. This adaptation is implemented in the Plastic Caching Model with synaptic strengths w_{fx}^{cache} (red in Fig. 4A) that undergo plasticity following two simple principles. First, the weights associated with available caching sites slowly increase for all food types as a simulated bird feels hungry for, which explains findings in experiments without retrieval trials during training¹⁰. Second, the retrieval attempts of caches cause synapses w_{fx}^{cache} (Fig. 4B) to decrease for birds that found degraded food or were unsuccessful in finding cached food and to increase for successful birds via a neoHebbian plasticity rule²⁸. Because associative recall of memories reactivates the relevant pre- and post-synaptic neurons, such a rule enables delayed reinforcement learning of those weights that determined caching decisions potentially long ago (Fig. 4A). We compare the Plastic Caching Model to a formalization of the mental-time-travel hypothesis: the Planning-By-Replay Model. This model adapts the caching strategy with an explicit planning

module that records the sequence of available caching trays, successful and unsuccessful retrieval attempts and the hunger levels that were perceived when these trays were available (Fig. 4C). Before caching a food item in a certain tray, those positions in memory are searched that match best the current context. Starting at these positions, the sequence of events is replayed for a few steps and the preference of caching is influenced by the outcome of the replayed retrieval attempts ("Methods"). Despite similarities with hippocampal replay^{17,29}, which would be consistent with the replay-and-plan module and models of mental-time-travel in general, we have not yet found a simple implementation of the replay-and-plan module (orange box in Fig. 4C) in terms of neural network dynamics and plasticity rules. In fact, a precise hypothesis of neural implementations of mental-time travel requires much more than hippocampal replay, as it would have to specify which aspects of the detailed multi-sensory processing stream are stored in the hippocampal replay memory, how the memory system can efficiently be queried, and how the outcome of multiple replayed episodes are combined to reach a decision for the next action ("Methods").

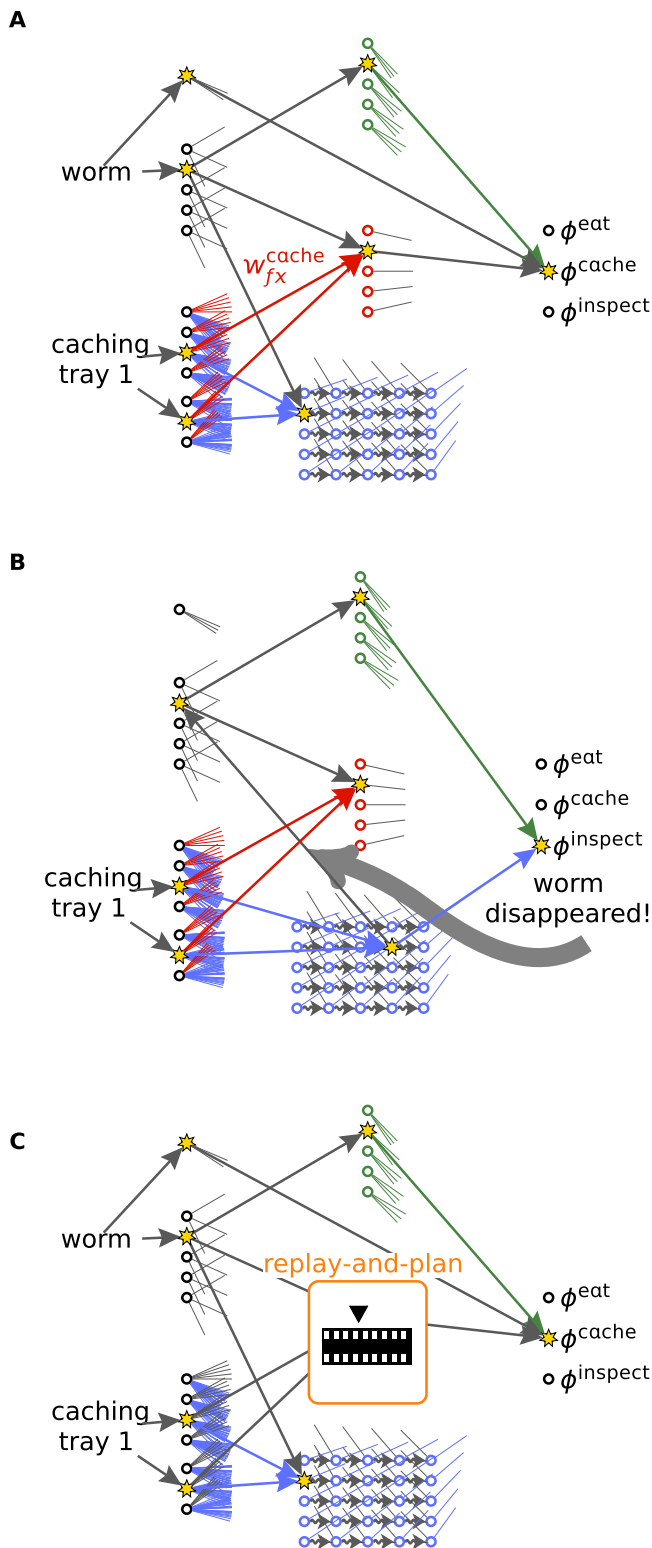


Fig. 4 | Adaptation to anticipated future needs. A The preference of caching worms in a given tray depends on the red caching weights w_{fx}^{cache} . **B** One day later, when the bird inspects the caching tray, the memory of the cached food is reactivated and, in particular, the pre- and postsynaptic neurons of the caching weights. After discovering that the cached food item has disappeared, a feedback signal acts as a third factor in a neoHebbian plasticity rule, thereby decreasing the caching weight. **C** The Planning-By-Replay Model memorizes the sequence of events (indicated with film strip). The caching preference is determined by searching for positions in memory that match best the current context (indicated with black triangle) and evaluating the preference of a caching action under the assumption that the sequence of future events resembles the sequence of events following the matched positions. All other components of the Planning-By-Replay Model are the same as in the Plastic Caching Model.

tray B, whereas birds in the control group cached preferentially in tray A.

In the Plastic Caching Model, the unsuccessful retrieval attempts of the caches in tray A cause a decrease of the caching weight (Fig. 4B) of tray A for the birds in the pilfered group, whereas for the control group the successful retrievals from tray A lead to a strengthening of this caching weight (Eqs. (11)–(14), “Methods”).

In the Planning-By-Replay Model, the retrieval trials are stored in the replay memory together with the information of whether the trial was successful or not. When the birds decide where to cache, previous caching events with context similar to the current one are accessed in the replay memory. From these accessed caching events, the replay memory is played forward until retrieval events from the same caching tray are encountered in memory. For birds that encounter unsuccessful retrieval events in memory the probability of caching is lower than for birds with successful retrieval events.

Because the above explanations of the experiments require plasticity of the caching weights or explicit planning, a model with lesions performs worse than the Plastic Caching Model or the Planning-By-Replay Model (cf. Fig. 2D, entry deKort07 exp4).

Discussion

The Plastic Caching Model is sufficient to reproduce the experimental findings in 28 behavioral experiments with food-caching corvids. It is a memory-augmented reinforcement learning model in continuous time, where action selection and reward signals in simulated birds depend on external stimuli, motivational states and associative memories of past caching events. Plastic retrieval and caching weights enable adaptation to anticipated future needs.

The individual modules of the Plastic Caching Model are deliberately kept simple; the available data does not sufficiently constrain more sophisticated models. For example, little is known about how hunger sensitive neurons modulate behavior³⁰. Interestingly, however, the mere visual perception of a food item transiently reduces the activity of hunger sensitive neurons in the hypothalamus of mice³¹, which could potentially influence the action preferences similarly to how the motivational state modulates the preferences in our models. Further experiments on how motivational states influence behavior are needed before more refined models can be formulated.

The associative memory module with systems memory consolidation represents the time since a caching event with ‘chronological organization’³²: the identity of the memory layer that contains an active neuron at the moment of retrieval indicates the age of a memory. The specific implementation with moving synaptic connections illustrates the essential idea of how the age of memories can be retrieved in an associative memory with systems memory consolidation, but other implementations are possible (ref. 33, Methods). Alternative associative learning models like ‘associative chaining’, ‘time tagging’ or ‘strength coding’ would require a sophisticated neural mechanism to determine the age of memories³³ and to account for the

In the fourth experiment in ref. 9, birds were allowed to cache worms in tray A, whereas they could not cache in Perspex-covered tray B. One day later the Perspex covers were removed and the birds were allowed to inspect all trays. For birds in the pilfered group (blue in Fig. 1D) the experimenter removed the caches from tray A and placed them in tray B, whereas the caches of the control group were not manipulated. After two such training trials, when birds had on a later day the choice to cache either in tray A or B (not covered), birds in the pilfered group preferentially cached in

generalization observed in Clayton03 exp2⁶. Although there is evidence for sharp-wave ripples during sleep in the hippocampus of the food-caching bird species tufted titmice³⁴ and sharp-wave ripples are believed to be important for systems memory consolidation³⁵, future experiments are needed to determine if the time information is memorized with systems memory consolidation and an associative learning mechanism and, if so, which one of several possible associative learning mechanisms³³ is actually implemented in the brains of food-caching birds.

The multi-step consolidation mechanism in the associative memory module leads to sparse and non-overlapping activity patterns of recalled memories, which enables flexible learning of food-specific degradation^{4–6} and ripening intervals⁷. This model is also consistent with experiments where magpies were trained to retrieve objects of one color for one retention interval and objects of another color for another retention interval³⁶. The current model is flexible enough to learn, for example, that food of a certain type is palatable after retention intervals of 2 and 4 days and degraded after 1 and 3 days. A modified multi-step memory consolidation process may lead to more distributed and less flexible neuronal representations of retention intervals. To discriminate between different consolidation processes and neural representations of the age of a memory, future experiments should probe the limits of learnability and generalization to untrained retention intervals.

The associative memory module in the Plastic Caching Model encodes what kind of food was cached where and how long ago. It can thus be seen as a model of episodic-like memory³⁷, also called what-where-when memory³⁸. It is, however, not a model of an episodic memory systems that enables autonoetic consciousness or mental time travel³⁹. In the simulated birds, the associative memory module serves the sole purpose of coping with partial observability; it does not participate in constructive planning processes^{17,22}.

What looks like planning in the experiments is achieved in the Plastic Caching Model with model-free reinforcement learning, implemented with synaptic changes at the moment of retrieval. These synaptic changes follow neoHebbian three-factor learning rules which are supported by experiments in several brain areas^{28,40,41}. In the birds' brains the plastic caching module (red in Fig. 1C) could be implemented by the same anatomical structure as the associative memory module (blue in Fig. 1C), because both modules receive inputs from the same neurons, they influence the action preferences and their synapses undergo neoHebbian plasticity. There is, however, a key difference between these two modules: the associative memory module keeps track of time whereas the plastic caching module does not. Our implementation of a what-where-when memory in the associative memory module allows simulated birds to learn, for example, that unripe berries cached at a warm place are palatable after a few days, or that little pieces of meat are better preserved at cold and dry places than at warm and humid places, if the warmth and humidity of a cache site are part of the perceived cache-site features. For the plastic caching module, time is irrelevant, because the birds only need to learn, for example, that little pieces of meat should preferably be cached at cold and dry places or pilfered sites should be avoided.

Computational modeling is increasingly recognized as a complementary tool to statistical hypothesis testing in experimental neuroscience and psychology⁴². Although we used the reported p-values in our likelihood-free model fitting procedure ("Methods"), we treated them as summaries of the raw data without special meaning, similarly to other summaries of the data like means and standard deviations. Fitting computational models would be simplified with more detailed raw data than usually collected in these behavioral experiments: knowing for each bird the point in time when each action was taken would enable likelihood-based model fitting ("Methods").

Our approach of formalizing experimental protocols in a domain-specific and model-independent language is an example of how

Table 1 | Properties of the three types of objects Food, Tray and InspectionObserver

Food	Tray	InspectionObserver
freshness $\in [0, 1]$	appearance $\in \mathbb{N}$	tray_appearances $\in$ lists($\mathbb{N}$)
eatable $\in \{\text{true}, \text{false}\}$	position $\in \mathbb{N}$	
cacheable $\in \{\text{true}, \text{false}\}$	open $\in \{\text{true}, \text{false}\}$	
amount $\in \mathbb{N}$	eatable_items $\in$ lists(Food)	
type $\in \{\text{mealworm}, \text{waxworm}, \text{peanut}, \text{suet_pellet}, \text{pinenut}, \text{kibble}, \text{cricket}, \text{pine-apple}, \text{salami}, \text{stone}, \text{maintenance_diet}\}$		

The set lists(X) denotes all possible lists of arbitrary length with elements in X. In the actual experiments, caching trays are usually made visually distinct by attaching them to a unique structure of Lego Duplo. We model this feature with different appearance values. Sometimes the caching trays were made inaccessible by covering them with Perplex. In this case we set open = false.

experimentalists and modelers could communicate to foster computational modeling. Experimentalists who publish raw data together with experimental protocols in a formal domain-specific language enable modelers to test existing models on new evidence. Existing models help to design new experiments and estimate the number of subjects needed to reach a desired statistical power, because the models allow to scrutinize novel experimental protocols in simulations prior to actually performing the experiment. As an example we propose a behavioral experiment, where we expect a moderate number of subjects to be sufficient to discriminate between the Plastic Caching Model and the Planning-By-Replay Model (Figs. S6 and S7). With an extension of the domain-specific language to formalize future experiments that measure physiological quantities and behavior simultaneously, the Plastic Caching Model could also be probed at the neuronal level.

Although the Plastic Caching Model has all the features to reproduce the experimentally observed behavior, some simulated repetitions of the experiments fail to reach significance on the key statistical tests with the low number of subjects typically used in the experiments (Fig. 2A, B). This suggests an alternative explanation for the recent failure of reproducing the breakfast planning experiments with Canada jays⁴³: the sample number $N_{\text{experiment}} = 6$ in this experiment may have simply been too small.

With the available 28 experiments, the Plastic Caching Model reaches at least the same performance as the Planning-By-Replay Model, indicating that planning in the form of mental time-travel is not necessary to explain the experiments. Even though the Planning-By-Replay Model is just one model of mental time-travel and other mental time-travel models are conceivable, the important point is that the Plastic Caching Model can reproduce the outcomes of the planning experiments without an explicit mental time-travel strategy. The simulated birds in the Plastic Caching Model are memory-augmented stimulus-response machines that do not perform any off-line planning¹⁷ or imagination of what it would feel like to be in an alternative situation to the currently perceived one. We conclude that memory-augmented model-free reinforcement learning methods implemented with traditional concepts of computational neuroscience such as neoHebbian learning rules^{28,40,41} augmented by multi-step memory consolidation²⁷ are sufficient to explain the existing large body of experiments on food-caching behavior in birds.

Methods

All considered experiments study the caching and cache recovery behavior of jays. A typical experiment repeats the following steps for N individual birds, where N is between 4 and 24. (1) A few hours of food

Table 2 | Actions for formalizing the experimental protocols

Action	Object	Description
add	Food, Tray, InspectionObserver	Adds an object to the simulated bird cage.
remove	Food, Tray, InspectionObserver	Removes an object to the simulated bird cage.
cover	Tray	Covers a tray, prohibiting caching and inspection from this tray.
uncover	Tray	Uncovers a tray.
degrade	Tray	Degrades all food items in a caching tray.
pilfer	Tray	Removes all food items from a caching tray.
count_cached_items	Tray	Count food items in a caching tray.
count_food_items		Count food items in a simulated bird cage.
count_inspections	InspectionObserver	Count the numbers of inspections to the different trays as registered by the inspection observer.
first_inspections	InspectionObserver	Determine which tray was inspected first as registered by the inspection observer.
wait		Wait for a specified number of minutes.

deprivation raise the motivation of the bird. (2) The experimenter adds some food items and visually distinct caching trays at specific positions in the bird's cage and waits for some time, e.g. 15 minutes. (3) The caching trays and remaining food items are removed and counted. (4) While trays are outside the cage, the experimenter may or may not remove (pilfer) or degrade the food items cached in some of the trays. (5) After some waiting interval, the caching trays are returned to the cage and the number of times the bird inspects the returned trays are counted during a recovery interval. (6) After another waiting period, steps 2–5 are repeated with some variations.

Domain specific language for experimental protocols

The domain specific language to describe all experimental protocols consists of three types of objects, Food, Tray and InspectionObserver with different properties (see Table 1) and eleven actions (see Table 2). With this language and some standard control commands of programming languages, like assignments, loops and if-else-clauses, all experimental protocols can be written formally as a function that takes models as input and returns result tables (see Fig. S5).

Model description

Input and output. The simulated birds can perceive food items, caching trays and immediate outcomes of their actions. We do not simulate explicit visual scenes. Instead, we use a one-hot encoding of food items, i.e. the activity ϕ_f of food-type neuron f is equal to one, if a food item of type f is perceived or remembered; otherwise, $\phi_f = 0$ (see Table 1 for possible food types f). To distinguish perception from memory retrieval, there is a food indicator neuron with activity $\phi = 1$, if a food item of any type is actually perceived. For caching trays x we use a two-hot encoding: neurons x_a and x_p have activity levels $\phi_{x_a} = \phi_{x_p} = 1$, if the caching tray with appearance x_a is placed at position x_p . When there is no ambiguity, we will use the symbol x to denote caching trays and indices of caching-tray-feature neurons, e.g. the activity of caching-tray-feature neurons is $\phi_x = \delta_{x,x_p} + \delta_{x,x_a}$ when caching tray x is present.

Simulated birds take actions. The action set

$$\mathcal{A}^{\text{bird}} = \{\text{eat}_f, \text{cache}_{fx}, \text{retrieve}_x, \text{other}\} \quad (1)$$

consists of eating an item of food-type f (eat_f), caching an item of food-type f in caching tray x (cache_{fx}), inspecting and trying to retrieve food from caching tray x (retrieve_x) or doing something unrelated to the experiment (other).

There is no immediate outcome for all actions except retrieve_x . If a simulated bird tries to retrieve a food item from a caching tray, the food indicator neuron is active ($\phi = 1$) only if a food item could be retrieved from the caching tray. Furthermore, a freshness neuron signals the palatability of a retrieved food item, i.e. $\phi^{\text{fresh}} = 1$, if the food

Table 3 | Variables

Symbol	Description	Minimum	Maximum
$s_f(t)$	stomach	0	∞
$h_f(t)$	hunger or hunger	0	1
$M(t)$	associative memory		
$w_{fx}^{\text{cache}}(t)$	caching weights	0	1

$f \in \text{FoodTypes}, x \in \text{CachingTrays}.$

item is fresh and $\phi^{\text{fresh}} = 0$, otherwise. The immediate outcome signals are used to modulate synaptic plasticity (c.f. Associative memory with systems consolidation and Plasticity of the caching policy in the Plastic Caching Model).

Variables and parameters. The state of each simulated bird is characterized by some variables that change over time (Table 3) and some constant parameters (Table 4). Variable are the neural activities $\phi(t)$, the motivational control states $s_f(t)$ (stomach) and $h_f(t)$ (hunger), synaptic weights of an associative memory system $M(t)$, and the caching weights $w_{fx}^{\text{cache}}(t)$, that connect caching-tray-feature neurons x to food-type neurons f (Table 3). Constant parameters (Table 4) affect the dynamics of the variables as described in Motivational control, Associative memory with systems consolidation, Plasticity of the caching policy in the Plastic Caching Model and Decision making. For each individual simulated bird the fixed parameters are sampled from a distribution adjusted to the data (c.f. Model Comparison and Fitting).

Event-based integration. We perform event-based integration of the variables. At each event, the variables of the system can jump to new values, whereas they change smoothly between events. Events $e_k = (t_k, a_k, o_k)$ are given by the event time $t_k \in \mathbb{R}$, the event type $a_k \in \mathcal{A}^{\text{bird}} \cup \mathcal{A}^{\text{experimenter}}$, and the immediate outcome $o_k \in \mathcal{O}$, where the birds' action set is given in Eq. (1) and the experimenters' action set

$$\mathcal{A}^{\text{experimenter}} = \{\text{add}_f, \text{add}_x, \text{remove}_f, \text{remove}_x\},$$

consists of addition and removal of food items and caching trays. Note that the domain specific language to describe the experiments contains more actions, like degrade, pilfer, count_items, that are performed out of sight of the birds and do not affect their state variables directly. There is a small set of immediate outcomes $\mathcal{O} = \{\text{pilfered}, \text{fresh_food_item}, \text{degraded_food_item}\}$ that influence synaptic plasticity through the food indicator neuron and the freshness neuron, i.e. $\phi \cdot (t_k^+) = 1$ if and only if $o_k \neq \text{pilfered}$ and $\phi^{\text{fresh}}(t_k^+) = 1$ if and only if $o_k = \text{fresh_food_item}$, where t_k^+ indicates the time point after t_k at which the outcome is known. We do not simulate any events when

Table 4 | Parameters

Symbol	Description	Minimum	Maximum
τ_s	stomach time constant	0.5 min	10.0 min
τ_d	digestion time constant	0.0 min	20.0 min
τ_h	appetite increase time constant	50.0 min	300.0 min
n_f	nutritional value	0.1	1.0
v_f^{eat}	eating preferences	0.1	1.0
v_f^{cache}	caching preferences	0.0	1.0
η^{eat}	eating bias	-1.0	1.0
η^{inspect}	inspection bias	-2.0	0.5
s^{inspect}	inspection preference slope	0.0	5.0
η^{cache}	caching bias	-1.0	1.0
α^{fresh}	freshness learning rate	0.0	1.0
α^{reward}	retrieval reward learning rate	0.0	0.9
α^{pilfer}	pilfer learning rate	0.0	0.2
α^{degrade}	degradation learning rate	0.0	0.2
τ^{hungry}	hunger-caching time constant	100.0 min	500.0 min
w_0^{cache}	initial caching weights	0.0	1.0
δ^{eat}	eating time-out	10.0 s	200.0 s
δ^{inspect}	inspection time-out	10.0 s	200.0 s
δ^{cache}	caching time-out	10.0 s	200.0 s
δ^{other}	other action time-out	10.0 s	200.0 s
p^{other}	other action preference	0.0	1.0

The ranges [minimum, maximum] are manually set. The nutritional values, eating preferences and caching preferences are estimated for each food type $f \in \{\text{mealworm, waxworm, peanut, suet pellet, pinenut, kibble, cricket, pineapple, salami}\}$; for stones only the caching preference is estimated.

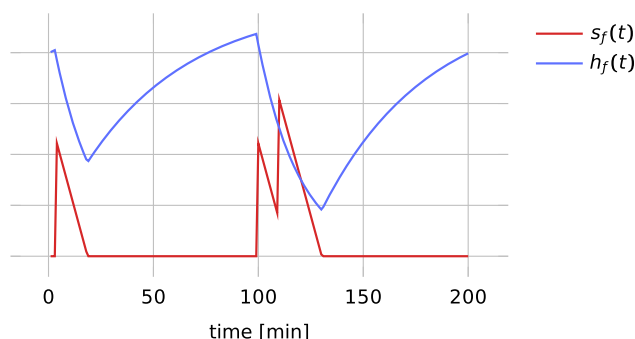


Fig. 5 | Example trace of stomach s_f and hunger h_f variables. At minutes 4, 100 and 110 the simulated bird eats an item of food type f , thereby increasing the stomach variable and shortly thereafter decreasing hunger.

a bird has no food items and caching trays available that are relevant to the experiment.

Motivational control. We model motivational control with two variables per food-type: $s_f(t)$ indicates how much food of type f is in the stomach and $h_f(t)$ indicates how hungry the bird is for food of type f (Fig. 5). Instead of one stomach and hunger variable per food type one could assign one variable per nutrient class, e.g. carbohydrates, fats, fiber, minerals, proteins, vitamins, and water. However, the mapping between food types and nutrient classes is non-trivial and in the experiments the food types and quantities were chosen such that the birds satiated on one food clearly

continued desiring and eating another food. Therefore we work directly with food types.

The stomach variables evolve according to

$$\frac{ds_f}{dt} = -\frac{1}{\tau_s} [s_f > 0] + \sum_k n_f \delta(t - t_k) [a_k = \text{eat}_f], \quad (2)$$

where τ_s is a stomach time constant, the Iverson bracket $[P]$ is 1 if the statement P is true and 0 otherwise, n_f is the nutritional value of food of type f , $\delta(\cdot)$ is the Dirac delta distribution, and t_k, a_k are event times and event types, respectively. The hunger variables are coupled to the stomach variables through

$$\frac{dh_f}{dt} = -\frac{h_f}{\tau_d} [s_f > 0] + \frac{1 - h_f}{\tau_h} [s_f = 0] \quad (3)$$

where the digestion time constant τ_d determines how fast hunger decrease while there is some food in the stomach and τ_h determines how fast it increases otherwise. Whereas the stomach variable increases immediately with every eaten food item and decreases linearly as food is digested, hunger decreases slowly during digestion, because food absorption is not immediate and major hunger satiation signals arise from the gut³⁰. The value ‘zero’ of the stomach variable should not literally be understood as indicating a completely empty stomach; rather it is the emptiness level of the stomach at which a bird’s hunger feeling starts to increase again. If maintenance diet is available to the birds, we do not explicitly simulate eating events, but integrate Eq. (3) under the assumption $s_f > 0$ for all f . The hunger variables $h_f(t)$ critically influence decision making (see Decision making).

In the caching-modulated caching model of motivational control we use additionally the caching motivation variables c_f that evolve according to

$$\tau_d \frac{dc_f}{dt} = 1 - c_f(t) - c_f(t) c_0 \sum_k \delta(t - t_k) [a_k = \text{cache}_f] \quad (4)$$

where $c_0 \in [0, 1]$ is a fitted parameter that controls how strongly the caching motivation variables c_f decreases when caching an item of type f . In other words, whenever a simulated bird caches an item of type f , the caching motivation c_f for food f decreases by the amount $c_f(t) \cdot c_0 / \tau_d$ towards zero and in the absence of caching events it increases exponentially to one with time-constant τ_d . If the fitted parameter c_0 is large, few caching events suffice to bring the caching motivation close to zero.

Associative memory with systems consolidation. Memory about previous caching events is implemented in a simple associative (content-addressable) memory system $M(t)$. The memory system $M(t)$ consists of 7 memory sub-networks and readout weights $v_f^{(l)}$. Each sub-network $l = 1, \dots, 7$ contains neurons $\phi_f^{(l)}$ that are activated by actually perceived or remembered food of type f . Initially, all synaptic weights $w_{fx}^{(l)}$ from caching-tray-feature neurons ϕ_x to food-type neurons $\phi_f^{(l)}$ in layer l are zero. Each caching event increases a synaptic consolidation variable $\tilde{w}_{fx}^{(l)}(t)$ by 1. The effective synaptic strength of connections onto memory neurons $\phi_f^{(l)}$ on layer $l = 1$ is $w_{fx}^{(1)}(t) = H(\tilde{w}_{fx}^{(1)}(t - 1 \text{ hour}))$, with Heaviside function $H(x) = 1$ if $x > 0$ and $H(x) = 0$ otherwise. Implementing a simple systems consolidation process, active weights $w_{fx}^{(l)} = 1$ and variables $\tilde{w}_{fx}^{(l)}$ in networks $l = 1, 2, 3, 4$ are copied to networks $l + 1$ and erased (set to zero) in network l , every day, i.e. $w_{fx}^{(l+1)}(t + 1 \text{ day}) = w_{fx}^{(l)}(t)$ and $w_{fx}^{(l)}(t + 1 \text{ day}) = 0$ unless there was a caching event during t and $t + 1 \text{ day} - 1 \text{ hour}$. Active weights stay in network 5 for 3 days before moving to network 6, where they stay for 8 days before moving to the last layer of the memory network. In the experiments the birds never cached on different days in the same caching tray. Also in the wild, caching new items at a site where some

food items are already cached is unlikely, given that these birds are scatter hoarders. In our model, repeated caching at the same site on multiple days would lead to replacement of the old memory trace, e.g. caching peanuts at x on day 1 and caching again peanuts at x on day 3 would lead to the deletion of the weights targeting layer 3 and the growth of new weights to layer 1. Again, we choose a simple ‘one-hot code’ in time, where each memory about a caching event is encoded in exactly one neuron of the memory network at a time, but more distributed representations with gradual shifts would be possible.

Recall occurs when a caching tray is perceived at time t . Perception of caching tray x activates a cache-position and a cache-appearance neuron $\phi_{x_p}(t) = \phi_{x_a}(t) = 1$, which in turn activates those food neurons $\phi_f^{(l)}$ that have an active weight $w_{fx}^{(l)} = 1$, i.e.

$$\phi_f^{(l)}(t) = \sigma \left(\sum_x w_{fx}^{(l)} \phi_x(t) \right), \quad (5)$$

where $\sigma(x) = \min(1, \max(0, x))$ is the activation function. Each actual inspection of a cache site x reduces the variable $\tilde{w}_{fx}^{(l)}$ by 1, such that after sufficiently many inspection events the weight $w_{fx}^{(l)}$ becomes zero. With this recall mechanism, the identity l of the layer of the memory network that contains an active food neuron codes for the age of the memory: if the only caching event of food of type f at site x occurred three days ago, only $\phi_f^{(3)}$ becomes active. With the systems consolidation process described above, the corresponding age intervals of the memory networks are: layer 1: [0, 1) day, 2: [1, 2) days, 3: [2, 3) days, 4: [3, 4) days, 5: [4, 7) days, 6: [7, 15) days and 7: [15, ∞) days. Each neuron in the memory network signals its activity through synapses of strength $v_f^{(l)}$ to a retrieval preference neuron $\phi_f^{\text{retrieval}}$. These weights $v_f^{(l)}$ encode the expected freshness of food of type f that was cached at a time that lies in the corresponding age interval of network l . These weights are subject to experience-dependent plasticity: if the experienced freshness during a successful retrieval event differs from the expected value, the weight changes according to

$$\Delta v_f^{(l)} = \alpha^{\text{fresh}} (\phi^{\text{fresh}} - v_f^{(l)}), \quad (6)$$

where $\phi^{\text{fresh}} = 1$ indicates that the retrieved food was palatable and $\phi^{\text{fresh}} = 0$ indicates the opposite.

To summarize, the state of the memory network at time t is given by the variables

$$M(t) = \left\{ \phi_f^{(l)}(t), \tilde{w}_{fx}^{(l)}(t), w_{fx}^{(l)}(t), v_f^{(l)}(t) \right\}_{l=1, \dots, 7}. \quad (7)$$

This associative memory system with systems consolidation is just one hypothesis of automatic processes that keep track of when an event was memorized³³. One alternative is to grow connections to all layers of the memory network at the moment of storage, while maintaining a time-dependant activity pattern at retrieval through synaptic connections that disappear at different rates, e.g. the connections to the first layer could disappear after one day, whereas those to other layers disappear later. In this case, a young memory would be characterized by many neurons being active during recall and an old memory by few neurons being active during recall. Another alternative, without multiple layers and systems consolidation, is to store the ‘what-where’ information together with a ‘when’ tag, like a time stamp, or a ‘context’ tag that allows to reconstruct the ‘when’ information. Implementing the ‘when’ information with a time tag or the number of active neurons during recall has computational disadvantages to the sparse code of the memory network $M(t)$, because quickly learning flexible rules based on the what-where-when of recalled events is easiest with linear readout, when the input to the linear readout is sparse, ideally, one-hot coded³³. But further

experiments are needed to discover the actual implementation of the what-where-when memory in food caching birds.

Decision making. At any given moment in time the simulated birds have $1 + n_f + n_f \times n_x + n_x$ actions available (1 other, n_f eat, $n_f \times n_x$ cache, n_x inspect), where n_f and n_x are the currently available numbers of food types and cache sites, respectively. Actions are selected by weighted sampling from the available actions with the weights given by the preferences of the available actions. This is implemented iteratively as follows:

1. Compute the preference p_a of a randomly selected available action a .
2. Sample z from a uniform distribution over the interval [0, 1].
3. Choose action a if $p_a > z$, otherwise repeat steps 1 – 3.

After every action the simulated birds wait for a random duration sampled from a uniform distribution over $[1, \delta^{a_k}]$ seconds before taking the next action.

The preference p_{other} is a bird-specific preference for doing something unrelated to the experiment. The computation of all other preference p_a could be implemented by a visual attention mechanism that focuses randomly on an available food item or caching tray. For example, focusing on a food item of type f would activate the food-indicator neuron and one food-type neuron $\phi = \phi_f = 1$; all other input neurons would be inactive. The output neuron with activity

$$\phi^{\text{eat}} = \sigma \left(\sum_f v_f^{\text{eat}} h_f \phi_f + \eta^{\text{eat}} + \phi - 1 \right) = \sigma \left(v_f^{\text{eat}} h_f(t) + \eta^{\text{eat}} \right) = p_{\text{eat}f} \quad (8)$$

would then compute the preference $p_{\text{eat}f}$ of eating this item (see Fig. 1C). In Eq. (8), $\sigma(x) = \min(1, \max(x, 0))$ is an activation function, h_f are the current values of the hunger variables (c.f. Motivational control) and v_f^{eat} are fitted eating preference parameters. The preferences for caching a food item of type f in caching tray x are computed as

$$\phi^{\text{cache}} = p_{\text{cache}f\text{at}x} = \sigma \left(w_{fx_p}^{\text{cache}} + w_{fx_a}^{\text{cache}} + v_f^{\text{cache}} h_f(t) + \eta^{\text{cache}} \right), \quad (9)$$

where x_p and x_a are the position and appearance of the caching tray, respectively, and $w_{fx_p}^{\text{cache}}$, $w_{fx_a}^{\text{cache}}$ are corresponding weights that undergo plasticity in the Plastic Caching Model (c.f. Plasticity of the caching policy in the Plastic Caching Model). In the unmodulated caching model we drop the term $v_f^{\text{cache}} h_f(t)$ in the equation above and in the caching-modulated caching model we replace it by $v_f^{\text{cache}} c_f(t)$ (see also Fig. S8).

The preferences for inspection of caching tray x are given by

$$\phi^{\text{inspect}} = p_{\text{inspect}x} = \max_{f,l} \sigma \left(\phi_f^{(l)} v_f^{(l)} + v_f^{\text{inspect}} h_f + \eta^{\text{inspect}} \right), \quad (10)$$

where $v_f^{\text{inspect}} = s^{\text{inspect}} v_f^{\text{eat}}$ to reduce the number of free parameters.

Plasticity of the caching policy in the Plastic Caching Model. The caching policy is a simple stimulus-response mechanism that depends on caching-tray and food-type specific synaptic weights w_{fx}^{cache} (red weights in Fig. 1C of the main text). These are the weights from pre-synaptic caching-tray feature neurons ϕ_x to post-synaptic caching preference neurons ϕ_f^{cache} . Their changes are governed by a modulated Hebbian plasticity term (first term in Eq. (11)) and a pre-synaptic term (second term in Eq. (11))

$$\begin{aligned} \frac{dw_{fx}^{\text{cache}}(t)}{dt} = & l(t) \left(m_1 w_{fx}^{\text{cache}} + m_2 (1 - w_{fx}^{\text{cache}}) \right) \phi_x(t) \phi_f^{\text{cache}}(t) \\ & + H_{fx}(t) (1 - w_{fx}^{\text{cache}}(t)) \phi_x(t) \end{aligned} \quad (11)$$

$$m_1 = -\alpha^{\text{pilfer}}(1 - \phi_*(t_k^+)) - \alpha^{\text{degrade}}\phi_*(t_k^+)(1 - \phi^{\text{fresh}}(t_k^+)) \quad (12)$$

$$m_2 = \alpha^{\text{fresh}}p_f^{\text{eat}}\phi_*(t_k^+)\phi^{\text{fresh}}(t_k^+), \quad (13)$$

where $H_{fx}(t) = \frac{1}{\tau_{\text{hunger}}}[h_f(t) > \theta^{\text{hunger}}]$ is active when a bird's hunger for food of type f exceeds some threshold $\theta^{\text{hunger}} = 0.99$, $l(t) = \sum_k \delta(t - t_k)$ [a_k = retrieve] and p_f^{eat} is the current preference for eating food of type f (see Decision making). The effect of the two modulation factors m_1 and m_2 is best understood by a case-by-case analysis. Indeed, the modulated Hebbian part of the plasticity rule in Eq. (13) can be written as a change of the caching weights after attempting to retrieve a food item of type f at cache site x

$$\Delta w_{fx}^{\text{cache}} = \begin{cases} -\alpha^{\text{pilfer}} & \times w_{fx}^{\text{cache}} & \text{if } o_k = \text{pilfered} \\ -\alpha^{\text{degrade}} & \times w_{fx}^{\text{cache}} & \text{if } o_k = \text{degraded_food_item} \\ +\alpha^{\text{fresh}}p_f^{\text{eat}} & \times (1 - w_{fx}^{\text{cache}}) & \text{if } o_k = \text{fresh_food_item} \end{cases} \quad (14)$$

The gated presynaptic term in Eq. (13) leads to behavior consistent with the Compensatory Caching Hypothesis⁴⁴: birds learn to compensate for a lack of food experienced at a certain place by caching more at this place, because the caching preference weights increases at this place.

The caching policy in the Planning-By-Replay Model. The Planning-By-Replay Model maintains a list $\mathcal{T} = ((m_1, t_1), (m_2, t_2), \dots, (m_l, t_l))$ of l memory items m_i with their time-of-last-change t_i . The memory items are collections of tray-hunger-outcome associations

$$m_i = \left\{ (x_i^{(1)}, h_i^{(1)}, o_i^{(1)}), (x_i^{(2)}, h_i^{(2)}, o_i^{(2)}), \dots, (x_i^{(j)}, h_i^{(j)}, o_i^{(j)}) \right\}, \quad (15)$$

where $x_i^{(j)}$ identifies a caching tray, $h_i^{(j)}$ is a vector of hunger levels and $o_i^{(j)}$ is an outcome value in $\mathcal{O}' = \mathcal{O} \cup \{\text{not_inspected}\}$. The outcome not_inspected is used when a simulated bird perceives the presence of a caching tray but never inspects its content. If a simulated bird interacts at time t with a caching tray x and observes outcome o_t the memory gets updated in the following way. If the caching tray x is identical to x_i^j and $t - t_i < 1$ hour, $h_i^{(j)}$ becomes $\frac{1}{2}h_i^{(j)} + \frac{1}{2}h(t)$, where $h(t)$ is the currently perceived vector of hunger levels, $o_i^{(j)}$ becomes o_t and t_i becomes t . If the caching tray x is different from all x_i^j and $t - t_i < 1$, a new tray-hunger-outcome associations $(x, h(t), o_t)$ is appended to m_i . If the last change of the memory dates back more than one hour, i.e. $t - t_i > 1$ hour, a new memory item $m_{i+1} = (x, h(t), o_t)$ is created and appended to $\mathcal{T}$. This grouping of individual interactions simplifies the search for replay positions, which would be difficult in a naive memory list $((x_1, h_1, o_1, t_1), (x_2, h_2, o_2, t_2), \dots)$ that keeps track of all individual interactions with all caching trays.

The set of replay indices $\mathcal{I}^*$ is determined by comparing the tray identifiers in the last three memory items to those at different positions in the memory list $\mathcal{T}$ and picking the closest matches, i.e.

$$\mathcal{I}^* = \arg \max_{i < l} \sum_{k=0}^2 [x_{l-k}^{(1)} = x_{i-k}^{(1)} \wedge x_{l-k}^{(2)} = x_{i-k}^{(2)} \wedge \dots \wedge x_{l-k}^{(j-k)} = x_{i-k}^{(j-k)}], \quad (16)$$

unless the closest match is zero, in which case we empty the set of replay indices $\mathcal{I}^* = \emptyset$. Using the last three memory items to query the memory list allows to search for patterns in memory with similar context. The similarity of the context is further quantified by a similarity weight

$$w_i = \frac{1}{3} \sum_{k=0}^2 \frac{1}{J_{i-k}^{j-k}} \sum_{j=1}^{J_{i-k}^{j-k}} 1 - \left(\|h_{l-k}^{(j)} - h_{i-k}^{(j)}\|_1 + [o_{l-k}^{(j)} = o_{i-k}^{(j)}] \right) / (\dim(h) + 1).$$

To determine the preference of caching a food item of type f in caching tray x , the set of replay indices $\mathcal{I}^*$ is determined and memory items are replayed. During replay from index $i^* \in \mathcal{I}^*$, memory items m_{i^*+k} for $k = 1, \dots, 6$ are searched for tray-hunger-outcome associations that contain the tray identifier x . The result of this search can be written as the set $Q(x, i^*) = \{(i, j) | x_i^{(j)} = x, i^* < i \leq i^* + 6\}$ of index pairs with matching tray-identifier. For each index pair (i, j) the weight $w_{ij}(f)$ for caching a food item of type f is determined by computing

$$w_{ij}(f) = \alpha^{\text{hunger}}h_{ij}^{(j)} + \alpha^{\text{fresh}}[h_{ij}^{(j)} \leq \theta][o_i^{(j)} = \text{fresh_food_item}] - \alpha^{\text{degrade}}[o_i^{(j)} = \text{degraded_food_item}] - \alpha^{\text{pilfer}}[o_i^{(j)} = \text{pilfered}].$$

The distributions over parameters α^{hunger} , α^{fresh} , α^{degrade} and α^{pilfer} are fitted (see Model Comparison and Fitting). Finally, the weight w_{fx} for caching is determined by a weighted sum

$$w_{fx} = \sum_{i^* \in \mathcal{I}^*} w_{i^*} \sum_{(i,j) \in Q(x, i^*)} \gamma^{i-i^*} w_{ij}(f), \quad (17)$$

where $\gamma \in [0, 1]$ is a discount factor. The tangent hyperbolic of this weight $\tanh(w_{fx})$ (instead of $w_{fx}^{\text{cache}} + w_{fx}^{\text{cache}}$) is used to determine the preference of caching in Eq. (9).

The algorithmic description of the Planning-By-Replay Model illustrates three problems that a neural implementation of mental-time travel for planning needs to solve: first, what is stored in memory (Eq. (15)), second, how is the memory queried (Eq. (16)); and third, how are replayed episodes used for decision making (Eq. (17)). If food caching birds use indeed mental-time travel for planning, they probably do not store in memory every single observation and action, as if everything was recorded on a videotape. Instead they may store some compressed representation of past experiences, like the available food or the average hunger level in the afternoon of a given day. Fast hippocampal replay of recent experiences may contribute to storing such compressed representations in long-term memory, but we do not yet have a detailed hypothesis of the underlying neural processes. Likewise, querying the memory system would involve probably non-trivial neural processing. The sight of a peanut should not trigger the memory system to retrieve the myriad of past experiences with peanuts. Instead, the memory system should be queried with information that is relevant in the given context. Finally, if a query leads to recall of multiple past episodes, their content should probably be held in some working memory for further information processing and decision making, which is likely to involve intricate neural processing.

Models with lesions

No-Plasticity-No-Memory-No-Motivational-Control Model. The simplest model is obtained by setting to zero at all times all state variables $h_f(t)$, $w_{fx}^{\text{cache}}(t)$ and $\phi_f^{(j)}(t)$ in the computation of the action probabilities Eq. (8), Eq. (9) and Eq. (10). In this case, the fitted parameters are the preference p^{other} , the biases η^{eat} , η^{cache} , η^{inspect} and the time-out constants δ^{other} , δ^{eat} , δ^{cache} , δ^{inspect} .

No-Plasticity-No-Memory Model. In this simplified model the state variables $w_{fx}^{\text{cache}}(t)$ and $\phi_f^{(j)}(t)$ are kept zero at all times, but the hunger variables evolve according to the dynamics of the full model. The fitted parameters for this model include, in addition to the parameters of the simplest model, all food parameters and the stomach, digestion and hunger-increase time constants.

No-Plasticity Model. In this simplified model only the state variables $w_{fx}^{\text{cache}}(t)$ are kept zero at all times, but the hunger dynamics and the associative memory dynamics with systems consolidation evolve as in

the full model. The additional parameters are the inspection preference slope and the freshness learning rate.

Average reproducibility

For each experiment we identify 1 to 3 relevant statistical tests (called ‘most relevant tests’ in the following) to support the major finding (see Supplementary Results). For each test we compare the simulated p-values p_{sim} to the experimental p-values p_{exp} . If they are both below 0.05 or both above 0.05 we say the simulation reproduces the test. We say a simulated experiment *reproduces* the experimental result, if all ‘most relevant tests’ are simultaneously reproduced. The results in Fig. 2A are obtained by simulating each experiment 1000 times and computing the fraction of simulated experiments that reproduce the experimental results. Given the intrinsic variability of the experimental data (see Supplementary Results) we consider average reproducibilities above 50% as high values.

To check whether the plotted differences in average reproducibility are significant, we note that the plotted numbers can be interpreted as empirical rates of 1000 Bernoulli trials. Therefore, differences of 3% average reproducibility have a one-sided p-value of at most 0.031.

Model comparison and fitting

For each experiment $\mathcal{E}$, the data $\mathcal{D}_{\mathcal{E}}$ collected in the experiments consists of numbers x_i , like means, standard errors of the mean or p-values for some statistical tests, i.e. $\mathcal{D}_{\mathcal{E}} = \{x_i | i \in \mathcal{I}_{\mathcal{E}} \cup \mathcal{P}_{\mathcal{E}}\}$, where $\mathcal{I}_{\mathcal{E}}$ is the index set of all quantities except p-values and $\mathcal{P}_{\mathcal{E}}$ is the index set of p-values for experiment $\mathcal{E}$. We extracted at least 5 (Raby07 planning) and at most 212 (Clayton99B exp1) observed quantities from figures and text of the respective publications (1568 observed quantities in total).

Models are characterized by their structure, i.e. the probability of actions given the parameters and past observations, and their distribution over parameters. This distribution describes a population of birds, rather than individual birds. Each model has a different number M of parameters θ_i that are constrained to be in an interval $[l_i, u_i]$, $i = 1, \dots, M$ (see Table 4). We parametrize probability densities over these intervals with the help of beta distributions

$$\theta_i = (u_i - l_i)z_i + l_i \quad p_Z(z_i; s_i, d_i) = \begin{cases} c(s_i, d_i) z_i^{f(s_i)} (1 - z_i)^{f(s_i + d_i)} & d_i < 0 \\ c(s_i, d_i) z_i^{f(s_i - d_i)} (1 - z_i)^{f(s_i)} & d_i \geq 0 \end{cases} \quad (18)$$

where $c(s_i, d_i)$ is the normalization constant of the beta distribution and $f(x) = \log(\exp(x) + 1) - 1$. We found empirically that this parametrization led to reasonably fast and robust optimization results. We write

$$p(\theta; \mathbf{d}, \mathbf{s}) = \prod_{i=1}^M p_Z((\theta_i - l_i)/(u_i - l_i); s_i, d_i) \quad (19)$$

for the density of parameter vectors θ given hyperparameters $\mathbf{d}, \mathbf{s}$. A simulated bird is characterized by a sample θ from this probability distribution. The hyperparameter vectors $\mathbf{d}$ and $\mathbf{s}$ are fitted. In Fig. 1B of the main text we display the parameters θ with a blue bird and use ϑ for the hyperparameters $\mathbf{d}, \mathbf{s}$.

For each simulated experiment $\mathcal{E}$, we sample $N_{\mathcal{E}}$ birds independently from the population distribution (Eq. (18)), where $N_{\mathcal{E}}$ is the number of birds used in the actual experiment $\mathcal{E}$. Although some birds participated in multiple experiments, we believe re-sampling is justified, because we do not explicitly model other effects that may influence behavior, like aging or the change of seasons. Excluded from re-sampling is exclusively the series of experiments described in^{5,6}. We treat this series denoted as Clayton01&Clayton03 in Fig. 2C as one very

long experiment, because some experiments rely on the experience the birds made in earlier experiments of the same series.

For the given data and the models described in Model Description we have an intractable likelihood function

$$\begin{aligned} \ell(\mathbf{d}, \mathbf{s}) &= \prod_{\mathcal{E}} p(\mathcal{D}_{\mathcal{E}} | \mathbf{d}, \mathbf{s}) \quad \text{with} \quad p(\mathcal{D}_{\mathcal{E}} | \mathbf{d}, \mathbf{s}) \\ &= \int p(\mathcal{D}_{\mathcal{E}} | \theta^{(1)}, \dots, \theta^{(N_{\mathcal{E}})}) \prod_{j=1}^{N_{\mathcal{E}}} p(\theta^{(j)}; \mathbf{d}, \mathbf{s}) d\theta^{(j)}, \end{aligned} \quad (20)$$

where $\theta^{(j)}$ denotes the parameters of bird j and the product runs over all experiments $\mathcal{E}$ under consideration. The source of intractability in Eq. (20) are the conditional probability densities $p(\mathcal{D}_{\mathcal{E}} | \theta^{(1)}, \dots, \theta^{(N_{\mathcal{E}})})$, which are marginal distributions over all possible action sequences giving rise to the reported summary statistics. Note that the likelihood would be tractable and a standard likelihood based fitting procedure could be used, if the action sequences of all birds that participated in the real experiments were known. For Fig. 2 of the main text the parameters are fitted to each experiment individually, except for the experiments in the Clayton01&Clayton03 series that are fitted jointly. See Supplementary Results for joint fits of all experiments or subsets thereof.

Because the dimensionality of the hyperparameters ($8 \leq \dim(\mathbf{d}) = \dim(\mathbf{s}) = M \leq 45$) and the data is rather high, we resort to approximate maximum likelihood estimation to find the parameters that characterize best the population of birds. We use k -nearest-neighbor density estimates⁴⁵ to approximate the logarithm of the likelihood function in Eq. (20). To do so, we repeat each experiment $\mathcal{E}$ with independent groups $k = 1, \dots, K$ of $N_{\mathcal{E}}$ birds, each characterized by parameters $\theta^{(k,1)}, \dots, \theta^{(k,N_{\mathcal{E}})}$, where $N_{\mathcal{E}}$ is the number of birds that participate in experiment $\mathcal{E}$. From the simulations with index k , we compute the same means, standard errors of the mean and p-values $x_i^{(k)}$ as in the actual experiment. This results in simulated data $\mathcal{D}_{\mathcal{E}}^{(k)} = \{x_i^{(k)} | i \in \mathcal{I}_{\mathcal{E}} \cup \mathcal{P}_{\mathcal{E}}\}$ for experiment $\mathcal{E}$. For each simulated data set we compute the distance

$$\Delta_{\mathcal{E}}^{(k)} = \sqrt{\sum_{i \in \mathcal{I}_{\mathcal{E}}} (x_i^{(k)} - x_i)^2 + \sum_{j \in \mathcal{P}_{\mathcal{E}}} (s(x_j^{(k)}) - s(x_j))^2}, \quad (21)$$

where x_i is the experimentally observed value, $\mathcal{I}_{\mathcal{E}}$ and $\mathcal{P}_{\mathcal{E}}$ are the index sets defined above and s is a quantization of p-values given by

$$s(p) = \begin{cases} 1 & p < 0.001 \\ 2 & 0.001 \leq p < 0.01 \\ 3 & 0.01 \leq p < 0.05 \\ 4 & 0.05 \leq p < 0.1 \\ 5 & \text{otherwise} \end{cases} \quad (22)$$

This corresponds roughly to computing differences of p-values on a log-scale, but it does not emphasize difference between highly significant p-values, e.g. $s(10^{-11}) = s(10^{-5}) = 1$, which reduces the variance and helps during fitting. Let us assume that these distances are ordered such that $\Delta_{\mathcal{E}}^{(1)} \leq \Delta_{\mathcal{E}}^{(2)} \leq \dots \leq \Delta_{\mathcal{E}}^{(K)}$. With this we can estimate the log-likelihood function as⁴⁵

$$\hat{\ell}(\mathbf{d}, \mathbf{s}; n, K) = - \sum_{\mathcal{E}} d_{\mathcal{E}} \log \left(s_n \left(\mathcal{D}_{\mathcal{E}}; \mathcal{D}_{\mathcal{E}}^{(1)}, \dots, \mathcal{D}_{\mathcal{E}}^{(K)} \right) \right) + c(d_{\mathcal{E}}) \quad (23)$$

where $c(d_{\mathcal{E}})$ is a constant, $d_{\mathcal{E}} = |\mathcal{I}_{\mathcal{E}}| + |\mathcal{P}_{\mathcal{E}}|$ is the dimensionality of data set $\mathcal{D}_{\mathcal{E}}$, and $s_n(\mathcal{D}_{\mathcal{E}}; \mathcal{D}_{\mathcal{E}}^{(1)}, \dots, \mathcal{D}_{\mathcal{E}}^{(K)}) = \Delta_{\mathcal{E}}^{(n)}$ is the distance to the n th neighbor of $\mathcal{D}_{\mathcal{E}}$.

We optimize the approximate likelihood function in Eq. (23) with the CMA evolutionary strategy (CMA-ES)⁴⁶. For optimization and evaluation we use the $n = 5$ th nearest neighbor. Because the variance of

the approximate log-likelihood function in Eq. (23) decreases with increasing K , the noise handling strategy of CMA-ES selects $K \in \{5, 6, \dots, 500\}$ adaptively, such that K is small in regions where the direction of improvement in the approximate log-likelihood is obvious. This adaptivity saves computation time. To compute the performance $\Delta \log \hat{p}$ in Figs. 2 and S2 and S3, we compute $\log \hat{p}$ 10 times with $K=10^4$ and $n=5$ and take the average performance; the standard error of the mean is below 1.5 for all models and experiments, which is too small to be seen in the figures.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Source data for all figures are provided in the file ‘source_data.zip’ with this paper. The behavioral data extracted from published articles is available at <https://github.com/jbrea/FoodCachingExperiments.jl>.

Code availability

The source code is available at <https://github.com/jbrea/FoodCaching47>.

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Author contributions

J.B., N.S.C., and W.G. designed the study. J.B. developed the models, wrote the code, and ran the simulations. J.B., N.S.C., and W.G. wrote the text.

Competing interests

The authors declare no competing interests.

Additional information

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Computational models of episodic-like memory in food-caching birds

Supplementary Information

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Contents

1	Supplementary Results	2
1.1	Experimental Variability	2
1.2	Fitting All Experiments Jointly	2
1.3	Generalization	3
1.4	Frequently Asked Questions	3
2	Supplementary Figures	6
2.1	Experiments	11
2.1.1	Raby07 breakfastchoice	11
2.1.2	Raby07 planningforbreakfast	12
2.1.3	Cheke11 planning	14
2.1.4	Correia07 exp2	16
2.1.5	deKort07 exp2	18
2.1.6	deKort07 exp3	20
2.1.7	deKort07 exp4	22
2.1.8	Clayton05 exp1	24
2.1.9	Clayton05 exp2	29
2.1.10	Clayton05 exp3	34

2.1.11	Clayton05 exp4	37
2.1.12	Clayton99A exp1	39
2.1.13	Clayton99A exp2	42
2.1.14	Clayton99B exp1	44
2.1.15	Clayton99B exp2	55
2.1.16	Clayton01 exp1	65
2.1.17	Clayton01 exp2	76
2.1.18	Clayton01 exp3	81
2.1.19	Clayton01 exp4	86
2.1.20	Clayton03 exp1	89
2.1.21	Clayton03 exp2	92
2.1.22	deKort05	102
2.1.23	Cheke11 specsat	107
2.1.24	Correia07 exp1	109
2.1.25	Clayton99C exp1	111
2.1.26	Clayton99C exp2	113
2.1.27	Clayton99C exp3	116
2.1.28	deKort07 exp1	119

1 Supplementary Results

1.1 Experimental Variability

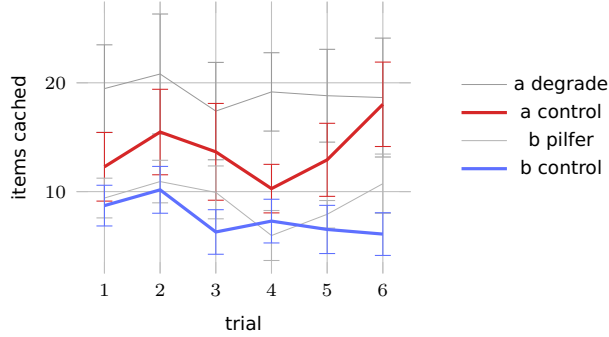


Figure S1: **Experimental Variability.** Evidence of variability across sessions and trials. In the experiment deKort07 exp1, the number of caches were counted in two sessions (a and b) of six trials under three conditions (degrade, pilfer, control; the control group is called “replenish group” in the original publication.). Under the same condition (control: red and blue) the counts in sessions a and b differ a lot. Even within the same session the counts can differ a lot between different trials (e.g. red curve trial 4 versus trial 6).

The birds’ exact behaviour as observed in the experiments may depend not only on the experimental protocol but also on other factors such as the subset of individual birds used for an experiment, the species (western scrub jays were used for all experiments but the Cheke11 and Amodio21 experiments that used Eurasian jays), the season and the weather, social interactions of the tested birds with each other or the experimenter. The strong dependence on other factors is evident for example in the experiment deKort07 exp1 (Figure S1), where under the same conditions very different numbers of items were cached in different trials and different sessions.

A similar variability has been observed across two experiments in Clayton01 and Clayton99B. In both experiments, birds in the control group (“replenish”) cached and inspected their caches under similar conditions, except that the inspection duration in Clayton01 exp1 was only 5 minutes compared to 15 minutes in Clayton99B exp1 and the spatial distance between the caches of different types of food was smaller in Clayton01 exp1 than in Clayton99B exp1. Despite a three-times shorter inspection period and on average less items cached, the birds in Clayton01 exp1 made almost twice as many inspections than the birds in Clayton99B exp1.

Our models capture some of the variability seen in the data with birds-specific parameters sampled from the fitted distributions (Figure 1B and Methods) and with stochastic action selection and random delays after each action (Methods). While more sophisticated noise models may capture even more variability we think that they are unjustified for the available data. Once future experiments provide more detailed data, modeling will be facilitated.

1.2 Fitting All Experiments Jointly

Our models can be fitted to all 28 experiments jointly, such that we have one set of hyper-parameters for all experiments. The main conclusions remain unchanged: first, the Plastic Caching Model is not worse than the Planning-By-Replay Model and, second, the lesioned models have lower reproducibility (Figure S2A) and likelihood (Figure S2B). The joint fits do not reproduce all experiments as well as the individual fits (Figure S2C).

Because of the observed variability in the experiments across trials, sessions and potentially seasons and sub-groups of birds (cf. Experimental Variability) a joint fit of all experiments forces the model to increase the intrinsic noise (i.e., the variance of the distribution of parameters from which individual birds are chosen from or the variance of the action selection distributions of individual birds) which in turn lowers the overall average performance. In other words, some of the exact experimental results seem to depend on factors not included in our models. Therefore, the performance in a joint fit is expected to be lower compared to individual fits. The individual fits performed in the main text explicitly acknowledge that data is not expected to be consistent

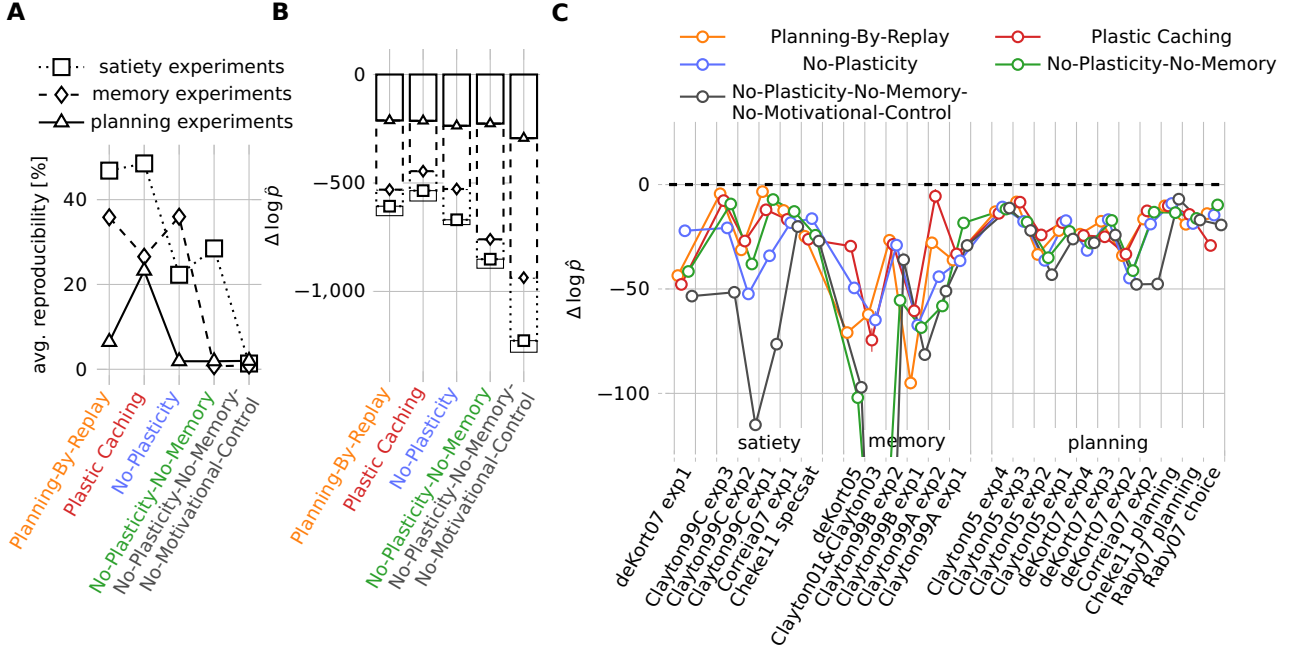


Figure S2: **Fitting all experiments jointly.** **A** Similarly to the results of the individual fits (Figure 2C, main text) the average reproducibility of planning, memory and satiety experiments drops almost to zero for lesioned models without plasticity, memory and motivational control, respectively. **B** The cumulative log-likelihood $\Delta \log \hat{p}$ relative to the same baseline as in Figure 2 of the main text (i.e. individual fits with the Planning-By-Replay Model) shows the same pattern with a drop of performance for the lesioned models. **C** The approximate log-likelihood $\Delta \log \hat{p}$ for each individual experiment reveals that some experiments are fitted almost as well as with individual fits (Figure 2E) whereas others are matched far worse. Which experiments are matched well depends on the random seed of optimization: because the optimization problem is non-convex, two different values for the fitted hyper-parameters may lead to the same cumulative performance but very different performances for the individual experiments.

across all 28 experiments.

1.3 Generalization

Fitting the models on a subset of experiments (training set) and evaluating them on test experiments leads to results consistent with previous findings: the Plastic Caching Model and the Planning-By-Replay Model generalize best and performance degrades for the models with lesions (Figure S3). Given the findings in [Experimental Variability](#) we expect that the Plastic Caching Model and the Planning-By-Replay Model capture the trend of the data but fail to match the exact numbers. This is confirmed by a higher average reproducibility (see Methods) for these models than for the lesioned models (Figure S3A1,A2 & B1, B2). Since the Plastic Caching Model and the Planning-By-Replay Model reproduce only the trend, but not absolute numbers, of the data, the values for $\Delta \log \hat{p}$ are low, despite the fact that qualitatively the main trends are captured (Figure S3A3 & B3).

1.4 Frequently Asked Questions

Are the differences between the models significant?

We do not quantify or claim significance for the difference between the Plastic Caching Model and the Planning-By-Replay Model. However, models with lesions fail clearly and qualitatively on some of the experiments (see Figure 2C-E in the Main Text and [Experiments](#)).

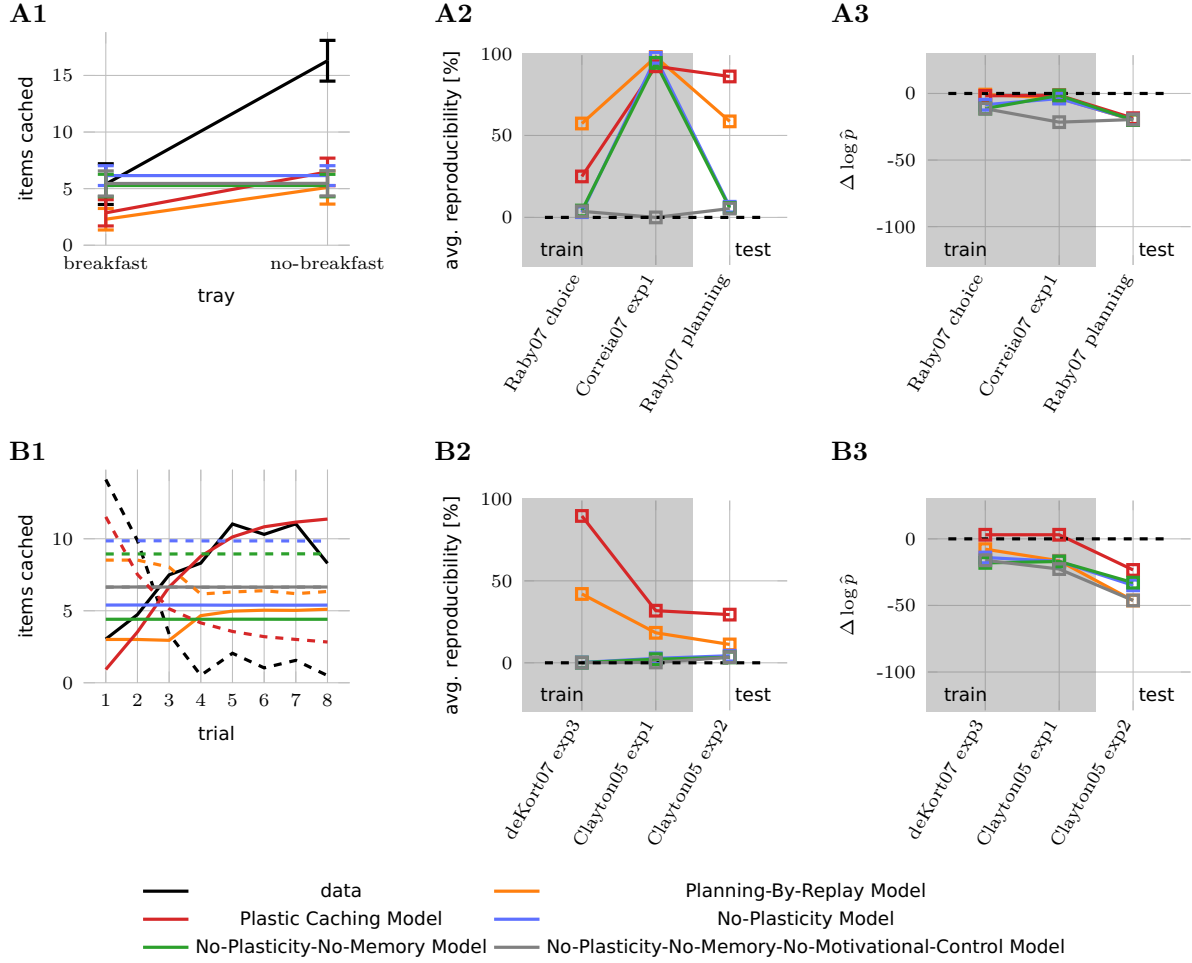


Figure S3: **Generalization.** The models were fitted on two training experiments jointly and evaluated on one test experiment. **A1, B1** The main trend in the data of the test experiment (black solid line: nuts cached and black dashed line: worms cached by birds that experienced degraded worms at recovery) is on average better captured by the full models (colored lines = simulation results averaged over 10^5 simulations, error bars = average SEM). **A2, B2** This is reflected by the average reproducibility of the most relevant statistical tests, which is higher for the Plastic Caching Model and the Planning-By-Replay Model than for the simpler models. **A3, B3** The performance $\Delta \log \hat{p}$ (measured relative to the same baseline as in Figure 2 of the main text, i.e. individual fits with the Planning-By-Replay Model) depends not only on the trends, but also on the absolute numbers; it is therefore lower than in the individual fits.

Do the models overfit the data?

Yes, our models potentially overfit some aspects (see [subsection 1.2](#) and [subsection 1.3](#)). Some unmeasured, relevant factors of variability are not included in our models (as explained above) and it is unlikely that the amount of available data assures that these factors average out.

However, our main conclusions do not depend on whether the models overfit the data or not: the Plastic Caching Model is not worse than the Planning-By-Replay Model on any experiment, thus explicit planning does not seem to be necessary to match the experimental results, and each model with lesions fails at reproducing on average some of the experiments (see Figure 2 in the Main Text and Supplementary Text).

What is the number of parameters?

Note that the numbers of free parameters are only upper bounds to the effective numbers of free parameters. For example, parameters that influence caching and inspection behaviour have no influence on the results

model	parameters per food type	other parameters	1 food type	2 food types
No-Plasticity-No-Memory-No-Motivational-Control Model	0	16	16	16
No-Plasticity-No-Memory Model	6	20	26	32
No-Plasticity Model	6	24	30	36
Plastic Caching Model	6	34	40	46
Planning-By-Replay Model	6	34	40	46

Table S1: Number of parameters for each model. The last two columns show the total number of parameters for an experiment with one or two food types, respectively. The numbers of parameters are multiples of two, because we parameterize each beta distribution with two parameters.

of experiments where the birds were not allowed to cache any food. Therefore, increasing the number of free parameters does not necessarily imply a higher number of effective free parameters and a higher degree of overfitting. We see clear evidence that the number of free parameters is less crucial than the available components of the model: all models with lesions make similar predictions for the deKort07 and Raby07 experiments, despite large differences in the number of free parameters, but none of these models captures on average the trends seen in the data (see [subsection 2.1](#)). In general, measuring the effective number of free parameters is not trivial.

What is the AIC/BIC value of the models?

The Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) depend on the number of free parameters of a model. Because of the above mentioned difficulties in counting the effective number of free parameters we did not compute AIC or BIC values.

2 Supplementary Figures

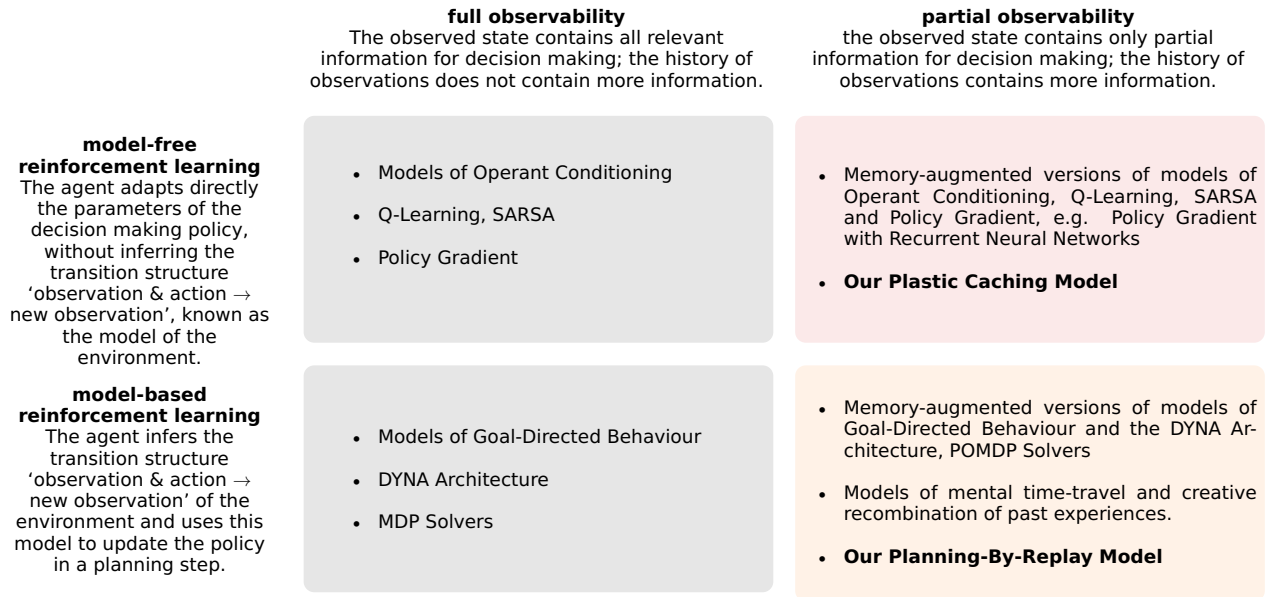


Figure S4: **Categorization of Reinforcement Learning Models.** In reinforcement learning we can classify methods based on their ability to learn a model of the environment (model-free versus model-based) and their ability to deal with partial observability (full observability versus partial observability). A model of the caching behavior of corvids needs to address the partial observability of the cache contents; it is thus in the right column. Our Plastic Caching Model is a model-free reinforcement learning model. Our Planning-By-Replay Model is a very simple model-based reinforcement learning model in that the experienced transitions are stored in memory and used at decision time to replay sequences that appeared in a similar context as the current one.

Procedure for the ‘planning for breakfast’ experiment.

Each bird received six training trials followed by one test trial. From 09:00–17:00 each day the birds maintenance diet was freely available. At 17:00 the test subject was separated from its companion into compartments A, B and C. Maintenance diet was removed for 90 min at 17:00 and caching trays were put in compartments A and C. At 18:30 a bowl containing 10 g of powdered pine nuts, which could be eaten but not cached, was put in the centre of compartment B for 30 min. At 19:00 the bowl of pine nuts and the caching trays were removed. The birds were deprived of food overnight so that they were mildly hungry in the morning. Each morning at 07:00 the experimental bird was confined in either compartment A or C for two hours. The caching trays were returned to both compartments A and C for this period. In one compartment the bird was given no food (‘no-breakfast’ compartment) and in the other the bird was given powdered pine nuts (‘break- fast’ compartment). Each bird experienced the two compartments on alternate days, with the compartment that was the ‘breakfast’ compartment (A or C) counterbalanced across birds. The order in which each bird experienced the two compartments was also counterbalanced. The test day followed exactly the same evening routine as training days with the exception that 30 (cacheable) whole pine nuts were given at 18:30 instead of powdered food. The jays did not get the opportunity to recover their caches on the morning after the test trial.

```
function run_experiment(::Experiment{Raby07_planningforbreakfast},
                        models; N = 8) # 8 birds in total
    results = DataFrame{tray = [], id = [], items = []}
    for id in 1:N
        m = models[id]
        trays = [Tray(i) for i in 1:2] # caching trays in compartments A and C
        for day in 1:7
            add!(m, MaintenanceDiet) # start at 9am on first day
            wait!(m, 8u"hr")
            remove!(m, MaintenanceDiet)
            add!(m, trays)
            wait!(m, 1.5u"hr")
            if day == 7
                add!(m, Pinenut, 30)
            else
                add!(m, PowderedPinenut, 30)
            end
            wait!(m, 30u"minute")
            if day == 7
                push!(results, ["breakfast", id, countcached(trays[1])])
                push!(results, ["no-breakfast", id, countcached(trays[2])])
                break # end of experiment
            end
            remove!(m, Any)
            wait!(m, 12u"hr")
            if (id + day) % 2 == 0 # counterbalancing
                add!(m, PowderedPinenut, 30)
                add!(m, trays[1])
            else
                add!(m, trays[2])
            end
            wait!(m, 2u"hr")
            remove!(m, Any)
        end
    end
    results
end
```

Figure S5: **Example of a formalized protocol.** The protocol as reported by Raby et al. [10] (reprinted on the left) is transformed into a function (on the right) that takes N models of a bird and its cage as input and returns a result in the form of a table (DataFrame). Highlighted in dark blue are the terms of the domain specific language and in light blue the control commands of the programming language Julia. For example the command `add!(m, PowderedPinenut, 30)` tells that an object of type Food with properties ‘freshness = 1’, ‘eatable = true’, ‘cacheable = false’ (because it is powdered), ‘amount = 30’ and ‘type = pinenut’ should be added to the simulated cage m (see also Methods).

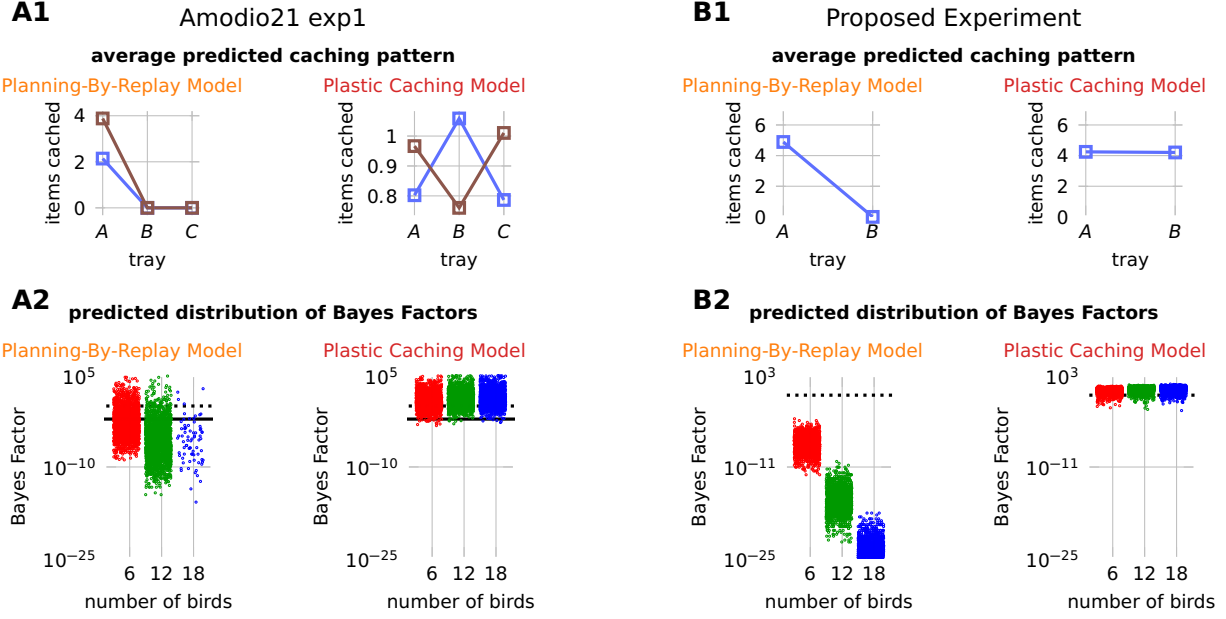


Figure S6: **Using the model and the domain-specific language to design new experiments.** Although the Plastic Caching Model matches some experimental results quantitatively better than the Planning-By-Replay Model, both models make qualitatively the same predictions on the 11 planning experiments so that a clear distinction between the models is not yet possible. To design more stringent tests, with experiments where the two models make qualitatively different predictions, we can leverage the domain specific language of experimental protocols and simulate expected outcomes before running the actual experiments. **A1** The Planning-By-Replay Model and the Plastic Caching Model make qualitatively different predictions on the caching pattern of two groups of birds (blue and brown) in the experimental protocol of Amodio et al. 2021 [44]. **A2** However, the effect is not strong as shown by the distribution of Bayes factors [44] after 2000 simulations of the experiment with 6 birds (one red dot per simulation) that overlap for both models with the experimental data (thick black line). Running the same experiment with 12 or 18 birds only slightly improves the situation. The Bayes Factor for 18 birds with the Planning-By-Replay Model is below 10^{-25} for most simulations, but there is a tail (blue dots) overlapping with the experimental data. **B1** Predicted average caching pattern of a proposed experiment (see supplementary Figure S4). **B2** The distributions predicted by the two models have little overlap, suggesting that it should be possible to discriminate the two models even in an experiment with only 6 birds.

```

function run_experiment (::Experiment{ :Hypothetical }, models)
  results = DataFrame{id = Int[], action = String[], A = Int[], B = Int[]}
  for id in eachindex(models)
    m = models[id]
    trays = [Tray(i) for i in 1:4] # C, D, A, B
    for day in 1:9
      cid = floor(Int, ((day-1) % 2 + 1))
      add!(m, trays[cid]) # compartment to spend the day from 10am onwards
      add!(m, MaintenanceDiet)
      wait!(m, 6.75u"hr")
      remove!(m, Any)
      if day == 9
        add!(m, trays[3:4])
        add!(m, Kibble, 10) # test
        wait!(m, 15u"minute")
        remove!(m, Any)
        push!(results, [id; "cache"; countcache.(trays[3:4])])
        break # end of experiment
      else
        add!(m, trays[3:4]) # access to caching trays in A and B
        add!(m, PowderedKibble, 10) # without option of caching
        wait!(m, 15u"minute")
        remove!(m, Any)
        add!(m, MaintenanceDiet)
        wait!(m, 3u"hr")
      end
      remove!(m, Any)
      wait!(m, 12u"hr") # morning
      add!(m, trays[cid == 1 ? 3 : 4]) # add (empty) caching trays to the compartment where the birds wakes up hungry
      wait!(m, 1u"hr")
      remove!(m, Any)
      wait!(m, 1u"hr")
      add!(m, MaintenanceDiet)
    end
  end
  results
end

```

Figure S7: **The proposed experiment in Fig S3B.** It is based on a modification of the experimental protocol of [10] (c.f. Figure S5). Birds spend their day after 10am in compartments C or D. If they spend their days in C they are transferred to A in the evening and stay there until 10am of the next morning; if they spend their day in D they will sleep and wake up in B. Neither in A nor in B they receive any food, such that both compartments are equally associated with being hungry. After sufficiently many repetitions of $C \rightarrow A$ and $D \rightarrow B$, the birds are given the chance to cache food in A or B, in the evening, after having spent their day in C or D. The Plastic Caching Model distributes caches equally between A and B, because both compartments are equally associated with being hungry and the model does not learn the transition structure of the world (Fig. S1). In contrast, the Planning-By-Replay Model takes into account the learned model of the world (associations $C \rightarrow A$ and $D \rightarrow B$) and caches more food in A after a day in C and more in B after a day in D. Hence this new protocol should enable falsification of one of the two models.

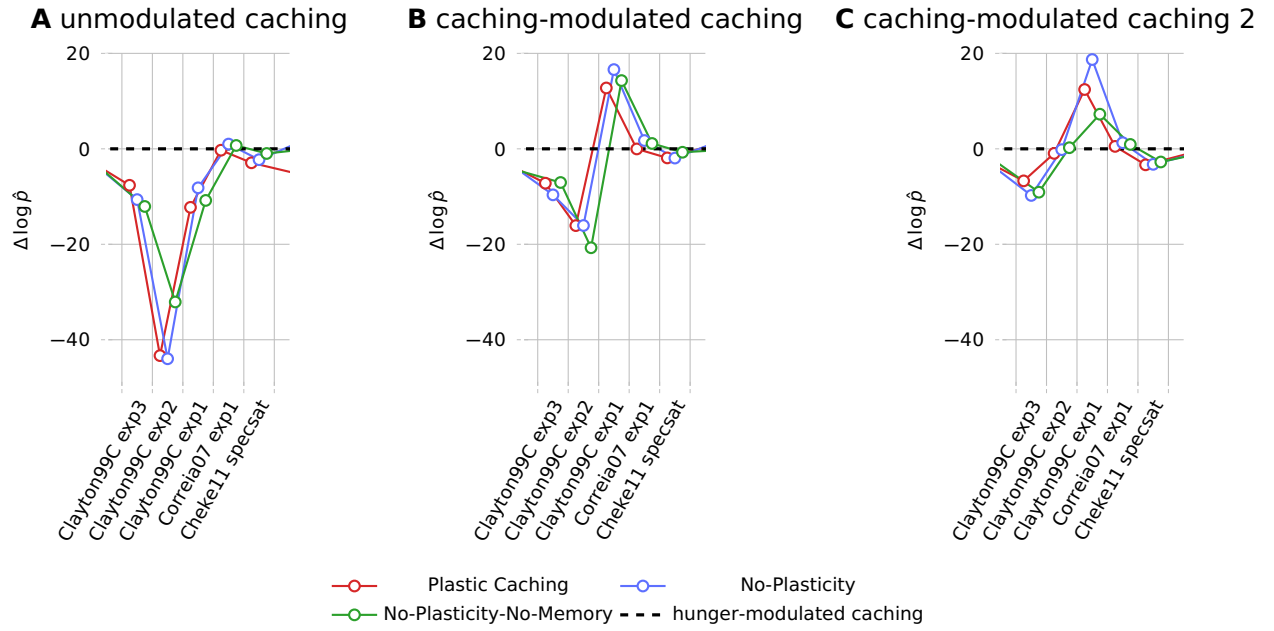


Figure S8: **Alternative models of motivational control of caching.** **A** The experiments in Clayton et al. (1999) [24] cannot be reproduced if caching does not depend on any motivational state and thus is independent of the recent eating and caching history. For each model the estimated log-likelihood is shown relative to a version of the same kind of model with hunger-modulated caching. **B** If caching depends on a low-pass filter of the recent caching history, the first experiment in Clayton et al. (1999) [24] is reproduced better than with hunger-modulated caching, but the other experiments are reproduced worse. In experiments Clayton99C exp3 and Cheke11 specsat, birds that ate to satiety on uncacheable, powdered food, cached subsequently fewer cacheable items of the same kind of food than birds under control conditions. This behavior is not reproducible with caching-modulated caching. **C** If caching-modulated caching is modelled with filters of the same kind as the hunger variables (instead of a simple low-pass filter), the results of Clayton99C exp2 can also be reproduced, but the experiments Clayton99C exp3 and Cheke11 specsat remain unreproducible.

2.1 Experiments

In the following we present brief summaries and result figures for all considered experiments. The figures in the center with title “Data” contain the measured values extracted from the published experiments. The figures on the right contain averaged simulated results, obtained by running the same simulated experiment 10^5 times and averaging the results. The error bars indicate average standard errors of the mean. The figures on the left contain the simulated result closest to the experimental result (closest out of 10^5 different simulated runs).

2.1.1 Raby07 breakfastchoice

Published in [10].

Major Finding

Birds cache more items of the food type that is different to the one they experienced to have available in the morning in a given compartment.

Most Important Tests

More different food cache (foodtype): $F(1, 8) = 2.29, p = 0.17$

More different food cache (tray): $F(1, 8) = 5.48, p = 0.047$

Total Number of Statistics

2 p-values, 8 means or sems.

Figures

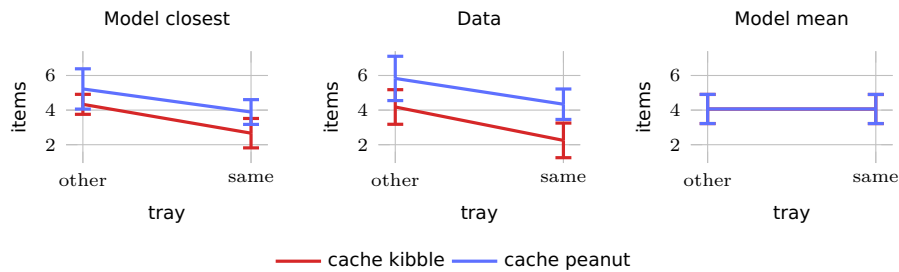


Figure S9: **Raby07 breakfastchoice: No-Plasticity-No-Memory-No-Motivational-Control Model**

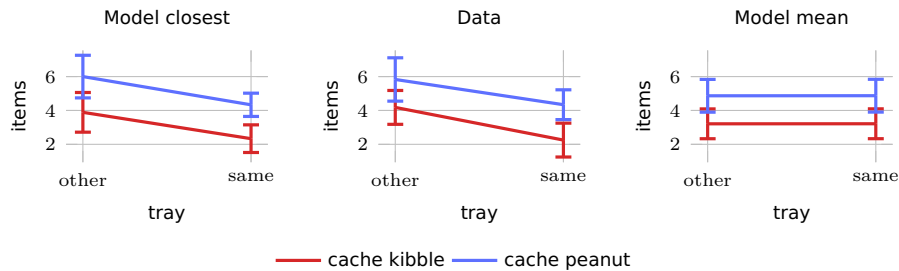


Figure S10: **Raby07 breakfastchoice: No-Plasticity-No-Memory Model**

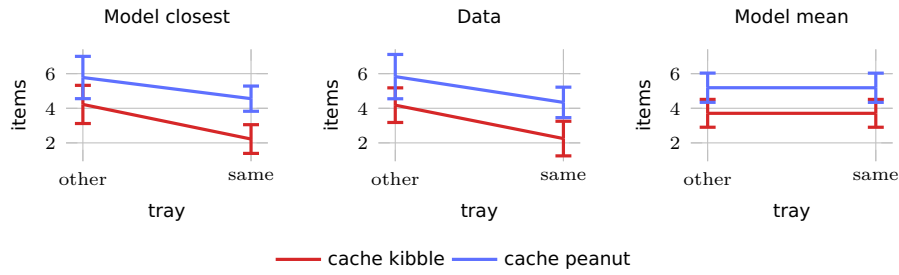


Figure S11: **Raby07 breakfastchoice: No-Plasticity Model**

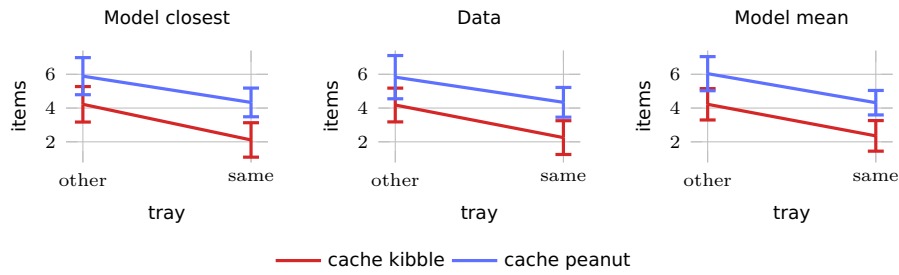


Figure S12: **Raby07 breakfastchoice: Plastic Caching Model**

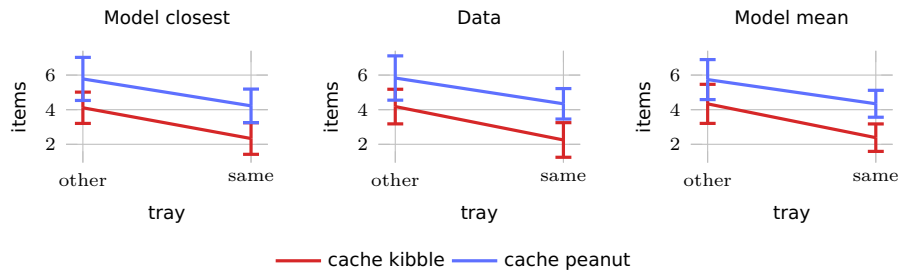


Figure S13: **Raby07 breakfastchoice: Planning-By-Replay Model**

2.1.2 Raby07 planningforbreakfast

Published in [10].

Major Finding

Birds cache more at sites where they experienced to be hungry in the morning.

Most Important Tests

More pine nuts in no-breakfast compartment (tray): $F(1, 7) = 9.06, p = 0.02$

Total Number of Statistics

1 p-values, 4 means or sems.

Figures

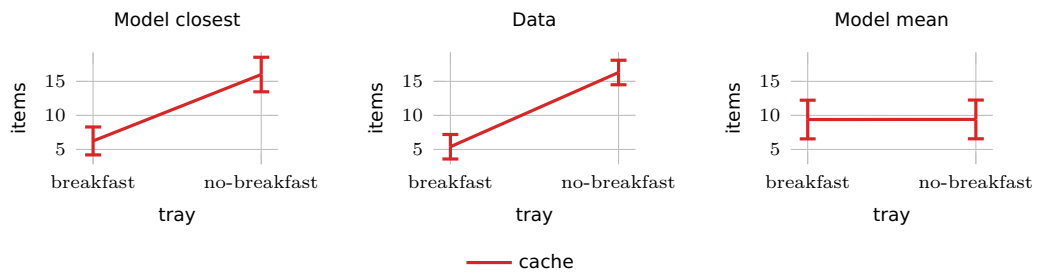


Figure S14: **Raby07 planningforbreakfast: No-Plasticity-No-Memory-No-Motivational-Control Model**

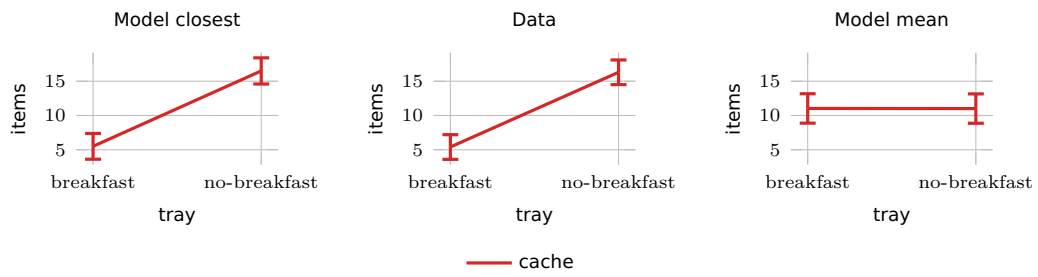


Figure S15: **Raby07 planningforbreakfast: No-Plasticity-No-Memory Model**

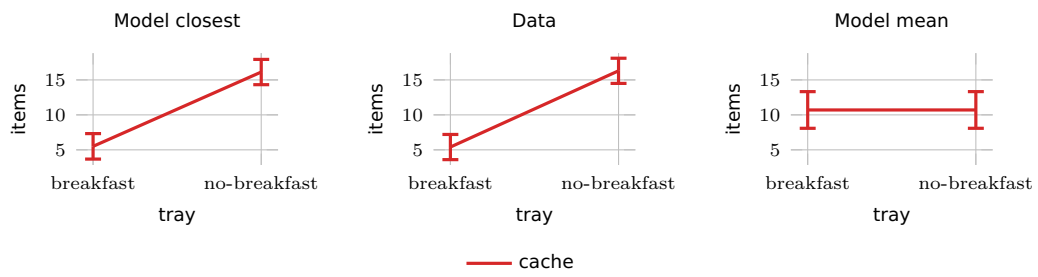


Figure S16: **Raby07 planningforbreakfast: No-Plasticity Model**

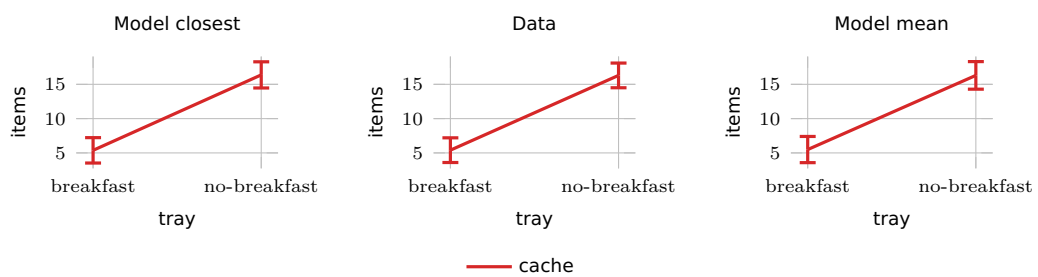


Figure S17: **Raby07 planningforbreakfast: Plastic Caching Model**

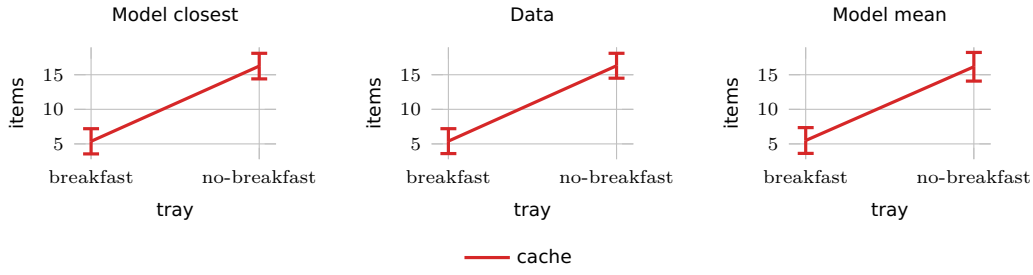


Figure S18: **Raby07 planningforbreakfast: Planning-By-Replay Model**

2.1.3 Cheke11 planning

Published in [12].

Major Finding

Birds cached both foods in equal amounts in both trays in trial 1 but then developed a differential preference between the trays, preferentially caching in each tray the type of food that they would desire when retrieving from it.

Comments

Since it is unclear how the tests for the overall reduction in caching and the trial $\times$ tray $\times$ foodtype were performed, we omitted these tests. This experiment used Eurasian jays.

Most Important Tests

Tray $\times$ foodtype interaction in trial 2 (tray & foodtype): $F(1, 3) = 14.24, p = 0.033$

Tray $\times$ foodtype interaction in trial 1 (tray & foodtype): $F(1, 3) = 3.0, p = 0.18$

Tray $\times$ foodtype interaction in trial 3 without hunter (tray & foodtype): $F(1, 2) = 15.429, p = 0.059$

Total Number of Statistics

6 p-values, 24 means or sems.

Figures

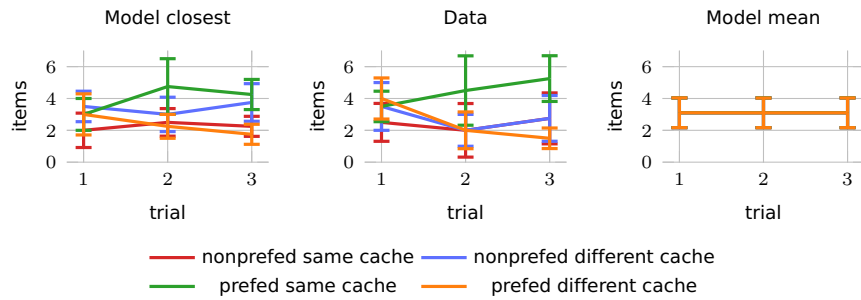


Figure S19: **Cheke11 planning: No-Plasticity-No-Memory-No-Motivational-Control Model**

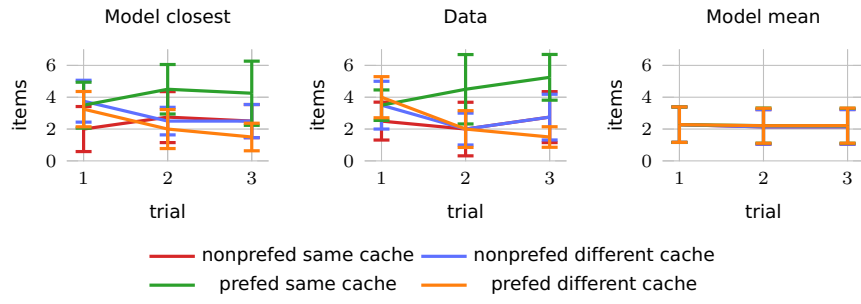


Figure S20: Cheke11 planning: No-Plasticity-No-Memory Model

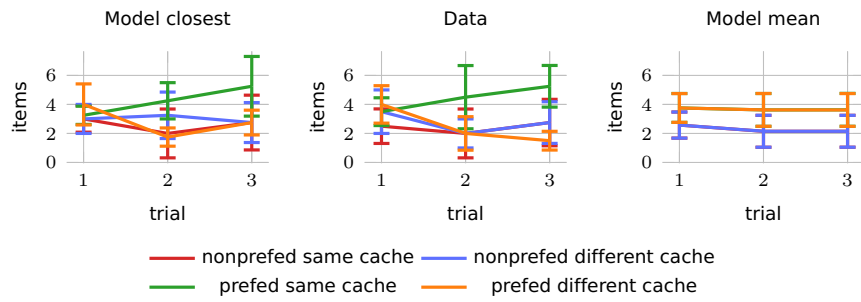


Figure S21: Cheke11 planning: No-Plasticity Model

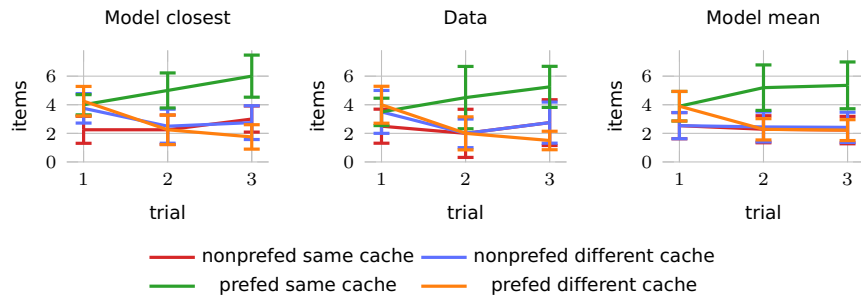


Figure S22: Cheke11 planning: Plastic Caching Model

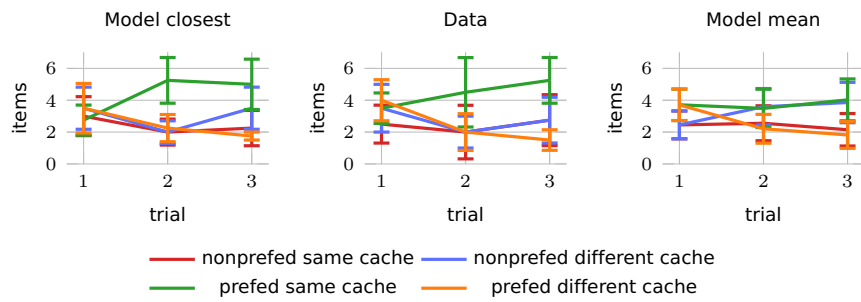


Figure S23: Cheke11 planning: Planning-By-Replay Model

2.1.4 Correia07 exp2

Published in [11].

Major Finding

The birds learn to cache more of the food type they will not be prefed on prior to recovery.

Most Important Tests

trial 1: proportion cache (group): $U(10) = .11, p = 0.11$

trial 3: proportion cache (group): $U(8) = 0., p = 0.0$

trial 2: proportion cache (group): $U(8) = .04, p = 0.04$

Total Number of Statistics

12 p-values, 48 means or sems.

Figures

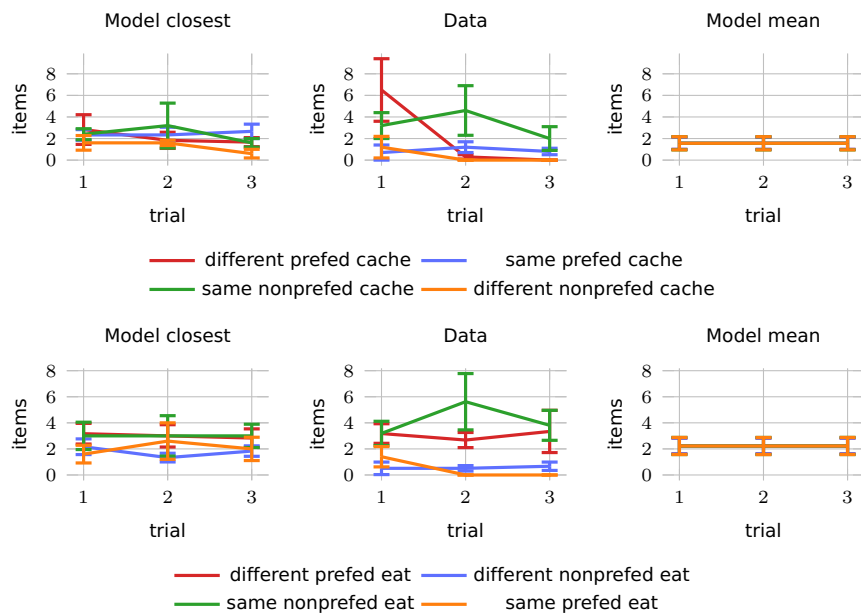


Figure S24: Correia07 exp2: No-Plasticity-No-Memory-No-Motivational-Control Model

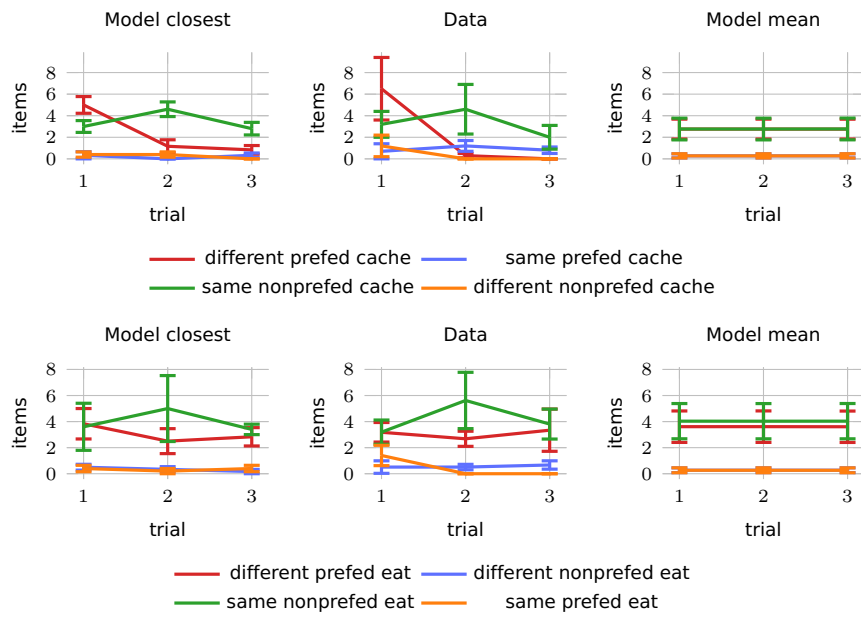


Figure S25: Correia07 exp2: No-Plasticity-No-Memory Model

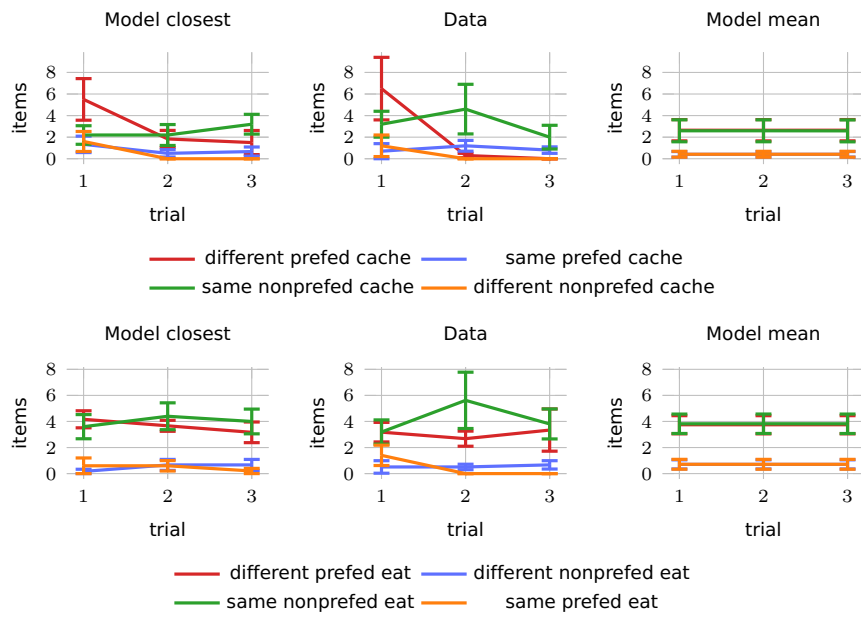


Figure S26: Correia07 exp2: No-Plasticity Model

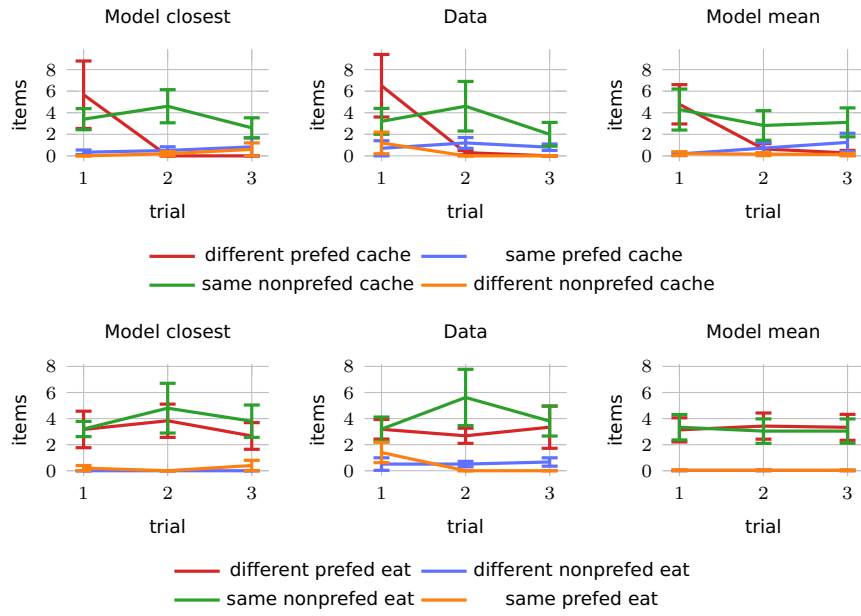


Figure S27: Correia07 exp2: Plastic Caching Model

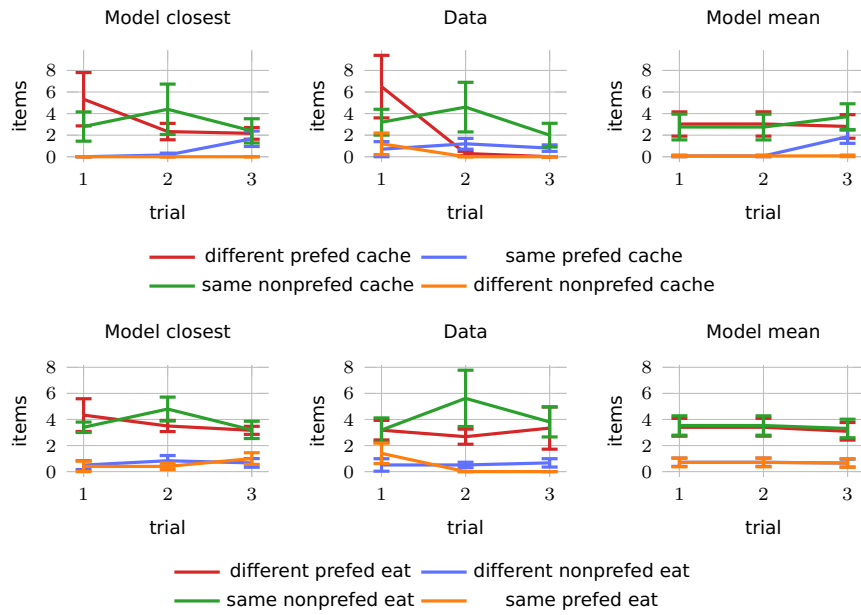


Figure S28: Correia07 exp2: Planning-By-Replay Model

2.1.5 deKort07 exp2

Published in [9].

Major Finding

Birds learn to reduce caching in sites where they always found their caches pilfered or the food items degraded, provided they had alternative caching locations.

Comments

The birds learned to reduce caching on one side of the cage despite trial-unique configuration of Lego Duplo blocks surrounding the ice cube tray, indicating that they generalized based on the side of the cage and not based on the local visual appearance of the caching site.

Most Important Tests

Experiment b tray $\times$ trial (tray & trial): $F(3, 42) = 4.39, p = 0.0089$

Experiment b, tray pilfer (trial): $F(3, 21) = 3.89, p = 0.023$

Experiment a, tray degrade (trial): $F(5, 45) = 2.46, p = 0.047$

Total Number of Statistics

10 p-values, 40 means or sems.

Figures

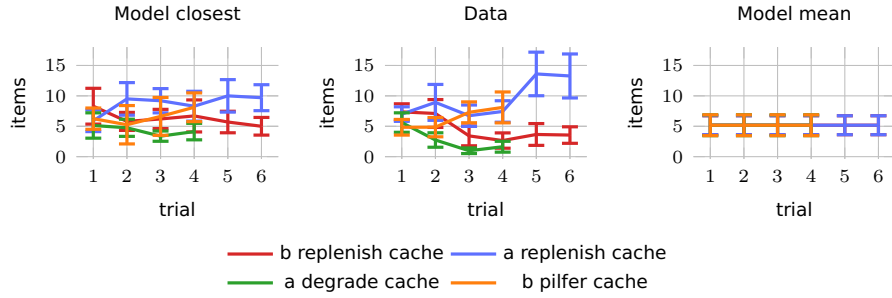


Figure S29: deKort07 exp2: No-Plasticity-No-Memory-No-Motivational-Control Model

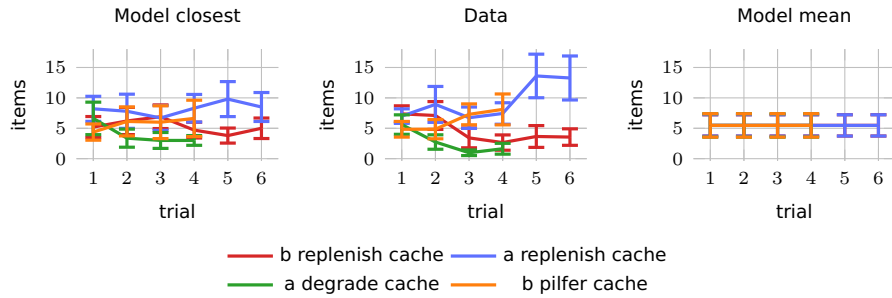


Figure S30: deKort07 exp2: No-Plasticity-No-Memory Model

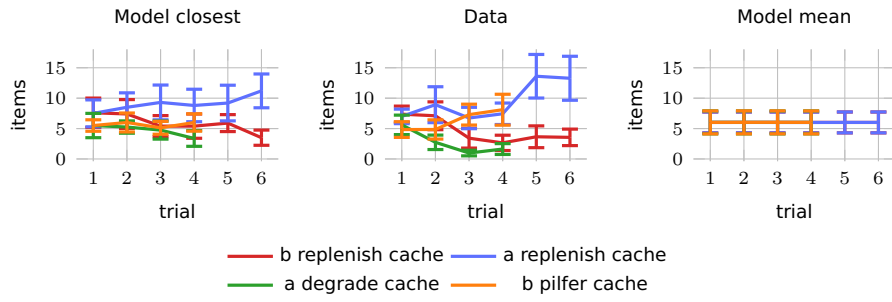


Figure S31: deKort07 exp2: No-Plasticity Model

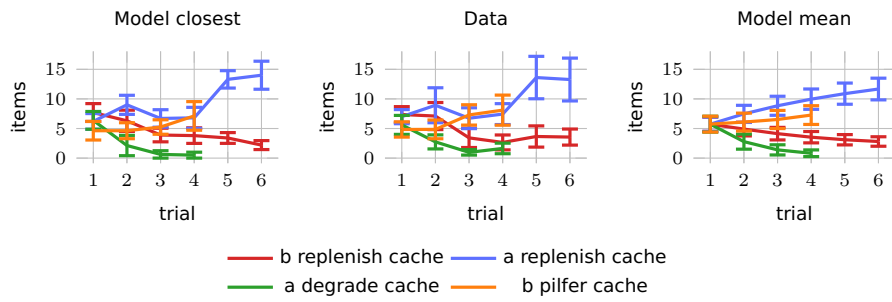


Figure S32: deKort07 exp2: Plastic Caching Model

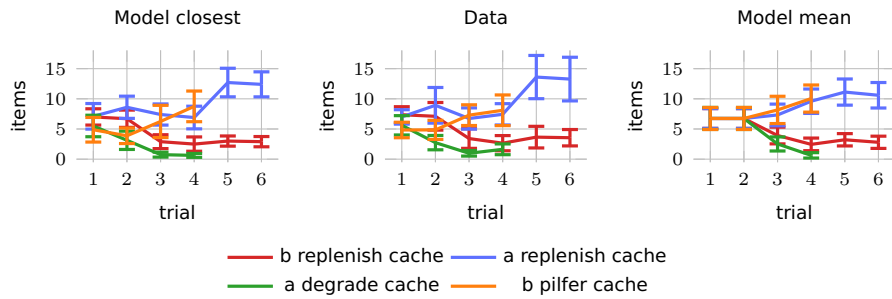


Figure S33: deKort07 exp2: Planning-By-Replay Model

2.1.6 deKort07 exp3

Published in [9].

Major Finding

Birds reduced caching in pilfered trays, even when the caching trays were available sequentially.

Most Important Tests

tray $\times$ trial (tray & trial): $F(9, 108) = 3.88, p = 0.00027$

Tray period2 (trial): $F(9, 54) = 3.18, p = 0.0037$

Tray period1 (trial): $F(9, 54) = 2.86, p = 0.0078$

Total Number of Statistics

3 p-values, 40 means or sems.

Figures

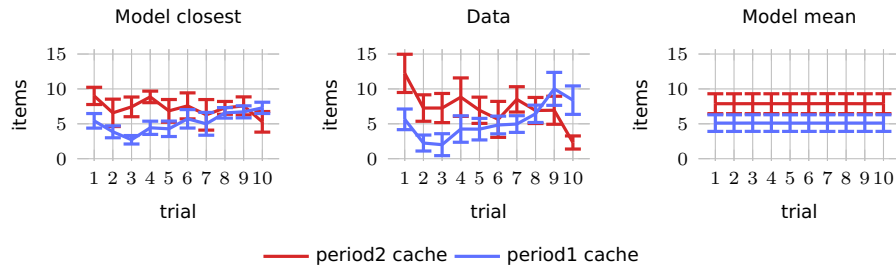


Figure S34: deKort07 exp3: No-Plasticity-No-Memory-No-Motivational-Control Model

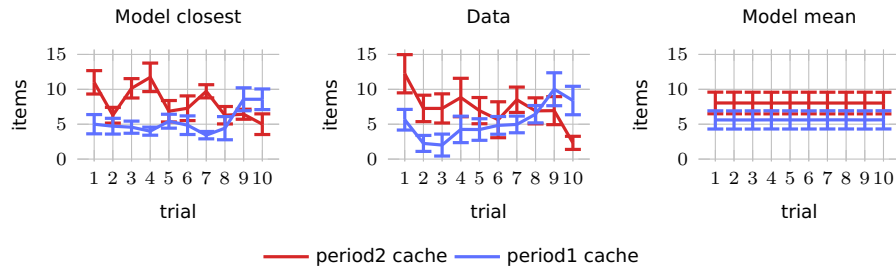


Figure S35: deKort07 exp3: No-Plasticity-No-Memory Model

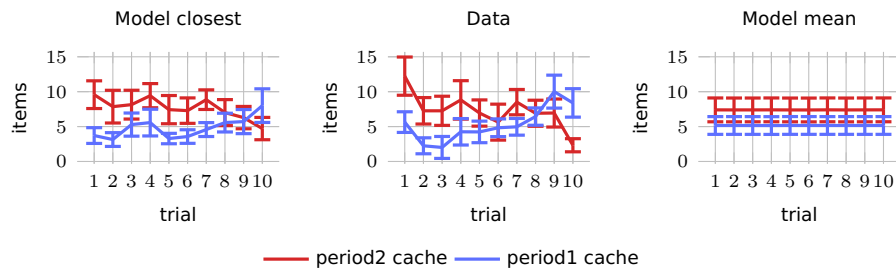


Figure S36: deKort07 exp3: No-Plasticity Model

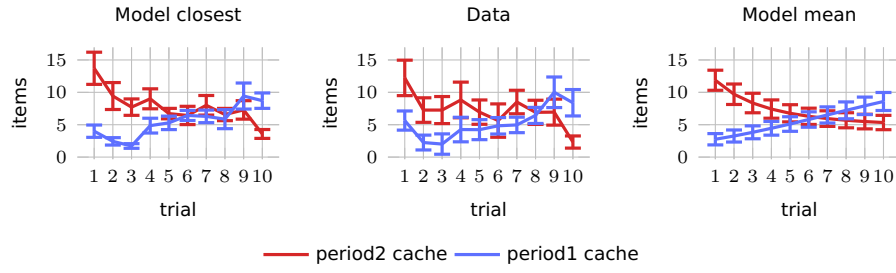


Figure S37: deKort07 exp3: Plastic Caching Model

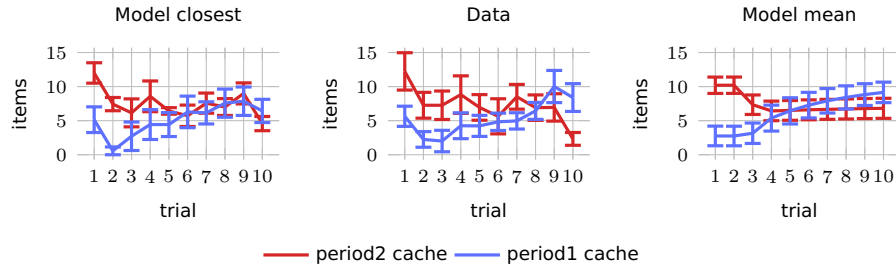


Figure S38: deKort07 exp3: Planning-By-Replay Model

2.1.7 deKort07 exp4

Published in [9].

Major Finding

Rather than caching in rewarded locations, the birds learned to avoid the unrewarded ones.

Most Important Tests

experimental group caches more in B than control (group): $F(1, 6) = 9.26, p = 0.023$

Effect of tray on the worms cache (tray): $F(2, 6) = 7.73, p = 0.022$

control group caches more in A than in B (tray): $F(1, 6) = 8.03, p = 0.03$

Total Number of Statistics

11 p-values, 14 means or sems.

Figures

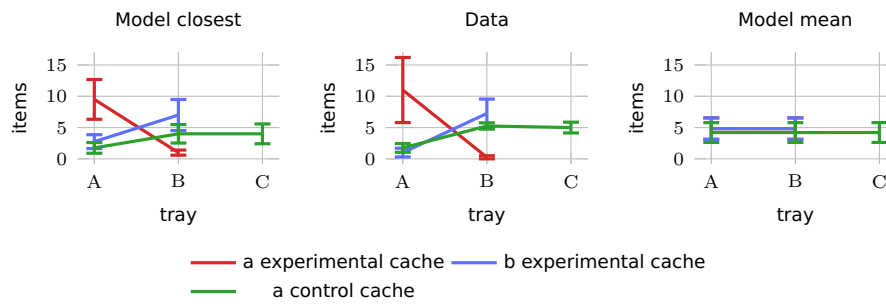


Figure S39: deKort07 exp4: No-Plasticity-No-Memory-No-Motivational-Control Model

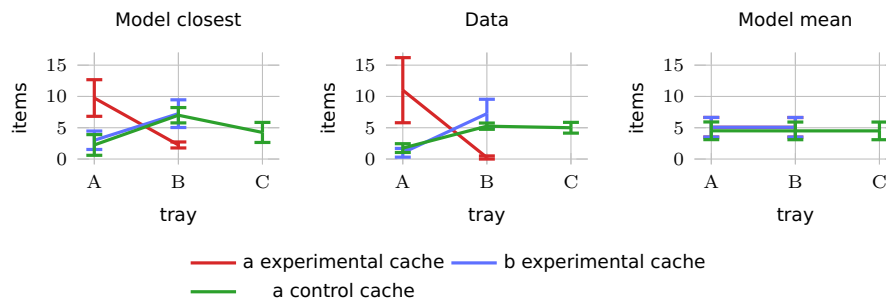


Figure S40: deKort07 exp4: No-Plasticity-No-Memory Model

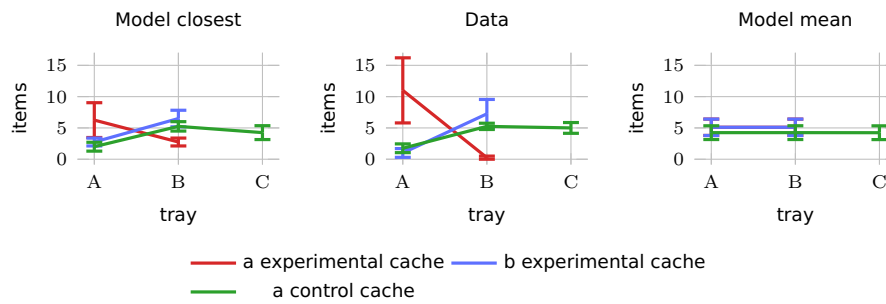


Figure S41: deKort07 exp4: No-Plasticity Model

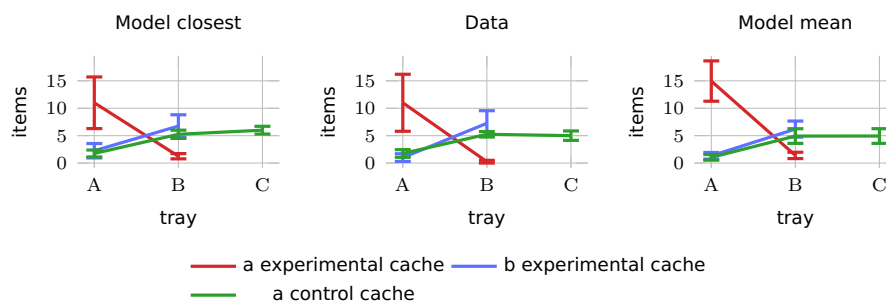


Figure S42: deKort07 exp4: Plastic Caching Model

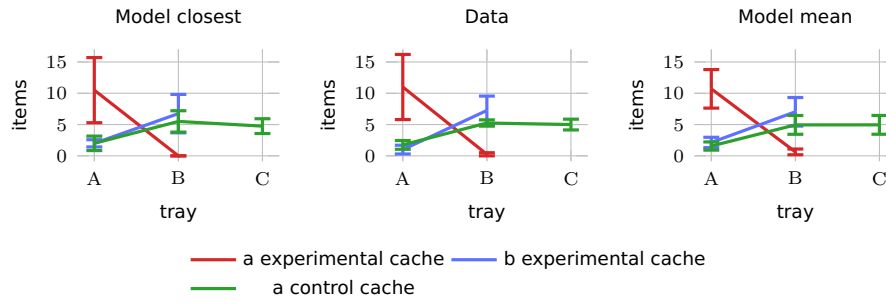


Figure S43: deKort07 exp4: Planning-By-Replay Model

2.1.8 Clayton05 exp1

Published in [8].

Major Finding

Birds reduce caching foodtypes that are always degraded at recovery, but keep caching when the foodtypes are fresh after some retention intervals.

Most Important Tests

Group partial degrade (foodtype): $F(1, 3) = 85.15, p = 0.0027$

Group consistent degrade (trial & foodtype): $F(5, 15) = 12.24, p = 7.4 \cdot 10^{-5}$

Group replenish (trial & foodtype): $F(5, 15) = 1.34, p = 0.3$

Total Number of Statistics

14 p-values, 36 means or sems.

Figures

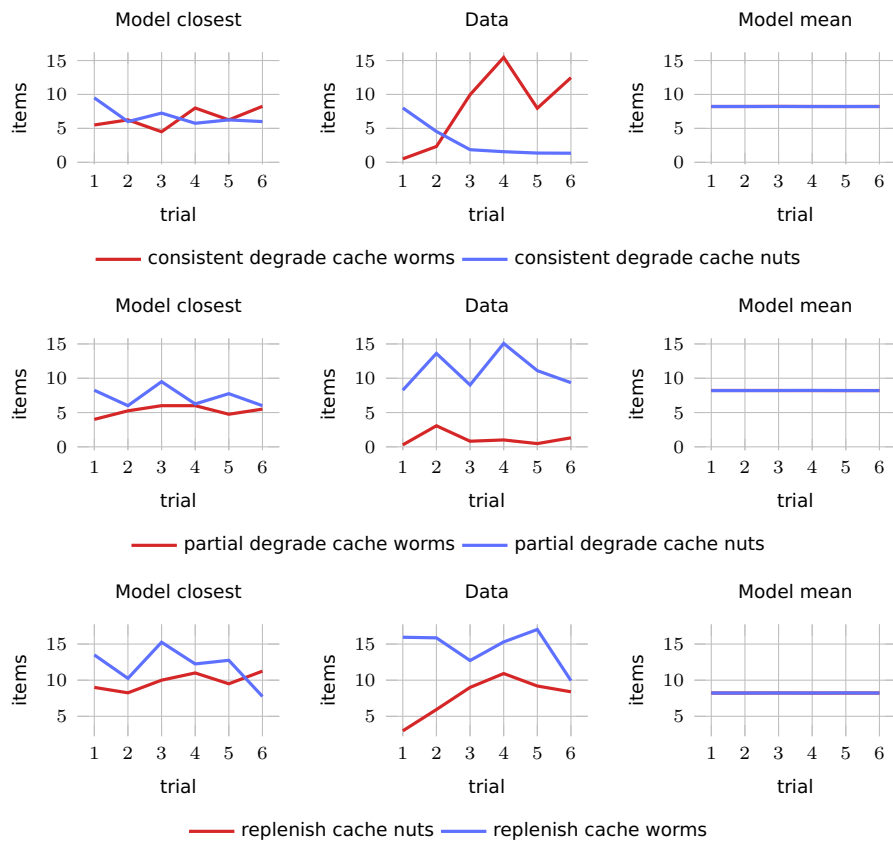


Figure S44: Clayton05 exp1: No-Plasticity-No-Memory-No-Motivational-Control Model

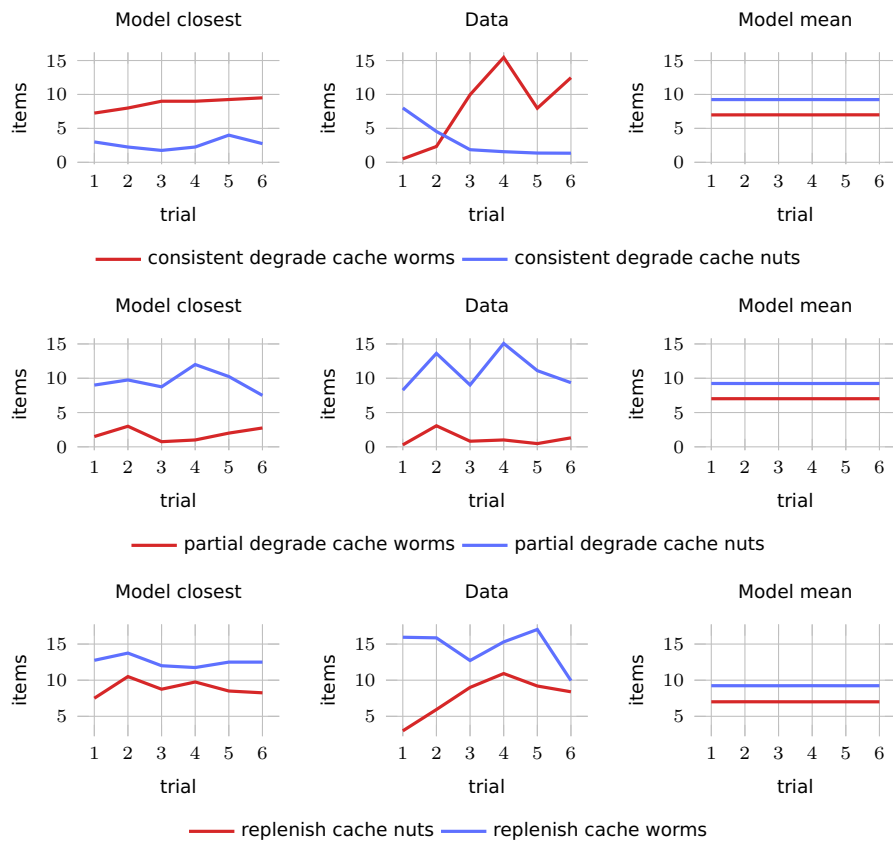


Figure S45: **Clayton05 exp1: No-Plasticity-No-Memory Model**

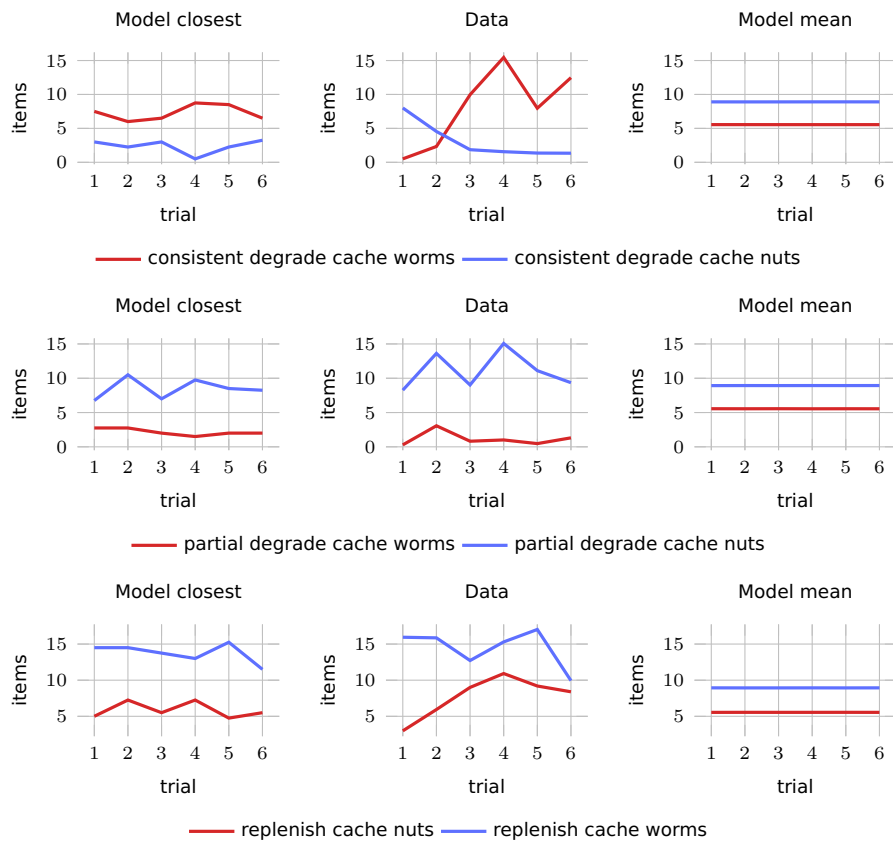


Figure S46: Clayton05 exp1: No-Plasticity Model

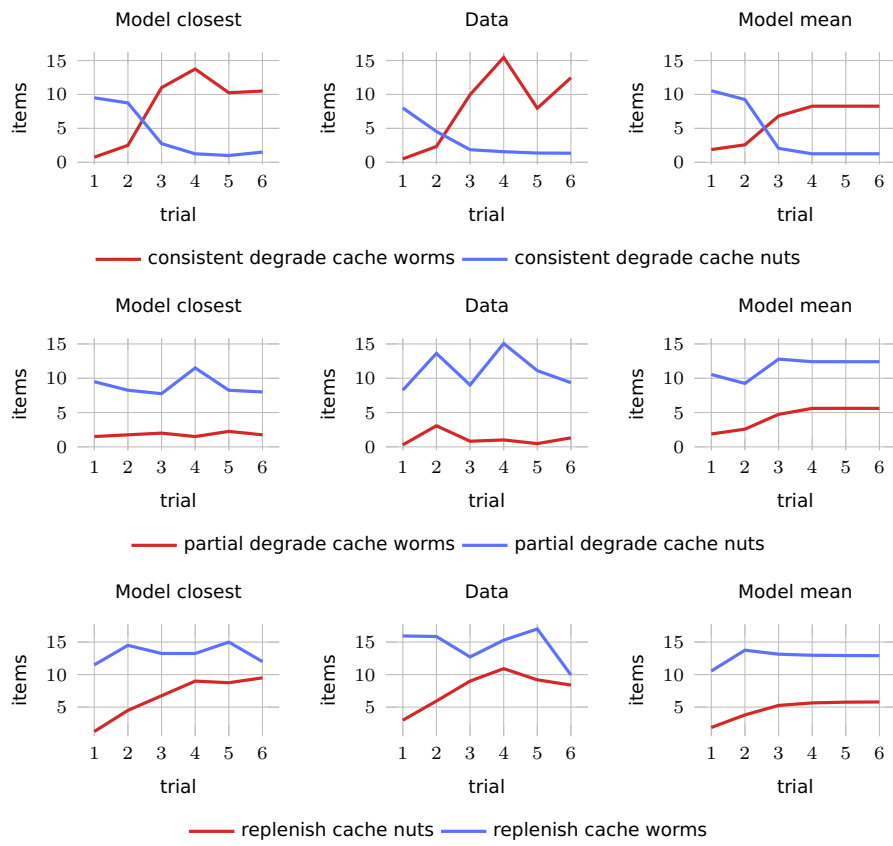


Figure S47: Clayton05 exp1: Plastic Caching Model

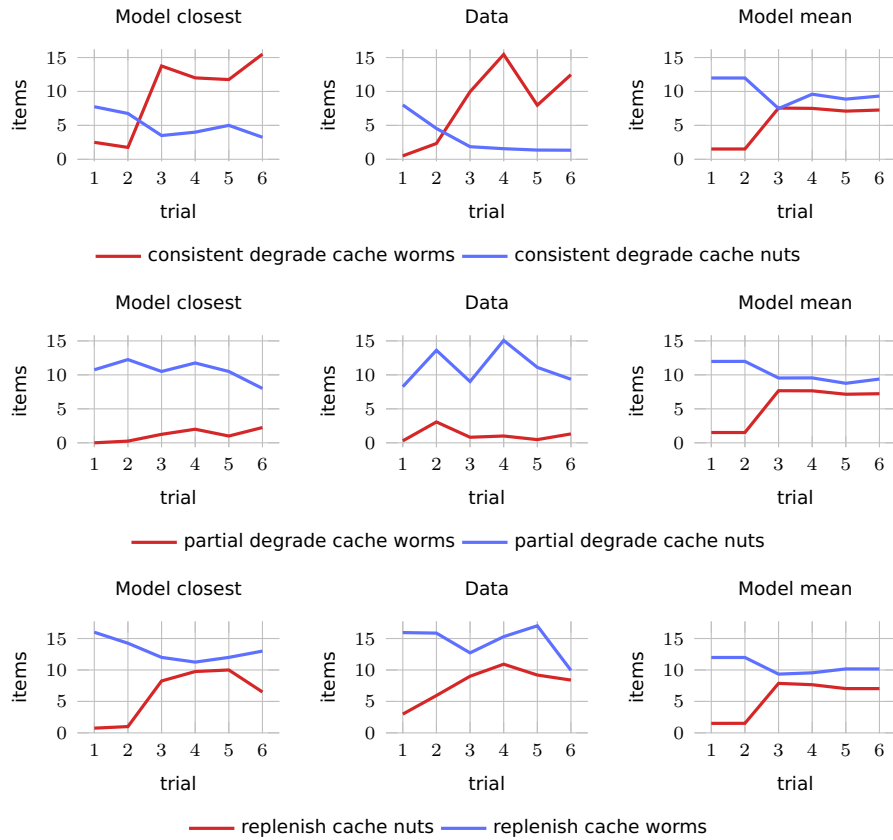


Figure S48: Clayton05 exp1: Planning-By-Replay Model

2.1.9 Clayton05 exp2

Published in [8].

Major Finding

Replication of experiment one showing additionally that consistent and partial pilfering has a similar effect on caching as degradation.

Comments

For quite a few statistical tests we were not able to reconstruct the reported degrees of freedom.

Most Important Tests

Change in caching worms replenish (trial): $F(7, 63) < 1.06, p = 0.41$

Change in caching worms consistent degrade (trial): $F(7, 63) = 4.39, p = 0.00051$

Change in caching worms partial degrade (trial): $F(7, 63) < 1.06, p = 0.41$

Total Number of Statistics

19 p-values, 80 means or sems.

Figures

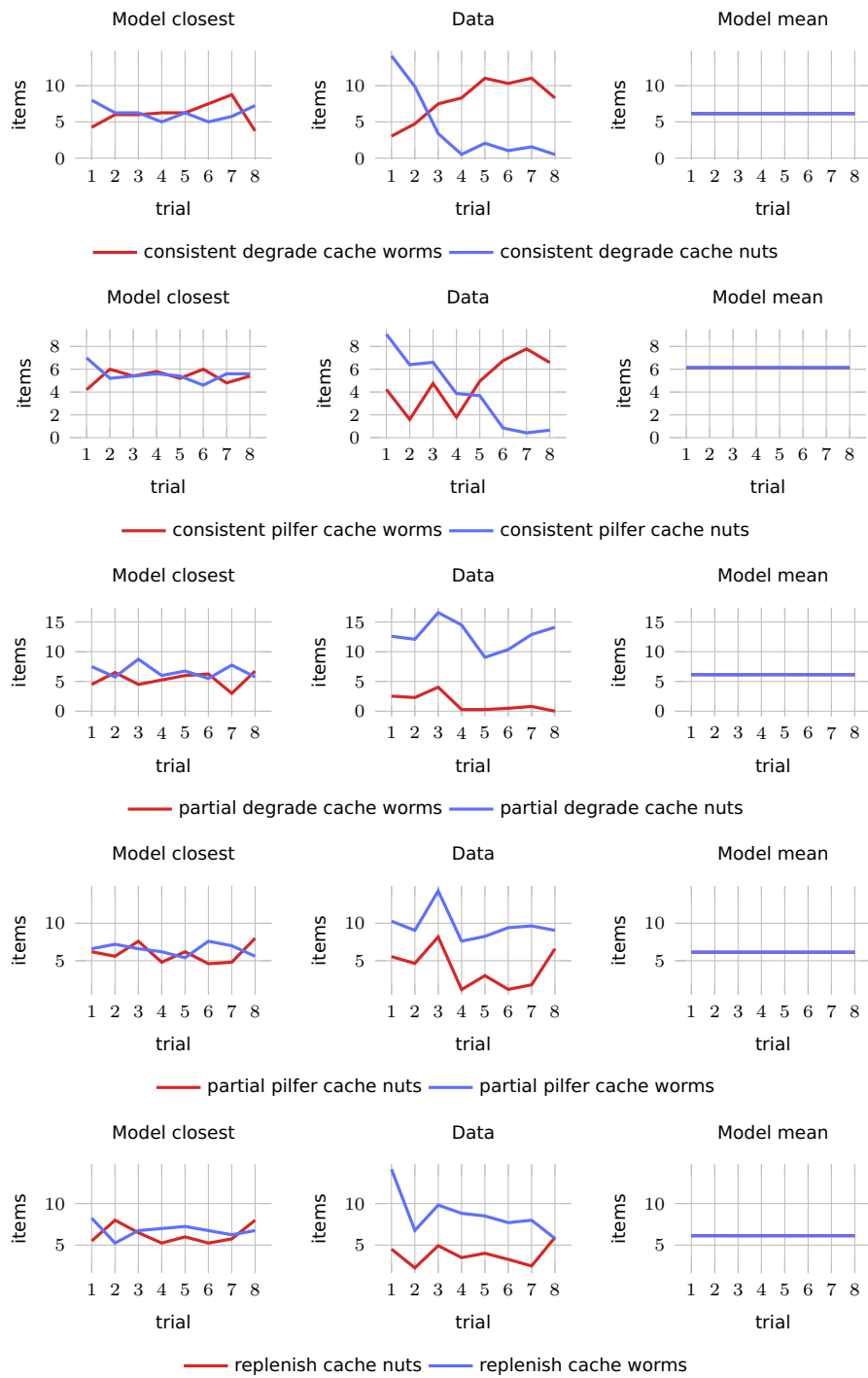


Figure S49: **Clayton05 exp2: No-Plasticity-No-Memory-No-Motivational-Control Model**

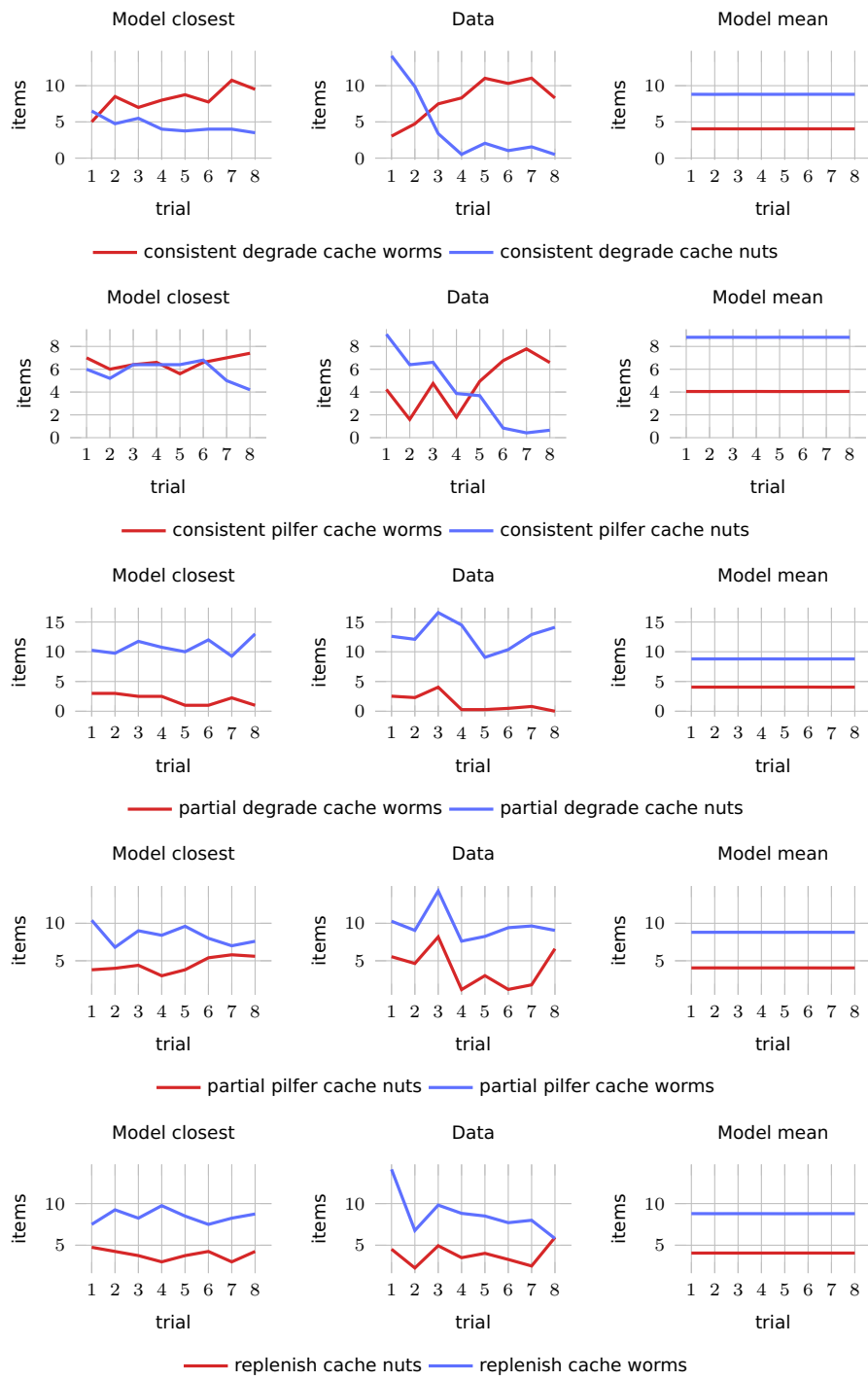


Figure S50: **Clayton05 exp2: No-Plasticity-No-Memory Model**

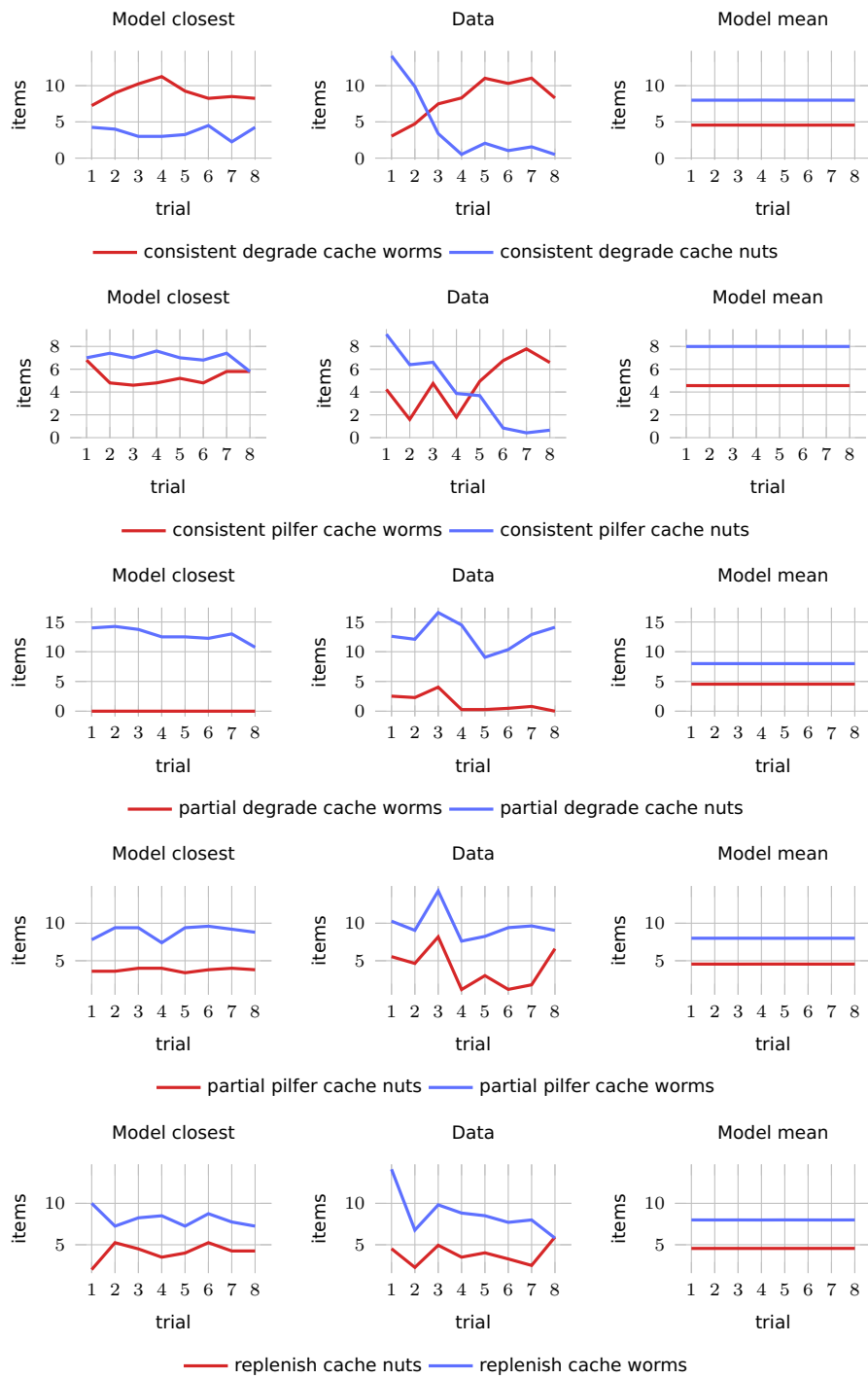


Figure S51: Clayton05 exp2: No-Plasticity Model

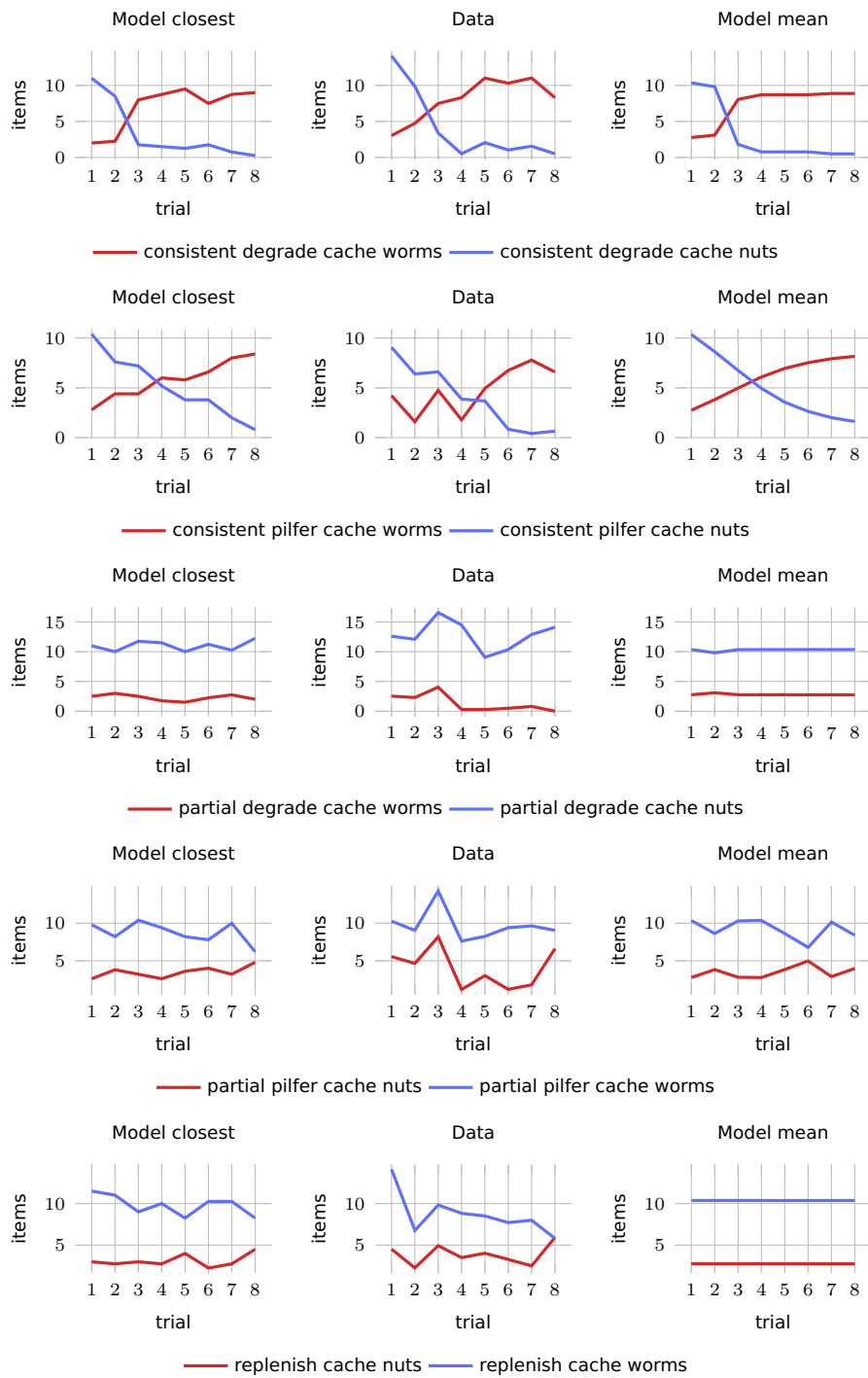


Figure S52: Clayton05 exp2: Plastic Caching Model

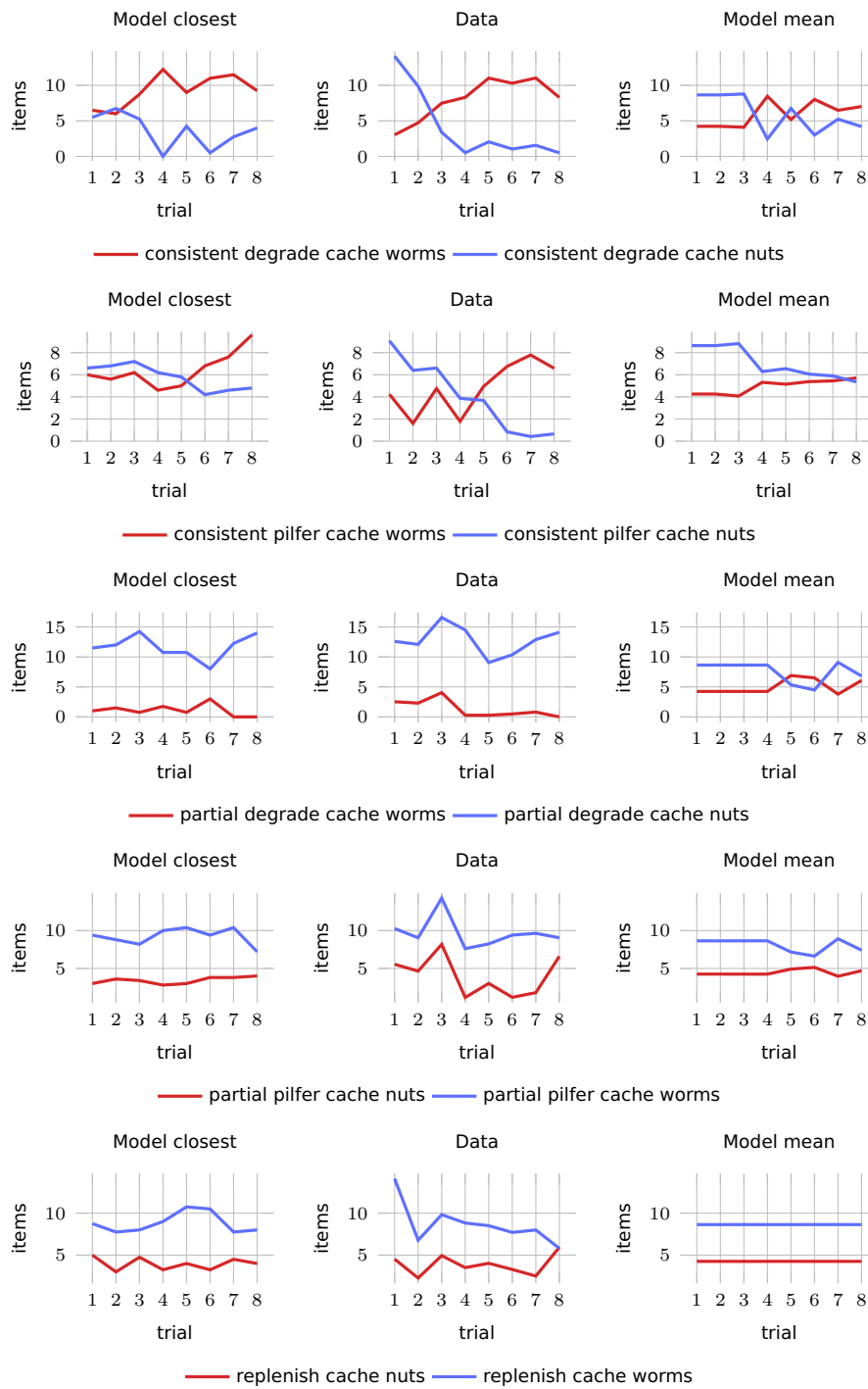


Figure S53: Clayton05 exp2: Planning-By-Replay Model

2.1.10 Clayton05 exp3

Published in [8].

Major Finding

Birds reduced caching of unfamiliar food types (Pineapple, Salami), if items of this type were consistently degraded at recovery.

Most Important Tests

partial degrade (foodtype & trial): $F(9, 54) < 1, p = 0.46$

overall (group & foodtype & trial): $F(9, 90) = 1.99, p = 0.049$

consistent degrade (foodtype & trial): $F(9, 36) = 2.75, p = 0.015$

Total Number of Statistics

20 p-values, 42 means or sems.

Figures

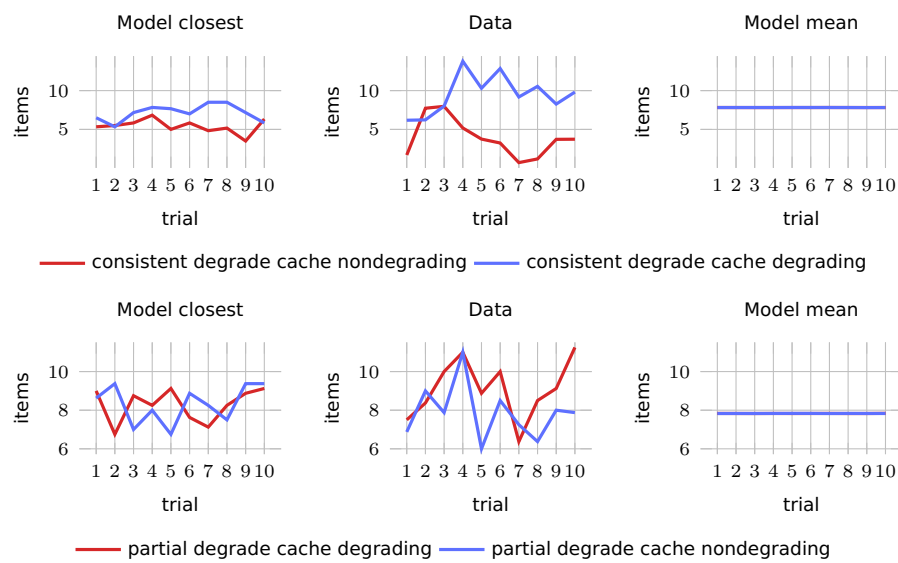


Figure S54: **Clayton05 exp3: No-Plasticity-No-Memory-No-Motivational-Control Model**

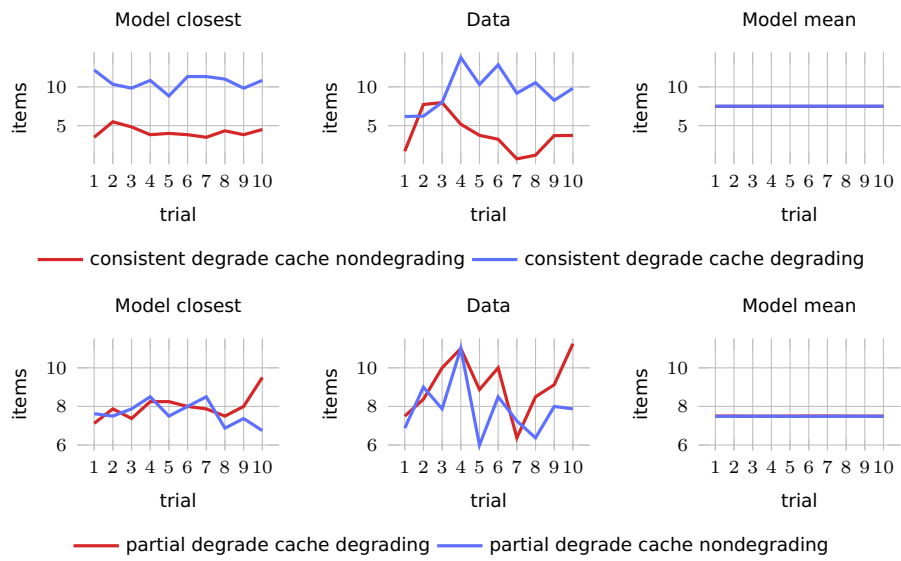


Figure S55: **Clayton05 exp3: No-Plasticity-No-Memory Model**

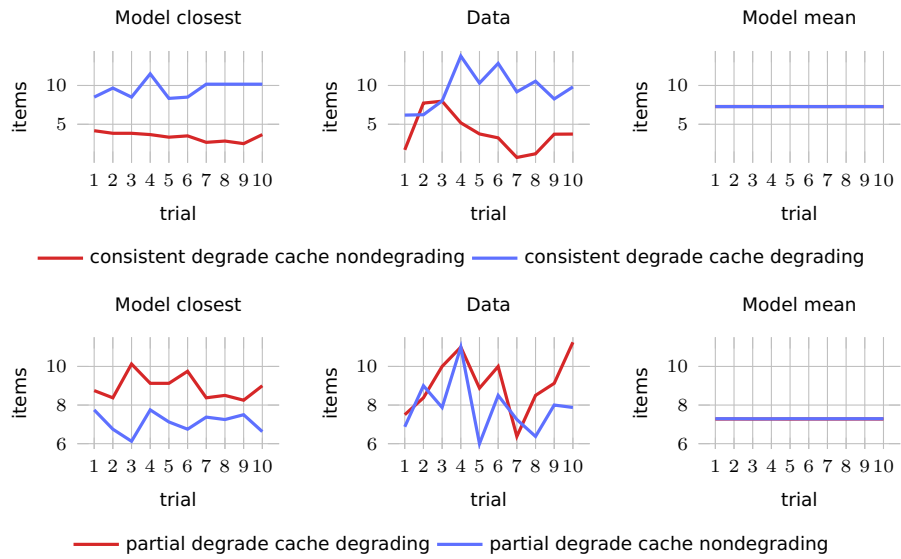


Figure S56: **Clayton05 exp3: No-Plasticity Model**

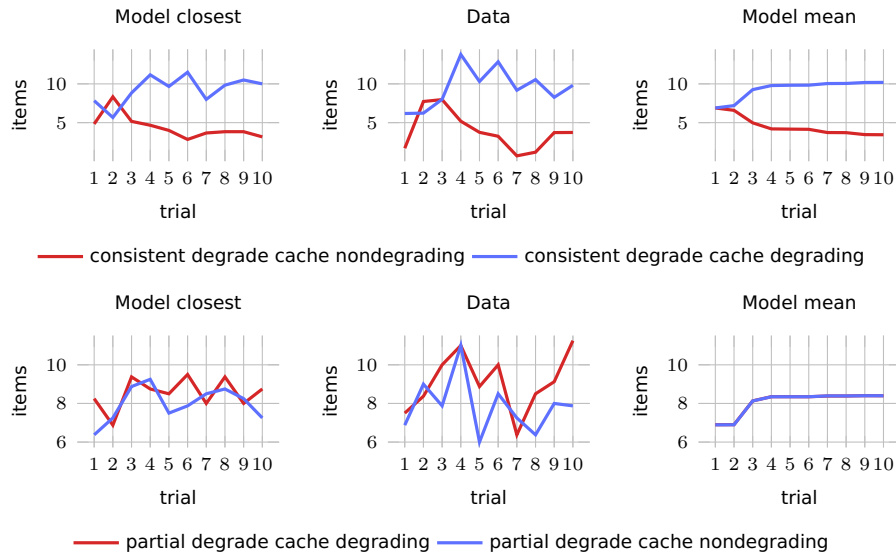


Figure S57: Clayton05 exp3: Plastic Caching Model

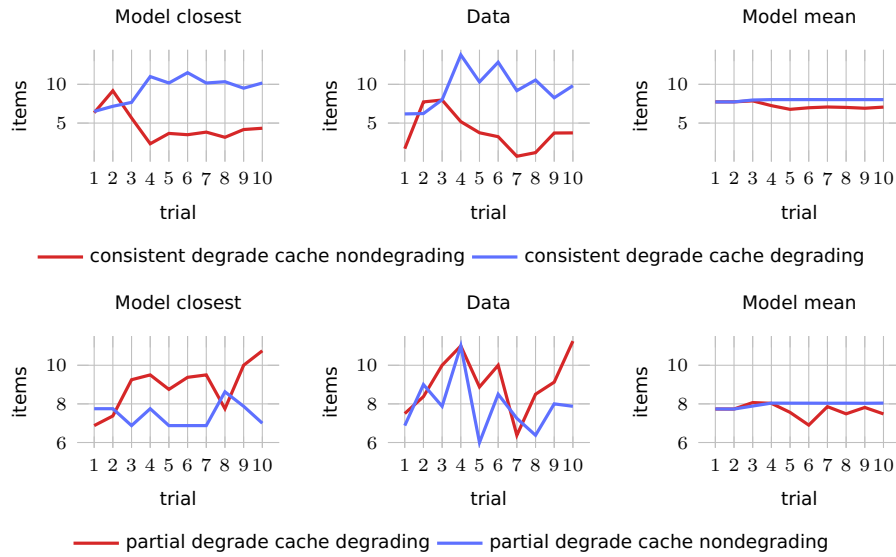


Figure S58: Clayton05 exp3: Planning-By-Replay Model

2.1.11 Clayton05 exp4

Published in [8].

Major Finding

The birds reduce caching items of the food type that was consistently degraded, even if the shortest retention interval was longer than two days.

Most Important Tests

overall (foodtype & trial): $F(3, 12) = 12.83, p = 0.00047$

trial 4 (foodtype): $F(1, 4) = 18.42, p = 0.013$

worms (trial): $F(3, 12) = 16.93, p = 0.00013$

Total Number of Statistics

5 p-values, 8 means or sems.

Figures

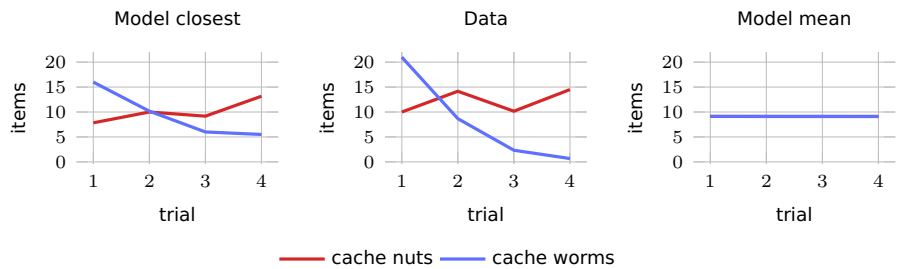


Figure S59: **Clayton05 exp4: No-Plasticity-No-Memory-No-Motivational-Control Model**

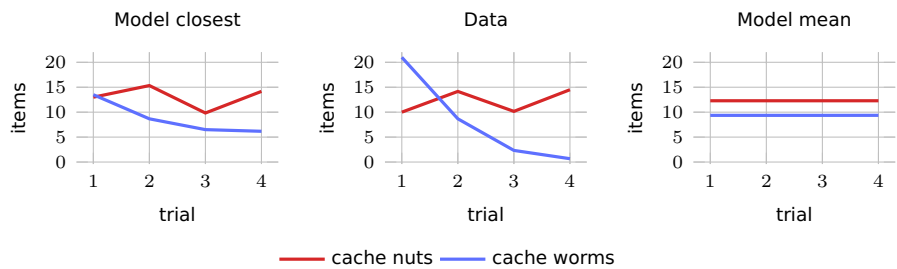


Figure S60: **Clayton05 exp4: No-Plasticity-No-Memory Model**

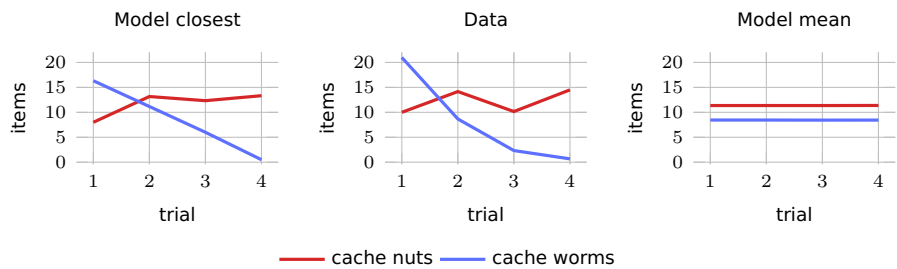


Figure S61: **Clayton05 exp4: No-Plasticity Model**

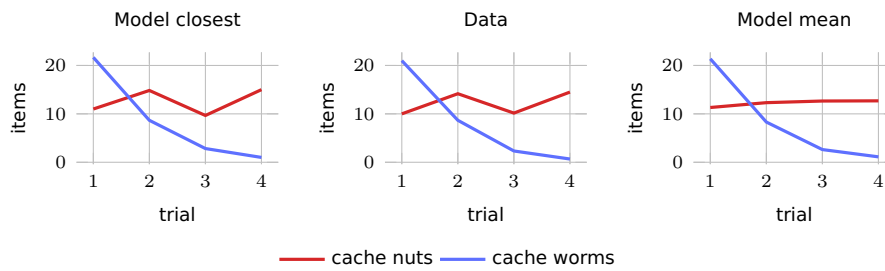


Figure S62: Clayton05 exp4: Plastic Caching Model

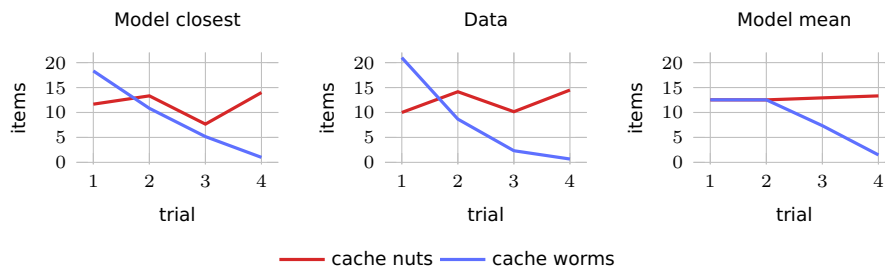


Figure S63: Clayton05 exp4: Planning-By-Replay Model

2.1.12 Clayton99A exp1

Published in [3].

Major Finding

Birds inspect preferentially caching sites where the previously cached food is of different type than the one they had been satiated on prior to recovery.

Most Important Tests

172h overall inspect (tray): $F(1, 8) = 5.73, p = 0.044$

4h overall cache (tray): $F(1, 6) < 5.17, p = 0.064$

4h overall inspect (tray): $F(1, 6) = 11.32, p = 0.015$

Total Number of Statistics

44 p-values, 16 means or sems.

Figures

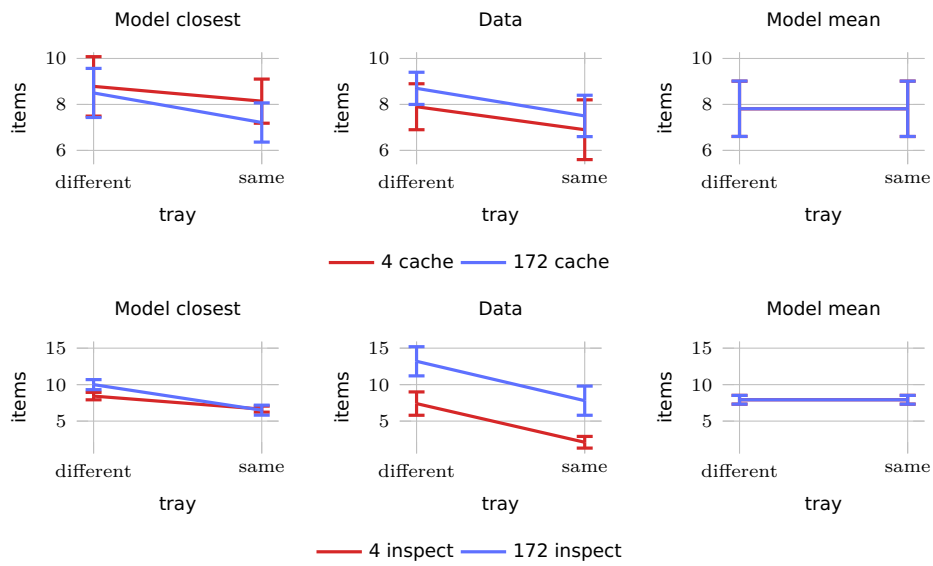


Figure S64: **Clayton99A exp1: No-Plasticity-No-Memory-No-Motivational-Control Model**

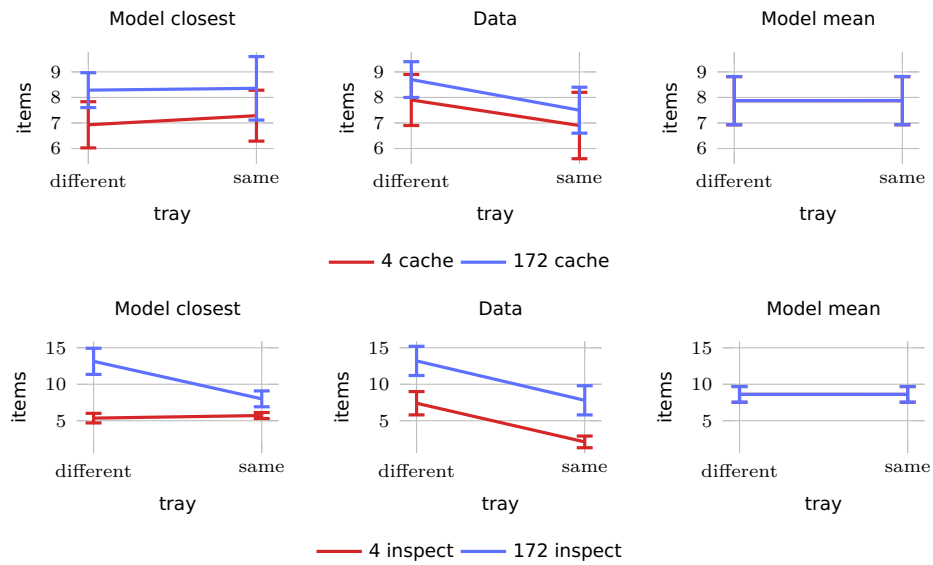


Figure S65: **Clayton99A exp1: No-Plasticity-No-Memory Model**

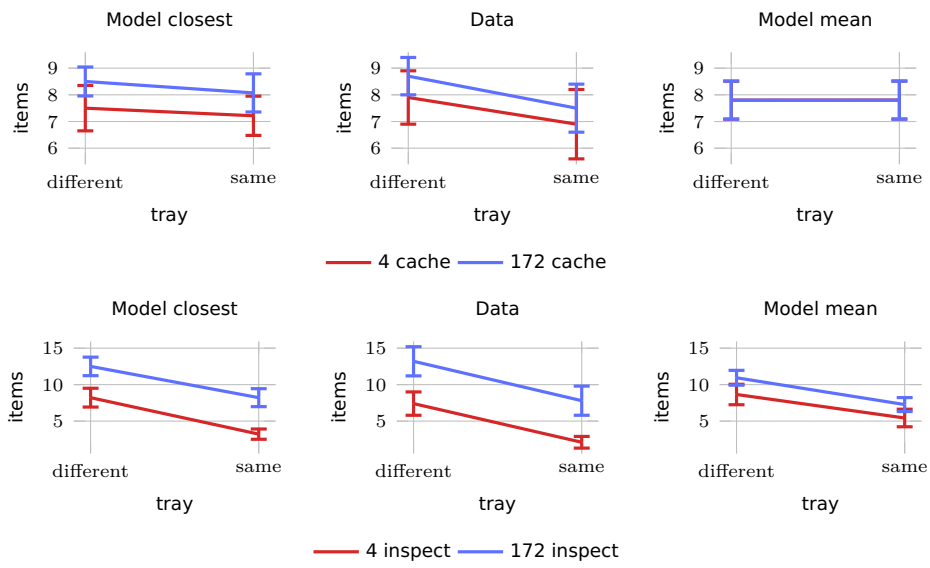


Figure S66: Clayton99A exp1: No-Plasticity Model

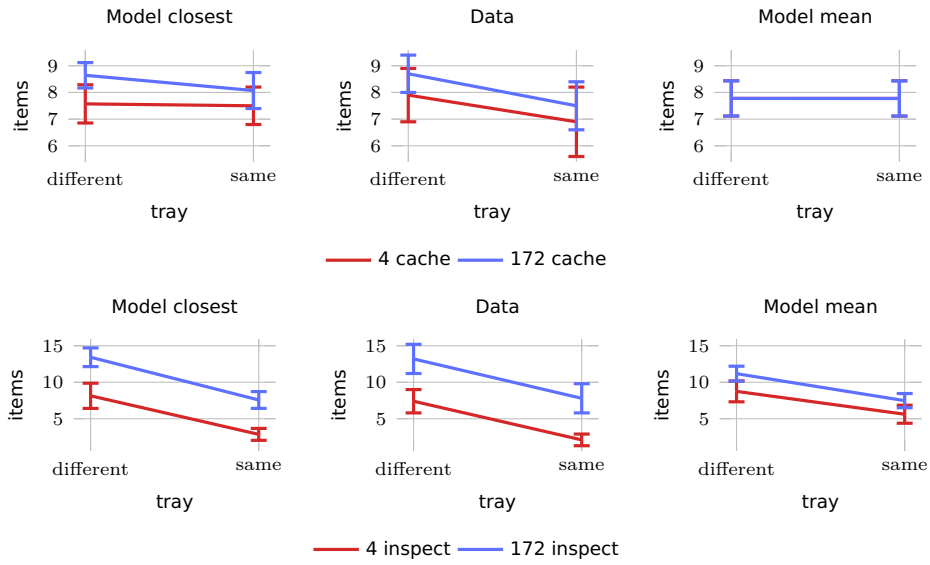


Figure S67: Clayton99A exp1: Plastic Caching Model

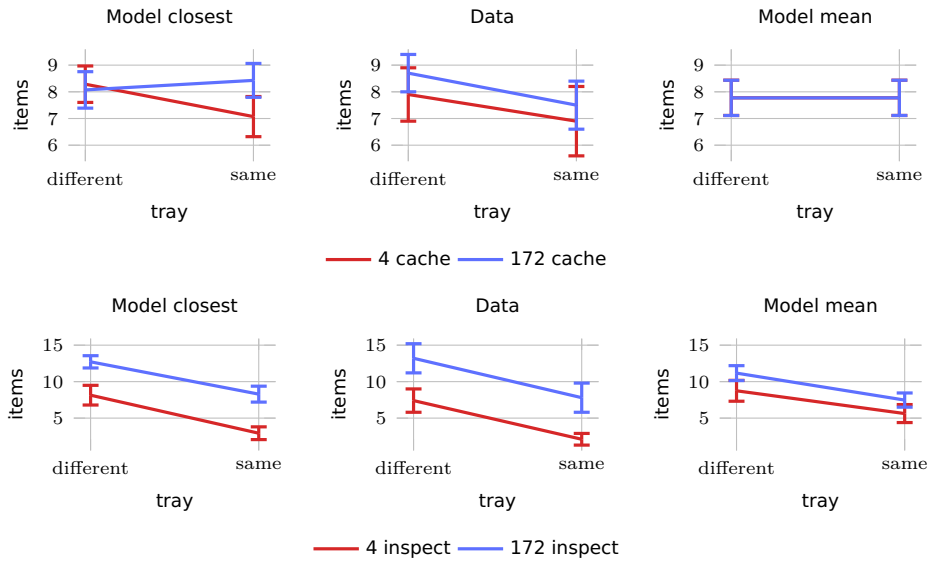


Figure S68: **Clayton99A exp1: Planning-By-Replay Model**

2.1.13 Clayton99A exp2

Published in [3].

Major Finding

Birds direct their searches to sites from which they did not yet recover the food items and contain food that is different from what they had just been prefed on.

Most Important Tests

inspect (tray): $F(1, 10) = 21.59, p = 0.00091$

inspect (side): $F(1, 10) = 74.39, p = 6.1 \cdot 10^{-6}$

inspect (tray & side): $F(1, 10) = 20.27, p = 0.0011$

Total Number of Statistics

15 p-values, 8 means or sems.

Figures

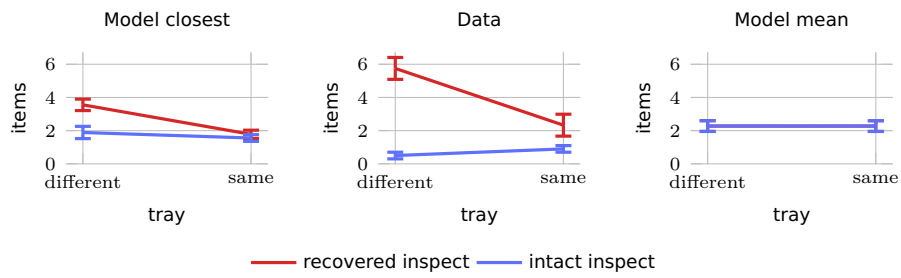


Figure S69: **Clayton99A exp2: No-Plasticity-No-Memory-No-Motivational-Control Model**

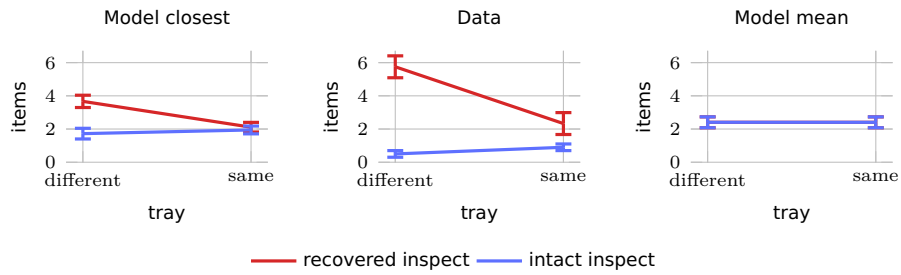


Figure S70: **Clayton99A exp2: No-Plasticity-No-Memory Model**

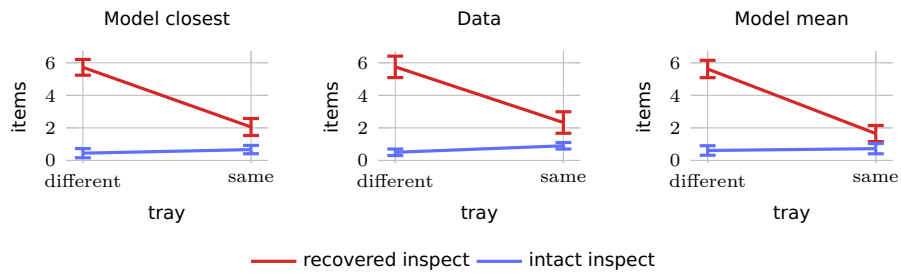


Figure S71: **Clayton99A exp2: No-Plasticity Model**

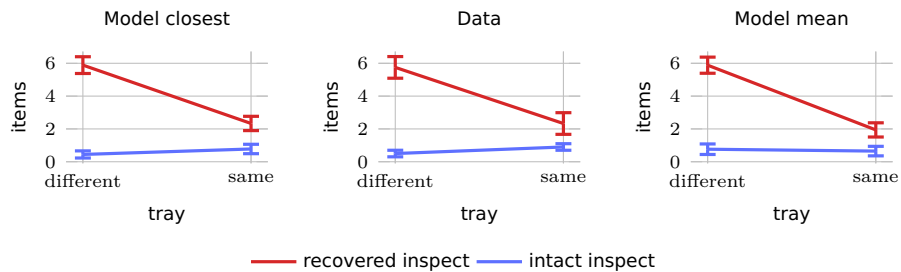


Figure S72: **Clayton99A exp2: Plastic Caching Model**

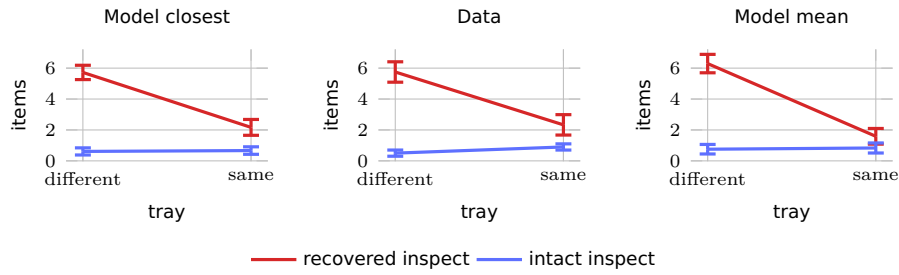


Figure S73: Clayton99A exp2: Planning-By-Replay Model

2.1.14 Clayton99B exp1

Published in [4].

Major Finding

The birds learned that their preferred food perished in the length of time since caching whereas their less preferred food did not: after some training trials, birds in the degrade group searched more at sites where they had previously cached their preferred food than at sites where they had previously cached their less preferred food after a retention interval of 4 hours and vice versa after a retention interval of 128 hours.

Comments

Birds in the degrade group cached progressively more wax worms than peanuts during training; a pattern not seen in Clayton05 experiment 1 where they continued to cache at roughly the same level.

Most Important Tests

test 4 (foodtype): $F(1, 17) = 18.21, p = 0.00052$

test 124 (group & foodtype): $F(2, 17) = 20.02, p = 3.4 \cdot 10^{-5}$

test (group & RI & foodtype): $F(2, 14) = 6.71, p = 0.009$

Total Number of Statistics

28 p-values, 184 means or sems.

Figures

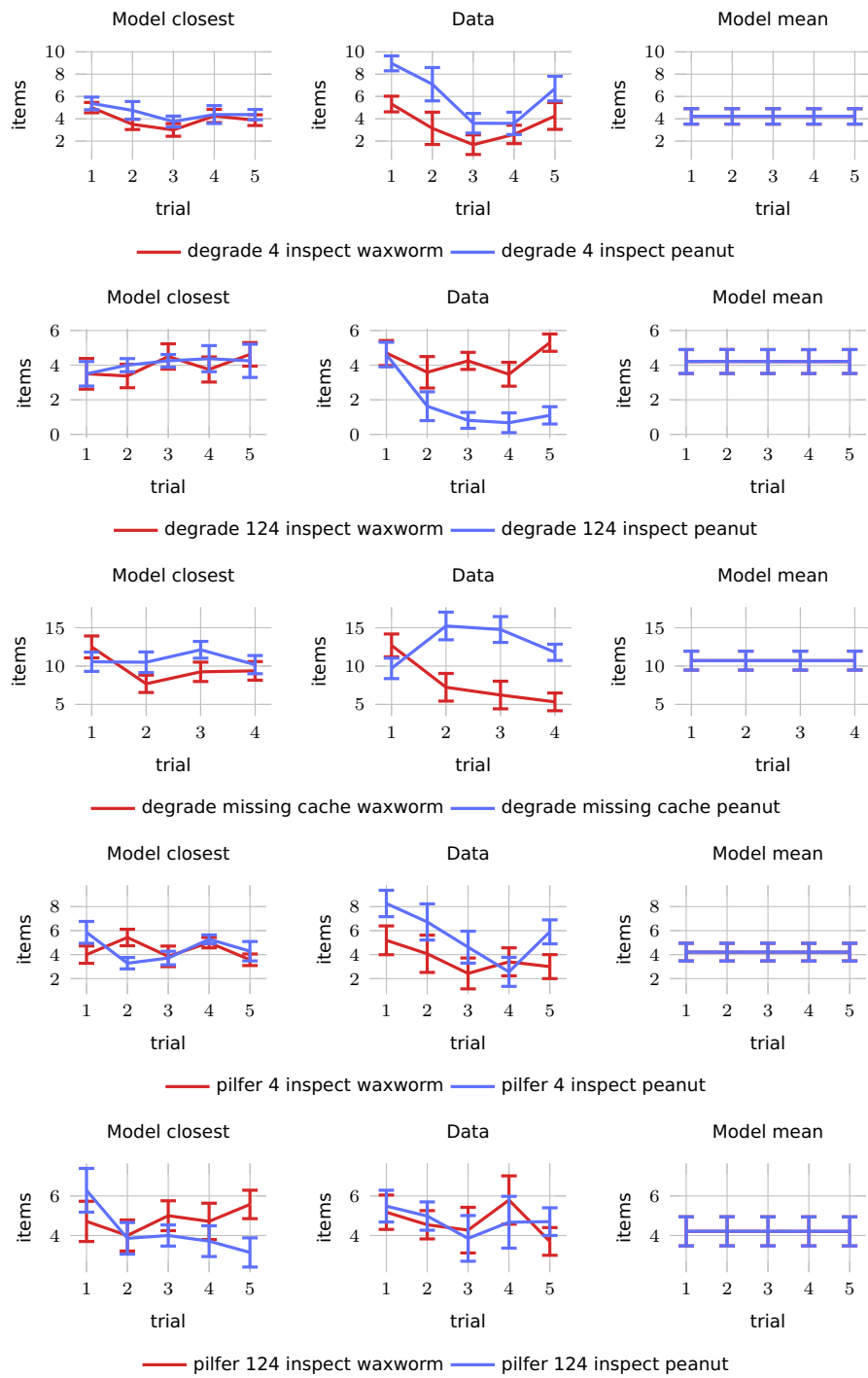


Figure S74: Clayton99B exp1: No-Plasticity-No-Memory-No-Motivational-Control Model (part1)

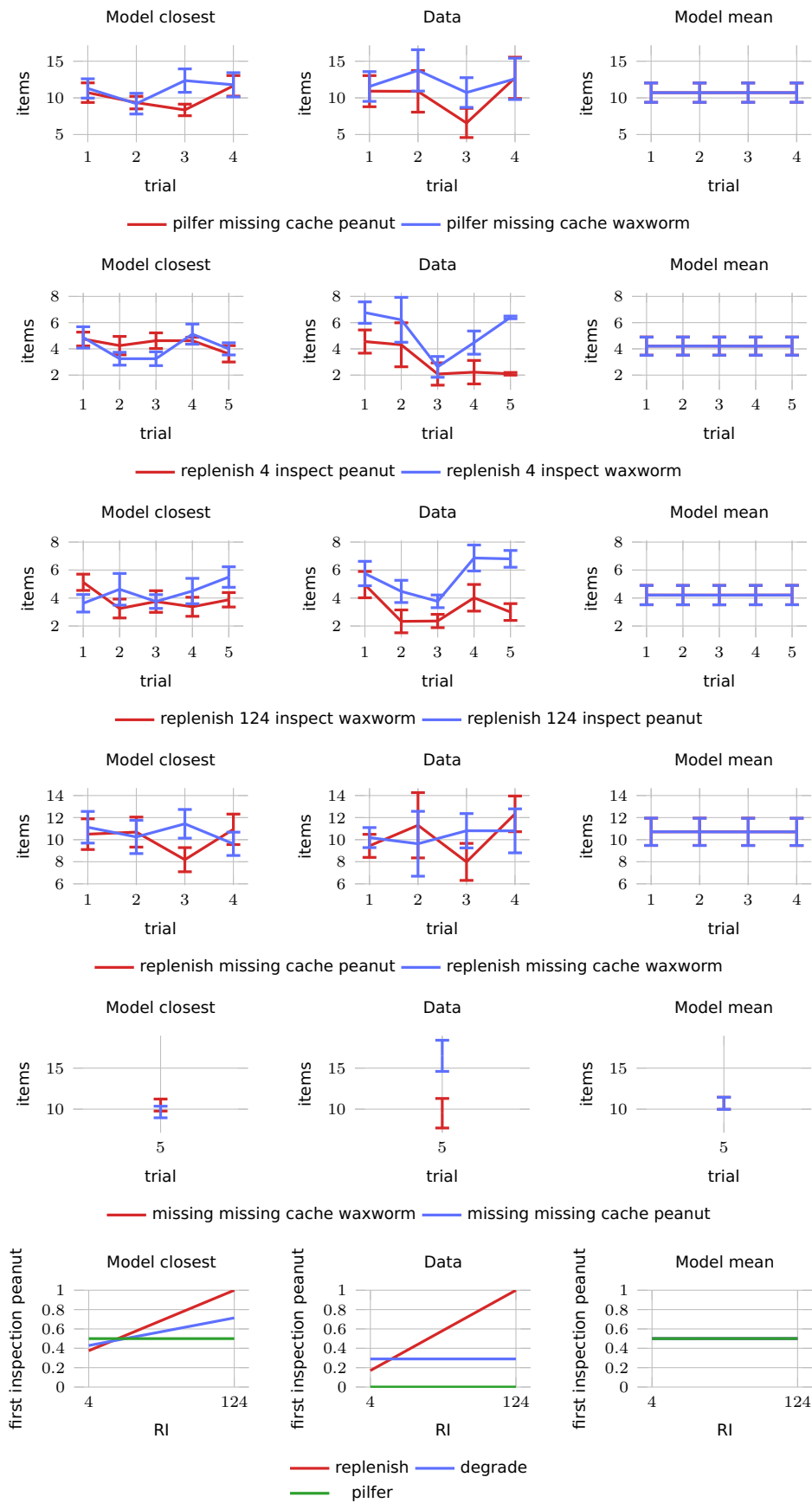


Figure S75: Clayton99B exp1: No-Plasticity-No-Memory-No-Motivational-Control Model (part2)

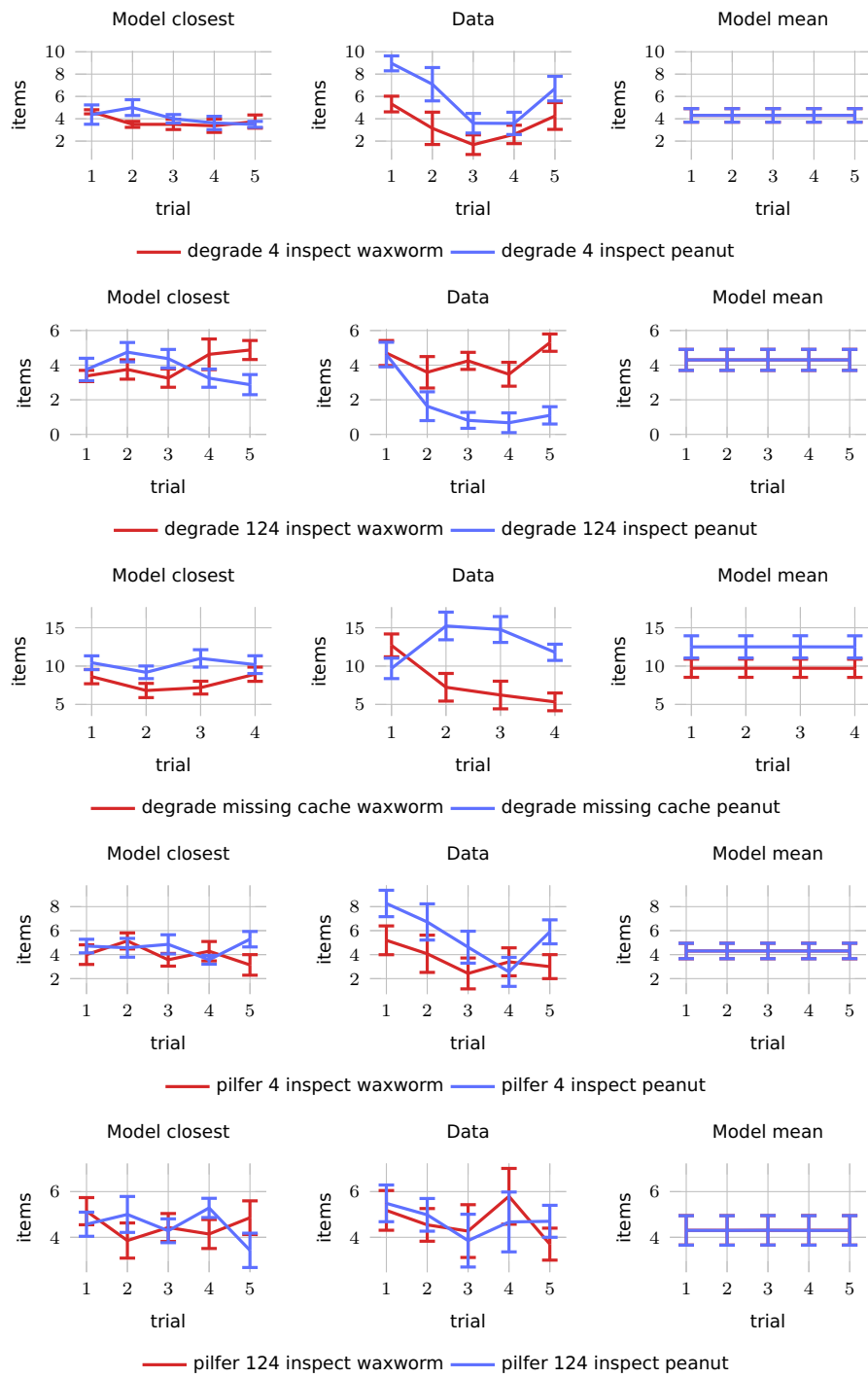


Figure S76: Clayton99B exp1: No-Plasticity-No-Memory Model (part1)

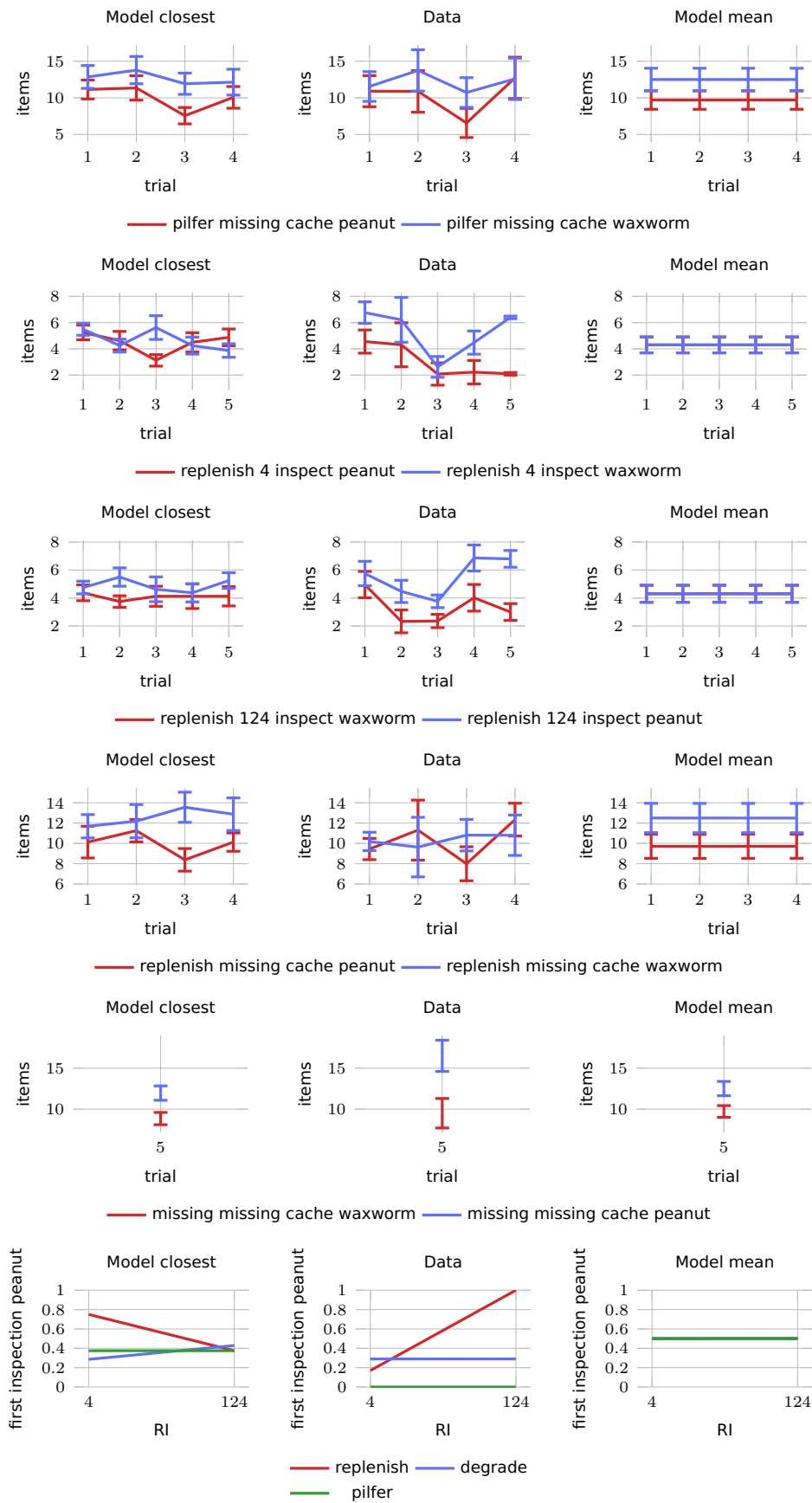


Figure S77: Clayton99B exp1: No-Plasticity-No-Memory Model (part2)

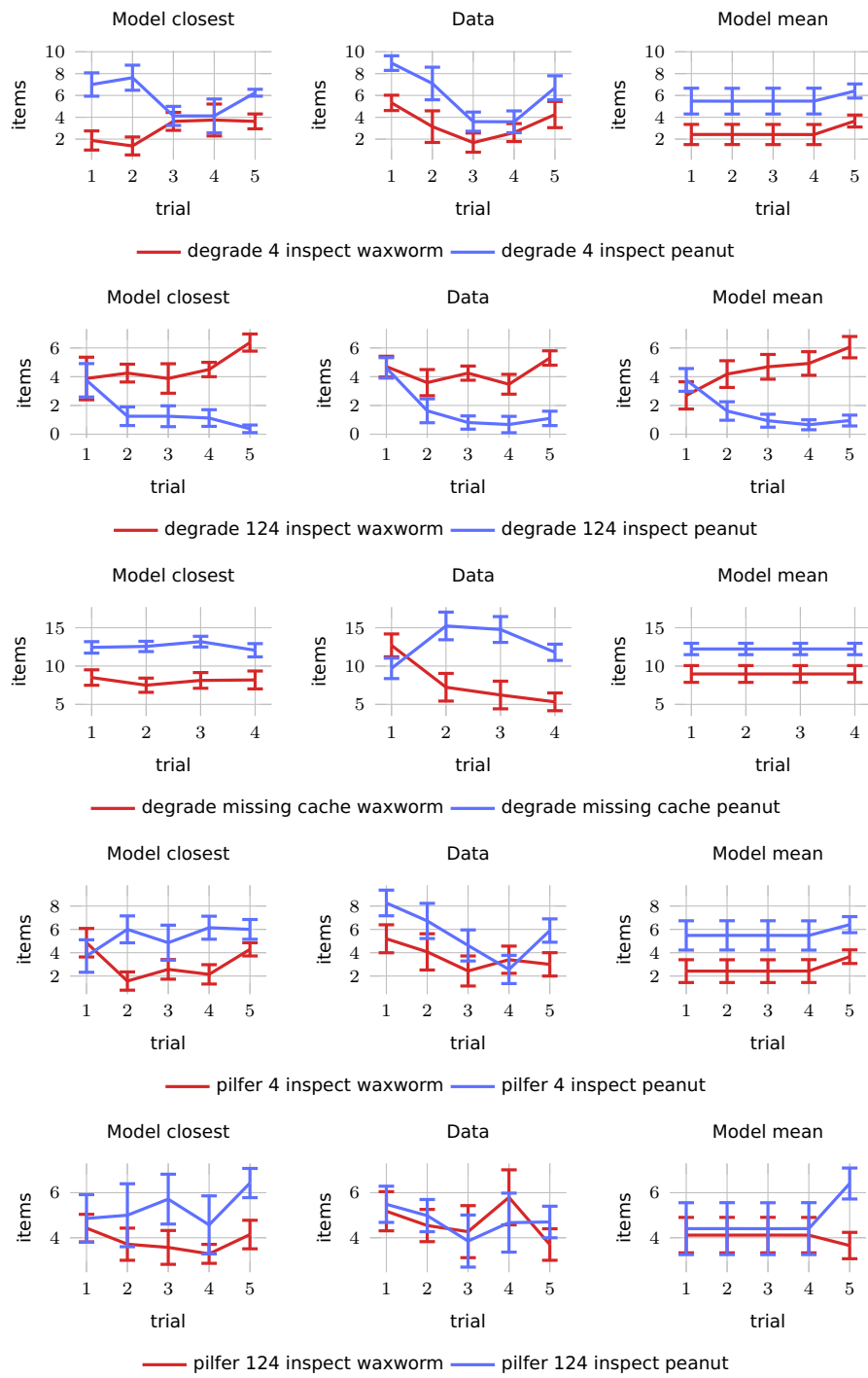


Figure S78: Clayton99B exp1: No-Plasticity Model (part1)

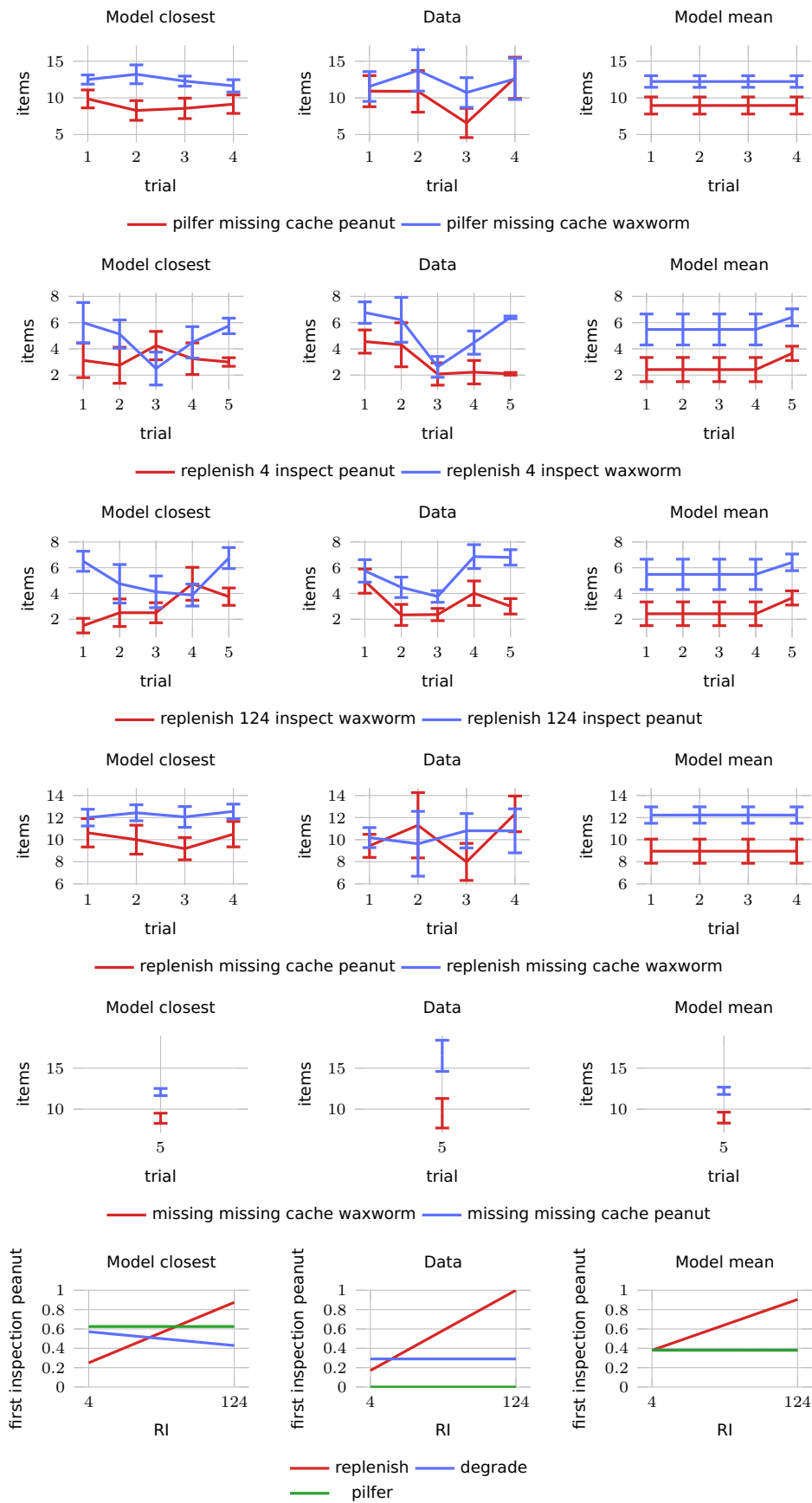


Figure S79: Clayton99B exp1: No-Plasticity Model (part2)

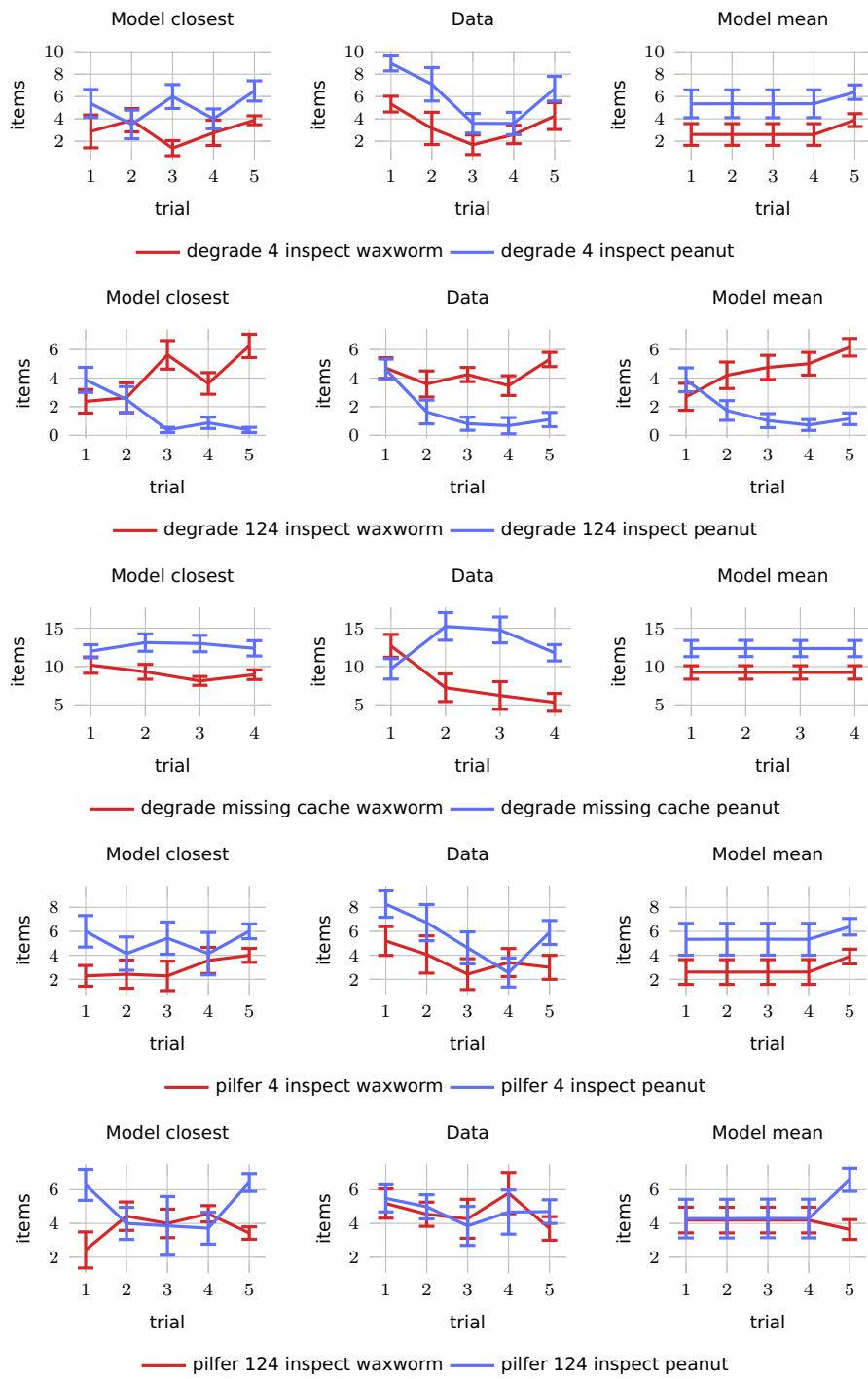


Figure S80: Clayton99B exp1: Plastic Caching Model (part1)

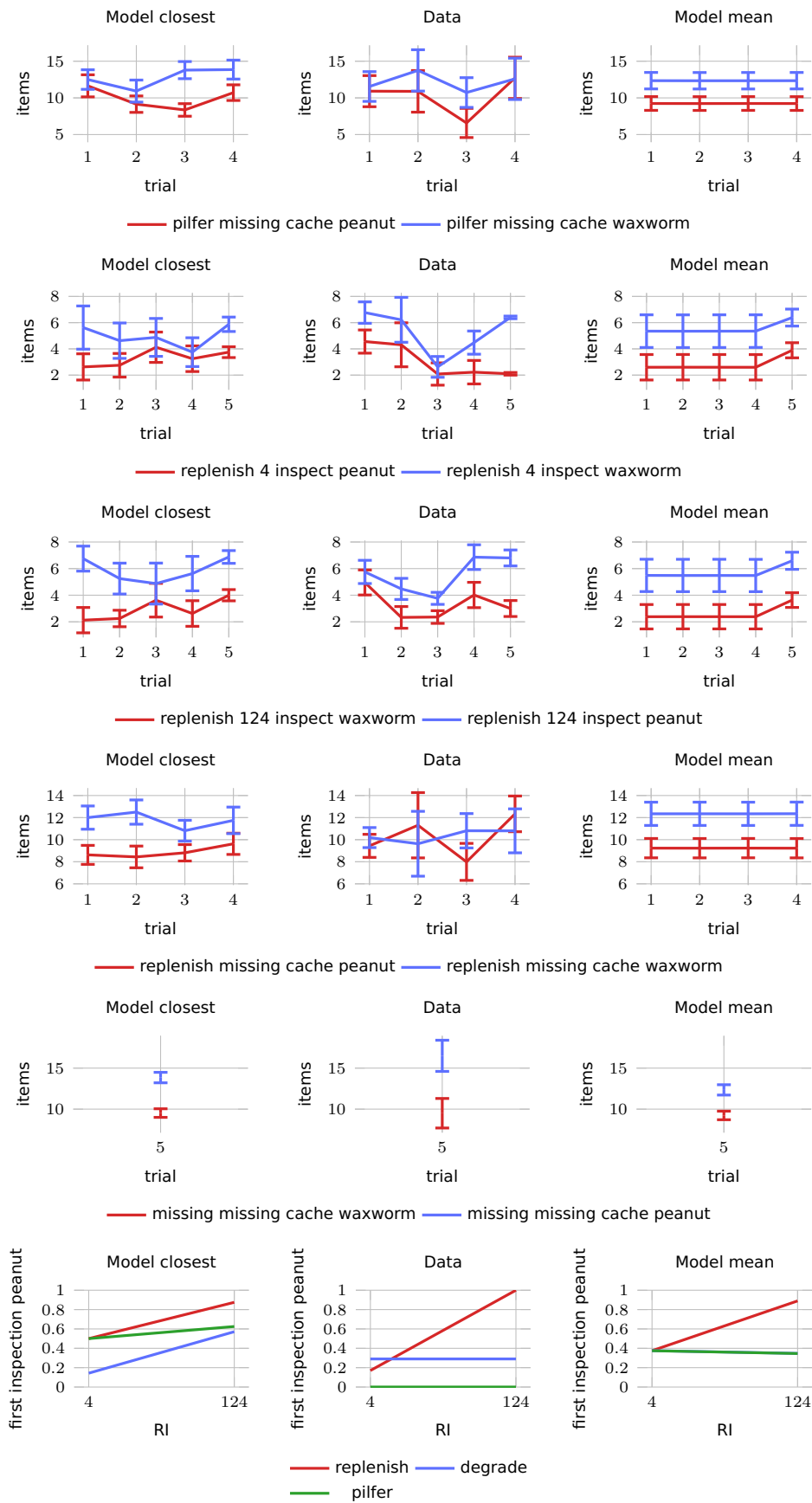


Figure S81: Clayton99B exp1: Plastic Caching Model (part2)

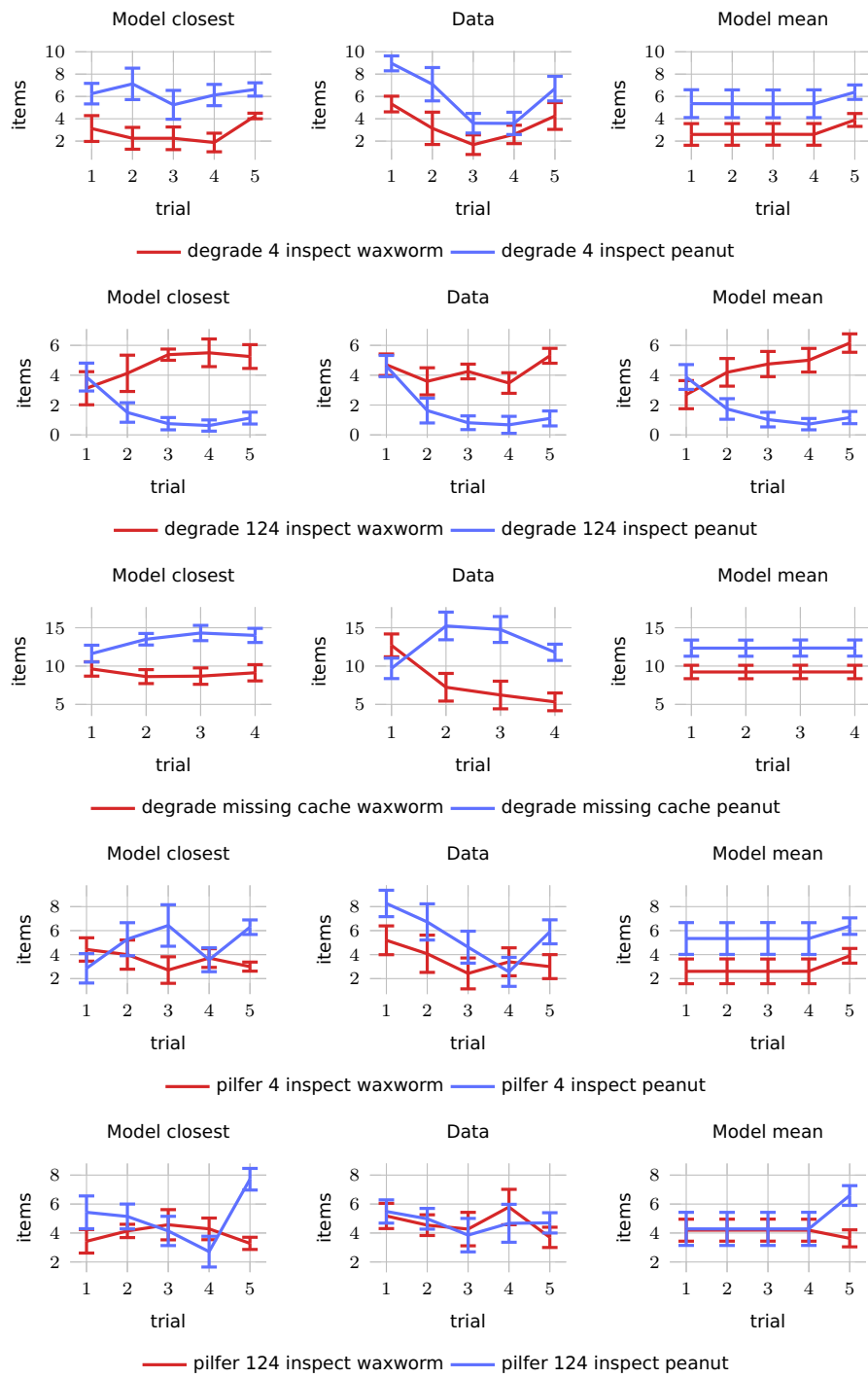


Figure S82: Clayton99B exp1: Planning-By-Replay Model (part1)

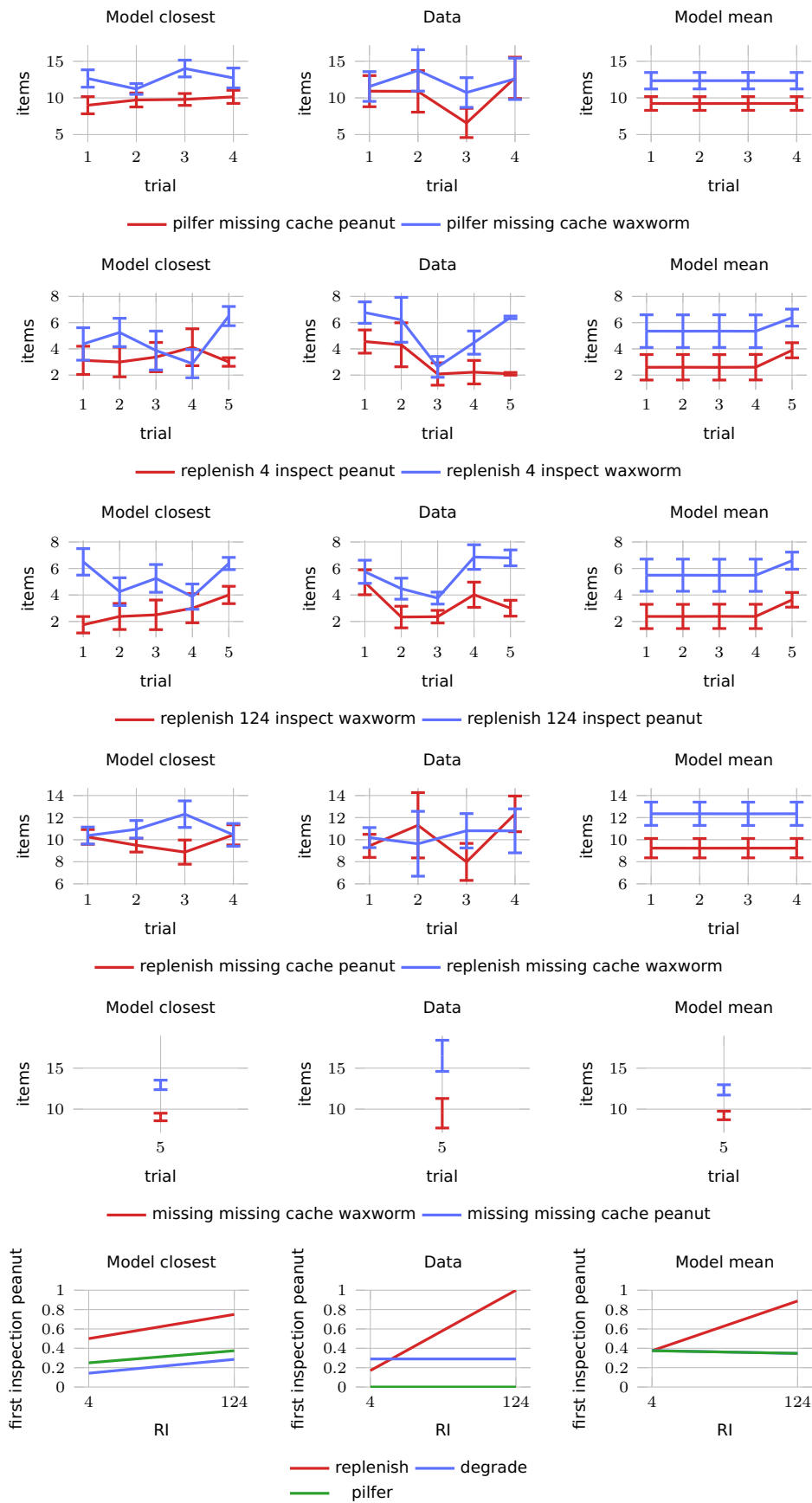


Figure S83: Clayton99B exp1: Planning-By-Replay Model (part2)

2.1.15 Clayton99B exp2

Published in [4].

Major Finding

Also when the birds cache in two sides of the same ice-cube tray they adapt their behaviour as in experiment 1.

Most Important Tests

test inspect (order): $F(1, 14) = 7.0, p = 0.019$

test inspect wp (group & foodtype): $F(2, 14) = 9.59, p = 0.0024$

test inspect pw (group & foodtype): $F(2, 14) < 1, p = 0.4$

Total Number of Statistics

31 p-values, 92 means or sems.

Figures

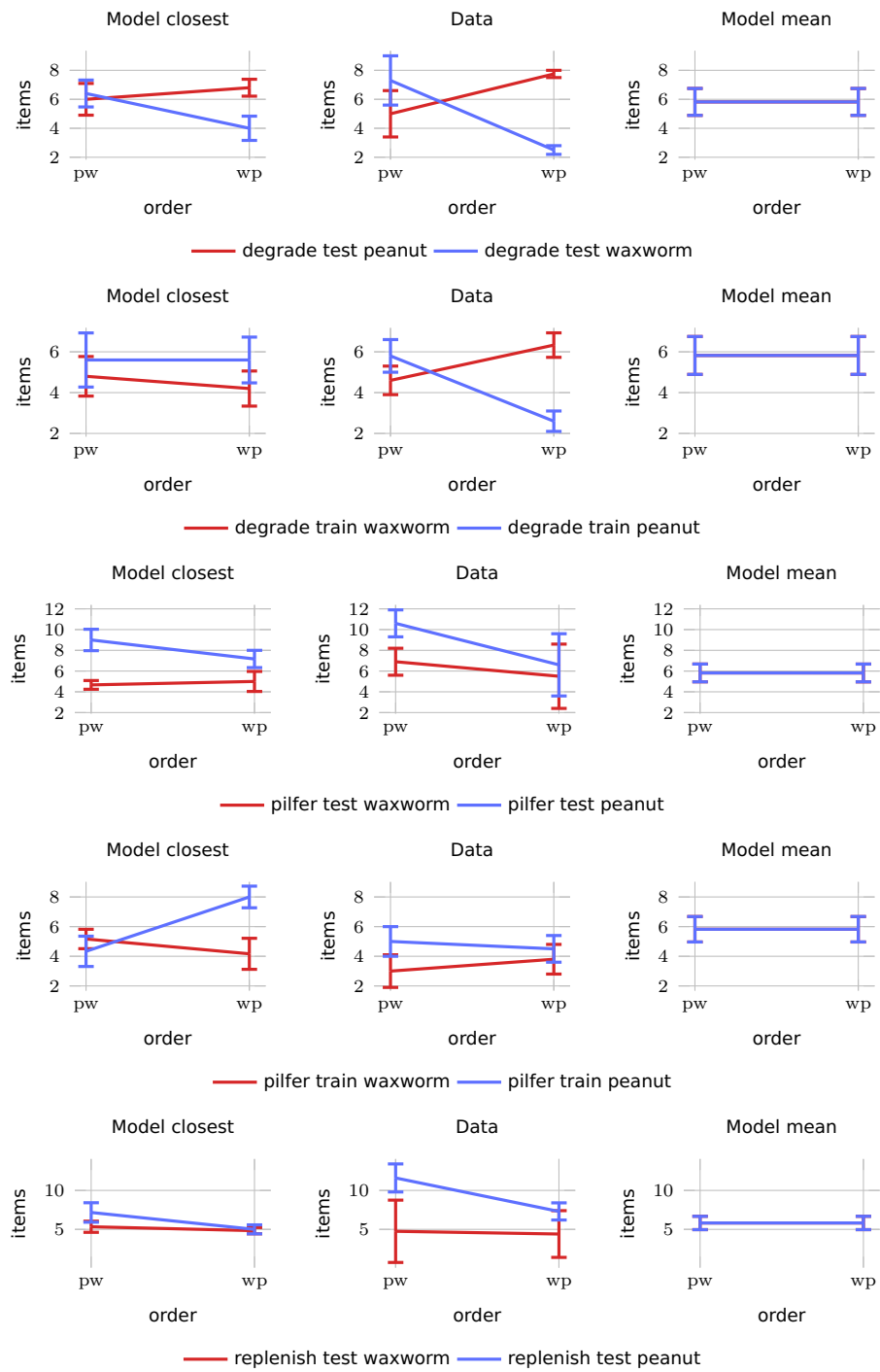


Figure S84: Clayton99B exp2: No-Plasticity-No-Memory-No-Motivational-Control Model (part1)

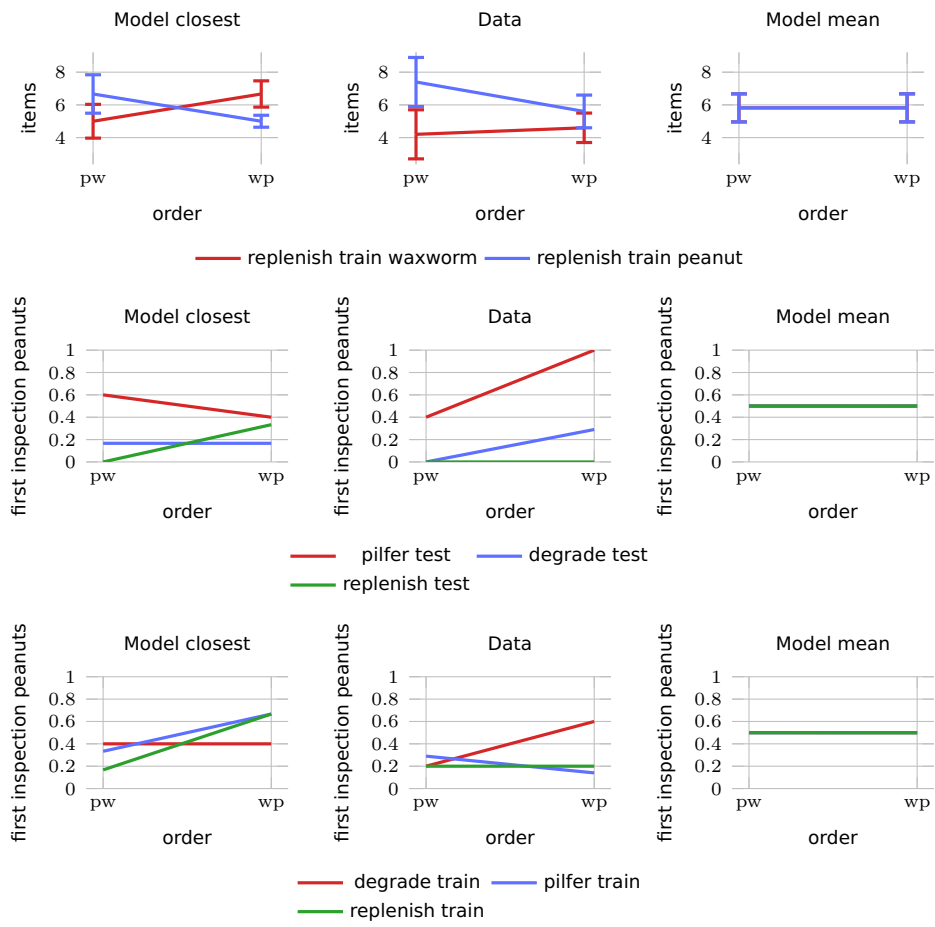


Figure S85: Clayton99B exp2: No-Plasticity-No-Memory-No-Motivational-Control Model (part2)

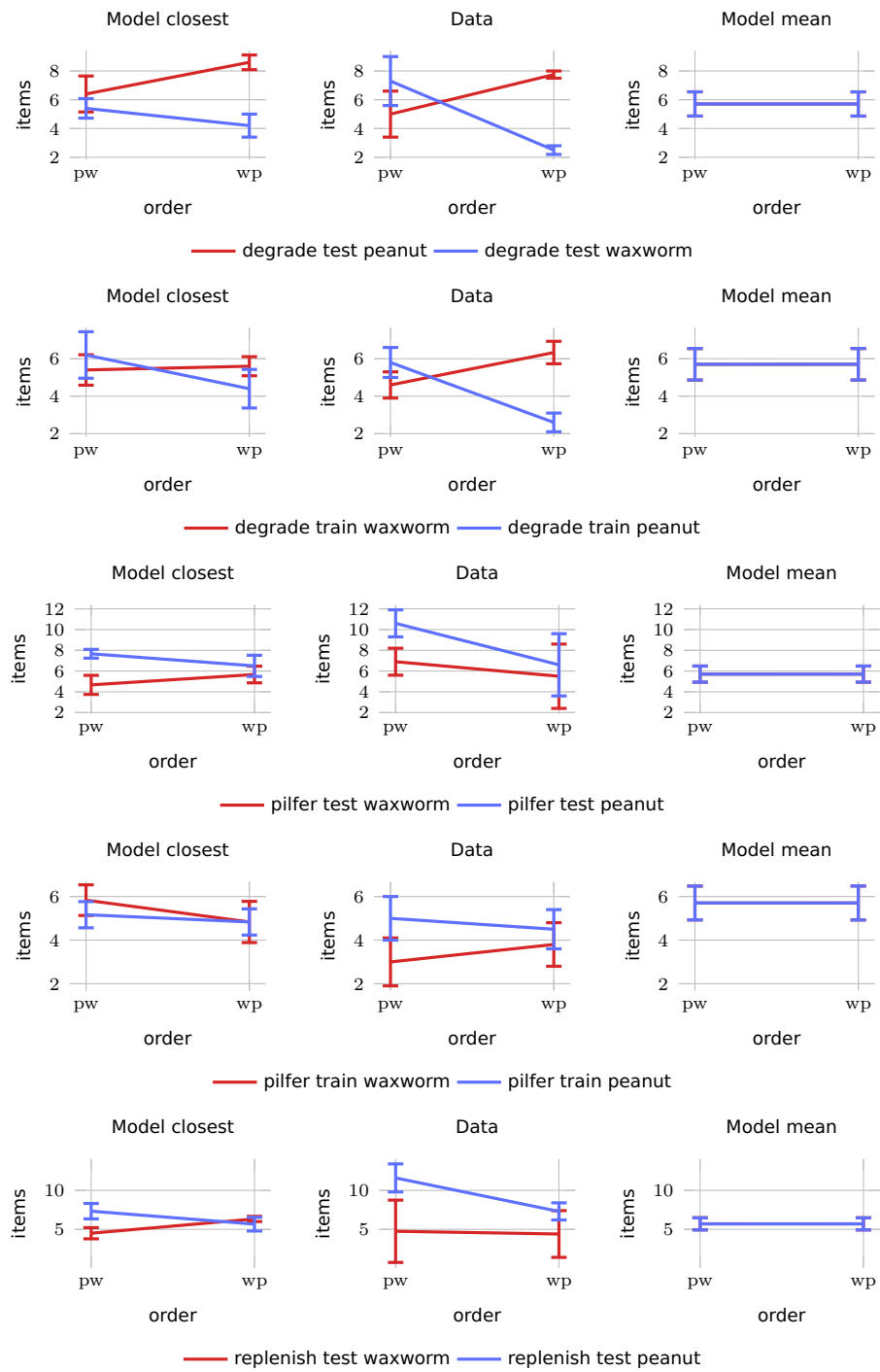


Figure S86: Clayton99B exp2: No-Plasticity-No-Memory Model (part1)

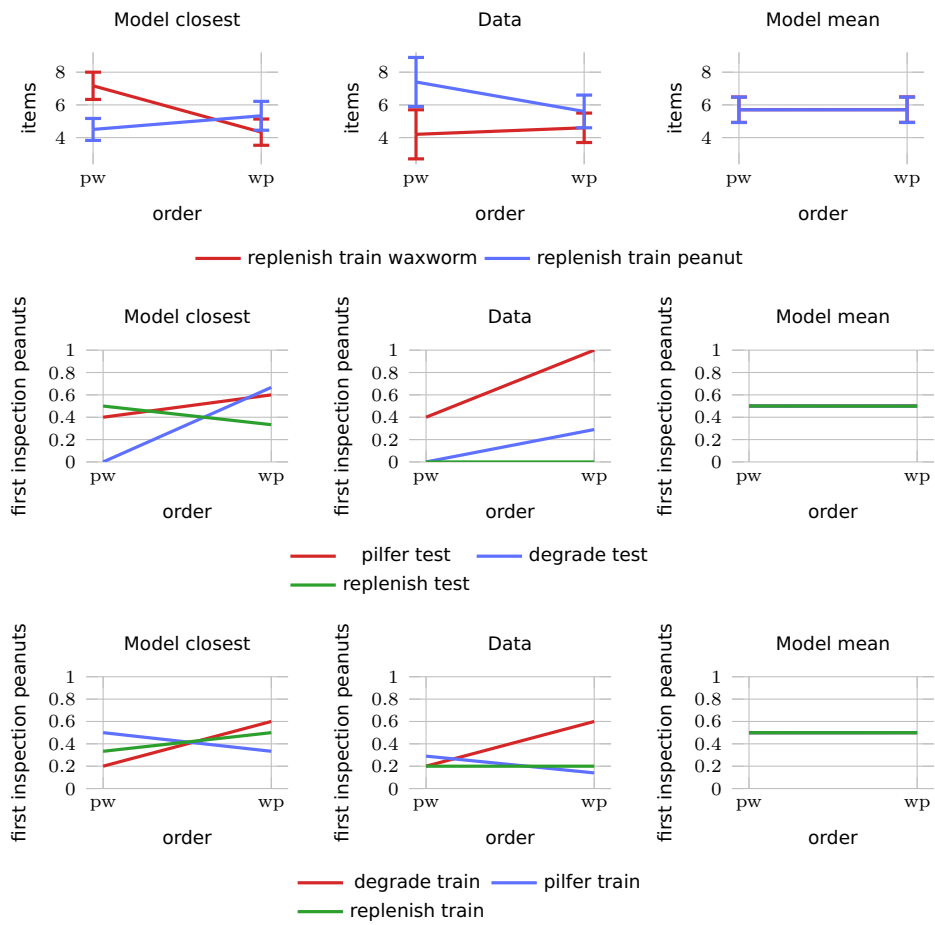


Figure S87: Clayton99B exp2: No-Plasticity-No-Memory Model (part2)

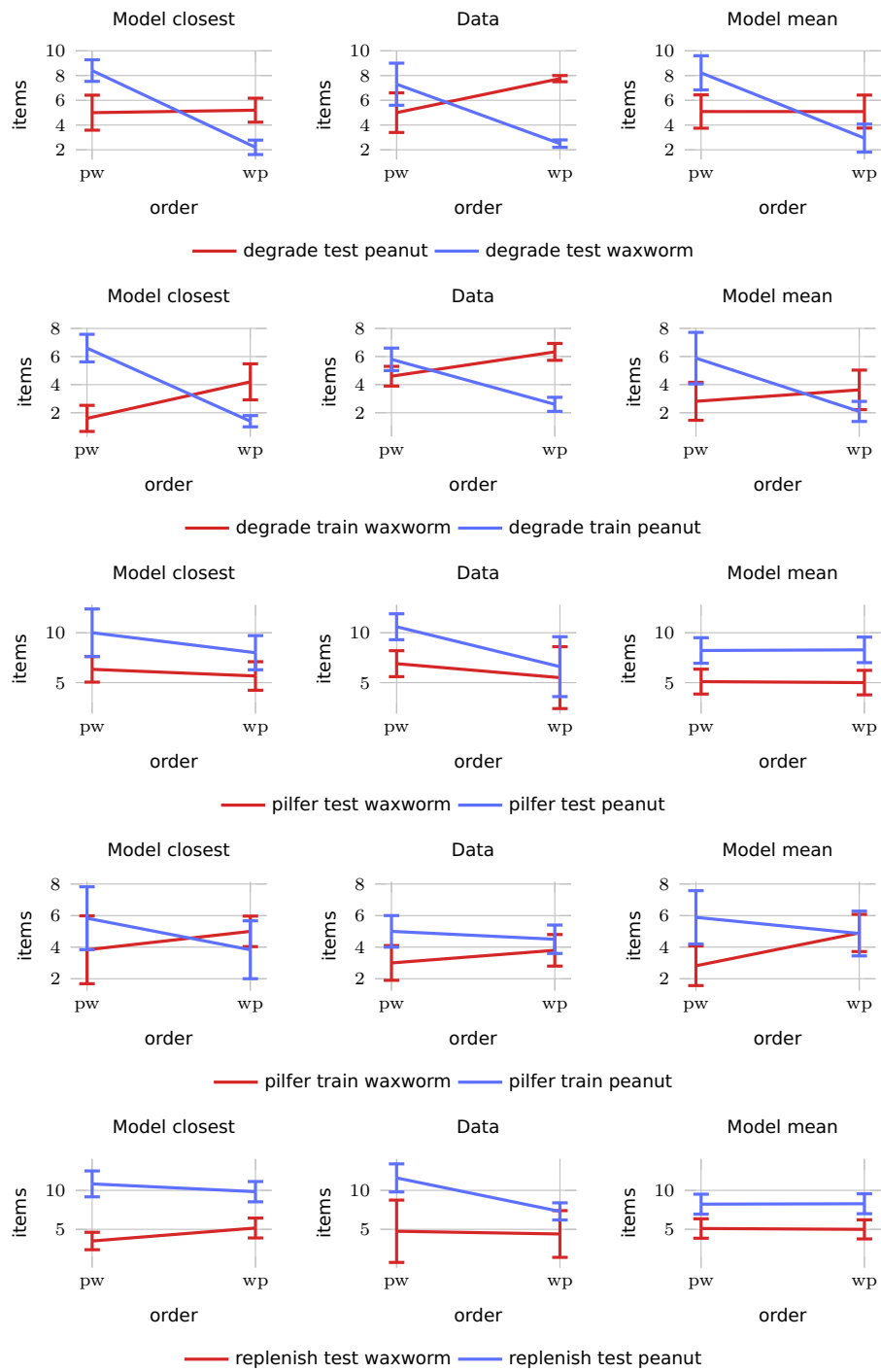


Figure S88: Clayton99B exp2: No-Plasticity Model (part1)

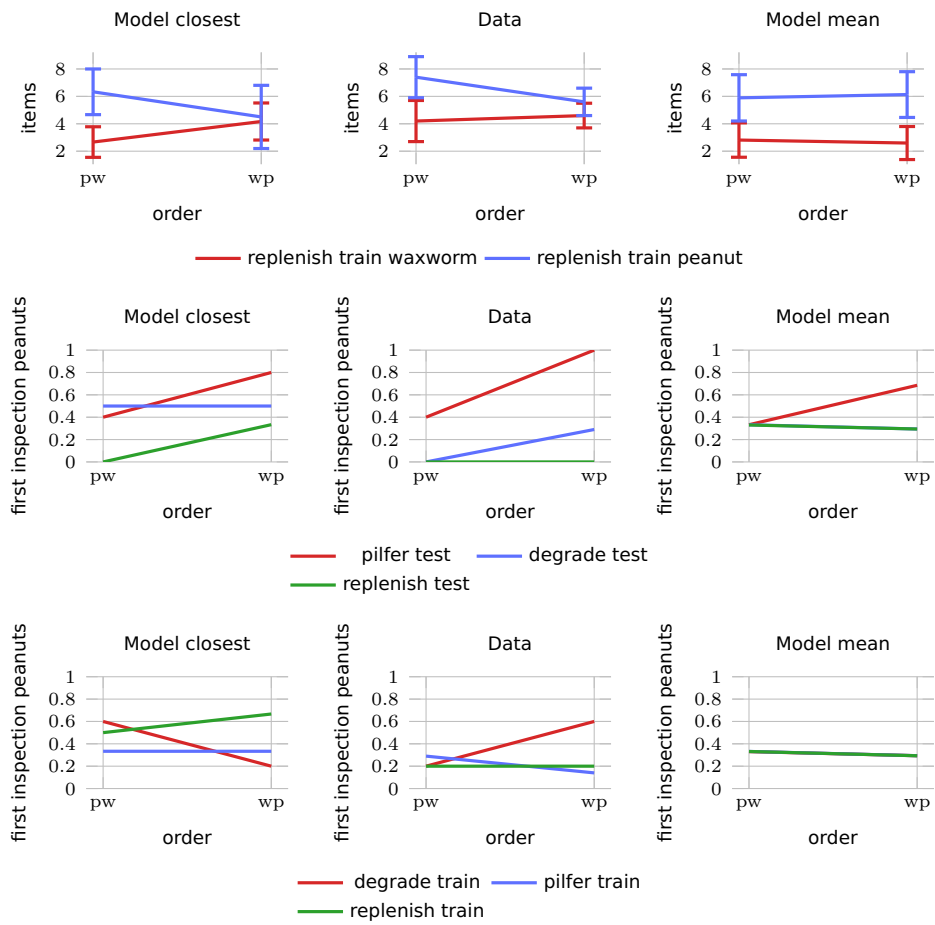


Figure S89: Clayton99B exp2: No-Plasticity Model (part2)

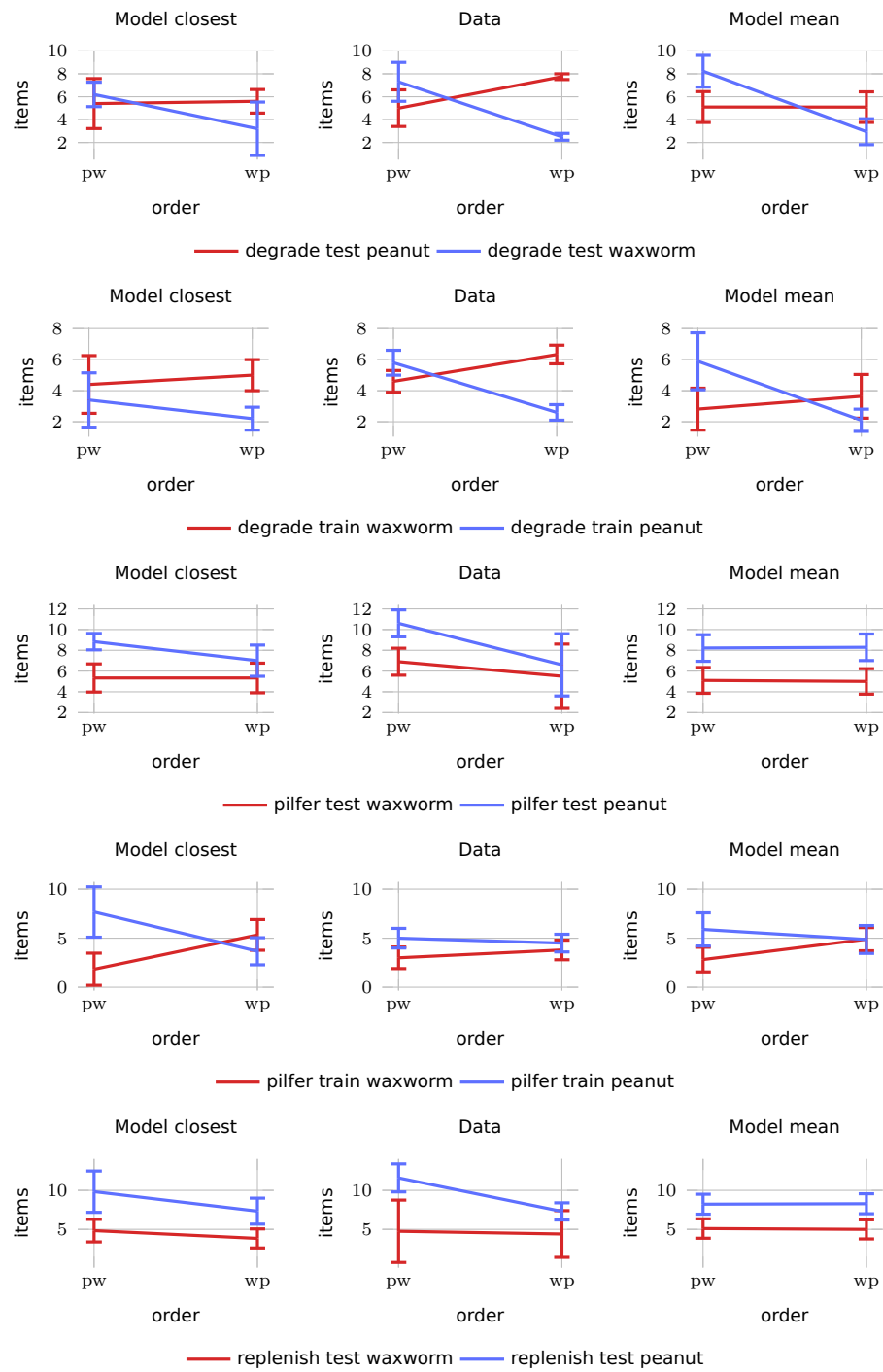


Figure S90: Clayton99B exp2: Plastic Caching Model (part1)

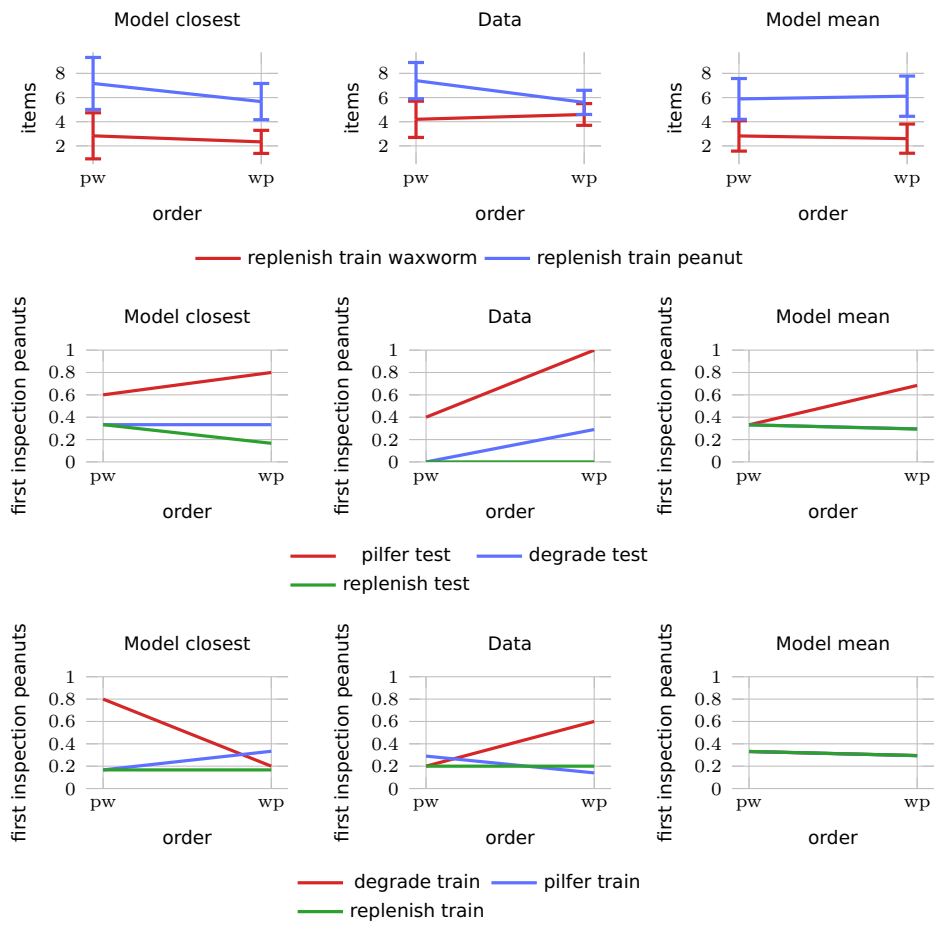


Figure S91: Clayton99B exp2: Plastic Caching Model (part2)

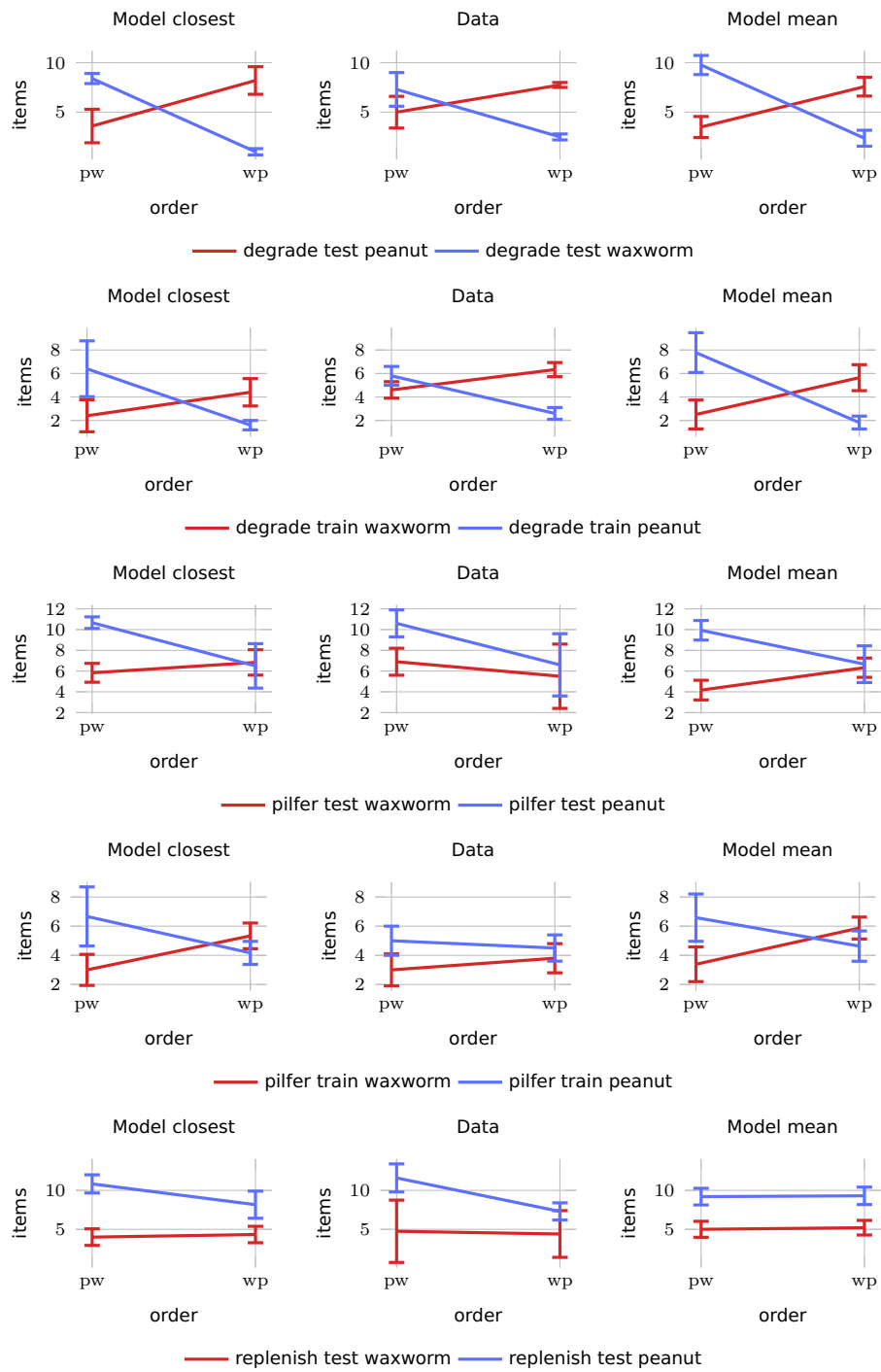


Figure S92: Clayton99B exp2: Planning-By-Replay Model (part1)

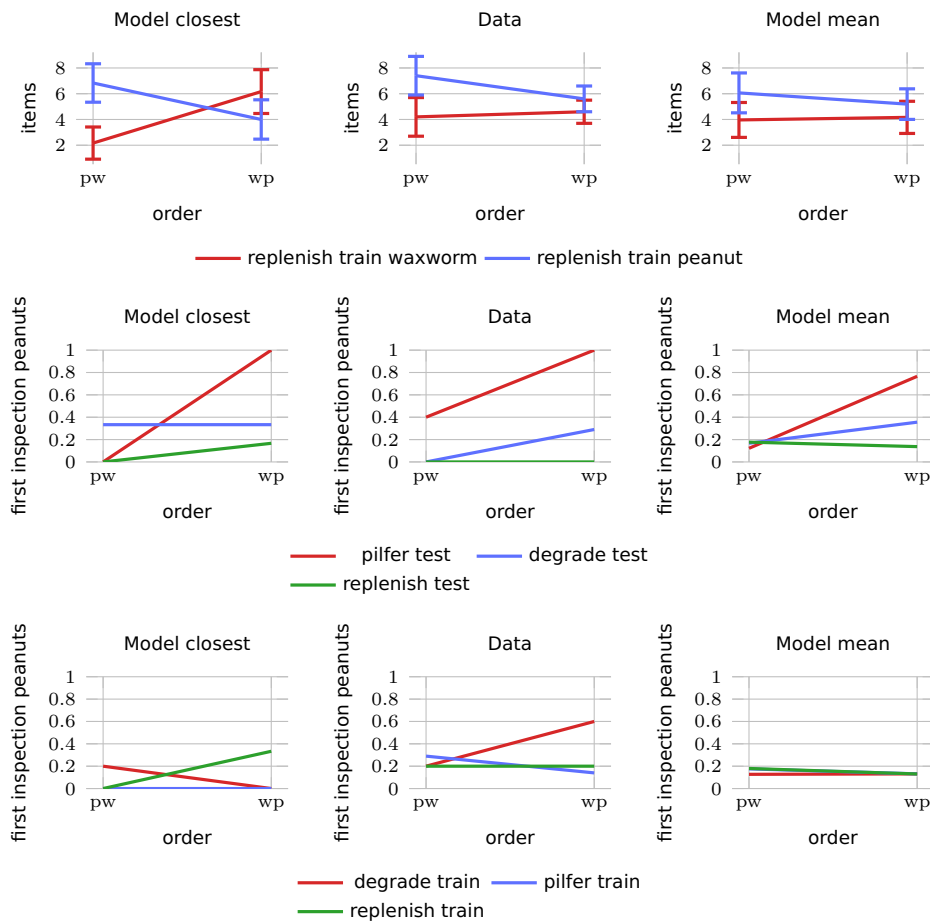


Figure S93: Clayton99B exp2: Planning-By-Replay Model (part2)

2.1.16 Clayton01 exp1

Published in [5].

Major Finding

The birds learn not to search for their preferred food after retention intervals where they experienced the preferred food to be degraded.

Comments

The choices between peanuts and waxworms or peanuts and crickets were investigated here. There was no significant effect in the analysis of the overall searches for the intermediate retention interval, but a significant difference in between the patterns of first searches shown by the two groups.

Most Important Tests

first inspect test RI 4 (group): $Chisq(2) = 0.27, p = 0.87$

first inspect test RI 100 (group): $Chisq(3) = 13.0, p = 0.0046$

first inspect test RI 28 (group): $Chisq(2) = 9.65, p = 0.008$

Total Number of Statistics

24 p-values, 122 means or sems.

Figures

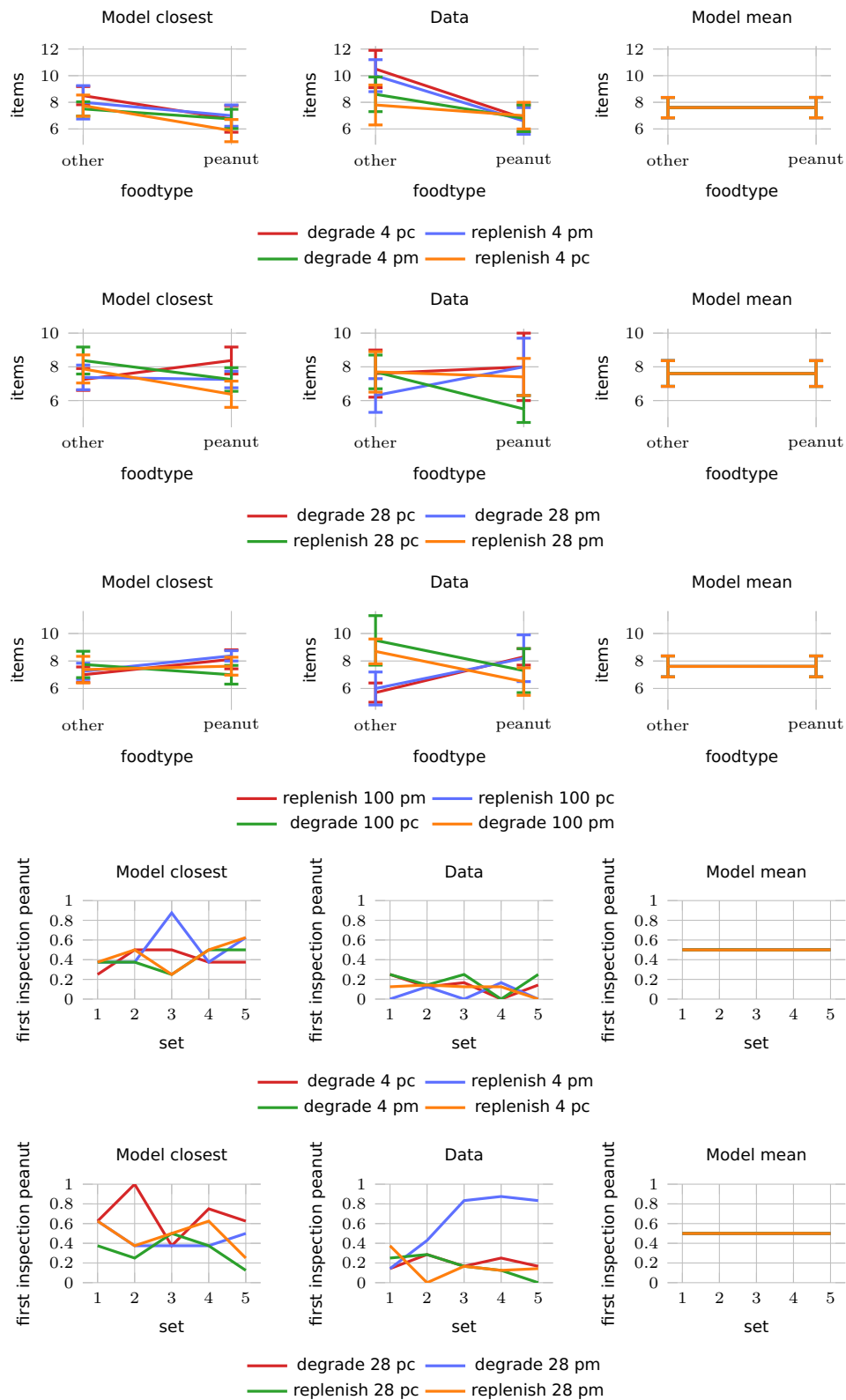


Figure S94: Clayton01 exp1: No-Plasticity-No-Memory-No-Motivational-Control Model (part1)

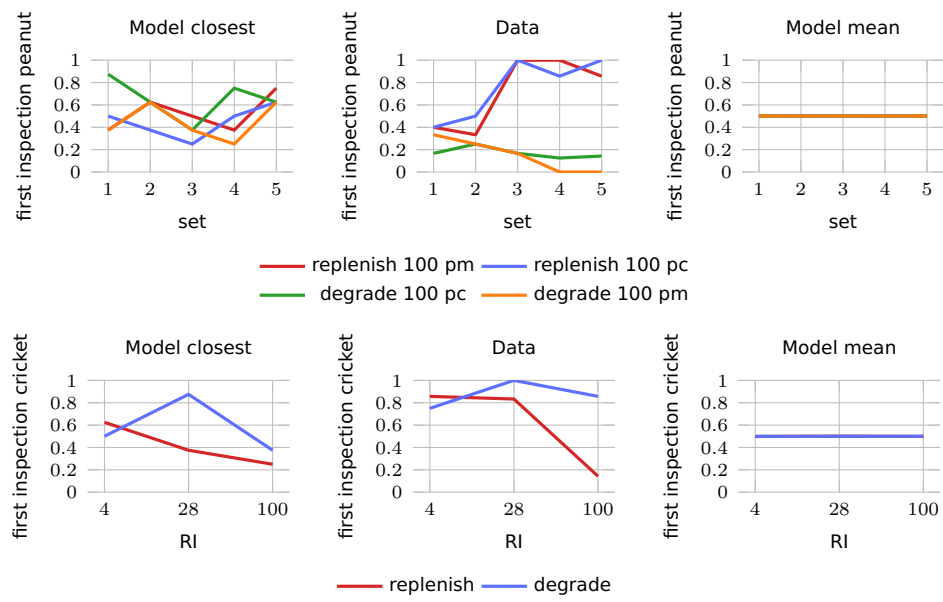


Figure S95: Clayton01 exp1: No-Plasticity-No-Memory-No-Motivational-Control Model (part2)

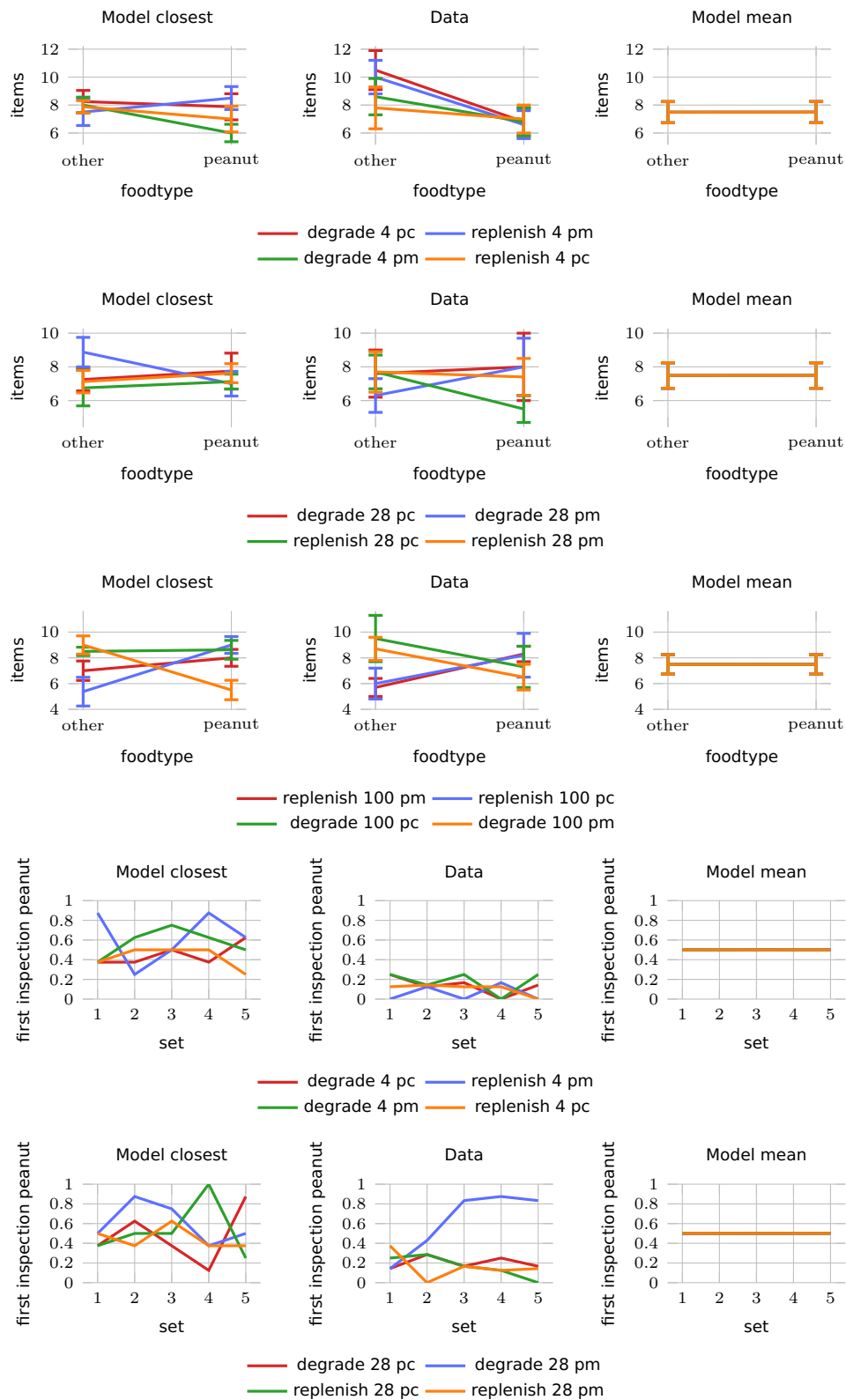


Figure S96: Clayton01 exp1: No-Plasticity-No-Memory Model (part1)

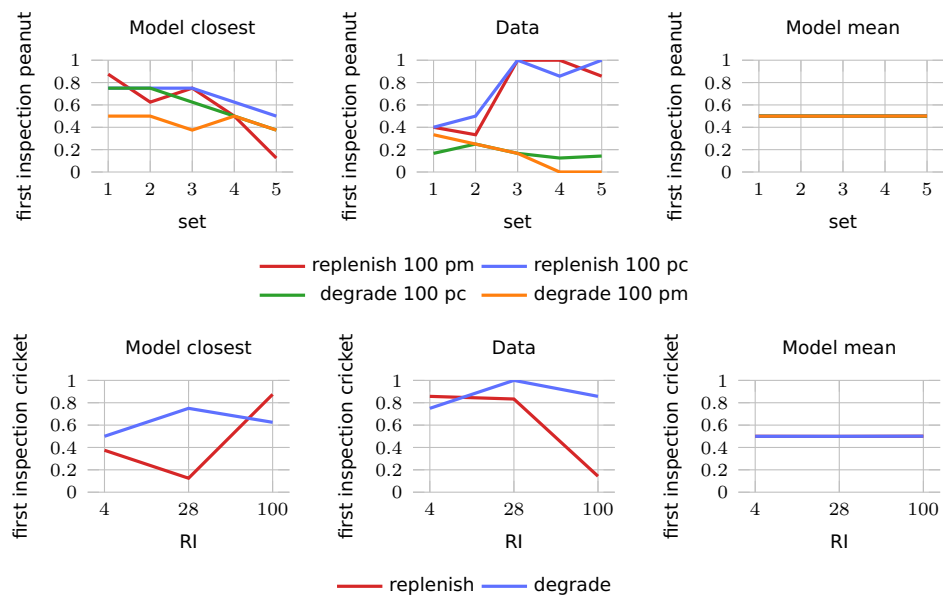


Figure S97: **Clayton01 exp1: No-Plasticity-No-Memory Model (part2)**

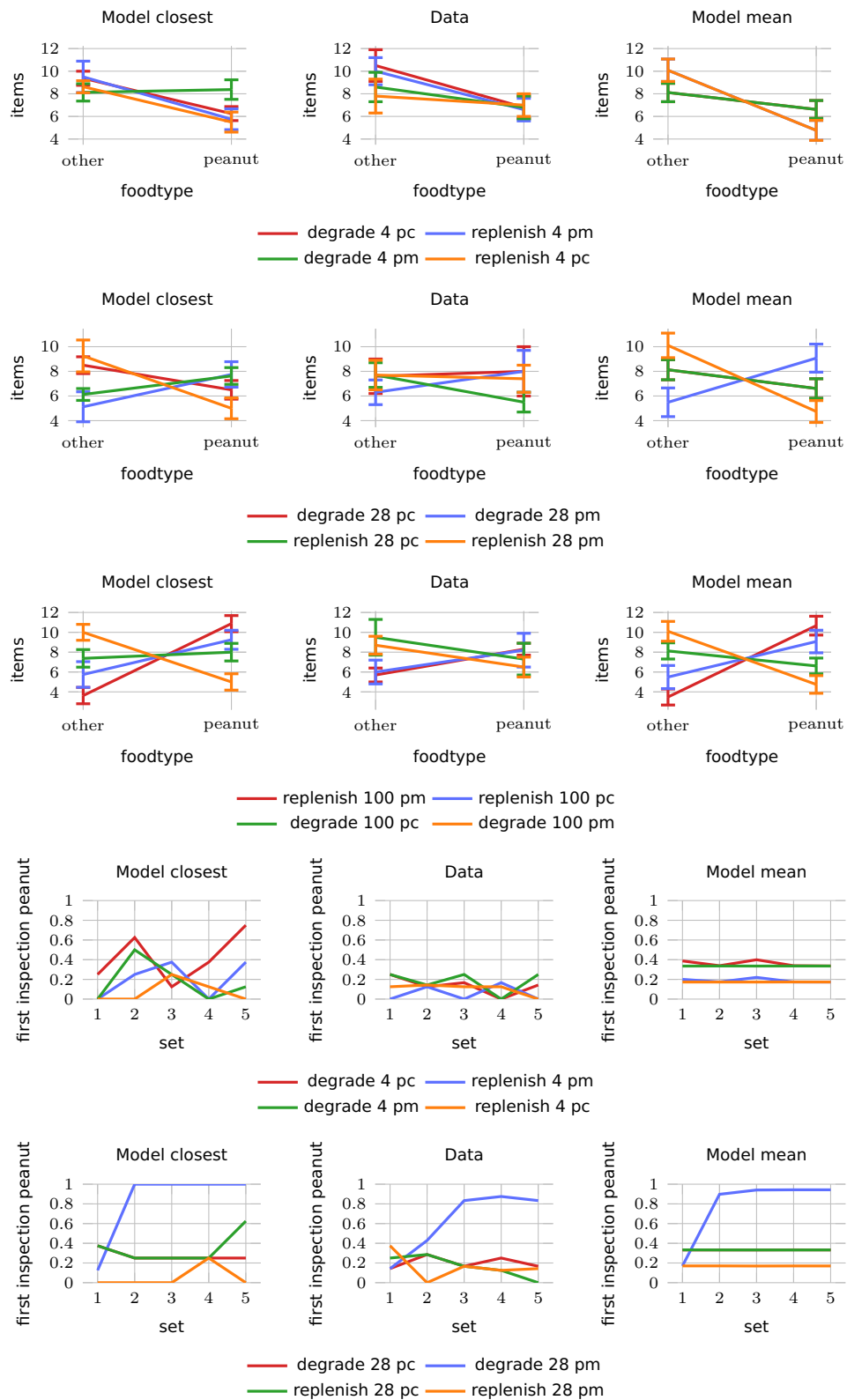


Figure S98: Clayton01 exp1: No-Plasticity Model (part1)

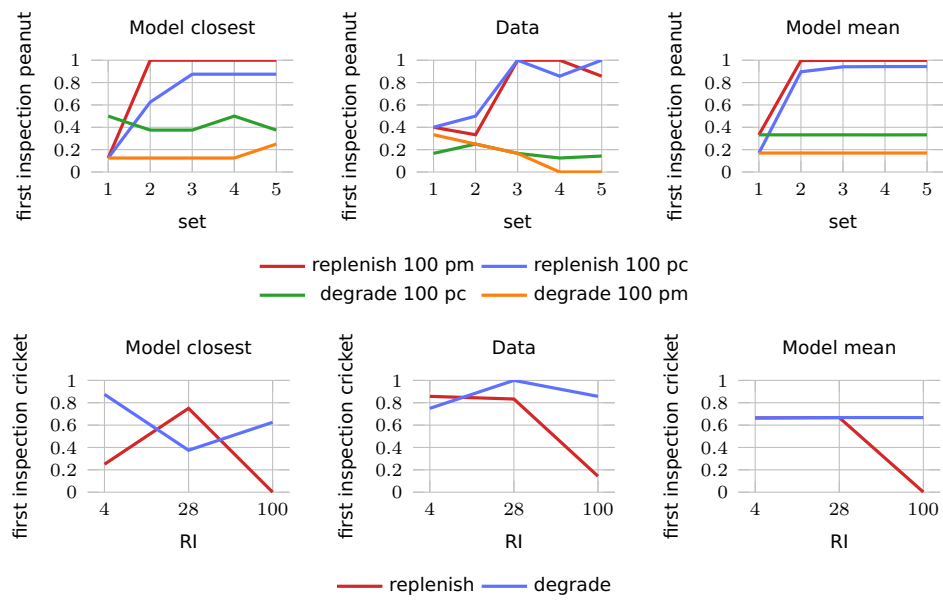


Figure S99: Clayton01 exp1: No-Plasticity Model (part2)

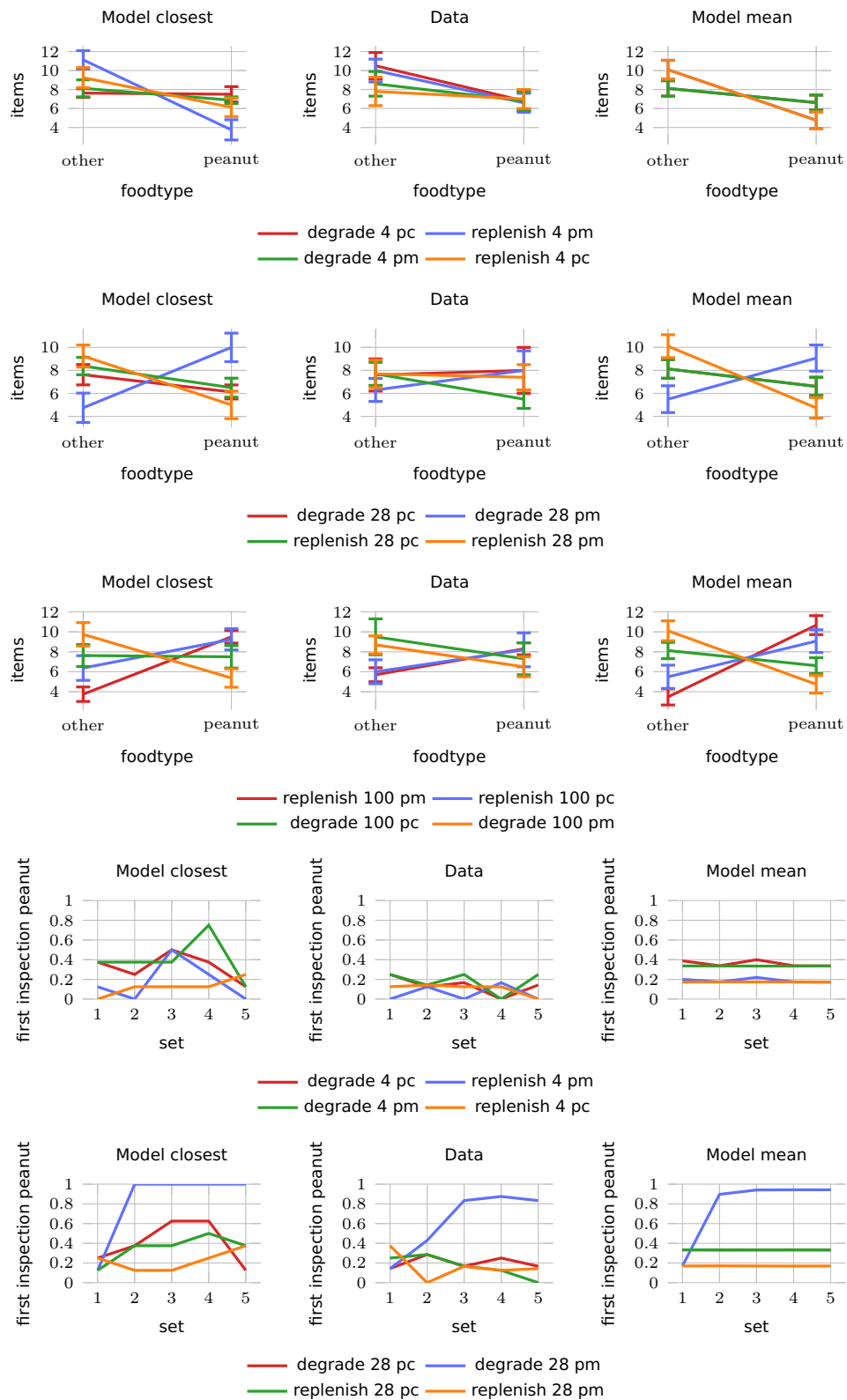


Figure S100: Clayton01 exp1: Plastic Caching Model (part1)

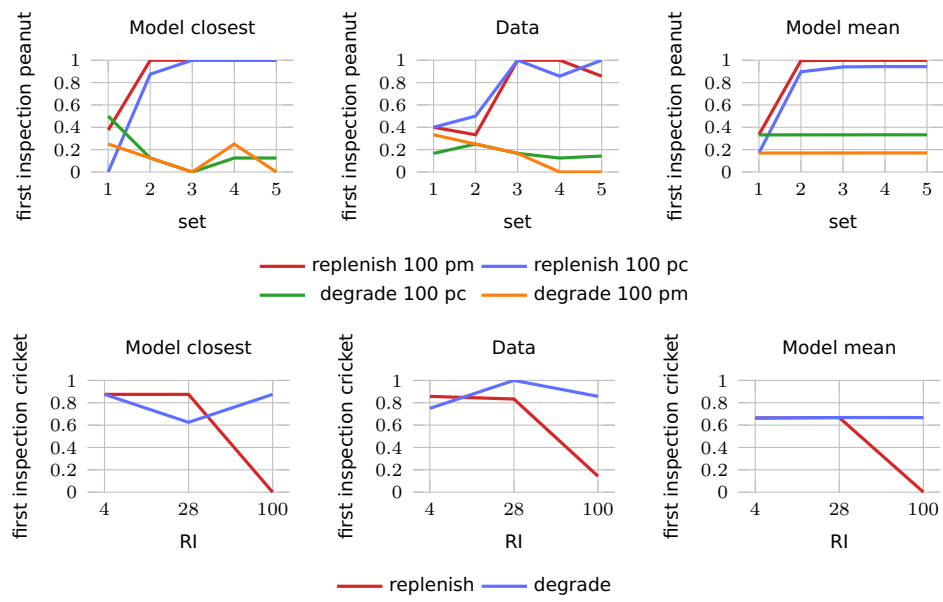


Figure S101: Clayton01 exp1: Plastic Caching Model (part2)

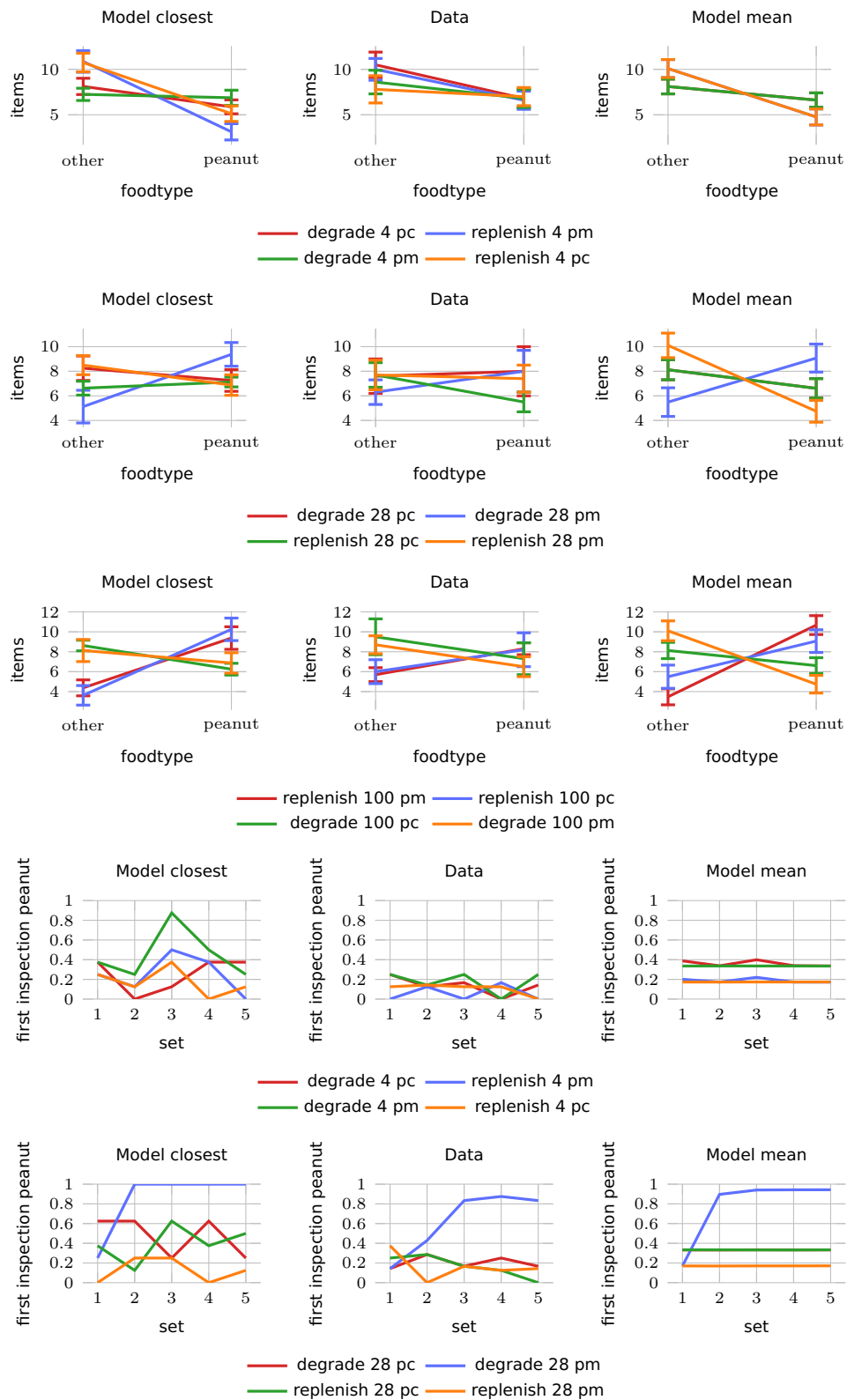


Figure S102: Clayton01 exp1: Planning-By-Replay Model (part1)

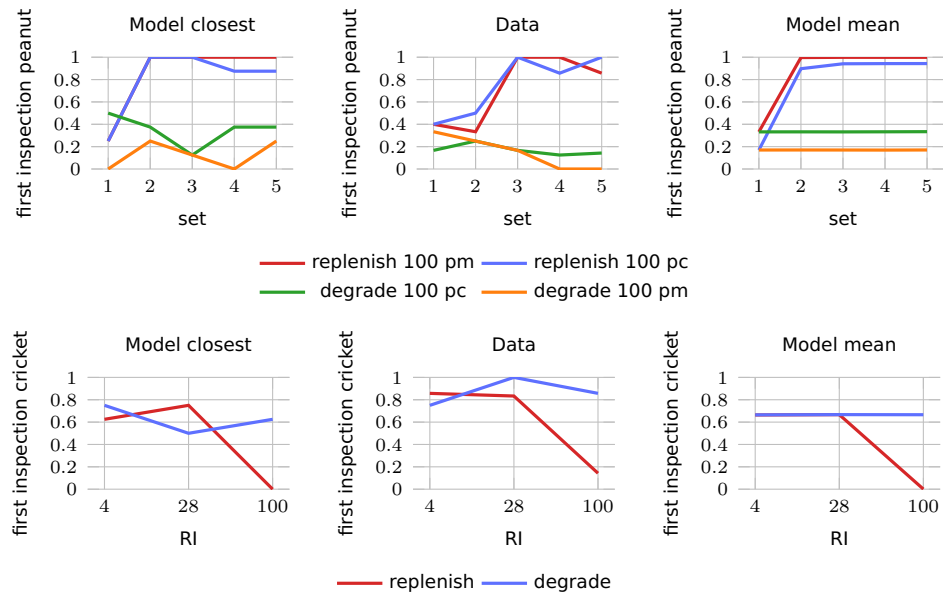


Figure S103: Clayton01 exp1: Planning-By-Replay Model (part2)

2.1.17 Clayton01 exp2

Published in [5].

Major Finding

Confirming experiment 1, birds search their preferred food after a short retention interval, where both food types are expected to be fresh, but they reverse this behaviour after a longer retention interval, where the preferred food is expected to be degraded.

Comments

The choice between mealworms (preferred) and crickets was investigated.

Most Important Tests

inspect test exp 2 RI 4 (foodtype): $F(1, 14) = 30.72, p = 7.3 \cdot 10^{-5}$

inspect test exp 2 RI 28 (group & foodtype): $F(1, 14) = 5.28, p = 0.038$

inspect test exp 2 (group & foodtype & RI): $F(1, 14) = 12.53, p = 0.0033$

Total Number of Statistics

17 p-values, 28 means or sems.

Figures

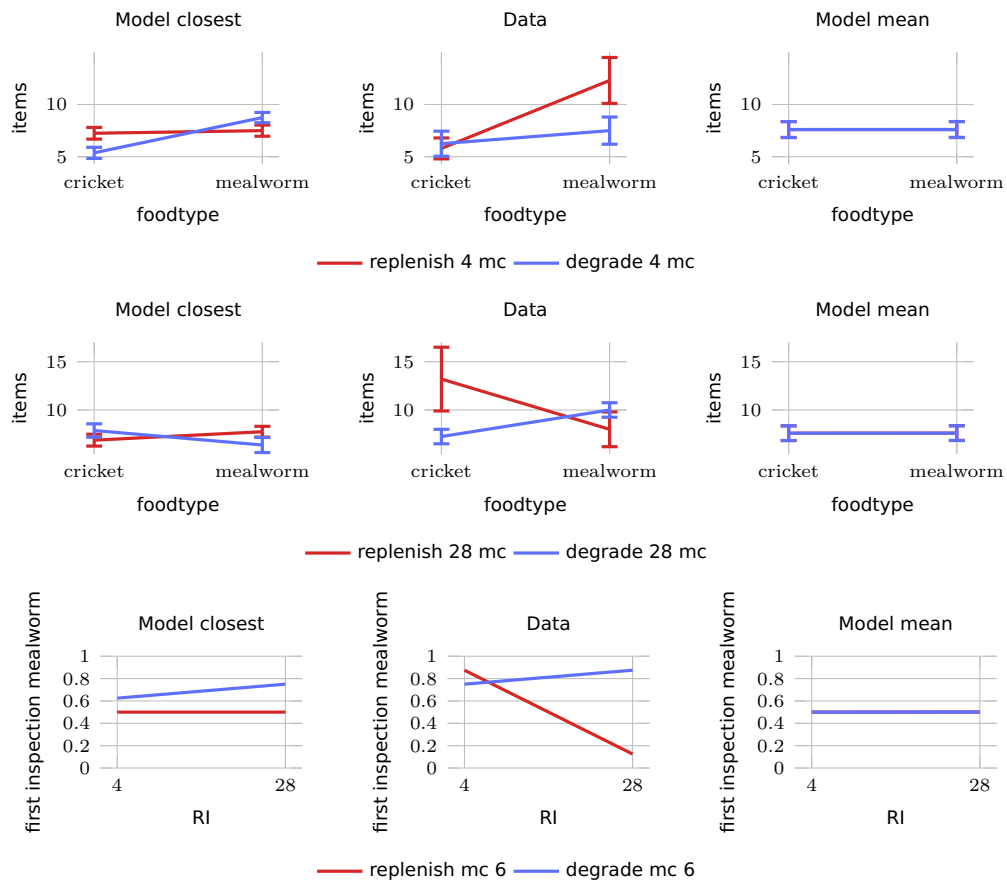


Figure S104: Clayton01 exp2: No-Plasticity-No-Memory-No-Motivational-Control Model

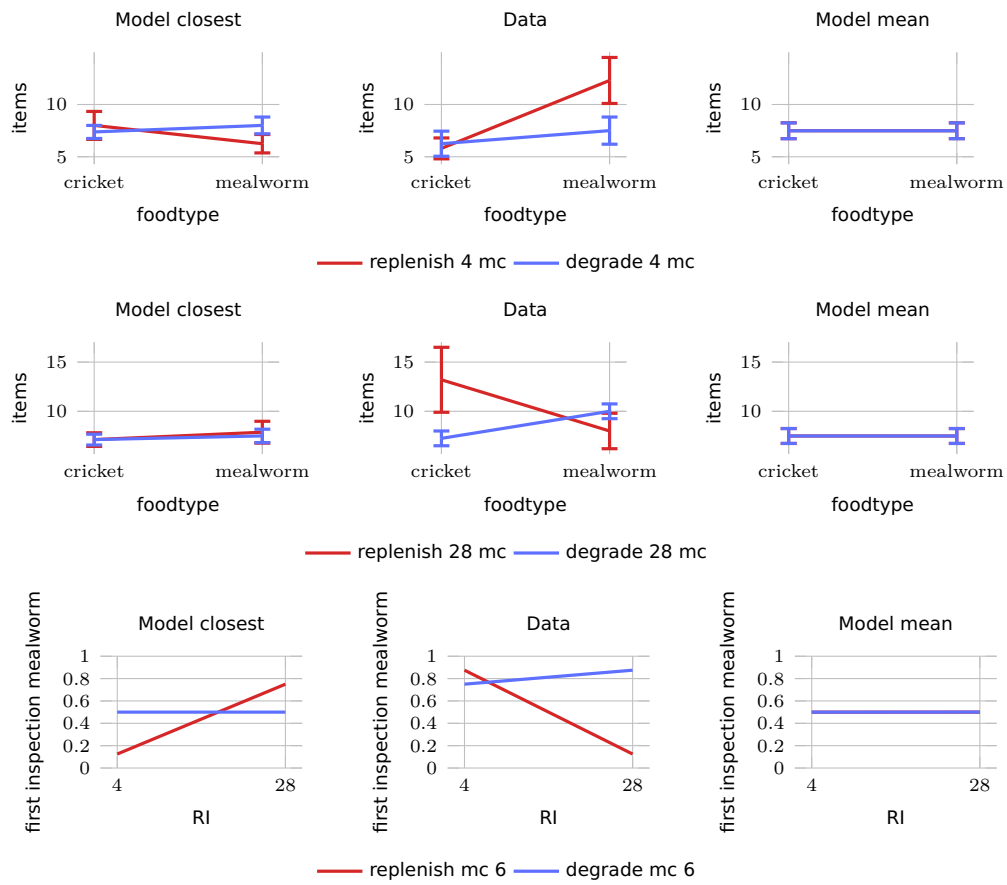


Figure S105: Clayton01 exp2: No-Plasticity-No-Memory Model

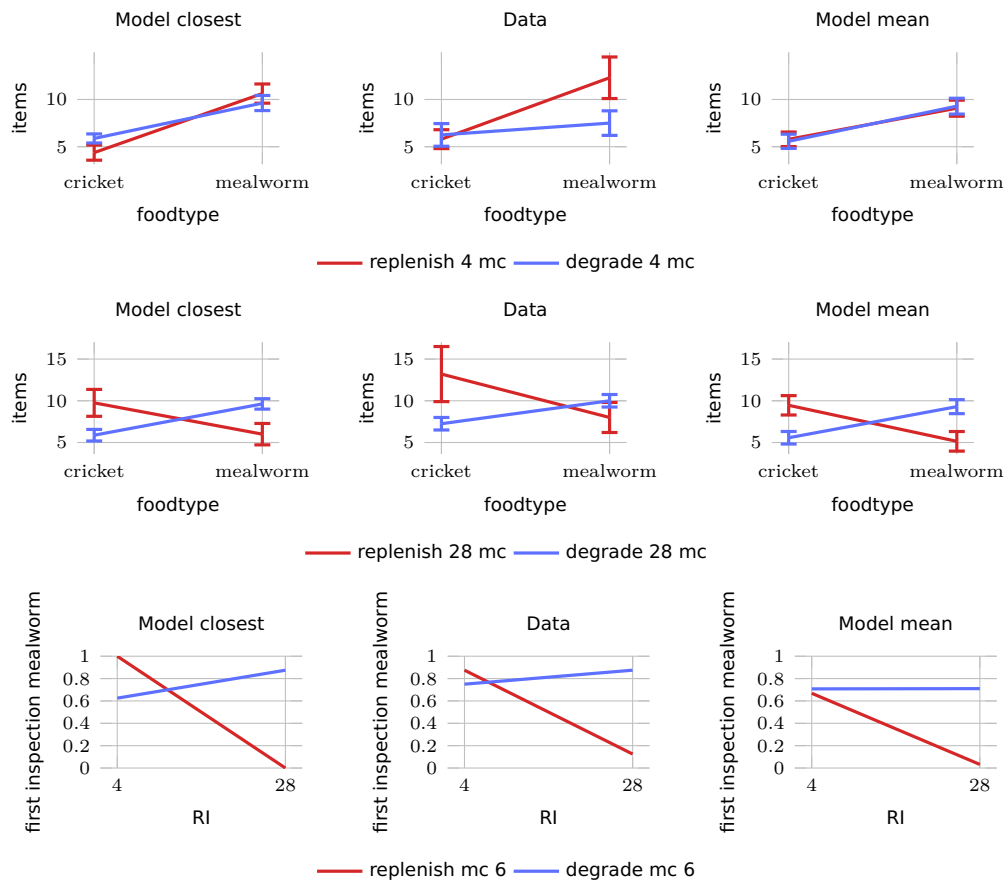


Figure S106: Clayton01 exp2: No-Plasticity Model

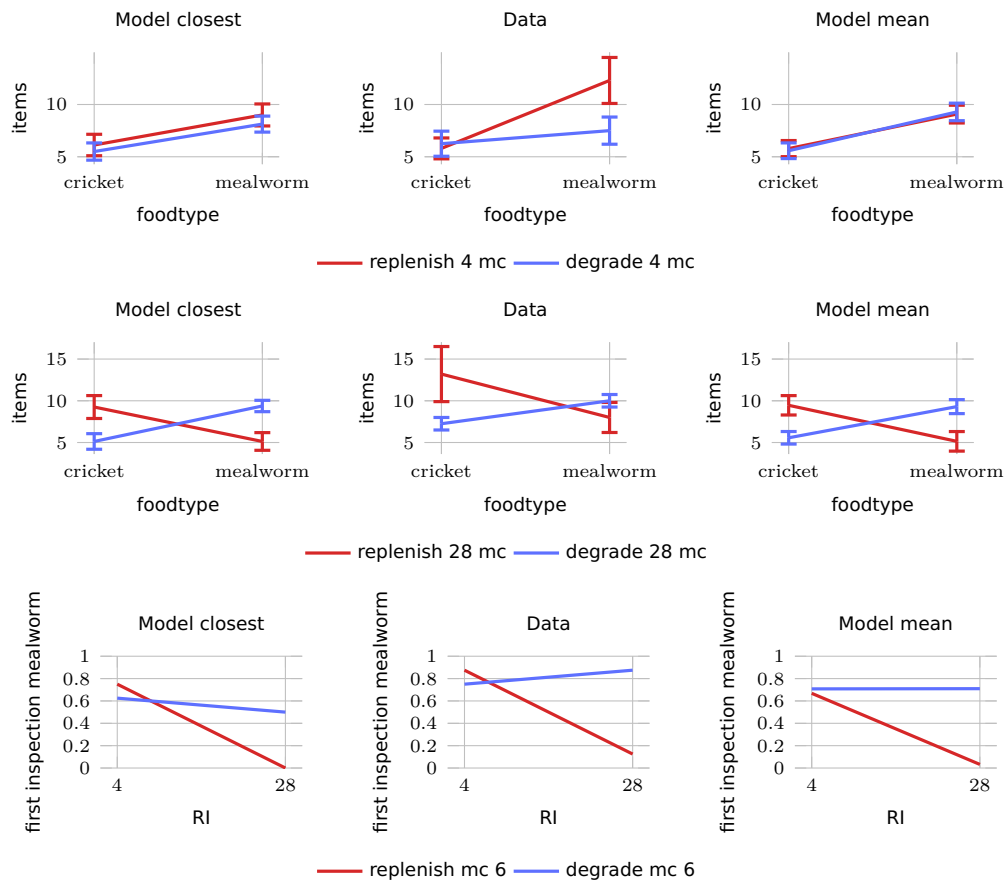


Figure S107: Clayton01 exp2: Plastic Caching Model

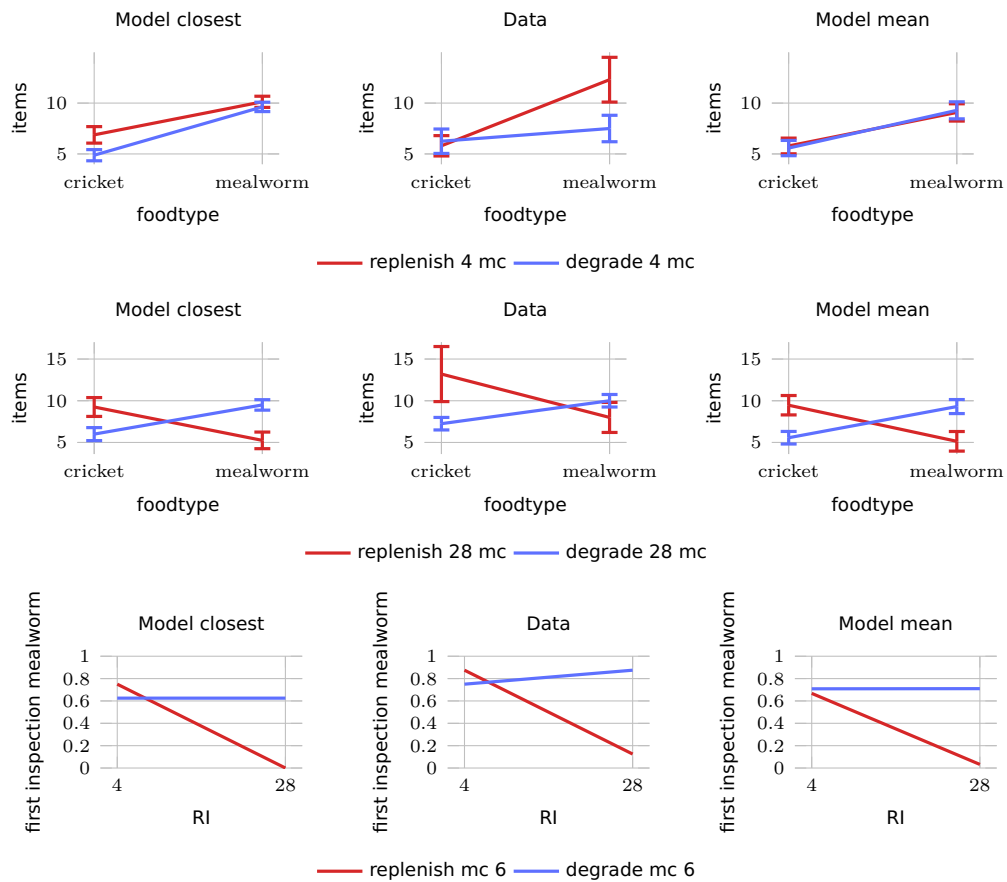


Figure S108: Clayton01 exp2: Planning-By-Replay Model

2.1.18 Clayton01 exp3

Published in [5].

Major Finding

The birds direct more searches to the unpreferred food after the long retention interval, even when the two types of food are cached in two sides of the same caching tray.

Most Important Tests

inspect cm (group & foodtype): $F(1, 14) < 1, p = 0.34$

inspect mc (group & foodtype): $F(1, 14) = 30.76, p = 7.2 \cdot 10^{-5}$

inspect degrade (foodtype & trial): $F(1, 7) = 59.94, p = 0.00011$

Total Number of Statistics

16 p-values, 28 means or sems.

Figures

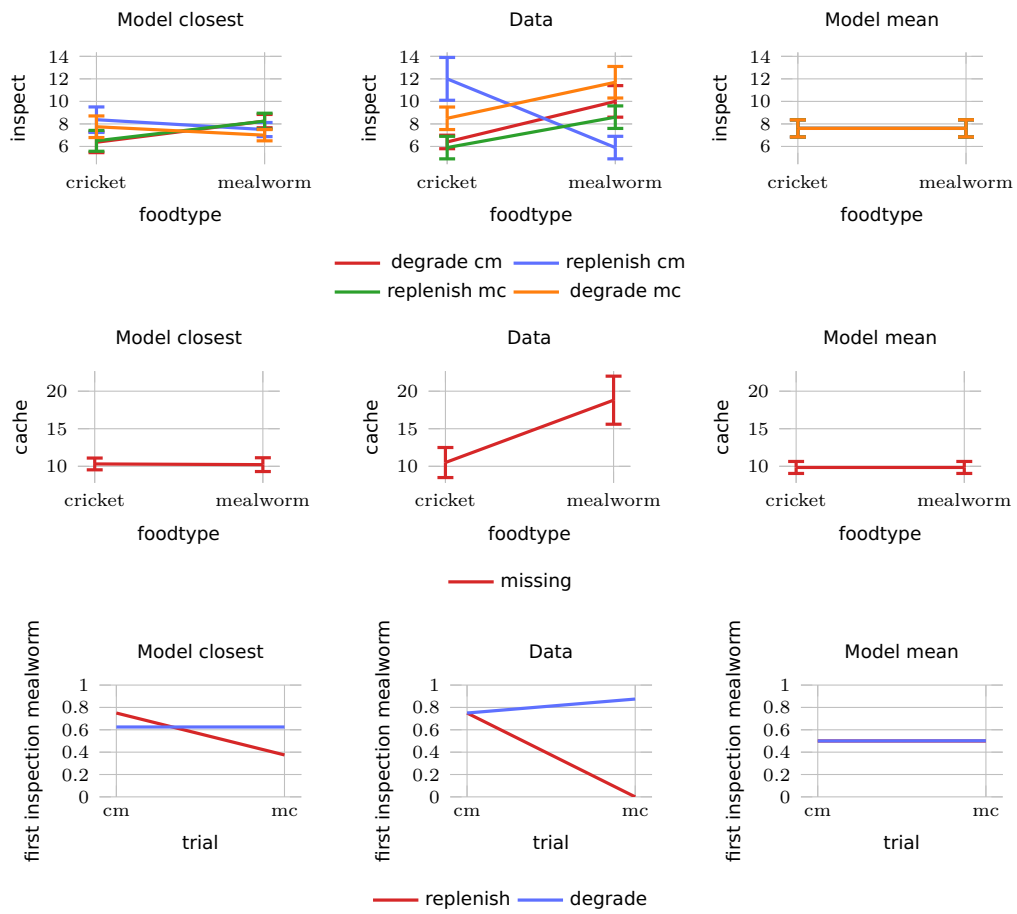


Figure S109: Clayton01 exp3: No-Plasticity-No-Memory-No-Motivational-Control Model

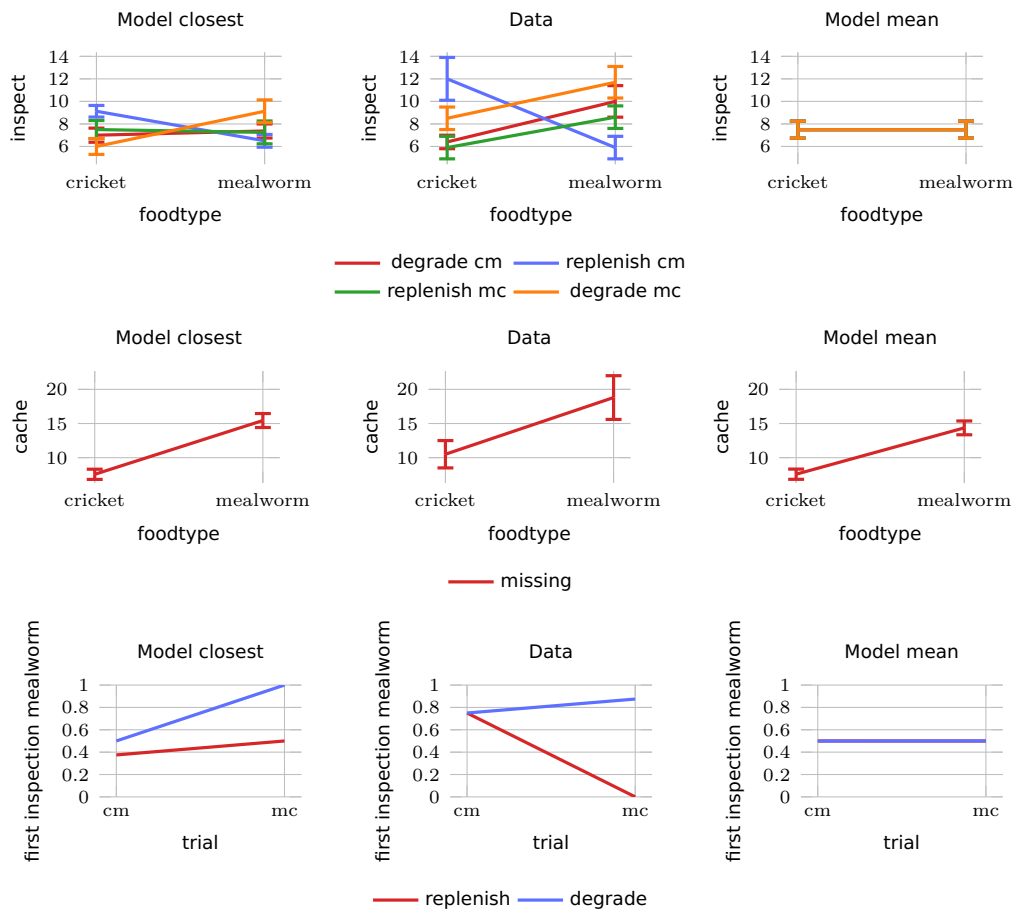


Figure S110: Clayton01 exp3: No-Plasticity-No-Memory Model

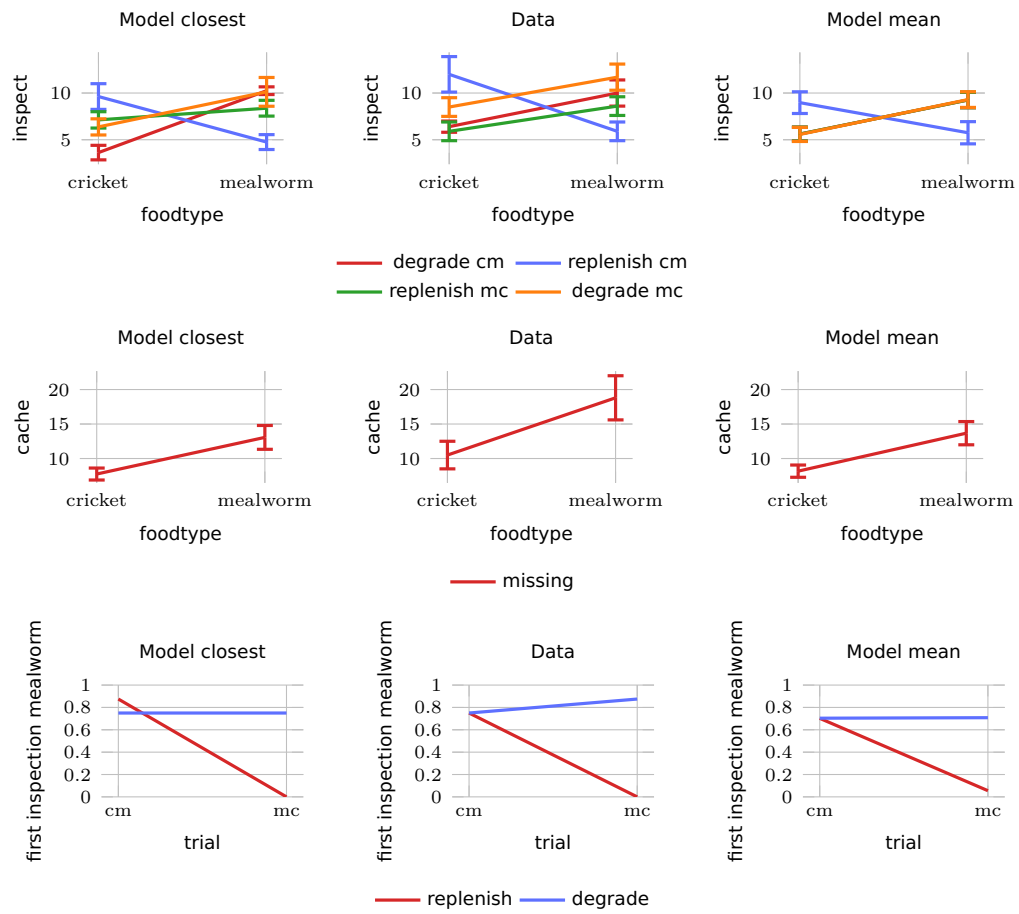


Figure S111: Clayton01 exp3: No-Plasticity Model

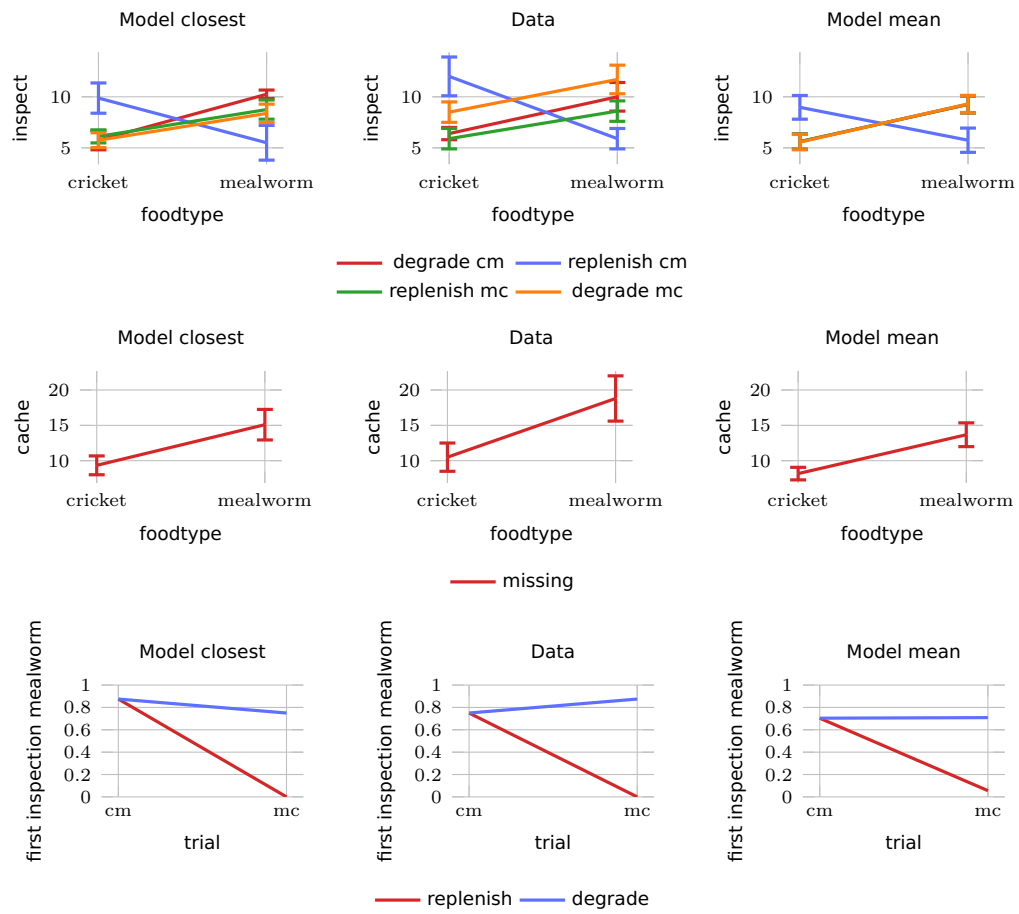


Figure S112: Clayton01 exp3: Plastic Caching Model

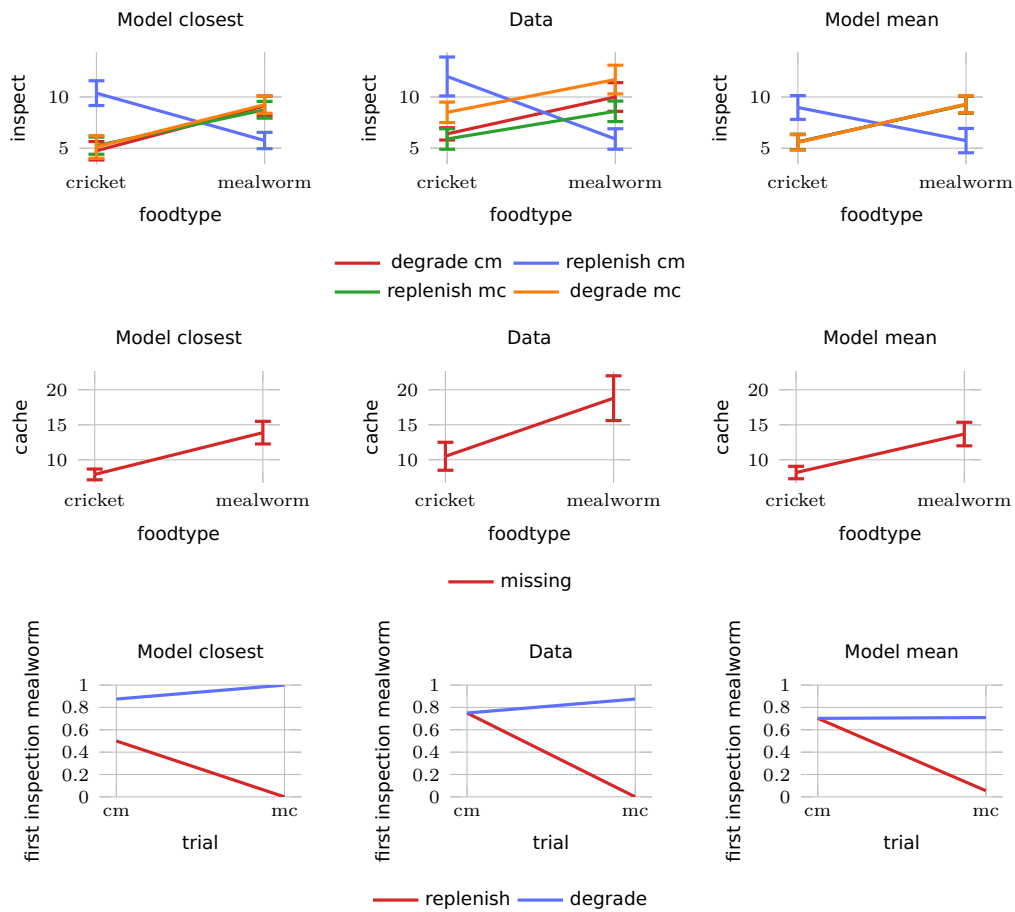


Figure S113: Clayton01 exp3: Planning-By-Replay Model

2.1.19 Clayton01 exp4

Published in [5].

Major Finding

The same search pattern as in experiments 1-3 is observed when the birds cache two types of food in the two sides of a caching tray at the same time: given the choice of a tray where the cached long ago they direct more searches to the unpreferred food and vice versa for short retention intervals.

Most Important Tests

inspect 4 (group & foodtype): $F(1, 14) = 3.51, p = 0.082$

inspect (group & foodtype & RI): $F(1, 14) = 9.53, p = 0.008$

inspect 28 (group & foodtype): $F(1, 14) = 10.45, p = 0.006$

Total Number of Statistics

14 p-values, 32 means or sems.

Figures

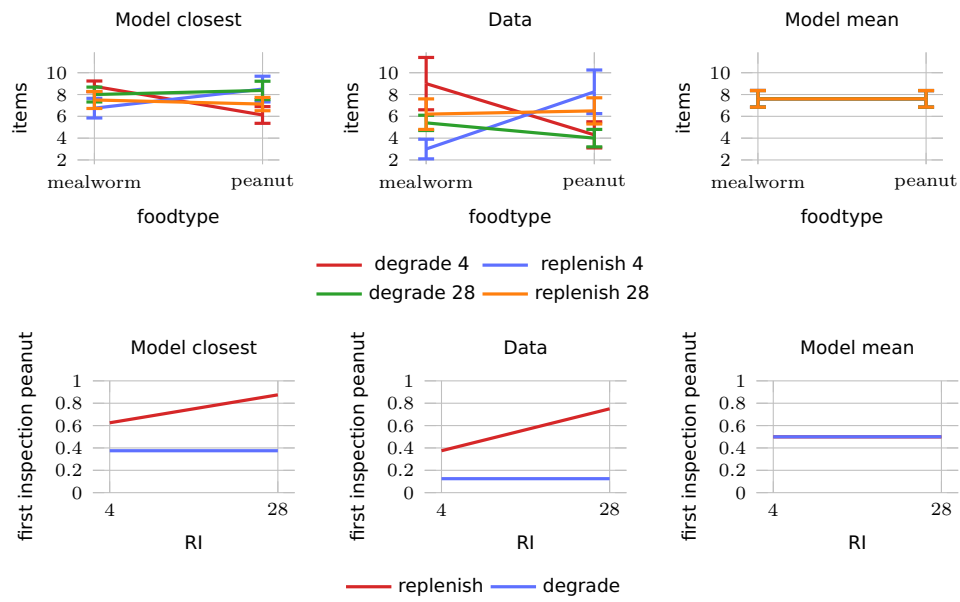


Figure S114: Clayton01 exp4: No-Plasticity-No-Memory-No-Motivational-Control Model

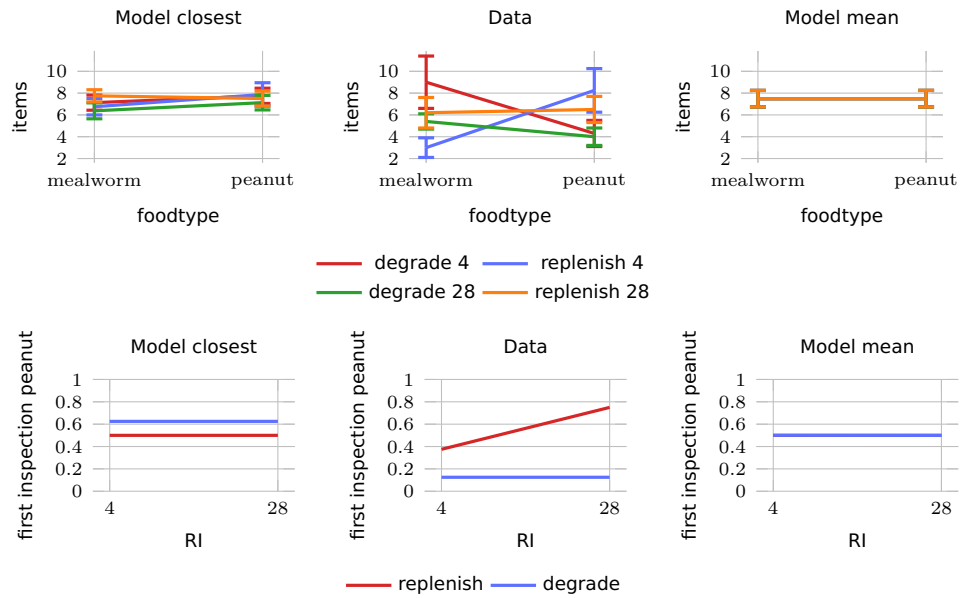


Figure S115: Clayton01 exp4: No-Plasticity-No-Memory Model

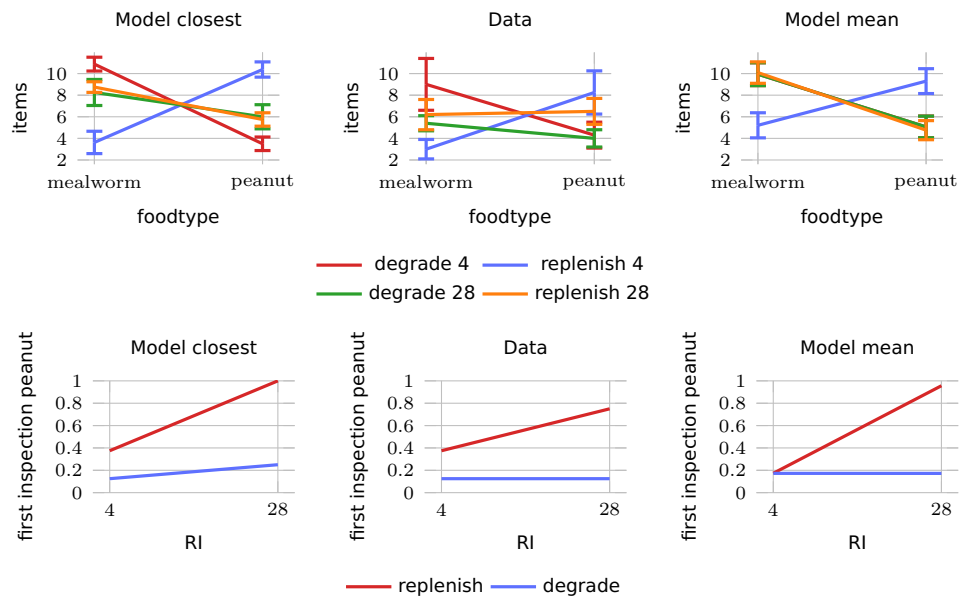


Figure S116: Clayton01 exp4: No-Plasticity Model

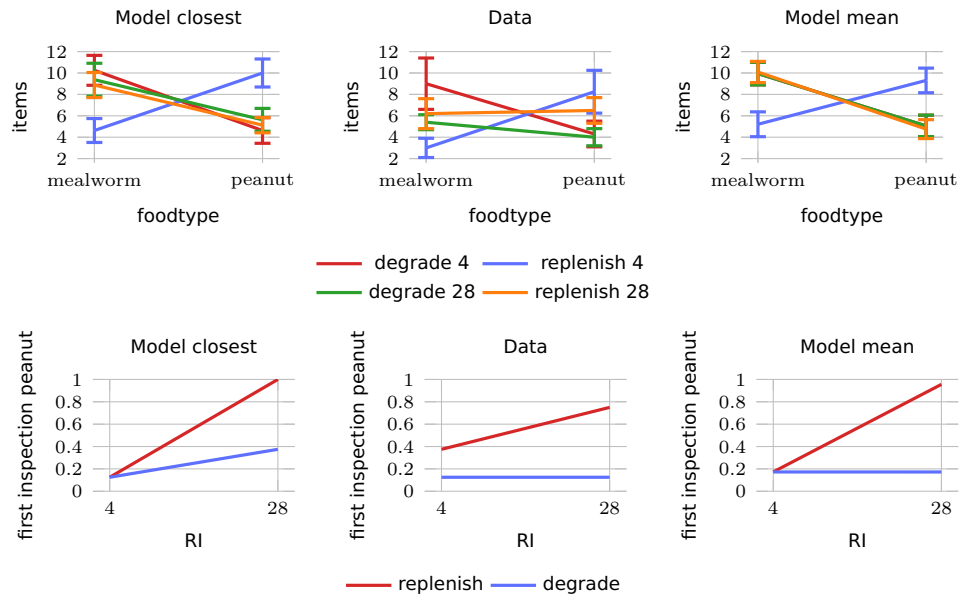


Figure S117: Clayton01 exp4: Plastic Caching Model

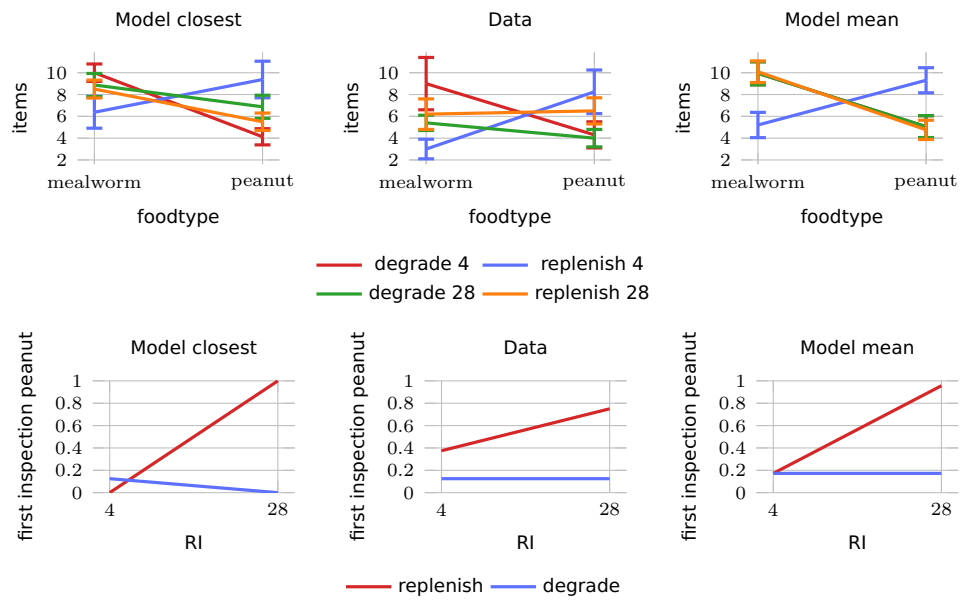


Figure S118: Clayton01 exp4: Planning-By-Replay Model

2.1.20 Clayton03 exp1

Published in [6].

Major Finding

Having experienced that crickets are fresh after 1 day but degraded after 4 days, the birds still search crickets after 2 or 3 days, but not after 5 days.

Most Important Tests

inspect RI 124 Clayton03 exp1 (group): $F(1, 11) > 7.01, p = 0.023$

inspect RI 76 Clayton03 exp1 (group): $F(1, 11) < 2.13, p = 0.17$

inspect RI 52 Clayton03 exp1 (group): $F(1, 11) < 2.13, p = 0.17$

Total Number of Statistics

16 p-values, 30 means or sems.

Figures

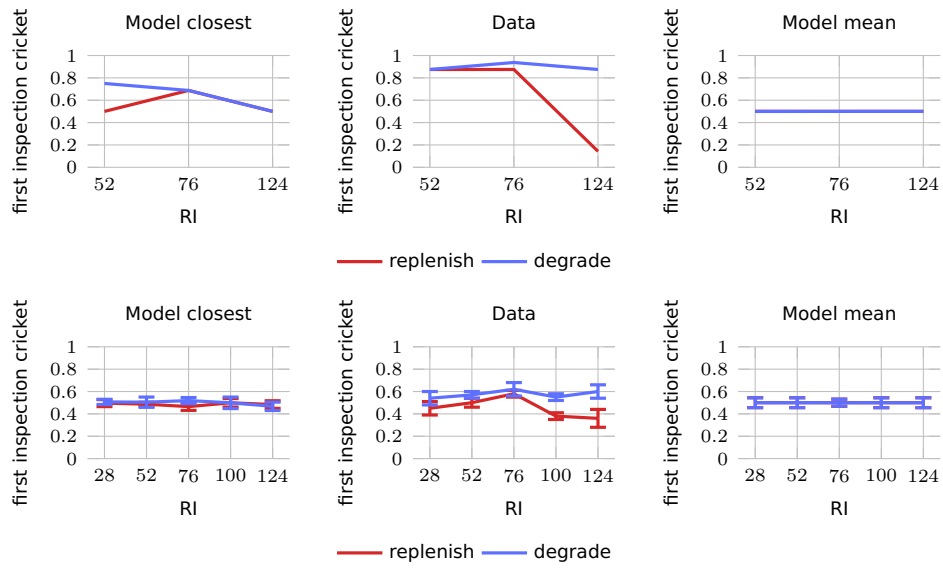


Figure S119: Clayton03 exp1: No-Plasticity-No-Memory-No-Motivational-Control Model

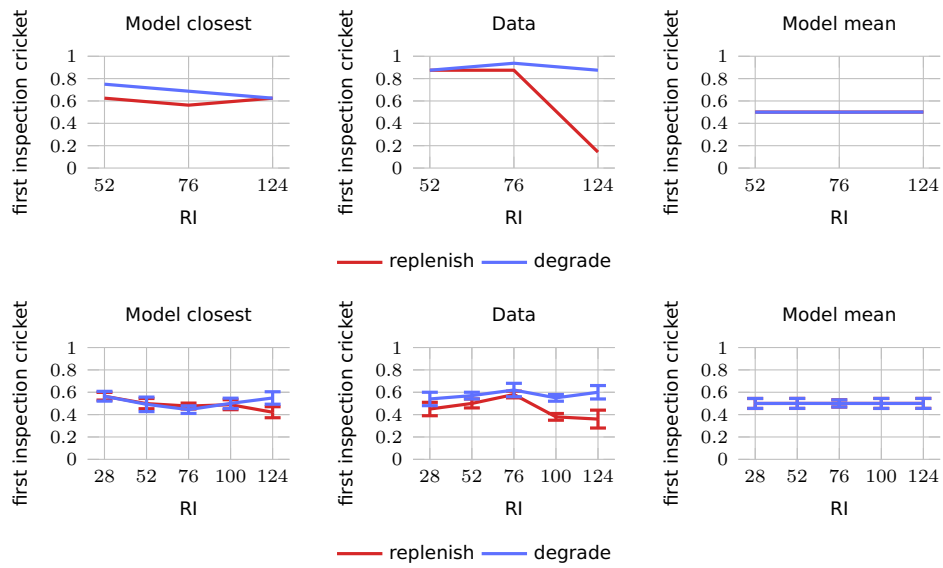


Figure S120: Clayton03 exp1: No-Plasticity-No-Memory Model

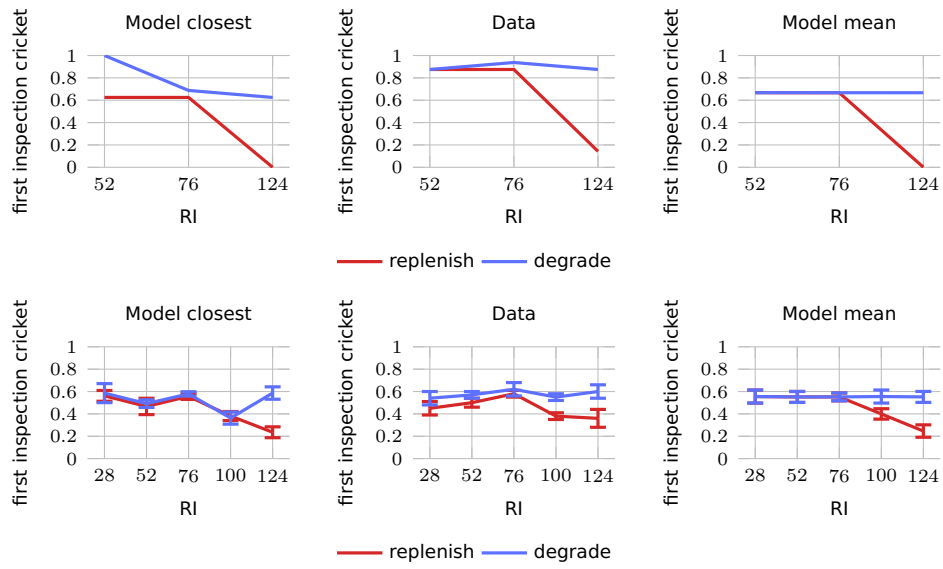


Figure S121: Clayton03 exp1: No-Plasticity Model

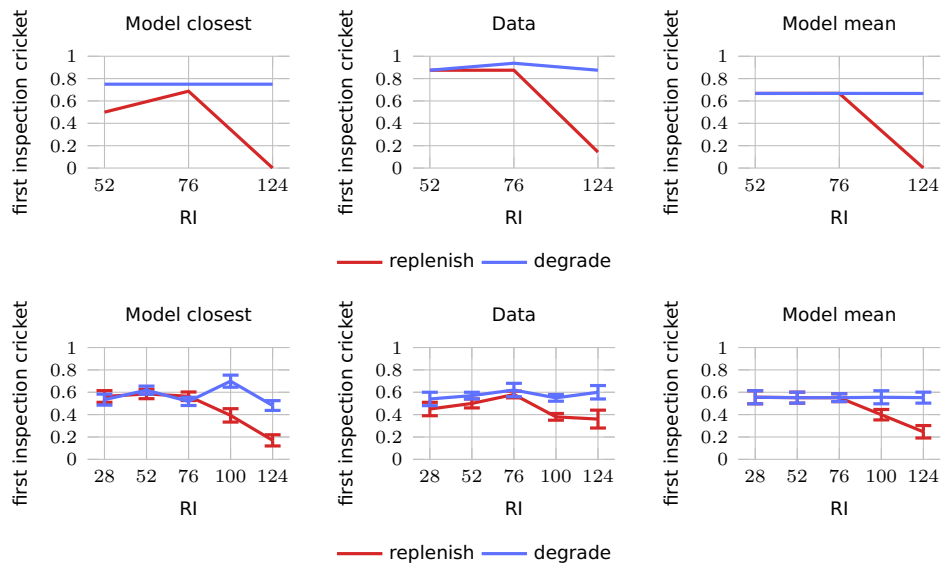


Figure S122: Clayton03 exp1: Plastic Caching Model

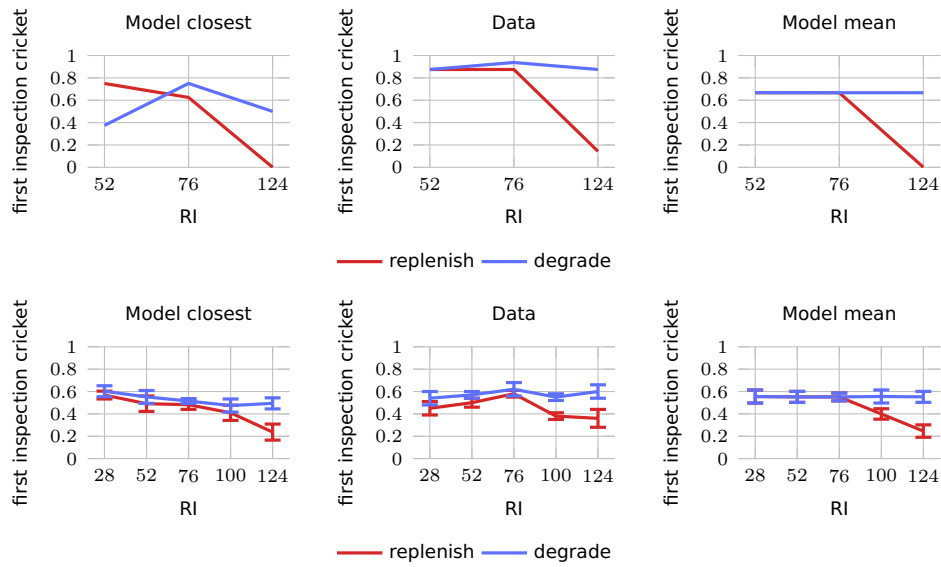


Figure S123: Clayton03 exp1: Planning-By-Replay Model

2.1.21 Clayton03 exp2

Published in [6].

Major Finding

Birds that experience degraded crickets after 3 days reduce searching for crickets after 3 day RIs even at sites where they cached prior to the unexpected experience of degraded crickets.

Comments

Some birds generalized to reduce searching for crickets after any retention interval (Replenish group, reversed, figure 4).

Most Important Tests

inspect degrade RI 76 (condition & foodtype): $F(1, 6) = 21.33, p = 0.0036$

inspect degrade (condition & foodtype & RI): $F(1, 6) = 7.44, p = 0.034$

inspect degrade RI 28 (condition & foodtype): $F(1, 6) < 1, p = 0.36$

Total Number of Statistics

25 p-values, 68 means or sems.

Figures

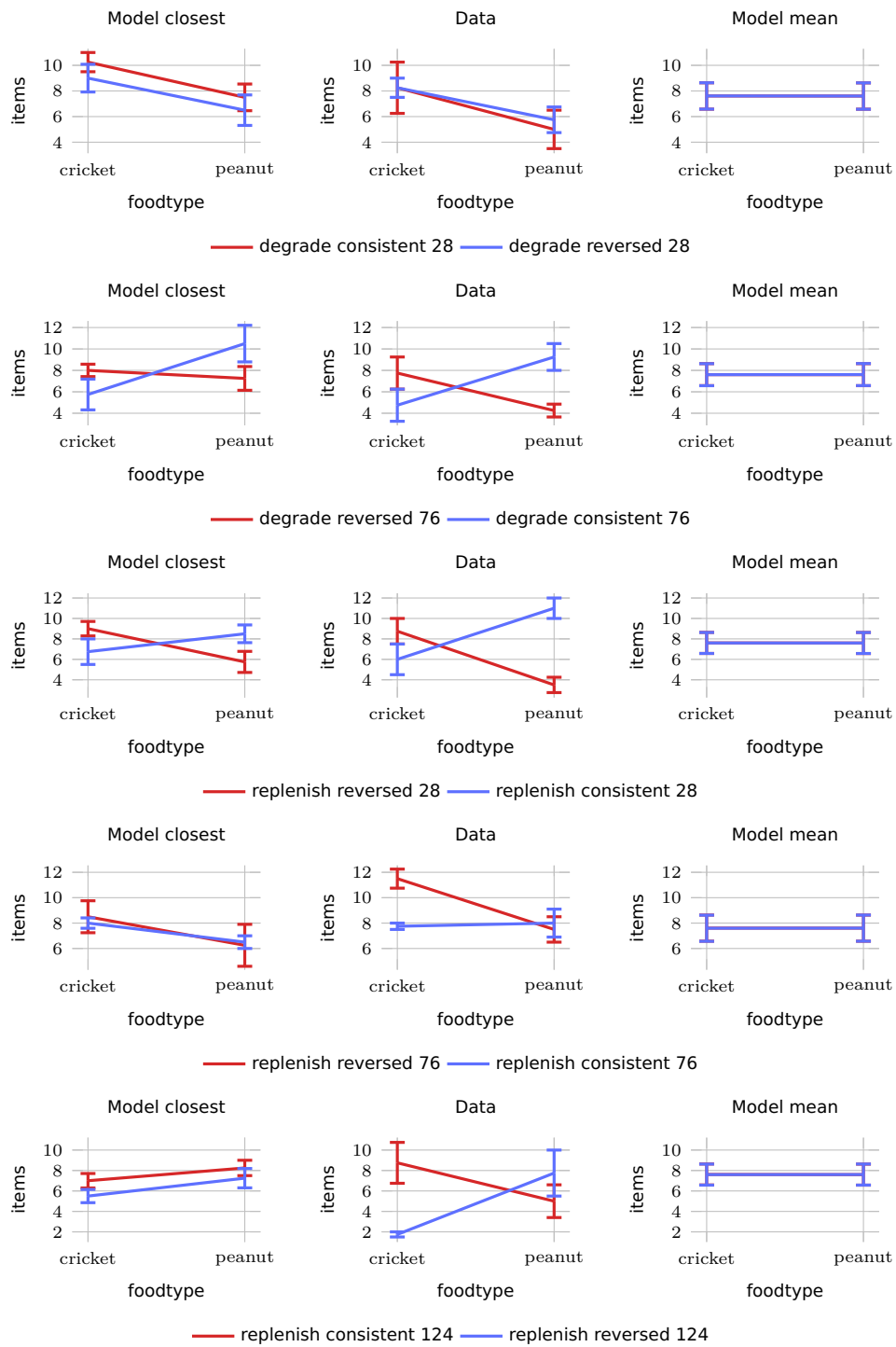


Figure S124: Clayton03 exp2: No-Plasticity-No-Memory-No-Motivational-Control Model (part1)

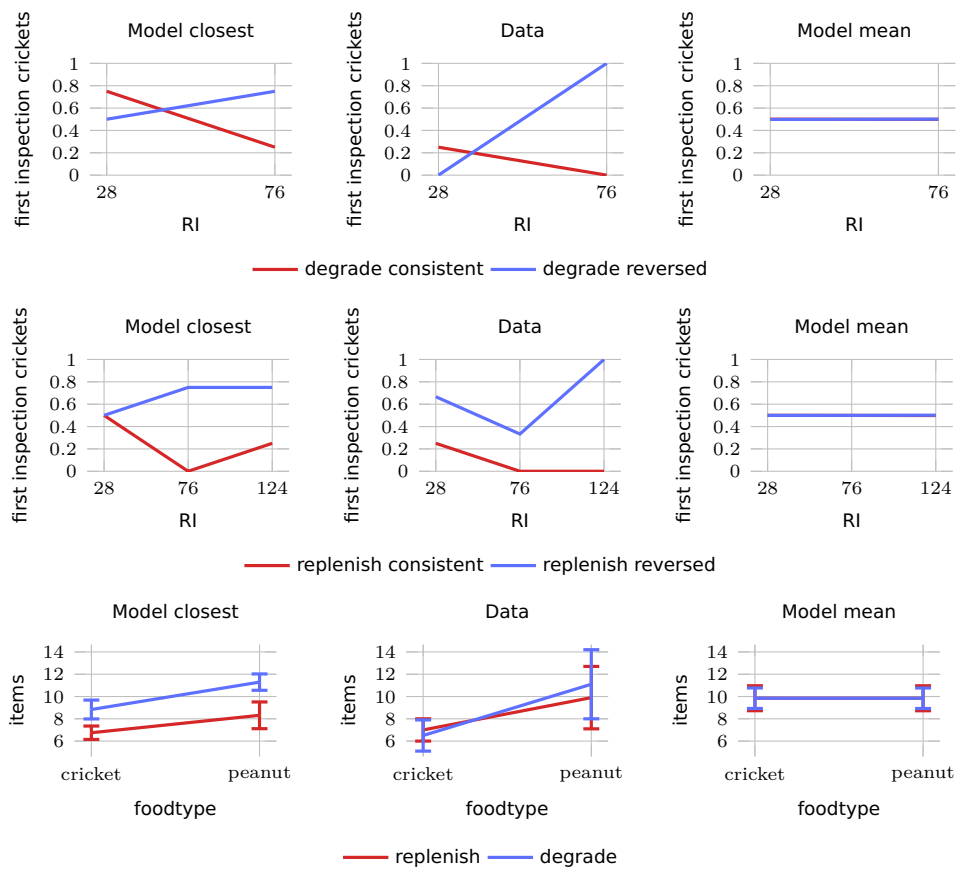


Figure S125: Clayton03 exp2: No-Plasticity-No-Memory-No-Motivational-Control Model (part2)

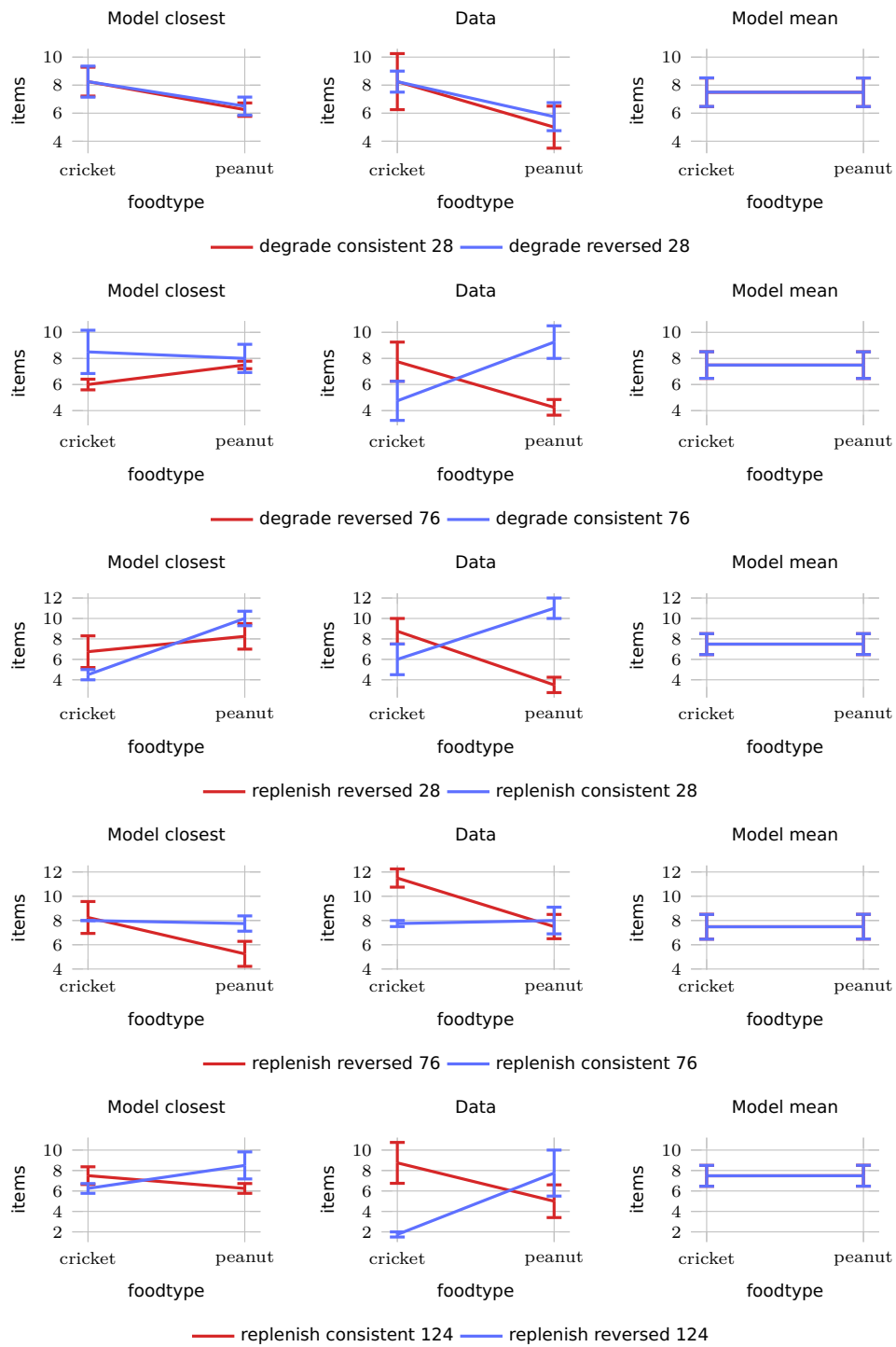


Figure S126: Clayton03 exp2: No-Plasticity-No-Memory Model (part1)

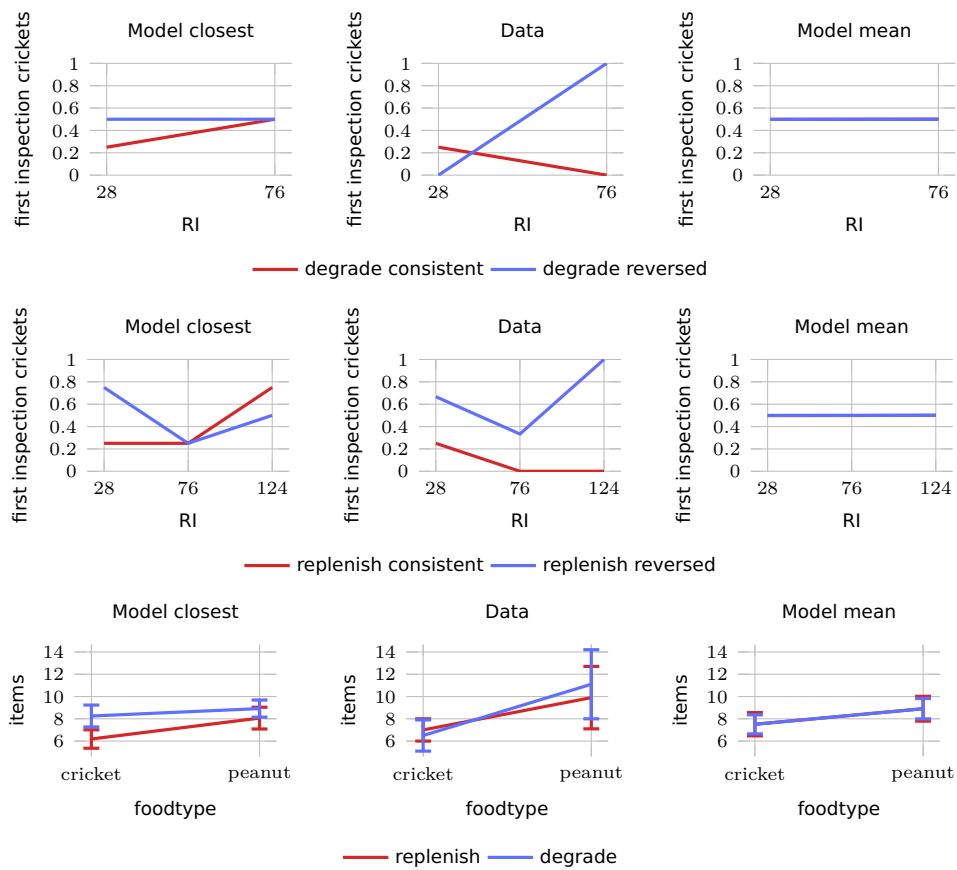


Figure S127: Clayton03 exp2: No-Plasticity-No-Memory Model (part2)

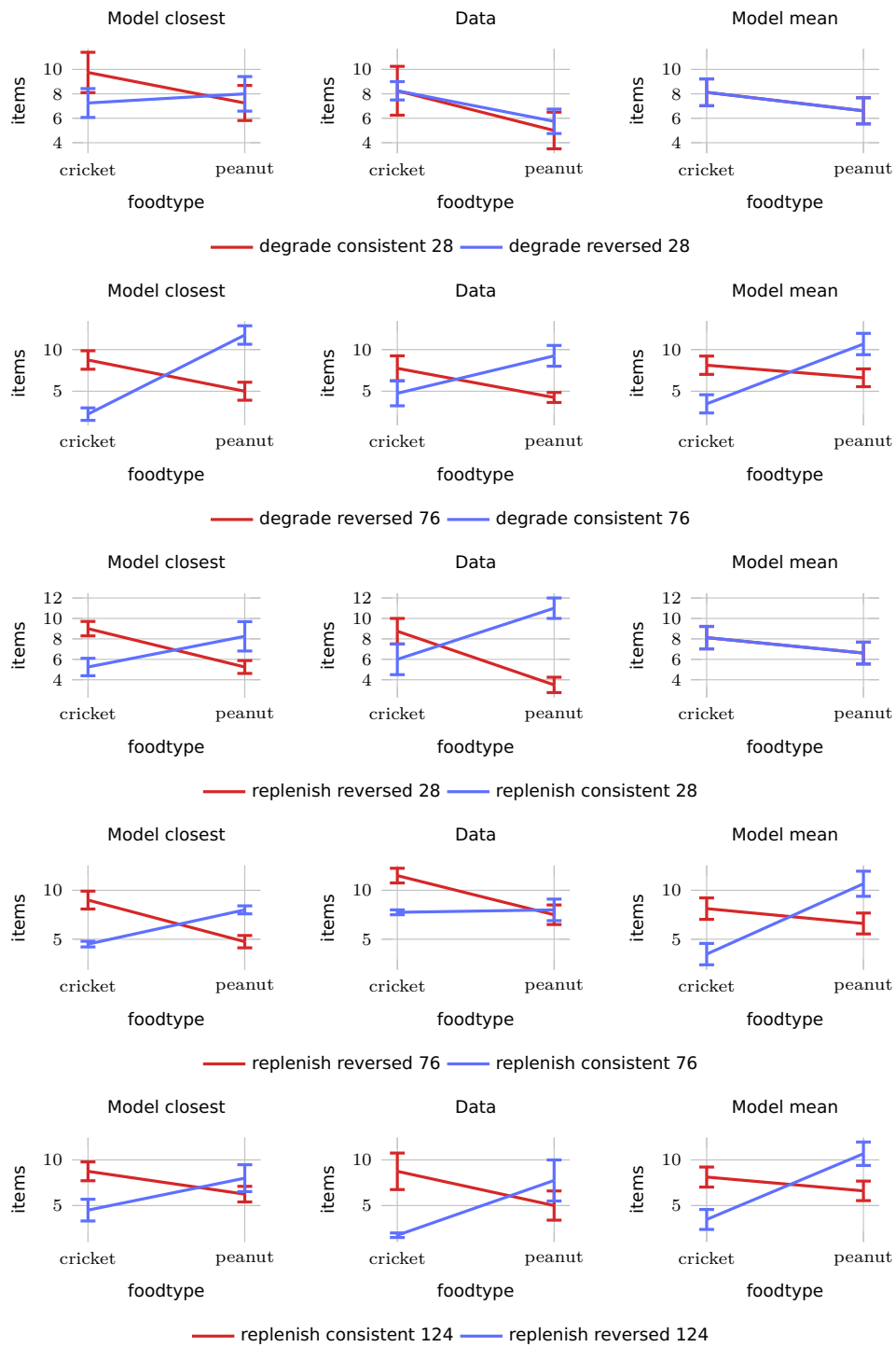


Figure S128: Clayton03 exp2: No-Plasticity Model (part1)

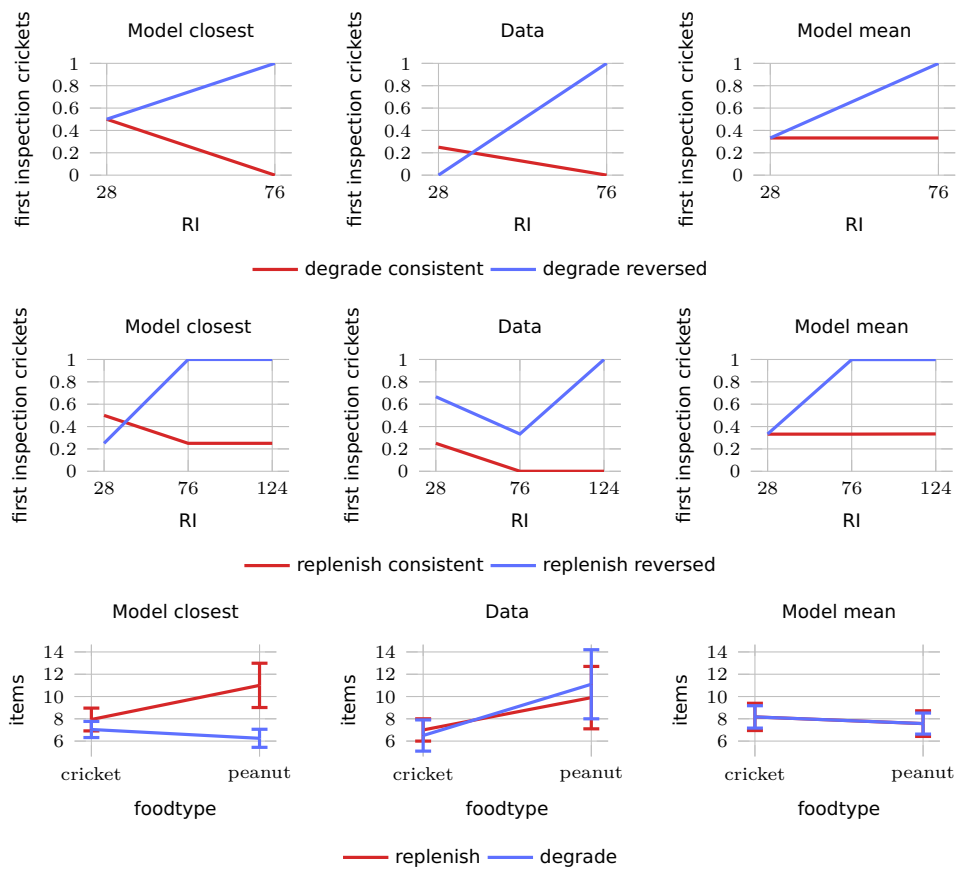


Figure S129: Clayton03 exp2: No-Plasticity Model (part2)

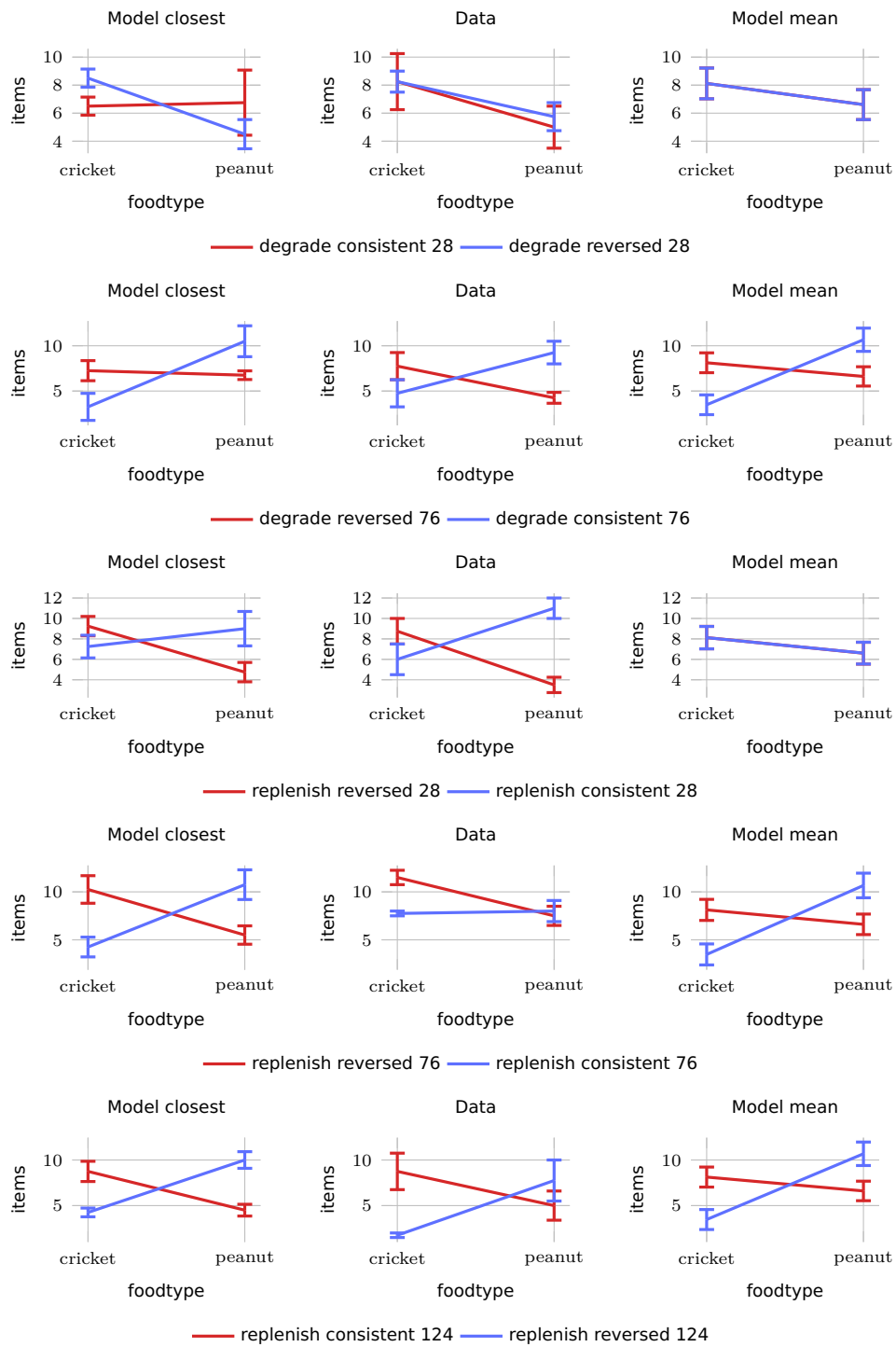


Figure S130: Clayton03 exp2: Plastic Caching Model (part1)

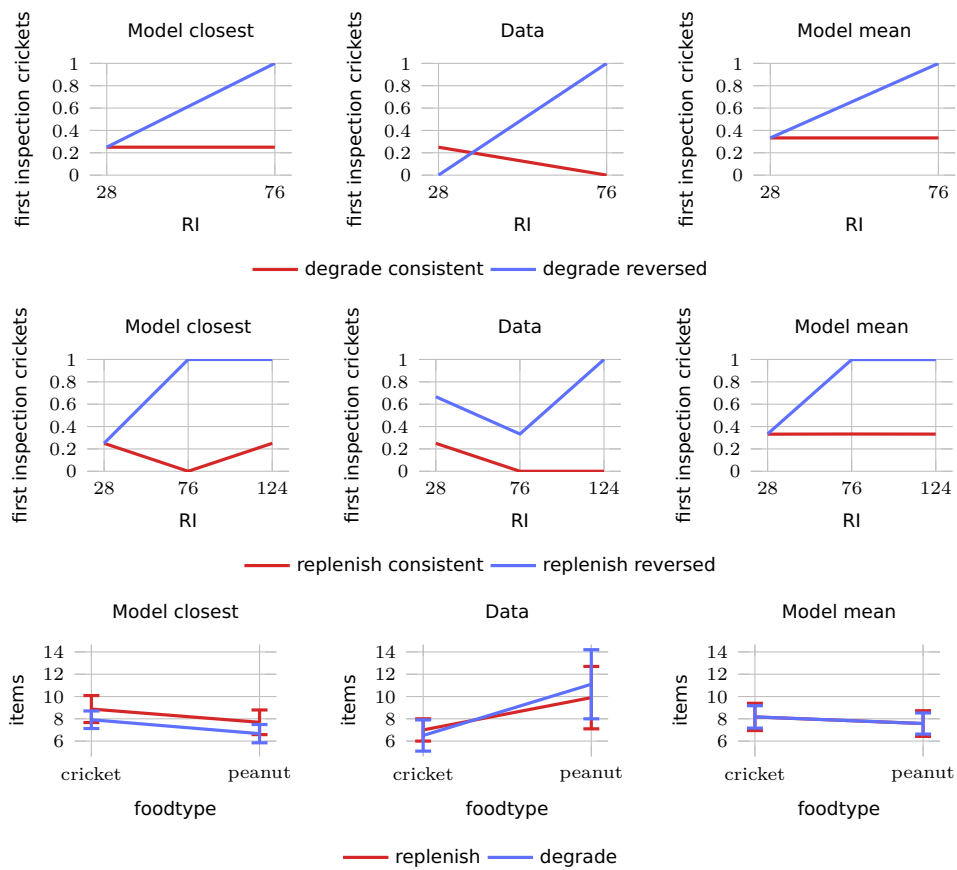


Figure S131: Clayton03 exp2: Plastic Caching Model (part2)

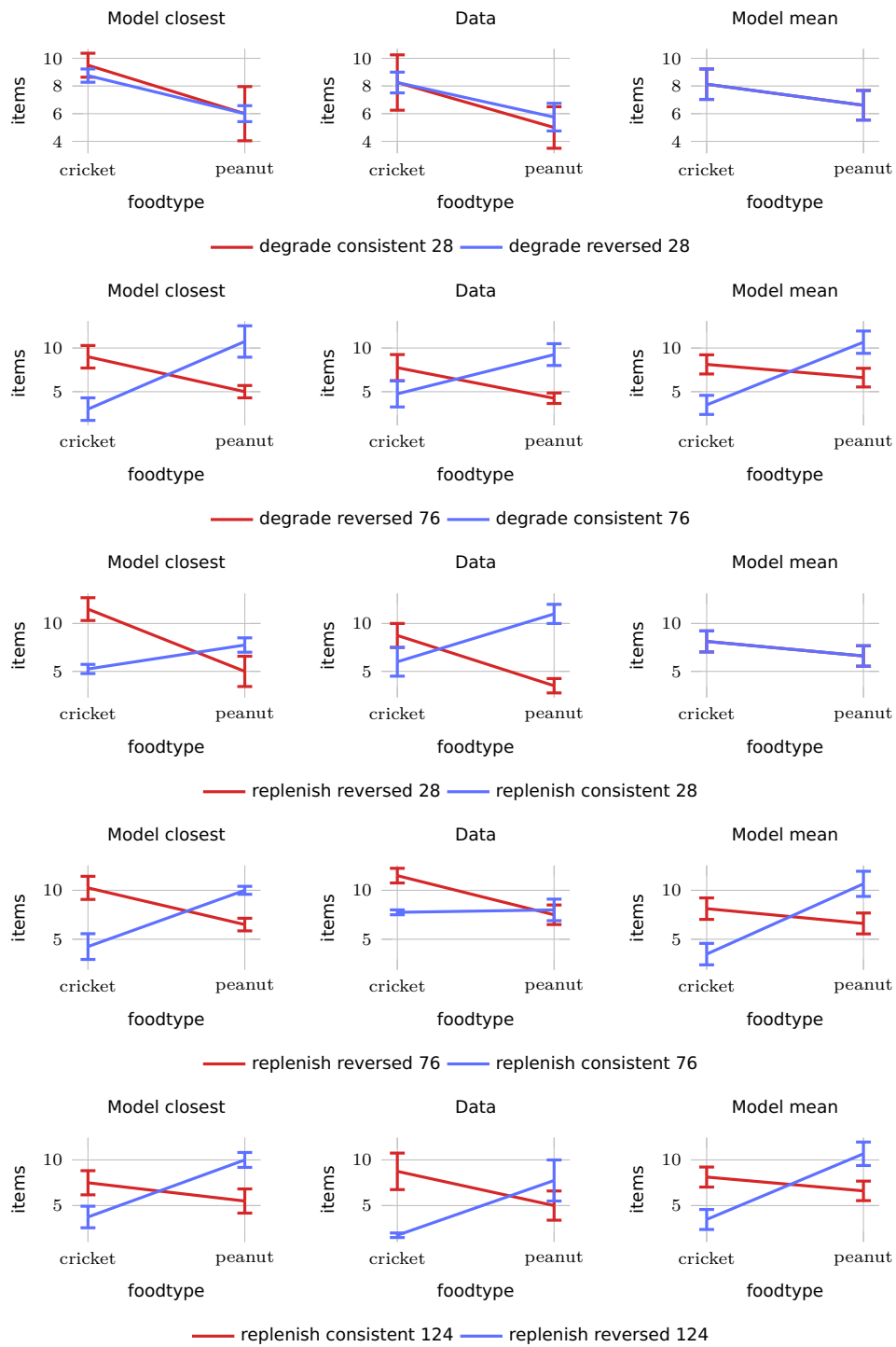


Figure S132: Clayton03 exp2: Planning-By-Replay Model (part1)

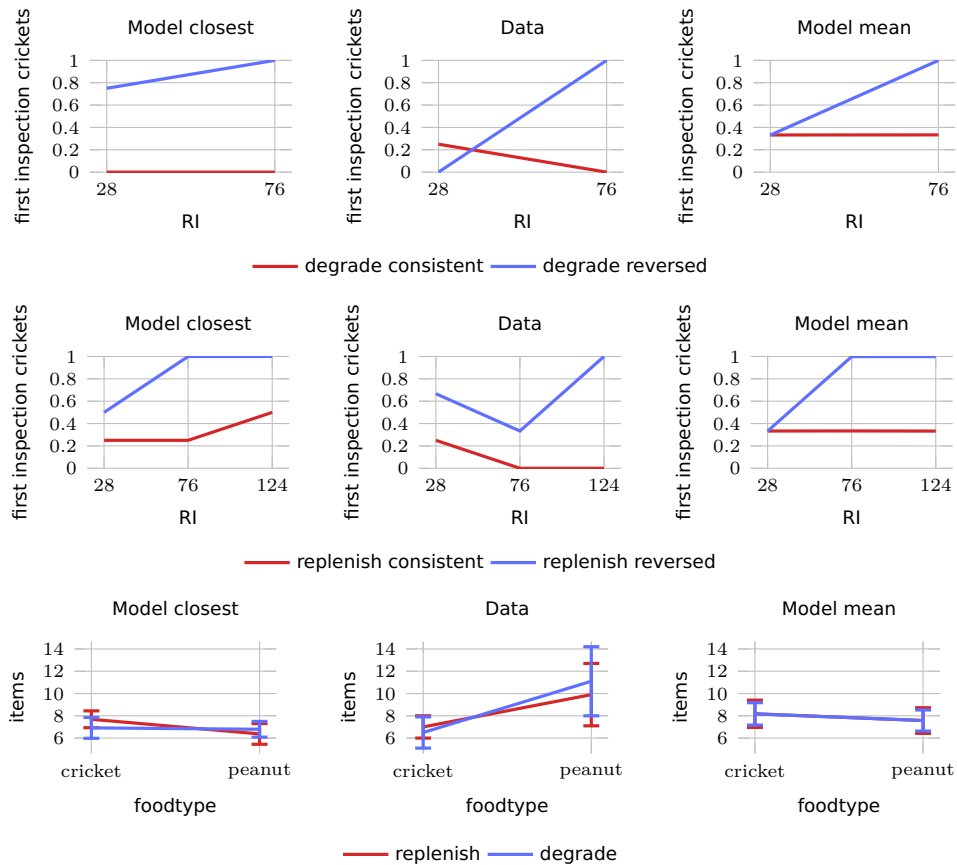


Figure S133: Clayton03 exp2: Planning-By-Replay Model (part2)

2.1.22 deKort05

Published in [7].

Major Finding

If palatability improves over time (ripening) the birds search more after the longer retention intervals than after the shorter retention interval and vice versa if the food items degrade over time.

Comments

For the Group x Retention Interval x Trail interaction we assume that the correct degree of freedoms would be 7,70.

Most Important Tests

first inspect trial 8 (group): $Chisq(3) = 9.08, p = 0.028$

inspect (group & RI): $F(1, 10) = 13.03, p = 0.0048$

inspect (group & RI & set): $F(1, 10) = 11.24, p = 0.0073$

Total Number of Statistics

3 p-values, 68 means or sems.

Figures

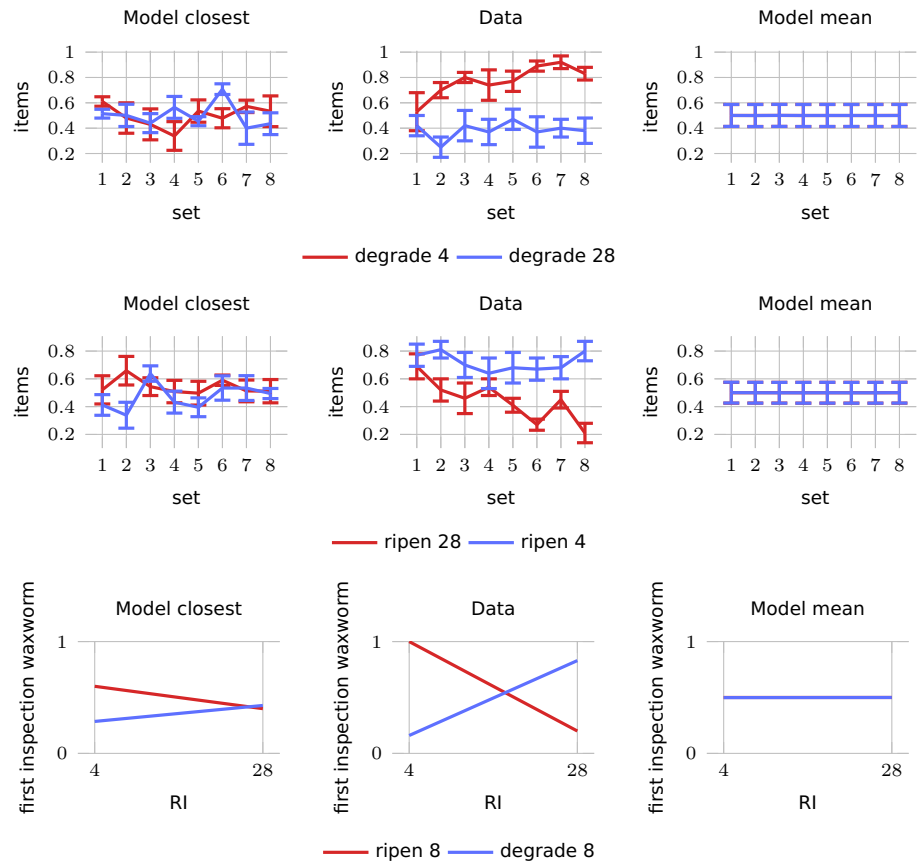


Figure S134: **deKort05: No-Plasticity-No-Memory-No-Motivational-Control Model**

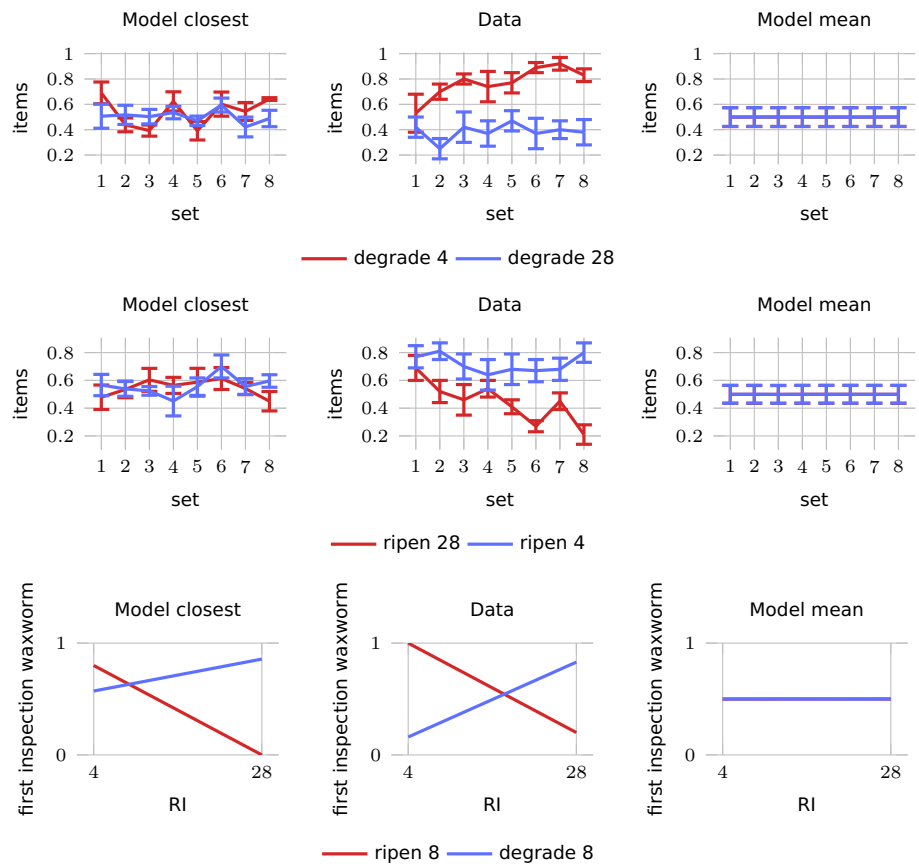


Figure S135: **deKort05: No-Plasticity-No-Memory Model**

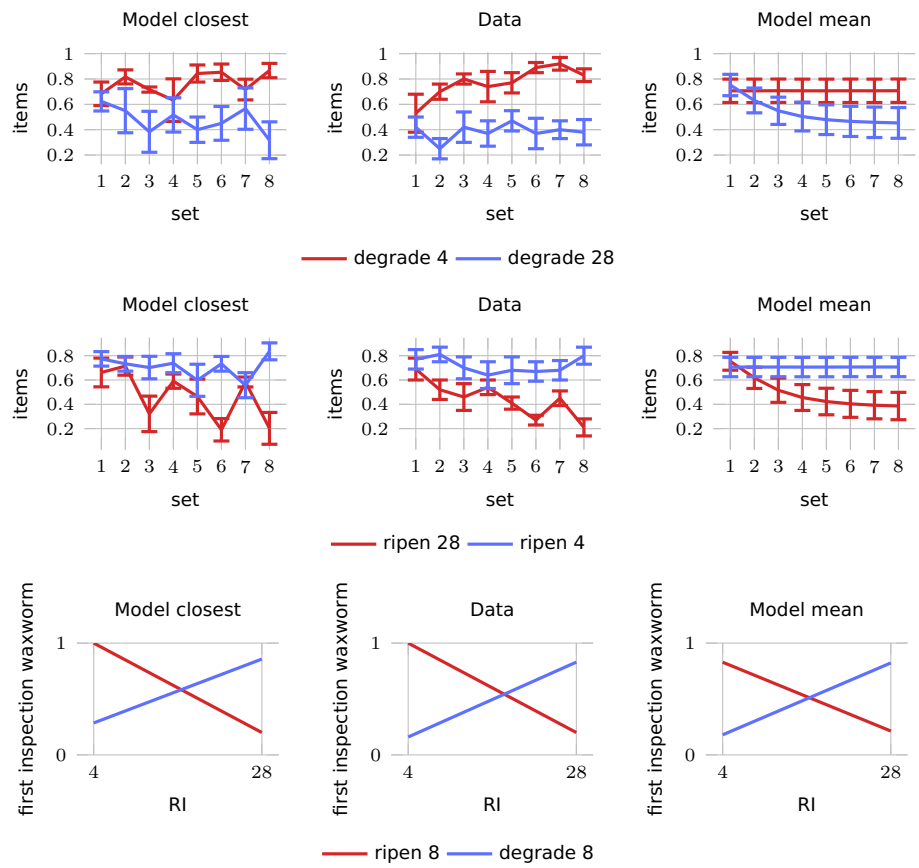


Figure S136: deKort05: No-Plasticity Model

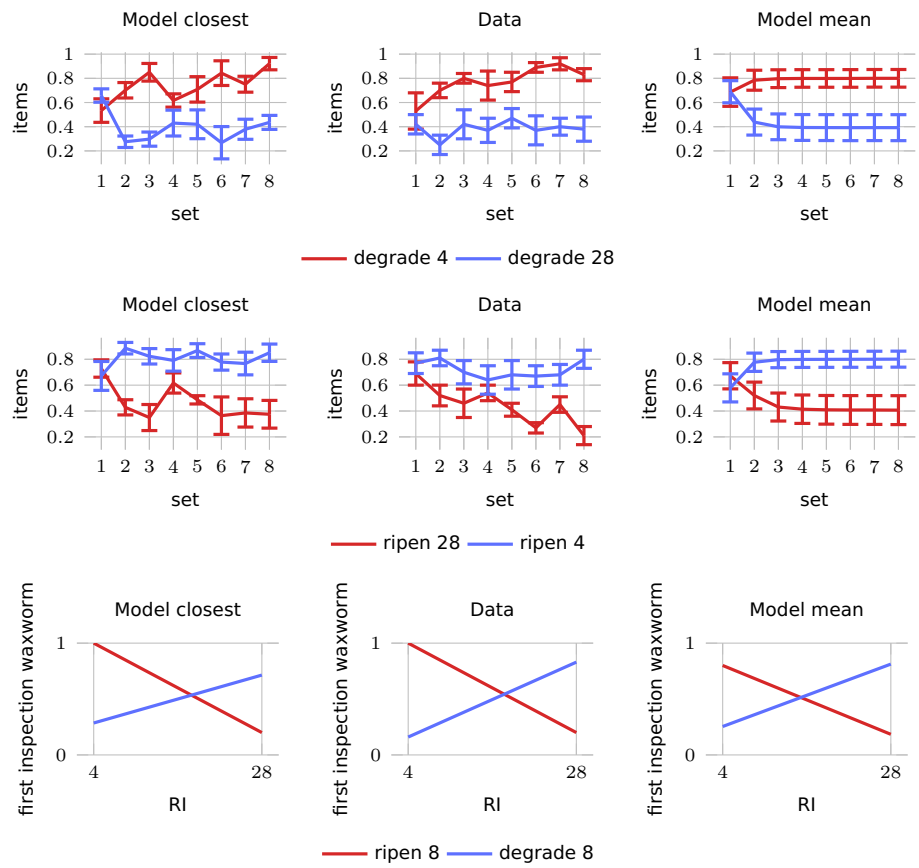


Figure S137: deKort05: Plastic Caching Model

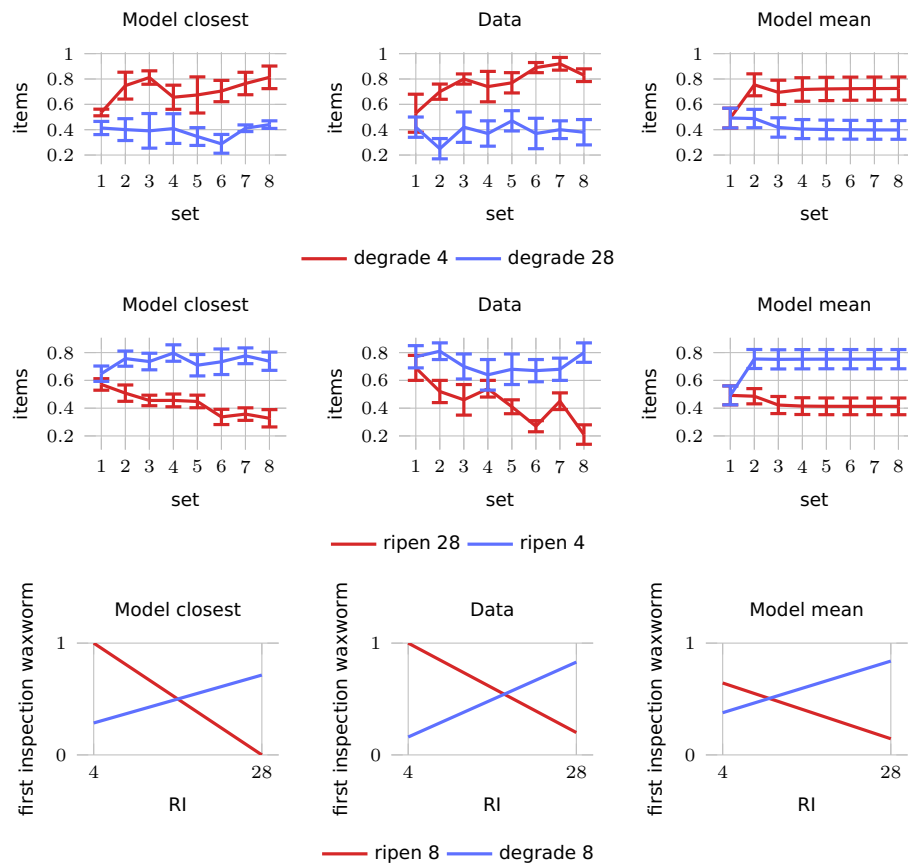


Figure S138: deKort05: Planning-By-Replay Model

2.1.23 Cheke11 specsat

Published in [12].

Major Finding

Birds showed specific satiety by eating and caching less of the pre-fed food than the non-prefed food.

Comments

This experiment is similar to Correia07 exp1 but with Eurasian jays instead of California scrub-jays.

Most Important Tests

cache (prefed & foodtype): $F(1, 3) = 10.45, p = 0.048$

overall (prefed & foodtype): $F(1, 3) = 12.4, p = 0.039$

eat (prefed & foodtype): $F(1, 3) = 9.8, p = 0.052$

Total Number of Statistics

3 p-values, 16 means or sems.

Figures

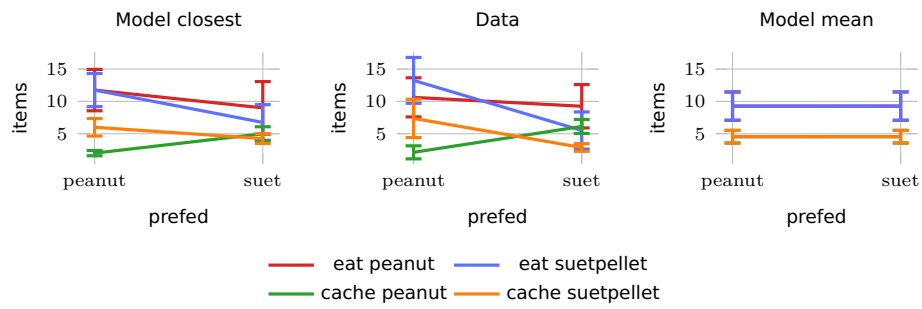


Figure S139: **Cheke11 specsat: No-Plasticity-No-Memory-No-Motivational-Control Model**

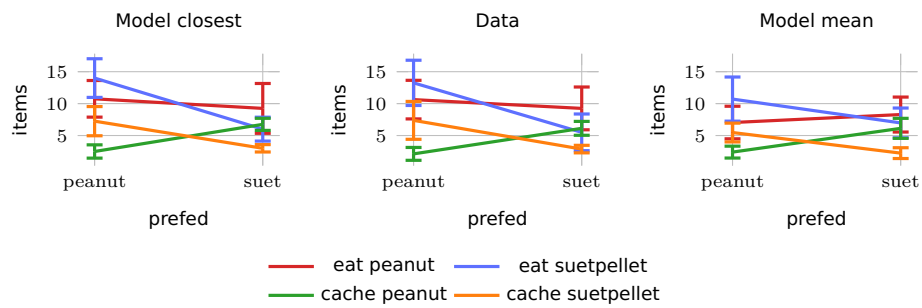


Figure S140: **Cheke11 specsat: No-Plasticity-No-Memory Model**

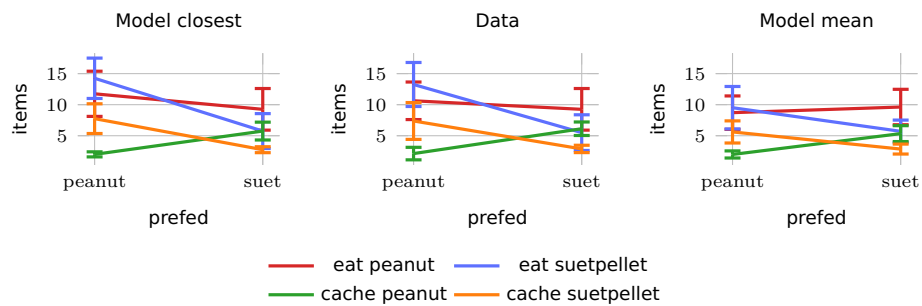


Figure S141: **Cheke11 specsat: No-Plasticity Model**

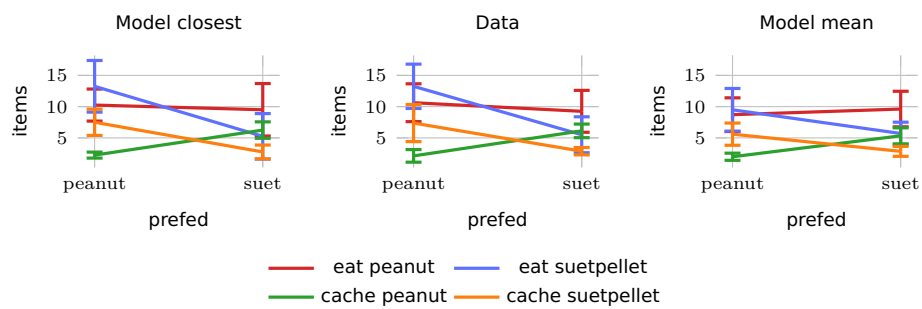


Figure S142: **Cheke11 specsat: Plastic Caching Model**

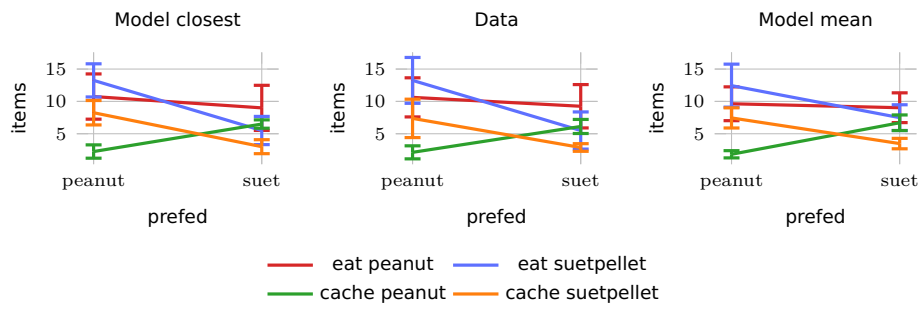


Figure S143: Cheke11 specsat: Planning-By-Replay Model

2.1.24 Correia07 exp1

Published in [11].

Major Finding

After eating to satiety a certain type of food, birds eat less of this type of food.

Most Important Tests

prefed pinenut (foodtype): $F(1, 10) = 6.96, p = 0.025$

prefed kibble (foodtype): $F(1, 10) = 5.12, p = 0.047$

interaction (prefed & foodtype): $F(3, 30) = 8.49, p = 0.00031$

Total Number of Statistics

5 p-values, 16 means or sems.

Figures

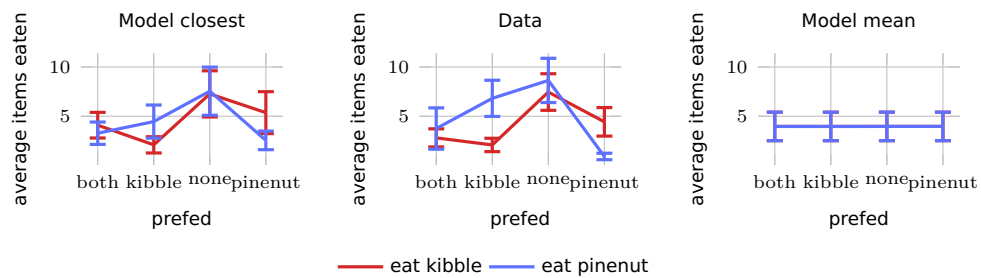


Figure S144: Correia07 exp1: No-Plasticity-No-Memory-No-Motivational-Control Model

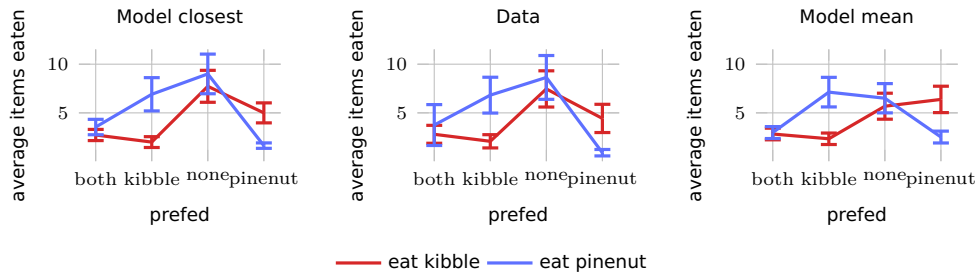


Figure S145: Correia07 exp1: No-Plasticity-No-Memory Model

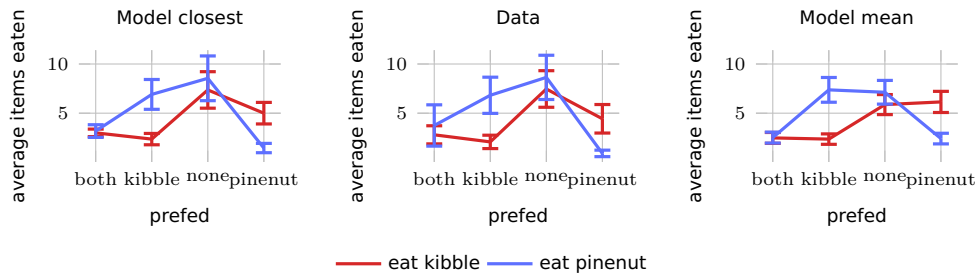


Figure S146: Correia07 exp1: No-Plasticity Model

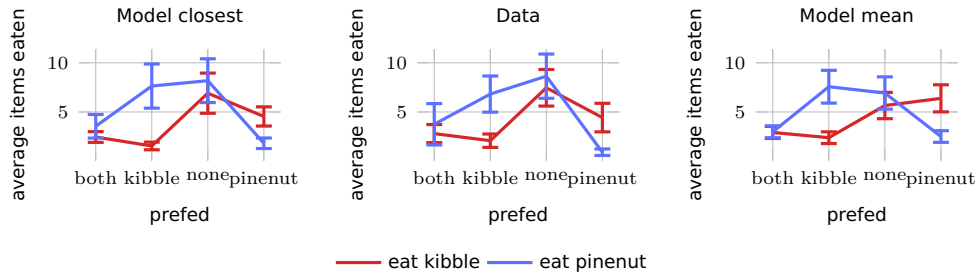


Figure S147: Correia07 exp1: Plastic Caching Model

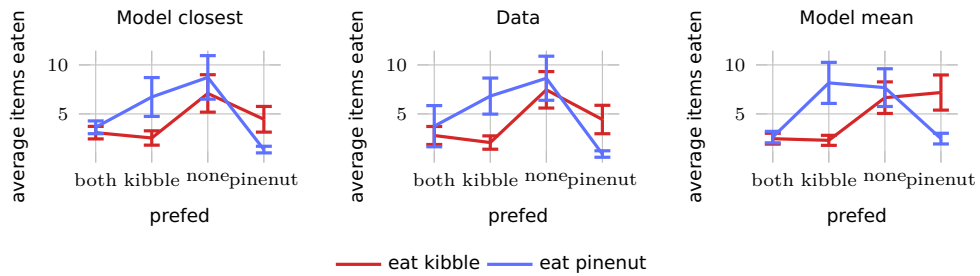


Figure S148: Correia07 exp1: Planning-By-Replay Model

2.1.25 Clayton99C exp1

Published in [24].

Major Finding

Neither prefeeding nor precaching alone had a major impact on the birds' subsequent propensity to cache peanuts or stones, but the combination of prefeeding and precaching in stage 1 significantly reduced the birds' propensity to cache in stage 2.

Most Important Tests

interactions (group): $F(3, 16) = 11.34, p = 0.00031$

interactions (foodtype): $F(1, 16) = 7.91, p = 0.013$

interactions (group & foodtype): $F(3, 16) = 12.64, p = 0.00017$

Total Number of Statistics

13 p-values, 30 means or sems.

Figures

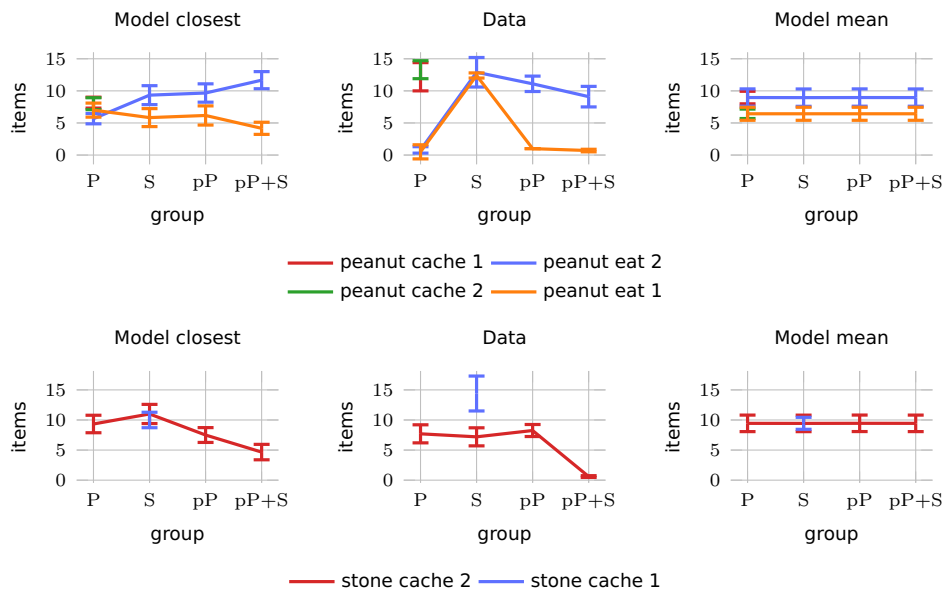
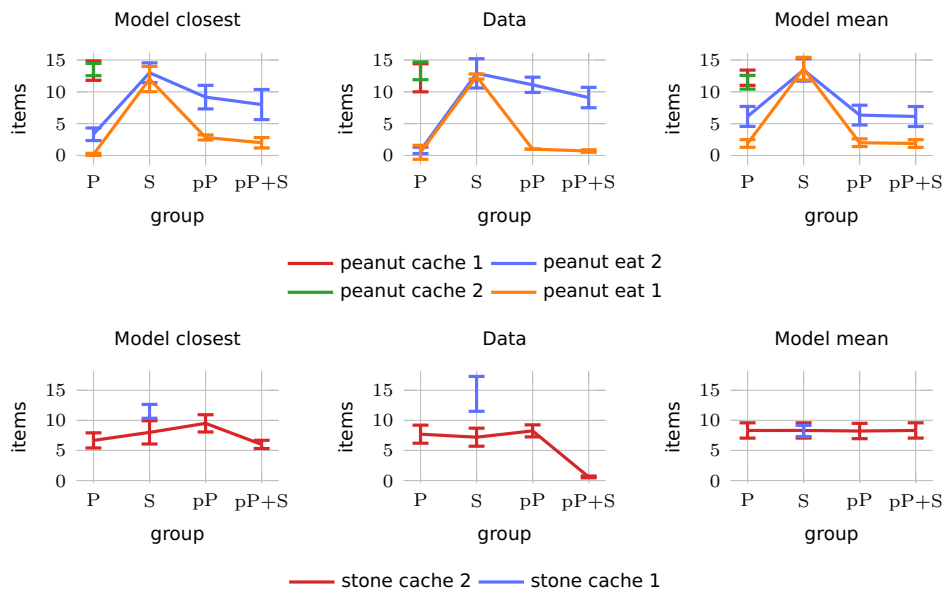
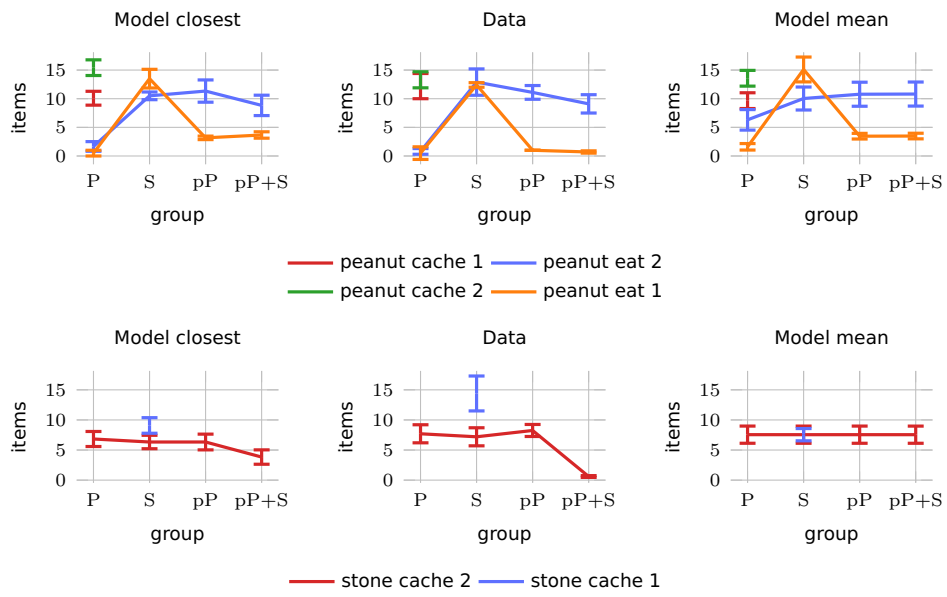
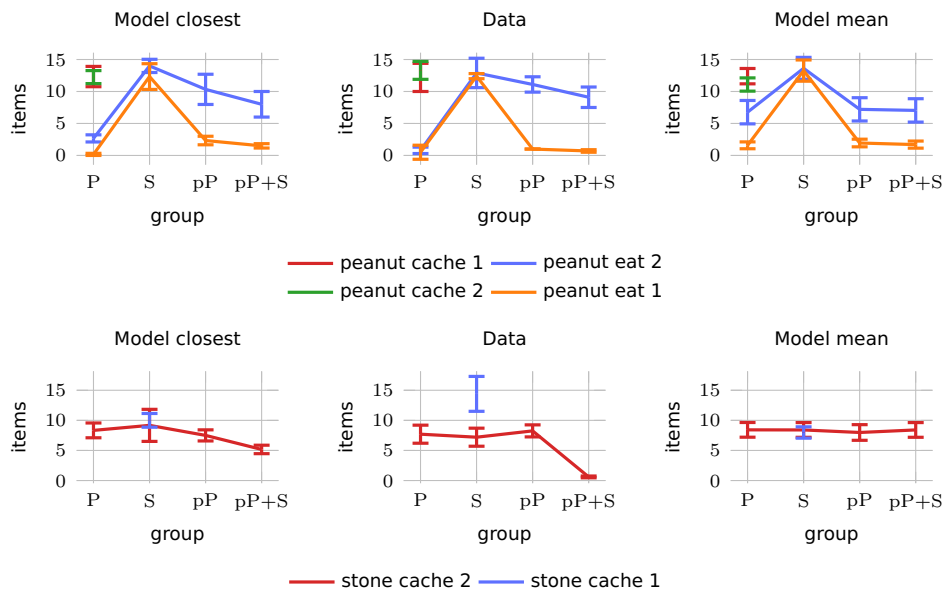
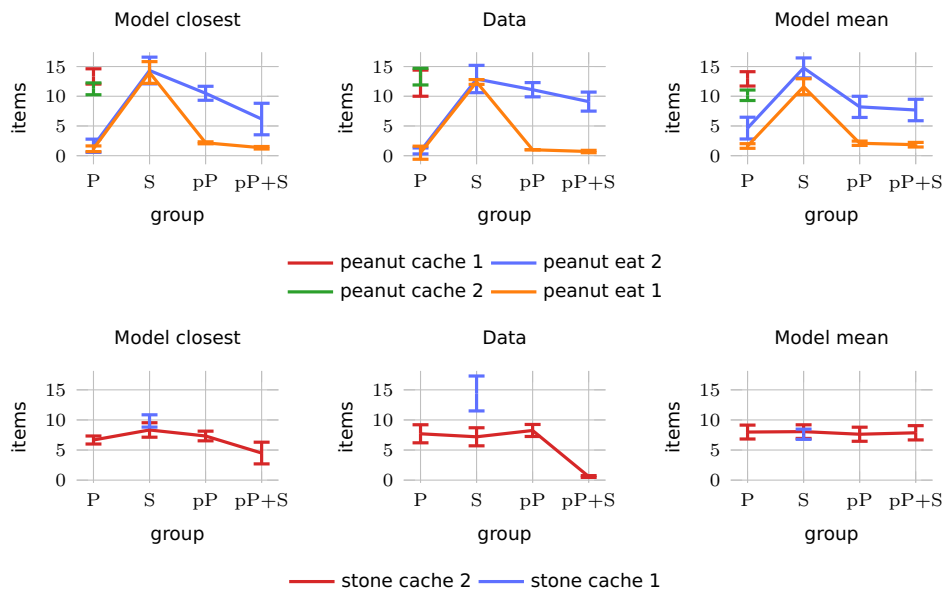


Figure S149: Clayton99C exp1: No-Plasticity-No-Memory-No-Motivational-Control Model





2.1.26 Clayton99C exp2

Published in [24].

Major Finding

Precaching and prefeeding produced a much greater reduction in subsequent caching and eating of the same food compared with a different food.

Most Important Tests

cache (stage): $F(1, 16) = 39.58, p = 1.1 \cdot 10^{-5}$

eat (stage): $F(1, 16) = 150.16, p = 1.5 \cdot 10^{-9}$

eat (group & stage): $F(1, 16) = 16.41, p = 0.00093$

Total Number of Statistics

38 p-values, 16 means or sems.

Figures

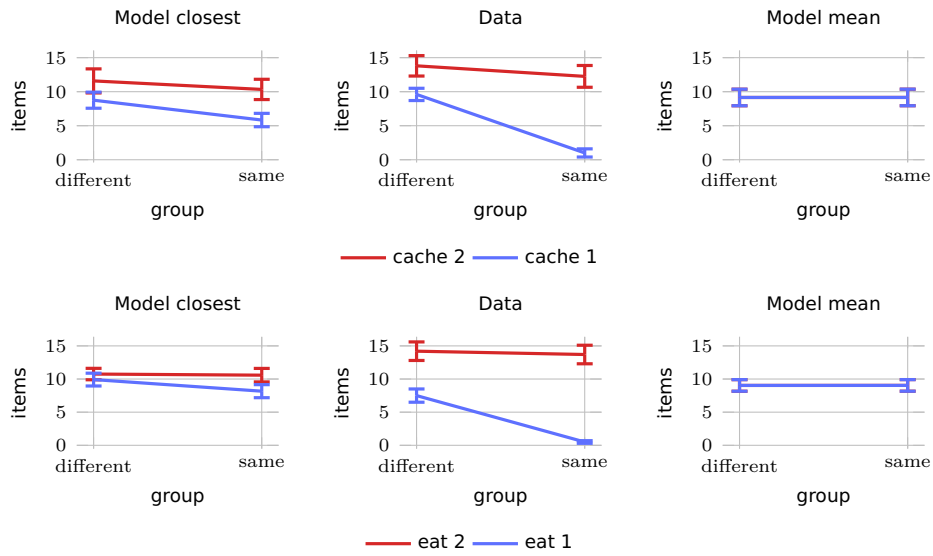


Figure S154: **Clayton99C exp2: No-Plasticity-No-Memory-No-Motivational-Control Model**

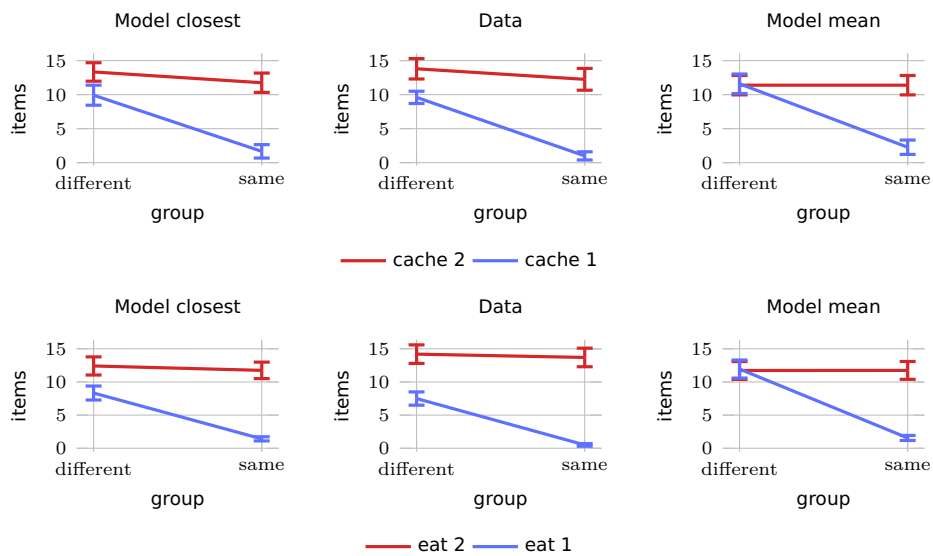


Figure S155: **Clayton99C exp2: No-Plasticity-No-Memory Model**

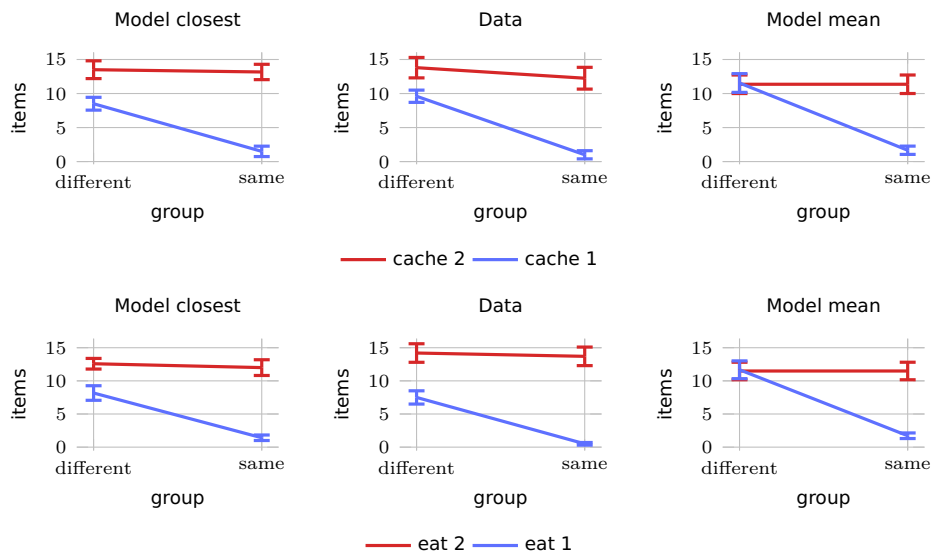


Figure S156: Clayton99C exp2: No-Plasticity Model

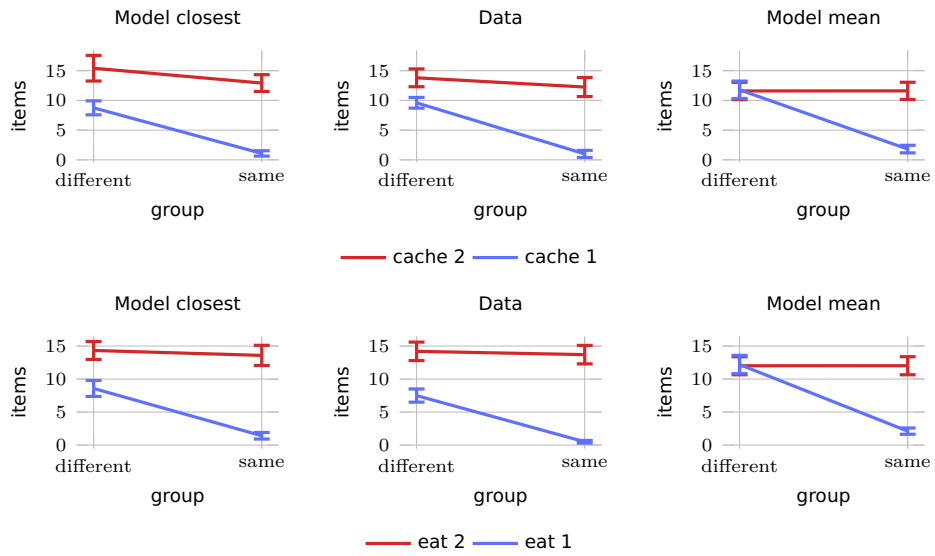


Figure S157: Clayton99C exp2: Plastic Caching Model

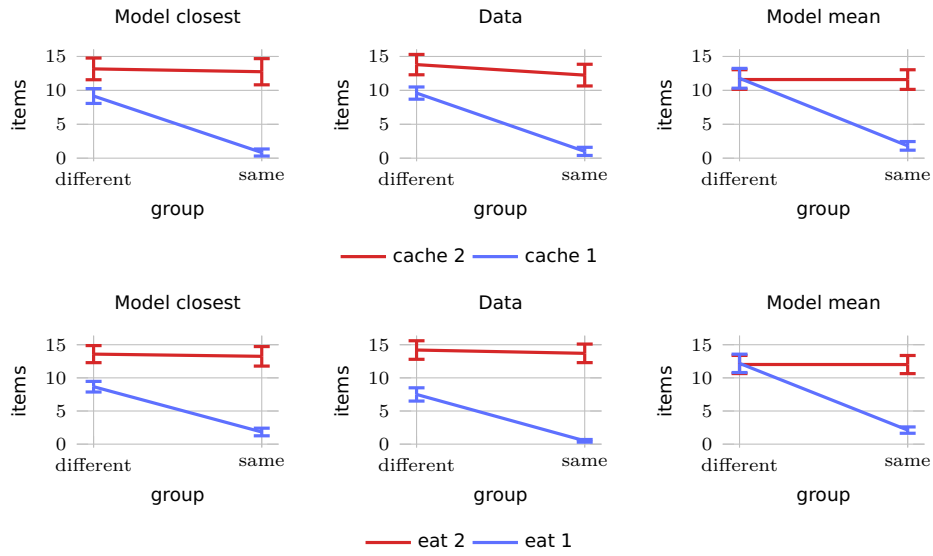


Figure S158: Clayton99C exp2: Planning-By-Replay Model

2.1.27 Clayton99C exp3

Published in [24].

Major Finding

Prefeeding of powdered, non-cacheable food leads to a strong reduction in eating of the same food, a slight reduction in eating different food and a small reduction in caching same food.

Comments

In contrast to experiment 1, pre-feeding powdered peanuts lead to a reduction in caching peanuts but not to a reduction in caching the other food.

Most Important Tests

cache (group): $F(2, 18) = 11.95, p = 0.0005$

within-subject prefeeding (group): $F(1, 22) = 2.33^2, p = 0.029$

eat (group): $F(2, 18) = 122., p = 3.4 \cdot 10^{-11}$

Total Number of Statistics

7 p-values, 16 means or sems.

Figures

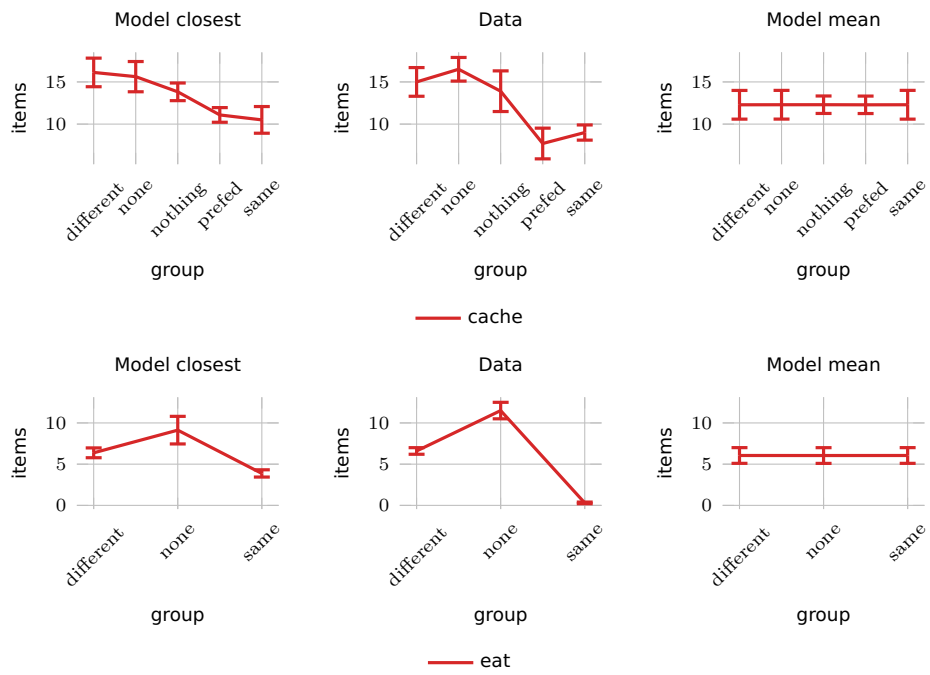


Figure S159: **Clayton99C exp3: No-Plasticity-No-Memory-No-Motivational-Control Model**

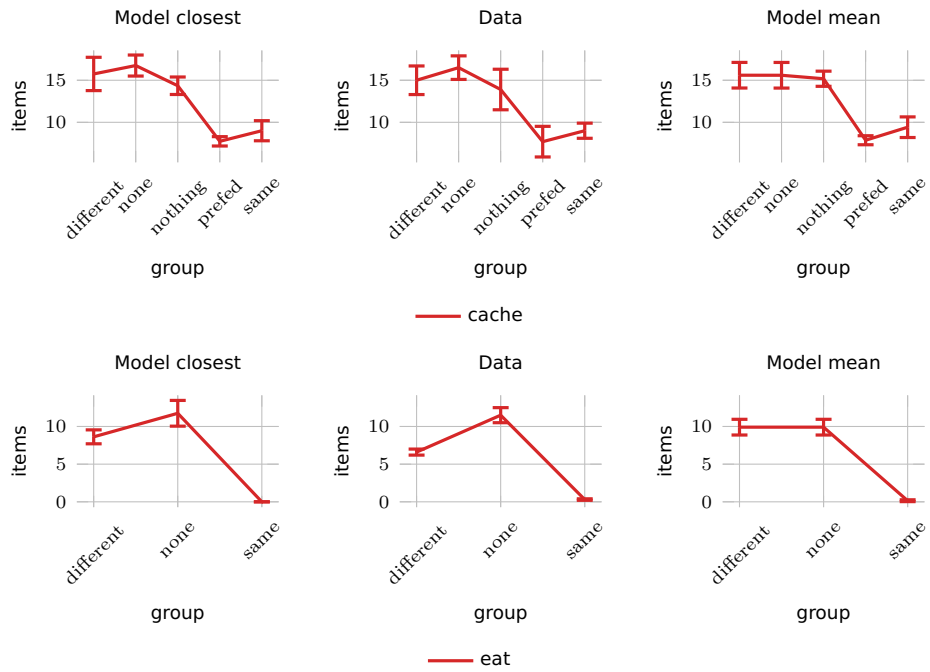


Figure S160: **Clayton99C exp3: No-Plasticity-No-Memory Model**

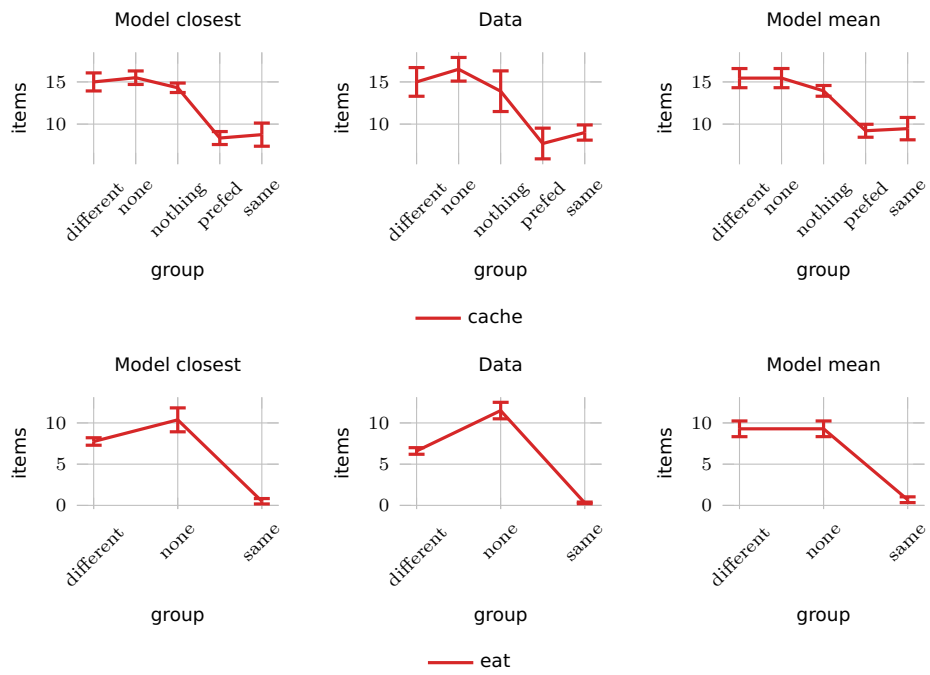


Figure S161: Clayton99C exp3: No-Plasticity Model

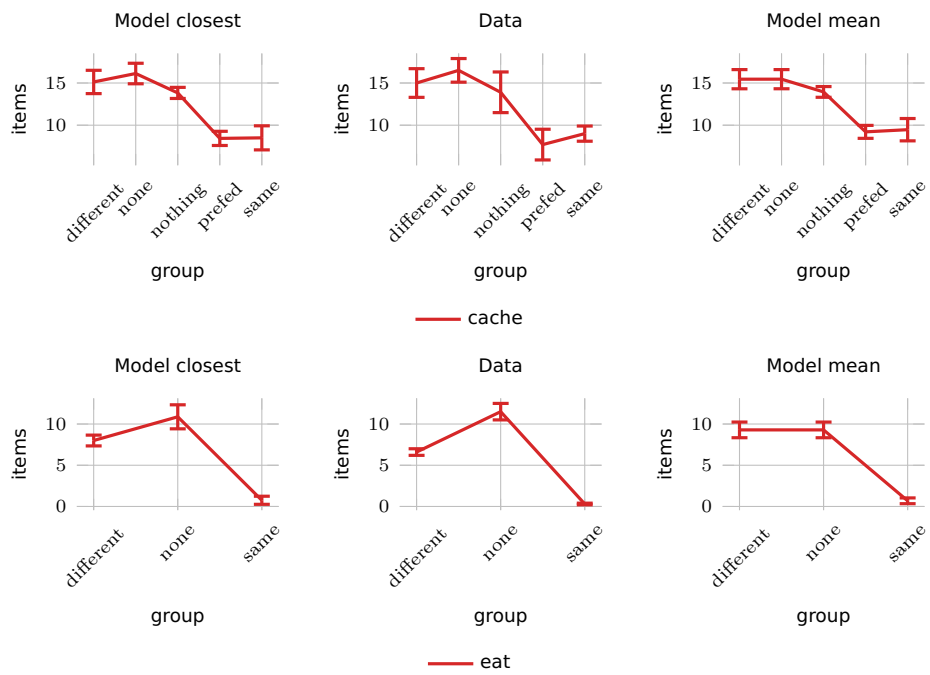


Figure S162: Clayton99C exp3: Plastic Caching Model

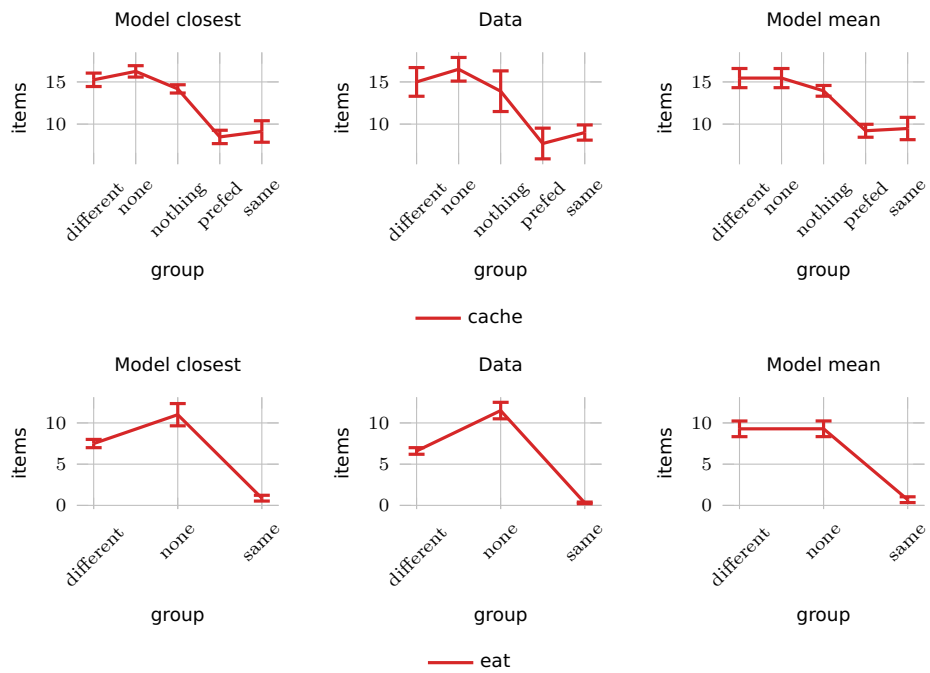


Figure S163: Clayton99C exp3: Planning-By-Replay Model

2.1.28 deKort07 exp1

Published in [9].

Major Finding

In the absense of alternative food items to cache, birds continue to cache mealworms, even if they are degraded or pilfered before recovery.

Comments

The caching rate during experiment 1b was much significantly lower than during experiment 1a.

Most Important Tests

Experiment b (group): $F(1, 14) < 1, p = 0.34$

Experiment a (group): $F(1, 14) = 1.26, p = 0.28$

Experiment a (trial): $F(5, 70) = 1.9, p = 0.11$

Total Number of Statistics

6 p-values, 48 means or sems.

Figures

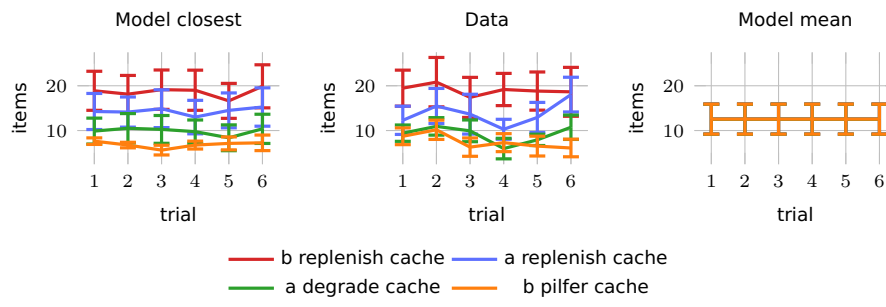


Figure S168: deKort07 exp1: Planning-By-Replay Model